



ENGINEERING STUDY

FOR

MOUNTAIN VILLAGE GONDOLA 1, 2 & 3

Order Number SAP0000008

AT

MOUNTAIN VILLAGE, CO

Type of Lift: DETACHABLE GRIP 8 PASSENGER GONDOLA

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Overview

Scope of Study:

Review the feasibility of increasing the gondola system capacity in addition to the feasibility of accommodating level walk-in cabins.

Original Design Function and Capacities:

The original design of the existing gondola Sections 1-2 & 3 utilized 'CWA Omega' 8 passenger 'step up' cabins.

All Sections were designed to an Ultimate capacity of 1200 PPH (People Per Hour) at an operating speed of 1000 FPM (Feet Per Minute). The initial ~1996 installed capacity for all sections was 350 PPH.

Capacity was increased in subsequent years and the system currently has three different models of CWA cabins. Cabin models currently on line are as follows.

- CWA Omega
- CWA Omega III
- CWA Omega III-XL

As of today the system operates in the following configuration.

Section 1-2, 48 Cabins, 920 PPH, @1000 FPM

Section 3, 11 Cabins, 660 PPH, @1000 FPM

Current and Future Condition of System:

The past and continued maintenance, along with the capital replacements and upgrades of the Mountain Village Gondola system could allow the system to run indefinitely. Note that the routine and preventative maintenance on the Gondola has been at a level of excellent. Continuing this level of maintenance to prevent emergency repairs results in less capital expenditure over time.

Included in Appendix B is a power point Doppelmayr began presenting 5 years ago in response to customers asking how long a lift can run. This is a very subjective issue and there is not a singular correct response that can be applied to all installations. This document is supplied to give further information about what affects the life cycle of a gondola. Please note that the descriptions, pictures and example issues are of other systems and NOT the Mountain Village Gondola.

1. Capacity Increase of Existing System

Additional 'step up' cabins may be added to the system to increase the current capacity. Due to upgrades, modifications and changing design codes over its life the original intended Ultimate Capacity of 1200 PPH cannot be reached on all sections. Sections 1-2 capacity has been reduced to an Ultimate Capacity of 1070 PPH. Section 3 Ultimate Capacity remains at the original design of 1200 PPH.

Utilizing original line, terminal equipment and adding additional 'step up' cabins:

Section 1-2, add 9 additional 'Step Up' cabins -> 1070 PPH

Section 3, add 9 additional 'Step Up' cabins -> 1200 PPH

2. Level Walk-In Cabin Implementation Retrofitting Existing System

Carriers:

Converting the existing gondola sections to LWI (Level Walk-In) loading style requires cabin replacement. The existing cabins were not engineered for the required cabin guides. The cabin door opening height is well below average passenger height. Modifying the cabins to adapt both of these is not practical or economically viable.

The current generation CWA Omega IV-LWI cabin with a special shortened hanger could be implemented with modification to the existing systems. Due to the increased weight, height and wind area of these cabins modification of the line equipment and existing gondola terminals is required to accommodate them. The existing grips in most cases may be reused with the exception on section 1-2. On Section 1-2 a grip for a larger rope is required to obtain a capacity of 2400 PPH.

Line (Tower Foundations, Masts, Sheave Trains & Rope):

The increased weight and wind area of these carriers will introduce larger wind loads, dead loads and tension loads on the gondola system even if current capacity is maintained. The load changes and adaptation of new LWI cabins would be classified as a Major Modification by the Colorado Passenger Tramway Safety Board (CPTSB). This classification requires using current code and best practices to perform a structural review of the line. The current foundations, tower masts and increased sheave loadings would not pass the current design requirements with the increased loads from LWI cabins. It is important to note that just because the system today possibly will not pass current code requirements it does not invalidate it as a safe system when operated within its original design parameters.

To accommodate the increased loading and meet current design requirements a full replacement of the tower equipment is required. Replacement of the tower equipment includes new foundations, towers, crossarms and sheave trains. The haul rope on each section depending on the chosen capacity may need to be replaced. For all sections multiple capacities were analyzed and the required equipment replacement tabulated and included in the cost in the pricing Matrix.

Terminals:

The existing terminal structures, grip coupling components and machinery that propel the cabins are compatible with the LWI cabins and may be reused. Portions of the machinery that touch the rope and drive each section will need to be reconfigured and upgraded with new components to handle the higher loads. The major items that would be replaced include all components of the drivetrain, bullwheels and their mounting platforms, rope deflection sheaves and their mounts, braking components, tensioning components and lift controls.

Floor cabin guides would be provided to accommodate the LWI cabins. Minor changes to the building would be required to integrate the new level loading surface at the provided guide elevation. This could be a combination of egress ramps and or steps to meet current building code requirements. This portion of building infrastructure is outside of this engineering review.

System Performance:

All of these component replacements and upgrades will result in a system that in addition to accommodating LWI cabins is functionally more reliable and in line with current 'best practices' of essential transportation systems. Most notably is replacing the current evacuation system with a ring and pinion system and bullwheels with emergency bearings. This type of evac system eliminates several single point failure modes on the existing system that could result in a not timely or impossible lift evacuation in the event of a critical component failure. Critical components that would now have an emergency backup include the drive shaft, gearbox, planetary and bullwheel bearings.

Replacement of the drivetrain, controls, bullwheels and deflection sheaves will put these items at the same reliability and performance level of these items in a new lift. The performance will remain unchanged for the reused existing terminal structure, grip coupling components and the machinery that propels the cabins through the terminal. Reliability of these items is dependent on their condition prior to the system upgrade.

3. LWI Implementation with Terminal Replacement

To accommodate CWA Omega IV-LWI cabins with the current standard stem complete terminal replacements would be required. In addition to all components listed in the retrofit option new terminal concrete, steel structure, grip coupling components and machinery that propels the cabins would be provided. Available grips would still be reused in this scenario. Terminal placement, lengths and optimization of space is possible with complete replacement.

Pricing in the cost matrix represents only the equipment required for the gondola systems. Removal of existing buildings and reinstallation or providing new enclosures is not represented.

System Performance:

Reliability and performance of all components will be that of a new lift. Optimum utilization and efficient use of standard machinery components is possible with a new standard terminal structure. A standard structure also does not compromise access to components requiring routine inspection and maintenance compared to a retrofitted structure. New terminal structures are galvanized to resist corrosion for longevity and will not require maintaining painted structural steel.

4. Options

Cabins:

Provisions to accommodate additional cabin features should be taken into account during the initial engineering of a system upgrade or replacement. As long as the weights of these items are taken into account they can more easily be added at a latter date if needed. A cost of requested features is included in the price matrix for consideration.

- Seat Heating
- Public Address System, One Way Communication
- Public Address System, Two Way Communication
- Bike Rack
- Cabin Lighting
- WIFI, this is a third party supplied item. Desired functionality of the system needs to be specified to allow for accurate pricing.

Additional System Redundancies:

The following additional items are not required but are typically are specified on transport gondola systems like The Mountain Village Gondola.

- Dual Loop Derail Circuit – The derail circuit consists of break forks (brittle metal bars) that brake if the haul rope derails from the sheave train. This opens an electrical circuit and the control system stops the lift. In the event the primary derail circuit has malfunctioned the lift can continue normal operation when switched over to the secondary derail circuit.
- Redundant Rope Position Detection (RPD) system – The RPD system constantly monitors the position of the rope in the sheave train. If it is detected that the rope has begun to migrate out of position then the control system stops the lift. A correctly functioning RPD system is required to operate the lift at speeds over 600 FPM. In the event one RPD system malfunctions the lift can be switched to the redundant system.
- Multi Drive and Motor Arrangements – This drive solution provides a high level of redundancy. Quantity and size of electric motors and drives are chosen so that in the event of the malfunction of one drive and or motor the remaining drive(s) and motor(s) are sufficient to operate the lift normally.

APPENDIX – A

PRICING MATRIX

APPENDIX – B

AGING LIFTS

1. 2015 Aging Lifts - Mountain Village Gondola, Power Point output with notes

Aging Lifts ~ the Iceberg Principle

A special presentation for...

Mountain Village Gondola

FALL, 2015

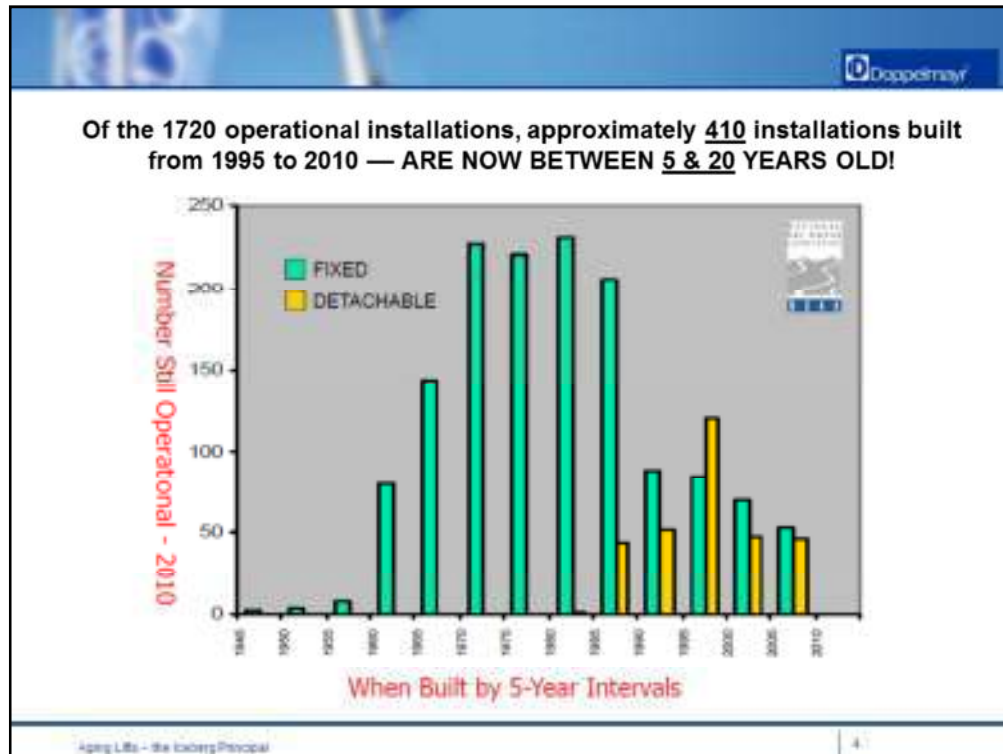
If you have questions or comments as you go through this presentation, jot them down and give to Jim Loebe. Jim can respond or pass them on to Doppelmayr for us to respond to.

Presentation Outline...what to expect.

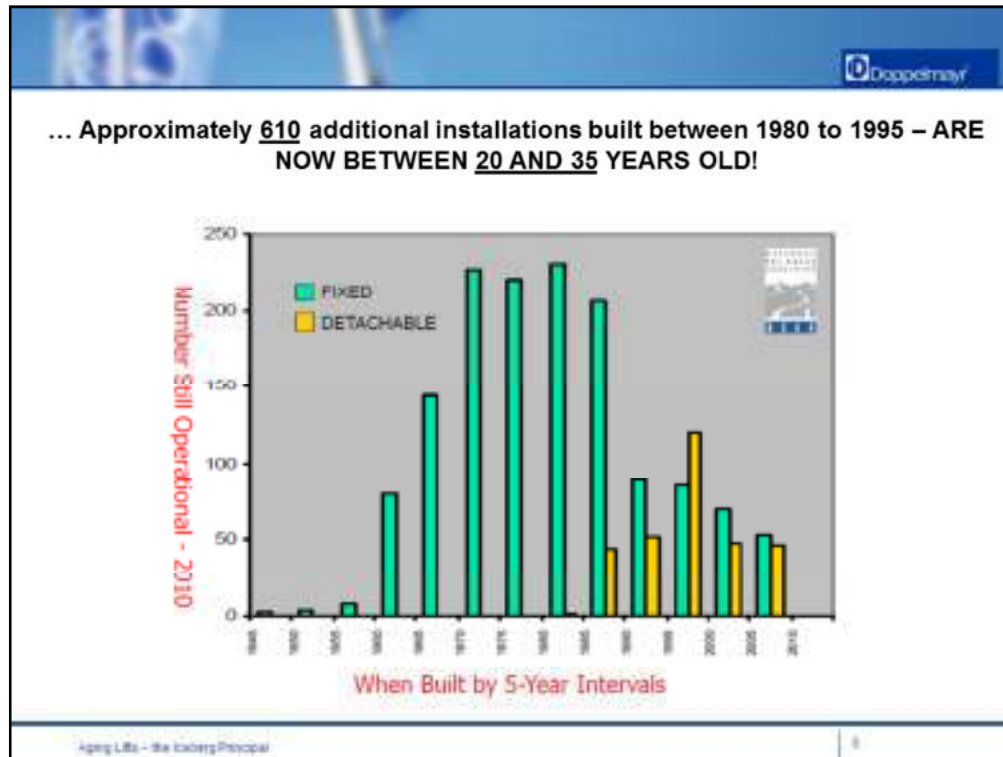
- Graph of Installations by type and 5-year periods
- Iceberg Principle
- Safety Concerns
- Support Network
- Original construction/design/installation
- Lift location/importance
- Wear & Tear – environment, usage, maintenance history
- Conclusion & Planning Document

As of 2010, there are a grand total of approximately 1720 operational installations in the U.S.

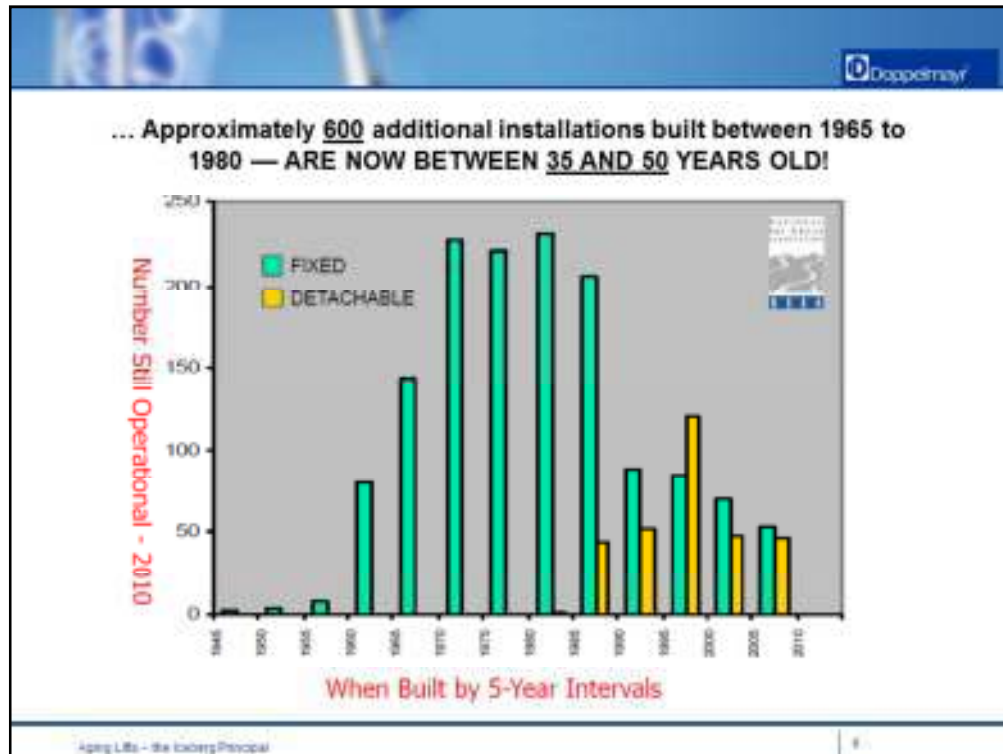
Here is a breakdown of types of installations and the 5-year period during which they were built!



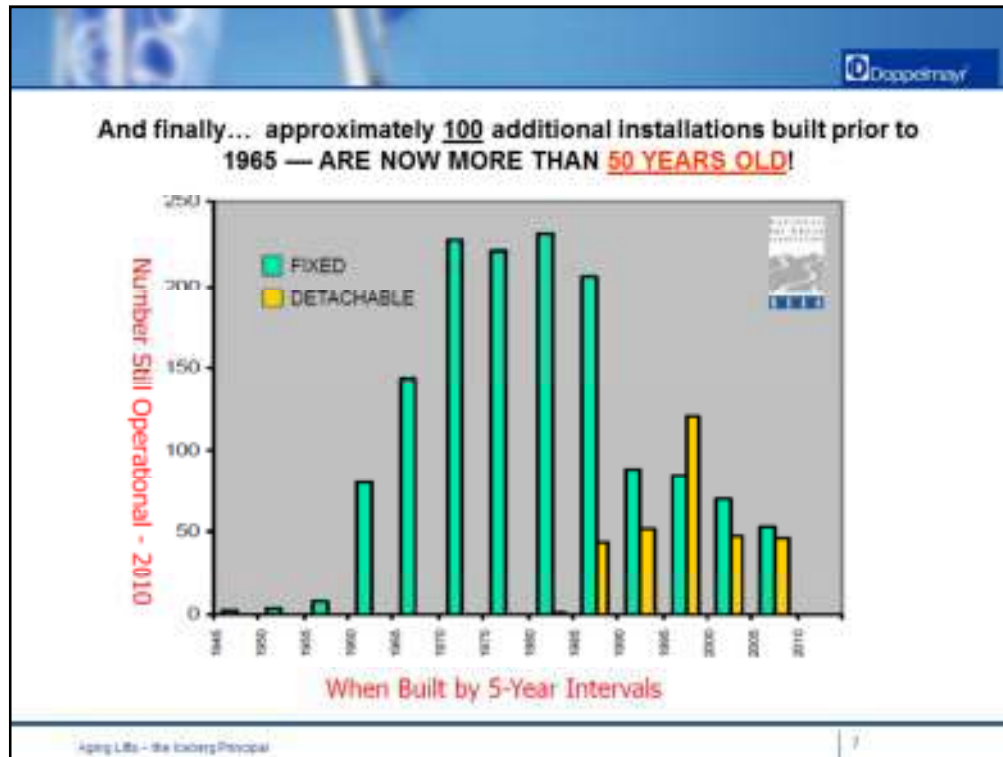
- Construction of Detachable Lifts peaked in 1997/1998 with about 130 for that period – but construction of new detachables continues and, in fact, outpaces construction of new fixed grips!
- When initially installed, the life expectancy of a Detachable lift (detach) was anticipated to be approximately 25 years!
- Now, operators are extending the operation to 30 years and more!



Point out the little blip at the bottom of the chart for the first detachable in 1983.



Point out the period between 1965 to 1980 that you are looking at!

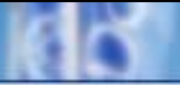


Point out the period before 1965 that you are looking at!

- 100 LIFTS – STILL IN OPERATION – THAT ARE NOW MORE THAN 50 YEARS OLD!



- What's lurking beneath the "surface"? We will attempt to show you...



The Iceberg Principal

Only one-seventh to one-ninth of the “whole picture” is visible above the surface.



- The largest part of an iceberg is underwater – only one-seventh to one-ninth is actually visible above the surface.
- The biggest ones lower as much as 400 feet above the surface of the ocean.

Aging Lifts – the Iceberg Principal

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Point out and accentuate the fact that – with an iceberg – only 1/7th to 1/9th is visible above the surface!



In relation to a ski lift,
what does the tip of
the iceberg
represent?

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Doppelmayr



The tip of the iceberg represents the *perception* of the perfect, carefree, turn-the-key and away up the hill to the happy Customers and a maintenance free* ski lift!

Ageis Lifts - the Iceberg Principle

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Key word is “perception”!

If it is turning, it must all be good. RIGHT!?



Upon each “click”, a new topic will fade in. Go SLOWLY. Mention the points below as you go...

- **Towers!** – after 10, 20, 30, or MORE years, what’s going on with your Towers? Are they full of water, have the bushings and pivot pins been replaced or at least checked? What about long term steel fatigue or corrosion?
- **Brakes!** – do your brakes work and can you rely upon them in an emergency? Yes, they may pass torque testing but have you confirmed the automatic application function required by Code?
- **Motor!** – who is checking resistance to ground

regularly? Who is considering the availability of a drop in replacement for an old motor?

- **Hydraulics! – check valves, solenoids, relief valves, etc. What do you have in stock? Who is regularly flushing, changing fluid, removing components, cleaning, and re-establishing operational parameters?**
- **Grips! – what is the condition of the jaws? Who's looking for dimensional changes in addition to fatigue? What is the availability of replacement components? How about the price!!**
- **Rope! – what are your inspectors telling you? What is the cost and lead time for a replacement rope?**
- **Bullwheels! – what is the recommended replacement frequency of bearings? Who's looking for fatigue? What is the availability of a replacement wheel?**
- **Gearbox! – dare we even go here? Bearings and seals are commonly available – not always! – but Gearing is another story! Who is regularly NDT'ing (visual & MPT) gear teeth? Who is scheduling routine rebuild? Vogel gearbox availability (24K)!**
- **Carriers! – corrosion & fatigue! Many are rotting out from the inside! Maybe they look good with a fresh coat of paint but are they?**
- **Low Voltage! – are the all the safety circuits still functional?**

When do you upgrade to improve safety and reliability (and support!)?

- **Drive! – technology doesn't wait! Drive refinements continue. DC to AC. What is the available parts & support for your drive?**
- **Drivelines! – belts, universal shafts, etc. Who is inspecting shafts for fatigue and what is the maintenance schedule for bearing replacement, rebalancing, etc.? OSHA is a higher authority than ANSI – who is confirming and retrofitting equipment guards to protect personnel? In the event of injury, OSHA WILL BE INVOLVED AND WILL LEVY FINES AND UPGRADE REQUIREMENTS AND MAY LOCK OUT YOUR EQUIPMENT!**
- **Commlines! – what are the remaining available pairs should existing ones go bad? What are the additional demands for data collection and mountain communication through your comlines? Do you know that you cannot just add a second or bigger comline without an engineering analysis on the effect that such an addition may have on the towers? Who is checking for messenger connection fatigue, re-doing boot connections, or confirming and correcting proper sag?**
- **Conveyors! – chains vs. tires. Available parts & OEM equipment? What's the internal wear on the pins, bushings, etc. on your conveyor chain and what is the replacement cost (40K) and lead time? Where do you get one?**



There are changes in design that occur on a regular basis – Are you keeping up?

SERVICE BULLETINS! – Manufacturers periodically release various levels of Bulletins! What system do you have in place to make certain that recommended or required maintenance or upgrades are completed?

What is the cost of an accident? You may be insured. That is fine....

What about the cost of spending 3, 4 years in litigation.



While this happened “over night” the contributing conditions did NOT happen “over night”!

This lift was “perfectly fine” at 4 PM... at 7 AM the next day...NOT so fine!

A close visual inspection on this tube would have shown early signs of the failure.

Due to a drive upgrade, line maintenance did not occur during the previous summer!

We may have a tendency to drop our guards, just

because a piece of equipment has required or received very little maintenance over a long period does not guarantee that it is not due or OVERDUE!

Aging lifts may hide some big flaws! Design issues, poor quality welds, fatigue and other factors sometimes only show up after many years of operation!



Support Network

**A concern ~
orphaned lifts &
technology**

Ageing Lifts - the Exoring Principal

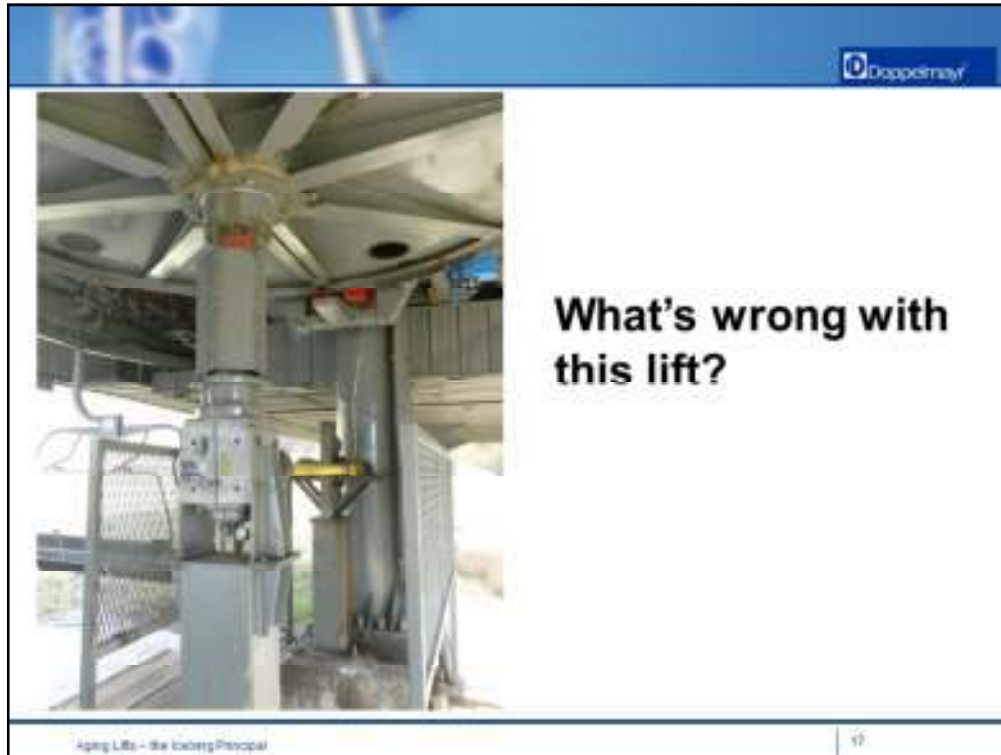
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The lack of an existing support network for parts & service is a concern regarding aging lifts.

"Orphaned Technology"

Technology that is no longer produced or available even though, at the time of installation, it may have been state-of-the-art.



ANSWER!... Nothing is wrong with this lift!

BUT technology has changed since the initial installation. A lot of moving parts.

Fortunately, this area has several installations from this vintage and has a ready source for spare parts should the need arise.

INVENTORY: do you know what you have and don't have? Are you holding to old parts and think your covered?

This resort is part of a larger group within the same Corporation. They have a large network of support (parts) if necessary so this resort is pretty well covered.

Are you?



**Slide title: HOW I SPENT MY NEW YEAR'S EVE
by Daniel Richer..**

**This failure is, coincidentally, from the same lift as
on the previous slide.**

**Lift is running fine... then, a noise noted during
operation. What do you do? INVESTIGATE!**

**Doppelmayr's standard response to "I have a
noise from my BW. Can I keep running?" is "NO.
YOU NEED TO SHUT DOWN OPERATION AND
REPAIR!" (reference Sunapee slide from
previous!)**

After an all night rig, BW removal, and bearing replacement operations were ready to go at 7:30 AM next morning!

NOW there was something wrong! Happy New Year!



A new gearbox to replace this one will cost you in the 200K range – but what is the potential revenue loss if this lift goes down for 2 weeks or longer?

Modifications might be needed to motor frame, service brakes, pump, cooling system.

Doppelmayr is getting all the older boxes they can get to bail you guys out.....what are you doing?



What are we looking at here?

How many potential failure points can you identify in this picture?

- Drivelines
- Gearbox (or how many individual parts within the gearbox?)
- Conveyor chain

Agas Lifts - the Iceberg Principle

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Note, this is a much older lift than the Mountain Village Gondola. The Mountain Village Gondola technology is new enough that it does not use this drive line and gearbox arrangement to power the tire banks.

Here is a typical Conveyor System in some of the “early days” (1983 to 1989).

Whether it is this lift – or your “new” lift, each and every component can break! Given enough time, without proper maintenance and upgrades – IT WILL BREAK!

- **Drive line**
- **Gearbox**
- **Conveyor chain**
- **Tensioner**

Ohh! What about equipment guarding? These conditions are NEVER ALLOWED TODAY! – how many of you are being proactive to protect yourselves and others from injury?

REMEMBER! OSHA SUPERCEDES ANSI! You cannot afford an industrial accident!

Just a note....every time you get some money to fix an emergency failure, your boss might assume that the machine has just got an extension to it's life.

It's a delicate situation.



Can anyone provide the definition for the term "Single Point Failure"?

How many single point failures can you identify in this photo?

- Chain
- Pillow block bearing (x4)
- Driveline
- Tensioner

Apogee Lifts - the Exberg Principal

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Have you ever heard “**YOU'RE** telling me that a 6 million dollar machine is down because of a 20 dollar part?”

That is a single point failure!

Think about the space shuttle Challenger disaster... while it may have been a single point failure, a series of events also contributed (cold weather, poor decisions, Corporate culture...)

Thousands of them are on your lift! (the best you can do is observe and prevent)



Can you buy replacement parts for this anymore or is an upgrade available?

Valves, pumps, accumulators... what service have you performed?

Ageing Lifts - the Exoring Principal 22

What type of preventative maintenance are YOU doing?

Don't wait till it fails!

Where do find replacement parts for this?

INVENTORY!! Do you know what you have in stock, where it is, and if it works?

Throw away junk and broken parts!

Nothing is more frustrating than troubleshooting

a hydraulic problem when parts – that were removed during a previous problem – are being re-installed to “correct” a current problem!

Contact your suppliers, see what your options are for support of the aging systems or available upgrades.

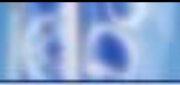
108 Doppelmayr

Not all grips are still supported. With your grips, can you still get all the necessary OEM rollers, bushings, pins, etc?

What about replacement jaws? Often the jaws are a custom order long-lead item and increasingly expensive.

Agar Lifts - the Exberg Principal 23

- **Mountain Village Gondola Agamatic 108 grips are still the current grips supplied on NEW installations today. All parts for them are in production today and for the foreseeable future.**



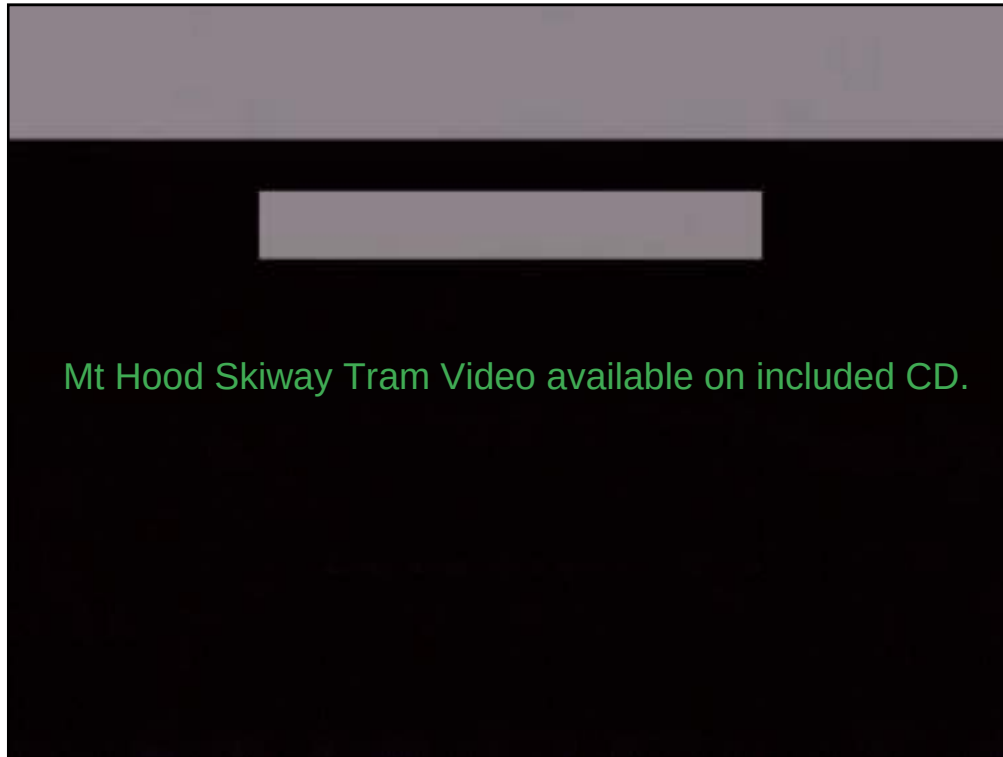
Original construction, installation and design



Ageing Lifts - the Exoring Principal

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The Skiway tram consisted of several modified city busses (\$40,000 each)

The system utilized a modified cable system used by Northwest loggers.

The tramway ride cost 75 cents while the shuttle bus only cost 50 cents and both rides took about a half hour.

The tramway proved to be a financial disaster.

The tram's cables had a tendency to ice up and a new road was opened between Government Camp and Timberline in 1950.

The Skiway Tram only lasted a few seasons. The towers were removed in 1961.

Kissling Doppelmayr



A very well built piece of equipment that was only 25 years old.

A preventive maintenance overhaul of this unit coupled to a V117 planetary found internal damage to both units...

Aging Lifts - the Iceberg Principal 27

Here is a quality Kissling gearbox - overbuilt for it's application in the ski industry's low RPM's , no wonder they last so long!

What are your options are when one of these fails on you? Not a question of "if" it will fail...



Due to the unavailability of direct replacement components, the preventive overhaul turned into a major redesign and fabrication of a new main drive platform!

Fortunately, there was sufficient time & funds available to respond to the emergency.

Will you?

Apex Lifts - the Exoring Principal

Doppelmayr

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This installation was removed in early Spring and only 50 miles from the Doppelmayr factory... a best case scenario...

and they already had snow on the ground when the new gearbox, drive frame, bullwheel, and braking went back in!

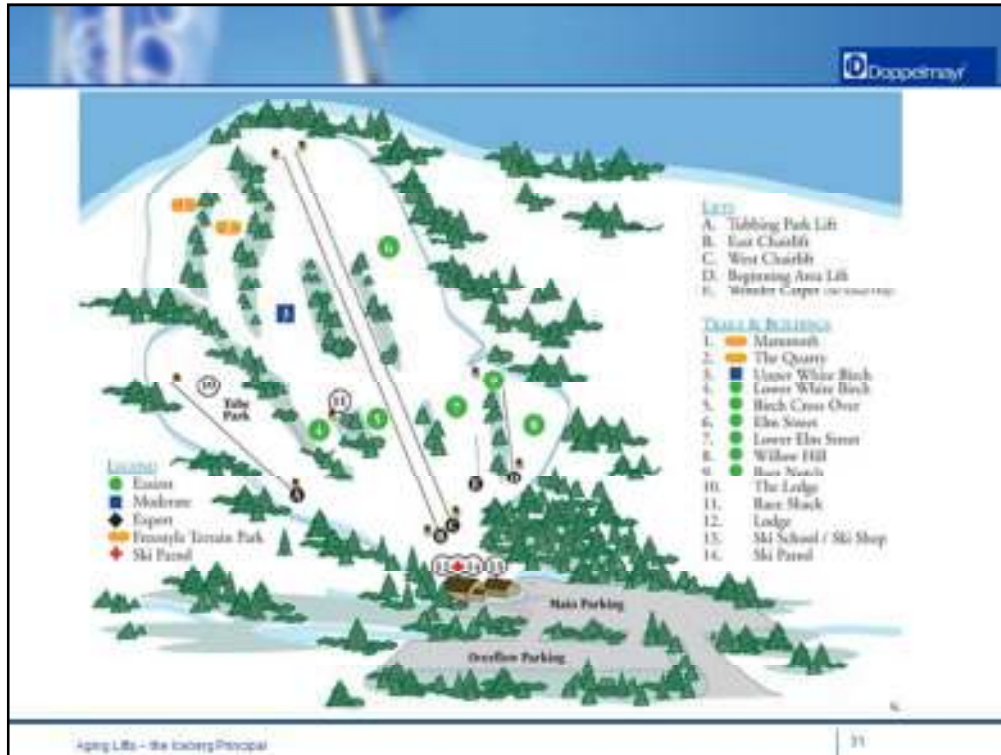


1985 - Sometimes our industry needs a wake up call in order to implement better practices and improved safety!

This lift was eventually removed by the original resort. Retired for scrap? NOPE! SOLD AND RE-INSTALLED AND STILL IN OPERATION NOW 30 YEARS AFTER THE ACCIDENT!



Not all lifts are created equal...



Does your resort depend on one or 2 main lifts?

How bad would it be for this resort to loose “B” or “C” lifts (the two in the center) during the Holidays?

Ask your manager to estimate the cost of each lift if it's out 2 weeks during Christmas holidays. What is the potential loss of revenues, loss of reputation, and cost of repair?

- **The cost to repair something after it fails is approximately 10x the cost of preventative maintenance!**

- Let management know that your request for parts (budget) is based on *realistic and necessary needs* for maintenance, upgrades, reliability, and Safety!
- Properly planned & presented, a budget should have little room for negotiation.



it's up to you to evaluate the risks you're willing to take based on your own resort.

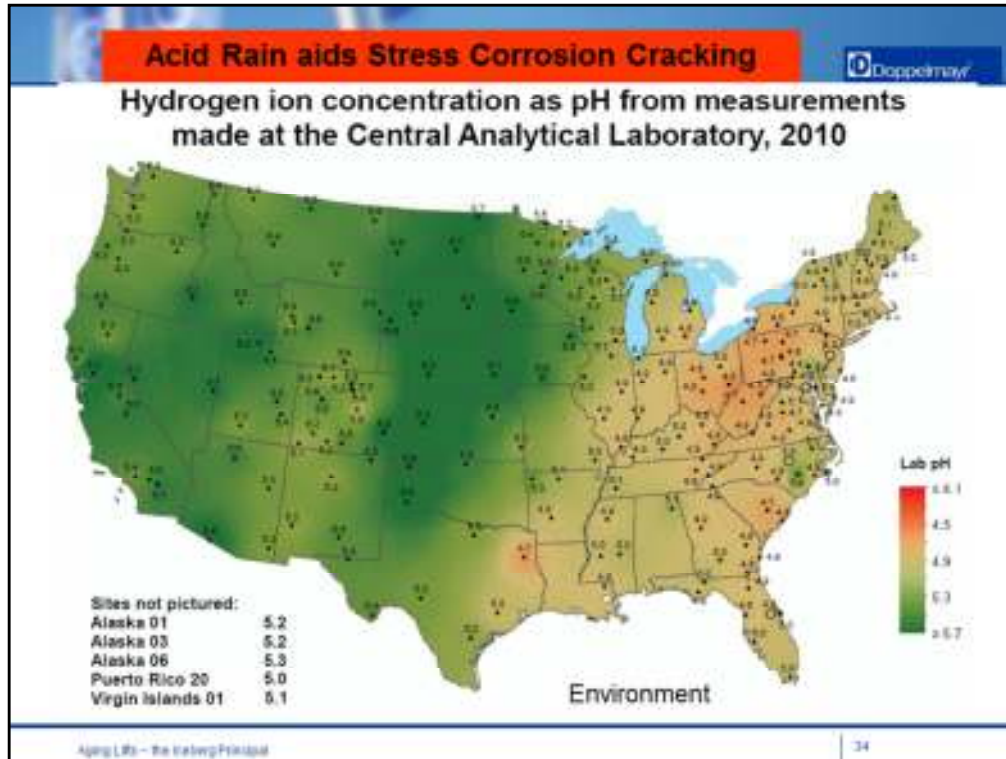
Not all lifts on this mountain are extremely critical....the ones at the base are.

34 lifts



Factors like :

- **where your lifts are located**
- **how many hours a day, a month, or a year are they accumulating**
- **how well have they been maintained – can you access the history?**



Here is a little map showing those areas of the country that are most affected by Acid Rain and where SCC concern is the highest.

A region in Vermont is in this corridor....never seen so much corrosion.

Telluride Mountain Village is in a good location.

KL8 **Doppelmayr**

Environment



If your lift looks like this on a regular basis, you may expect a slightly elevated rate of “aging” than a lift installed in a more temperate climate!

Aging Lifts - the Iceberg Principle 39

This is not exceptional – just another day at the office! FYI (Bachelor 4 CLD standing on 5 masts)

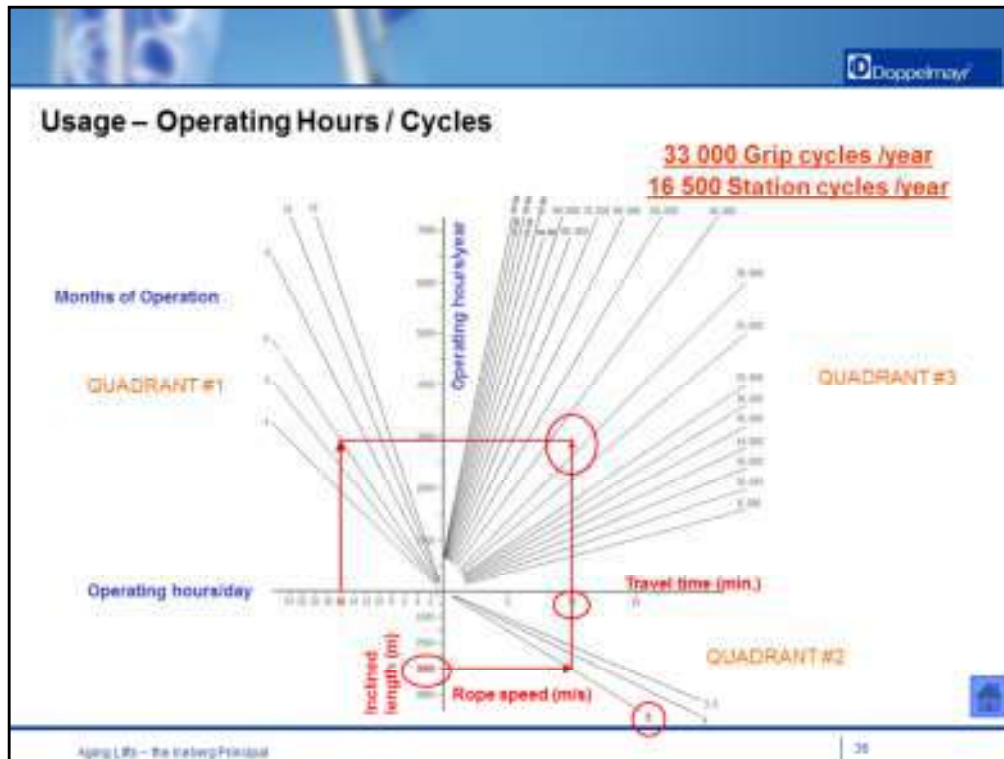
Just another day in the Pacific North West.

Another Aging Lift Consideration: “My guys are exhausted!”

Keeping an old lift like this requires a lot of time and energy. Roberts.BC

The older mechanics spend most of their time trying to keep up with those.

Telluride Mountain Village climate thankfully does not subject the Gondola to these sorts of weather events.



You may have seen this in a previous presentation?

This is important to consider the consequences whenever there is a planned increase of speed or hours of operation – it WILL cost more maintenance AND more money!

(Briefly explain & point out) Operating hours, months of operation, incline, travel time, & rope speed.

We don't need to spend too much time on this right now, but feel free later to enter you own numbers and see what kind of cycles you're accumulating on your installations each year.

Note! If this were an Agamatic type (non over center) grip, the grip cycles are doubled! Station cycles equals the total grip cycles divided by # of stations.

Mountain Village Gondola has an Agamatic grip so cycles would be doubled in the above chart.



There are several ways to organize Maintenance Records...

You need access to a full library of maintenance activities!

Here is one method that is clean and well organized for access by staff, authorities and, God Forbid – Lawyers!

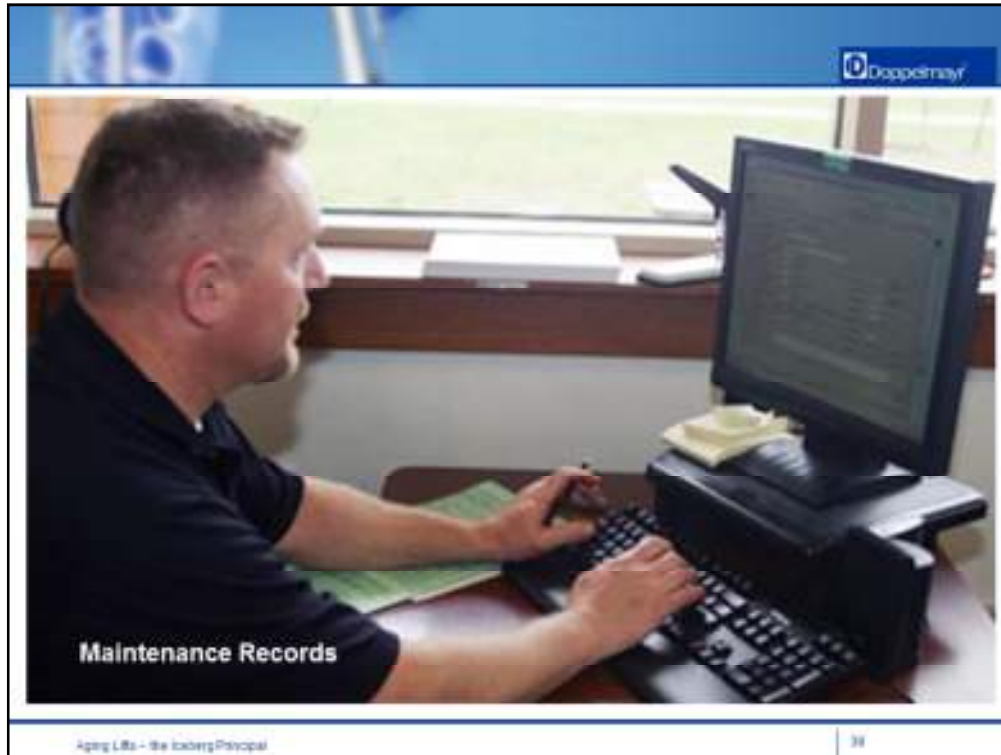
Where are your Service Bulletins?

What about a back up to your records? Do you

have a second copy, remotely stored, in case of damage by water, fire, or even an unhappy employee (Billy Bob took everything when we fired him!)

Have you developed a Quality Assurance Plan? Is it being audited by someone outside of the maintenance department to verify compliance?

Are you saying what you do... and doing what you say?



Another great alternative...

I see, you keep all your documentation on your computer....again, do you have back up?

How often is it updated?

Where is it stored? In the same room that may be subject to fire or flood?

Is the information accessible and retrievable by everyone on your team?

Do you know that ANSI requires a written schedule for systematic maintenance to be developed and followed for periodic lubrication, inspection, and adjustment?

The schedule shall include, but not be limited to the following:

- Wire rope
- Line sheave units, sheaves, bearings, and liners
- Bullwheels, bearings, and liners
- Tension systems
- Drive system incl. bearings & couplings
- Braking systems
- Electrical control systems
- Communications systems
- Carriers
- Structures
- Lift line



As Mechanics, how do you approach the concerns with aging equipment?

- **Communication** ~ you need to be certain that you understand the management's long term vision for maintenance support (parts, labor, upgrades, etc.) or replacement of the existing equipment
- **Planning** ~ based on the communication with management, it is important to develop a long term vision and to identify the maintenance, upgrade, or replacement needs for each major component of the lift
- The best way to support management's "vision" is to develop, document, and present a well prepared plan for each component that will help to assure the reliability of the installation into the future!
- Do not, do not under estimate the value and importance of your expertise and input to management

Aging Lifts - the iceberg Principal

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The blank template on the next page should help you put a short, medium and long term plan for your lifts .

		Additional Life Expectancy (in years)							
		Year 1		Year 2		Year 3		Year 4	
MAJOR COMPONENTS	WIRE MESSAGES								
	WIRE MESSAGES								
	CONSOLE								
	CONVEYOR								
	WIRE								
	WIRE MESSAGES								
	WIRE MESSAGES								
	WIRE MESSAGES								
	WIRE MESSAGES								
	WIRE MESSAGES								
	WIRE MESSAGES								
	WIRE MESSAGES								
	WIRE MESSAGES								
		TOTAL		TOTAL		TOTAL		TOTAL	

Aging Lifts - the Iceberg Principal

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This is just an example of the type of document that you could use to show your boss - or his boss – the plan and the associated costs to continue to assure the best possible reliability of the aging equipment.

(Point out...)

- Component list (add to it as they may need)
- 5 year plan, 5 year cost – and so on through the years
- Note “additional” life expectancy – could be applied to a one-year old lift or a 10 year old or a 20 year old lift
- Each lift should be analyzed in this manner

- It should be updated annually along with your budget

IMPORTANT!

DO NOT GET LULLED INTO DEFERRED OR LIMITED MAINTENANCE BECAUSE MANAGEMENT IS “THINKING” ABOUT REPLACING THE EQUIPMENT! THIS APPROACH MAY HAVE BEEN A FACTOR IN A FEW RECENT INCIDENTS!

MAINTENANCE & INSPECTIONS HAVE TO CONTINUE WITH THE SAME INTENSITY – MAYBE MORE! – UNTIL THE DAY THAT THE LIFT IS REMOVED!

Our “crystal ball” is no more clear than yours, but we may have more experience looking into it!



In conclusion, if you are hanging on to the tip of the iceberg without concern about what lies beneath...

...you may not be as secure as you think & you may be getting your feet wet a little sooner than expected!

**Thank you for your attention & support!
Your Doppelmayr Customer Support Team**

APPENDIX – C

ROPE LINE CALCULATIONS (RLC)

Section 1

1. RLC 1200pph
2. RLC 1800pph
3. RLC 2400pph

Section 2

4. RLC 1200pph
5. RLC 1800pph
6. RLC 2400pph

Section 3

7. RLC 1200pph
8. RLC 1800pph
9. RLC 2400pph

Ropeline calculation

Gondola Section 1

8-MGD

Project no.: SAA0002176

Index Text

Date

Name

created: 2015-07-05 RUSSO

vis.:

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

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Key to symbols and abbreviations

Alpha	Deflection angle
Ao	Suspension above
Au	Suspension below
BW	Bullwheel
D	Tower load (more exact: resulting force in the rope point)
DCG	Distance CTF to ground
DS	Load per sheave
EB	Entrance beam
Eps	Tower angle
F2	Distance sheave assembly main axle - rope point
HA	Suspension height of sheave assembly
HC	Distance bottom of yoke to bottom of connection plate
HG	Grout (more exact: distance bottom of shaft to CTF)
HS	Shaft length
HY	Yoke height
IO	In Operation
Lc	Load configuration
OO	Out of Operation
Phi	Rope angle
PhiH	Horizontal lead-on/lead-off angle
Psi	Tower load reaction angle
Q	Cross inclination
RP	Rope point
T	Rope tension
T0	Basic tension force
T0+	Basic tension force with overtension
T0-	Basic tension force with undertension
W	Wind
WS	Wind load per sheave
XA,ZA	Coordinates of sheave assembly main axle
XF,ZF	Coordinates CTF
XR,ZR	Rope point coordinates
f	Sag
fy	Horizontal rope deflection
h	Elevation difference
l	Horizontal length
l*	Inclined length
q	Wind pressure
r	Rope radius

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]Location: Telluride, CO
Client: Telluride Transit Authorities**Main data**

Installation type:	8-MGD
Direction of rotation:	Counter-clockwise
Uphill transportation side:	Right
Uphill capacity:	100 %
Downhill capacity:	100 %
Drive location:	Top
Tension location:	Bottom
Tensioning type:	Hydraulic tensioning
Tensioning force:	92846 lb
Basic tension force:	46423 lb
Regulation:	US design parameters <modified>
Haul rope:	44 mm 6x25 FW 1960 Fatzer
Carriers:	8 Omega IV-LWI CWAA108C
Number of carriers:	37 (on line: 29.6; in stations: 8.0)
Spacing:	400.00 ft

Geometrical data

Horizontal length:	5610.84 ft
Vertical rise:	1775.02 ft
Average grade:	31.64 %
Inclined length:	5914.59 ft
Transportation length:	6055.66 ft
Spliced rope length:	12056.56 ft
Rope gauge:	16.73 ft

Speed/Capacity

Main drive:	Rope speed:	1000 fpm	(backwards: -984 fpm)
	Capacity:	1200 persons/h	
	Interval:	24.00 s	
	Trip time:	7.51 min	
Evac drive:	Rope speed:	197 fpm	
	Capacity:	236 persons/h	
	Interval:	121.92 s	
	Trip time:	31.64 min	

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Carrier

Manufacturer: CWA
 Type: 8 Omega IV-LWI
 Dead weight: 1592 lb
 Pay load: 1358 lb or 8 persons
 Total weight: 2950 lb
 Length (L): 6.17 ft
 Dist. CL rope to top of seat (HS): 13.52 ft
 Dist. CL rope to bottom of carrier (H0): 13.52 ft
 Dist. CL rope to bottom of carrier (Hmax): 13.71 ft
 Projection area, in line: 49.5 sqft (Cf: 1.00)
 Projection area, perpendicular: 43.1 sqft (Cf: 1.00)
 Side swing 0.00 rad: Yi0 / Yo0: 3.97 / 3.38 ft
 Side swing 0.20 rad: Yi20 / Yo20: 5.94 / 5.54 ft
 Side swing 0.34 rad: Yi34 / Yo34: 7.55 / 7.12 ft
 Grip: A108C
 Grip pulling force: 5845.0 lb

Haul rope data

Manufacturer: Fatzer
 Rope construction: 6x25 FW compact type
 Diameter: 1.73 in
 Dia. outer wire: 0.116 in
 Metallic area: 1.349 sqin
 Rope weight: 5.09 lb/ft
 Tensile strength: 284274 psi
 Calculated breaking strength: 383299 lb
 Minimum breaking strength: 333616 lb
 Elastic modulus: 1450368 psi
 Thermal expansion factor: 1.2e-05 1/K
 Permanent elongation: 1.50 ‰
 Spliced rope length: 12056.6 ft
 Order length: 12329 ft (incl. Splice length: 173 ft; Surcharge: 98 ft)

Rope sheaves

Sheave	Side plate diameter	Base of Groove diameter
401	412 mm	382 mm
403	430 mm	400 mm
501	530 mm	485 mm

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Station: Bottom station

Type: Custom Return movable
Entrance beam: XR: 96.50 ft ZR: 8769.51 ft
Gauge at entrance beam: 16.73 ft
Rope angle at entrance beam: 0.00 °
Distance EB to loading/unloading: 70.54 ft
Rope length in station: 114.73 ft
Distance EB to grip checking: 0.46 ft
Min. distance EB to mid 1. line sheave T; N; T/N: 0.00 ft; 0.00 ft; 0.00 ft
Driving speed: 68.90 fpm
Deceleration anti-collision monitoring: 3.28 ft/s²
Station dwell time: 85.25 s (1426 Impulses)
Tire conveyor friction: 224.8 lb
Rope deflections per side: 5.00° vertical, 4 Sheaves (501)
Return bullwheel: Ø = 16.73 ft; I = 16500 kgm²
Tmax: 49166 lb (Lc: 3+)
Tmin: 43690 lb (Lc: 4-)

Station: Top station

Type: Custom Drive fixed
Entrance beam: XR: 5707.34 ft ZR: 10544.53 ft
Gauge at entrance beam: 16.73 ft
Rope angle at entrance beam: 0.00 °
Distance EB to loading/unloading: 70.54 ft
Rope length in station: 112.66 ft
Distance EB to grip checking: 0.46 ft
Min. distance EB to mid 1. line sheave T; N; T/N: 0.00 ft; 0.00 ft; 0.00 ft
Driving speed: 68.90 fpm
Deceleration anti-collision monitoring: 3.28 ft/s²
Station dwell time: 106.30 s (1778 Impulses)
Tire conveyor friction: 224.8 lb
Rope deflections per side: 5.00° vertical, 3 Sheaves (501)
1.35° horizontal, 2 Sheaves (501)
Min. wrap angle: 177.30 °
Drive bullwheel: Ø = 15.75 ft; I = 11000 kgm²
Tmax: 75381 lb (Lc: 3+)
Tmin: 50751 lb (Lc: 1-)

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Main drive:

Total inertia: 11.87 lbft²
Acceleration: 0.49 ft/s²
Deceleration: -1.31 ft/s²

Gearbox:

Manufacturer: Doppelmayr-Lohmann
Type: GPW 245
Nominal torque: 150831.5 ftlb
Max. torque: 211164.0 ftlb
Ratio: 94.25
Efficiency: 0.93
Inertia: 11.87 lbft²
Req. input speed: 1906 rpm
Utilisation: 70 %
Req. torque continuous: 104938.5 ftlb (Lc: 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO)
Req. torque maximum: 113563.0 ftlb (Lc: 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO)

Electric motor:

Manufacturer:
Type:
Rating: 0 HP
Speed: 0 rpm
Inertia: 0.00 lbft²
Req. power continuous: 434 HP (Lc: 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO)

Service brake(s):

Inertia: 0.00 lbft²
Max. braking torque: 0.00 ftlb (0.00 kN @ 0.00 m)
Req. braking torque: 646.36 ftlb (Lc: 0% UH Load, 100% DH Load; Braking (0.50m/s²); T0)

Emergency brake(s):

Max. braking torque: 0.0 ftlb (0.00 kN @ 0.00 m)
Req. braking torque: 60919.4 ftlb (Lc: 0% UH Load, 100% DH Load; Braking (0.50m/s²); T0)

Evac drive:

System: rim gear on bullwheel
Diesel engine:
Efficiency: continuous, starting: 0.80 braking: 1.00
Req. power continuous: 107 HP (Lc: 100% UH Load, 0% DH Load; Forward operation (1.0m/s); T0+; IO)
Req. power starting: 0 HP (Lc:)
Req. power braking: 0 HP (Lc:)
Req. M_{dmax} bullwheel: 0.0 ftlb (Lc:)

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Rope point coordinates - UH

Rope point	XR[ft]	ZR[ft]	Psi[°]	Alpha[°]	F2[ft]
BEB	96.50	8769.51	-0.7	1.4	0.00
1	109.89	8769.51	4.1	4.5	-1.21
2	124.04	8771.10	9.7	6.5	-1.34
3	212.18	8794.83	15.5	3.2	0.97
4	707.15	8934.61	16.1	6.1	1.09
5	1163.87	9054.22	14.5	6.5	1.17
6	1643.06	9164.49	14.5	3.7	1.04
7	2219.32	9316.72	14.9	4.1	1.03
8	2404.72	9366.14	15.1	3.6	0.97
9	2964.02	9519.27	15.5	6.2	1.18
10	3273.80	9592.00	13.6	3.8	1.03
11	3838.82	9728.33	13.9	3.1	1.00
12	4192.38	9820.92	21.7	9.6	-2.86
13	4364.50	9908.41	25.4	4.0	1.01
14	4824.82	10132.51	27.0	2.9	1.00
15	5257.20	10350.55	26.6	2.7	0.99
16	5608.31	10530.36	25.4	7.1	1.33
17	5625.03	10537.09	17.9	8.2	1.39
18	5640.28	10541.12	11.7	8.3	1.39
19	5658.96	10543.59	4.9	7.7	1.35
TEB	5707.34	10544.53	1.2	2.3	0.00

Rope point coordinates - DH

Rope point	XR[ft]	ZR[ft]	Psi[°]	Alpha[°]	F2[ft]
BEB	96.50	8769.51	-0.9	1.7	0.00
1	109.85	8769.50	2.2	4.5	-1.21
2	124.04	8771.09	9.7	6.5	-1.34
3	212.18	8794.83	15.5	3.4	0.97
4	707.18	8934.63	14.5	6.3	1.09
5	1163.90	9054.24	12.9	6.8	1.18
6	1643.09	9164.51	12.7	4.0	1.05
7	2219.32	9316.73	14.9	4.3	1.04
8	2404.72	9366.14	15.1	3.8	0.97
9	2964.02	9519.28	15.5	6.4	1.19
10	3273.80	9592.01	13.6	4.0	1.04
11	3838.83	9728.29	13.9	3.3	0.96
12	4192.32	9820.88	20.4	9.5	-2.88
13	4364.50	9908.41	25.3	4.2	1.02
14	4824.84	10132.52	25.8	3.2	1.00
15	5257.21	10350.52	26.5	2.9	0.96
16	5608.31	10530.36	25.5	7.2	1.34
17	5625.00	10537.08	18.9	8.3	1.39
18	5640.28	10541.12	11.7	8.5	1.40
19	5658.99	10543.60	3.7	7.9	1.36
TEB	5707.34	10544.53	1.2	2.4	0.00

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Span geometry - UH

Span	l[ft]	h[ft]	l*[ft]	Chord angle[°]	Chord angle[%]
BEB - 1	13.39	-0.00	13.39	-0.01	-0.02
1 - 2	14.15	1.59	14.23	6.42	11.25
2 - 3	88.14	23.74	91.28	15.07	26.93
3 - 4	494.97	139.78	514.33	15.77	28.24
4 - 5	456.72	119.61	472.12	14.68	26.19
5 - 6	479.19	110.27	491.72	12.96	23.01
6 - 7	576.26	152.23	596.03	14.80	26.42
7 - 8	185.39	49.41	191.87	14.92	26.65
8 - 9	559.31	153.14	579.89	15.31	27.38
9 - 10	309.78	72.72	318.20	13.21	23.48
10 - 11	565.02	136.33	581.24	13.57	24.13
11 - 12	353.56	92.59	365.48	14.67	26.19
12 - 13	172.12	87.49	193.08	26.94	50.83
13 - 14	460.32	224.10	511.98	25.96	48.68
14 - 15	432.37	218.04	484.24	26.76	50.43
15 - 16	351.12	179.81	394.48	27.12	51.21
16 - 17	16.72	6.73	18.02	21.94	40.28
17 - 18	15.26	4.03	15.78	14.78	26.38
18 - 19	18.68	2.47	18.84	7.55	13.25
19 - TEB	48.38	0.94	48.39	1.11	1.94

Span geometry - DH

Span	l[ft]	h[ft]	l*[ft]	Chord angle[°]	Chord angle[%]
BEB - 1	13.35	-0.01	13.35	-0.03	-0.06
1 - 2	14.18	1.59	14.27	6.40	11.22
2 - 3	88.14	23.74	91.28	15.07	26.94
3 - 4	495.00	139.79	514.36	15.77	28.24
4 - 5	456.72	119.62	472.12	14.68	26.19
5 - 6	479.19	110.27	491.72	12.96	23.01
6 - 7	576.23	152.22	596.00	14.80	26.42
7 - 8	185.39	49.41	191.87	14.92	26.65
8 - 9	559.30	153.14	579.89	15.31	27.38
9 - 10	309.78	72.72	318.20	13.21	23.48
10 - 11	565.03	136.28	581.24	13.56	24.12
11 - 12	353.49	92.59	365.42	14.68	26.19
12 - 13	172.18	87.53	193.15	26.95	50.84
13 - 14	460.34	224.11	512.00	25.96	48.68
14 - 15	432.37	218.00	484.22	26.76	50.42
15 - 16	351.09	179.85	394.48	27.12	51.22
16 - 17	16.70	6.72	18.00	21.93	40.25
17 - 18	15.28	4.04	15.80	14.79	26.41
18 - 19	18.71	2.48	18.87	7.54	13.24
19 - TEB	48.35	0.93	48.36	1.11	1.93

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Calculation mode and load configurations

Calculation method: Point load (increments of 3.28 ft)

Calculated load configurations:

- 1 Rope Only; Forward operation (5.1m/s); T0; IO
- 1+ Rope Only; Forward operation (5.1m/s); T0+; IO
- 1- Rope Only; Forward operation (5.1m/s); T0-; IO
- 2 Rope + Empty Carriers; Forward operation (5.1m/s); T0; IO
- 2+ Rope + Empty Carriers; Forward operation (5.1m/s); T0+; IO
- 2- Rope + Empty Carriers; Forward operation (5.1m/s); T0-; IO
- 3 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0; IO
- 3+ 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO
- 3- 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0-; IO
- 4 100% UH+DH Load; Forward operation (5.1m/s); T0; IO
- 4+ 100% UH+DH Load; Forward operation (5.1m/s); T0+; IO
- 4- 100% UH+DH Load; Forward operation (5.1m/s); T0-; IO
- 5 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0; IO
- 5+ 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0+; IO
- 5- 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0-; IO
- 6 100% UH+DH Load, 2 Spans Rope Only; Forward operation (5.1m/s); T0; IO

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Rope tension extremes, rope design factors, shear ratio

Rope tension extremes at nominal tension without dynamics

Tmax = 73000 lb (Rp: TBW; Lc: 3)

Tmin = 45631 lb (Rp: 2; Lc: 4)

Rope tension extremes at nominal tension with dynamics

Tmax = 73000 lb (Rp: TBW; Lc: 3)

Tmin = 45631 lb (Rp: 2; Lc: 4)

Rope tension extremes without dynamics

Tmax = 75381 lb (Rp: TBW; Lc: 3+)

Tmin = 43332 lb (Rp: 2; Lc: 4-)

Rope tension extremes with dynamics

Tmax = 75381 lb (Rp: TBW; Lc: 3+)

Tmin = 43332 lb (Rp: 2; Lc: 4-)

Rope design factor: min.: 4.57 max.: 7.70

Minimum shear ratio carrier: 15.47

Specific weights: rope only: 5.0868 lb/ft
rope + empty carriers: 9.0661 lb/ft
rope + loaded carriers UH: 12.4613 lb/ft
rope + loaded carriers DH: 12.4613 lb/ft

Max. angle change: 0.0681 rad; 3.90° (Rp: 3; to top; DH)

Greatest rope inclination: 59.17%; 30.61° (Rp: 15; to bottom; DH; Lc: 6)

Required min. pull force: 4506 lb

Bullwheels - SumTmax and Mdmax, extreme values with dynamic

	SumTmax [lb]	(Lc)	Md [lbft]	Mdmax [lbft]	(Lc)	SumT [lb]
Drive bullwheel	142905	(4+)	73907	113563	(3+)	137158
Return bullwheel	97488	(1+)				
	pmax [lb/in ²]	(Lc)				
Drive bullwheel	655	(4+)				
Return bullwheel	420	(1+)				

Bullwheels - SumTmax and Mdmax, values at nominal tension without dynamic

	SumTmax [lb]	(Lc)	Md [lbft]	Mdmax [lbft]	(Lc)	SumT [lb]
Drive bullwheel	138276	(4)	60572	103660	(3)	132527
Return bullwheel	92846	(1)				

Clearance envelope

Required line width: 46.54 ft

Required line width - UH - Right: 23.21 ft (RP: 6; Lc: Rope + Empty Carriers; Forward operation; T0-; IO)

Required line width - DH - Left: 23.33 ft (RP: 6; Lc: Rope + Empty Carriers; Forward operation; T0-; IO)

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Circumferential forces, torques, required power (min ... geo. mean ... max)

Lc	Circumferential force [lb]	torque [ftlb]	required power [HP]	μ nec
1	4983	41857	173	0.029
1+	5131	43131	178	0.029
1-	4835	40584	168	0.029
2	5503 ... 6221 ... 6965	46282 ... 51937 ... 57792	191 ... 215 ... 239	0.036
2+	5661 ... 6368 ... 7100	47634 ... 53208 ... 58970	197 ... 220 ... 244	0.035
2-	5344 ... 6073 ... 6830	44922 ... 50666 ... 56623	186 ... 210 ... 234	0.037
3	11625 ... 12770 ... 13885	94637 ... 103660 ... 112449	391 ... 429 ... 465	0.068
3+	11791 ... 12918 ... 14012	96051 ... 104938 ... 113563	397 ... 434 ... 470	0.066
3-	11458 ... 12621 ... 13759	93209 ... 102379 ... 111347	386 ... 423 ... 461	0.070
4	5805 ... 7282 ... 8842	48943 ... 60572 ... 72854	202 ... 251 ... 301	0.042
4+	5976 ... 7429 ... 8961	50402 ... 61839 ... 73907	208 ... 256 ... 306	0.041
4-	5632 ... 7135 ... 8724	47468 ... 59305 ... 71820	196 ... 245 ... 297	0.043
5	-318 ... 733 ... 1948	588 ... 8851 ... 18414	2 ... 37 ... 76	0.010
5+	-154 ... 879 ... 2072	1984 ... 10111 ... 19496	8 ... 42 ... 81	0.010
5-	-483 ... 587 ... 1827	-819 ... 7594 ... 17349	-3 ... 31 ... 72	0.009
6	2730 ... 7297 ... 11415	24655 ... 60618 ... 93055	102 ... 251 ... 385	0.055

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Maximum and minimum rope tensions - UH

Rope point	Tmax [lb]	Lc	Tmin [lb]	Lc
BBW	48890	(1+)	44234	(1-)
BEB	49166	(3+)	44443	(1-)
1	49262	(3+)	44444	(1-)
2	49764	(3+)	44561	(3-)
3	50522	(3+)	44844	(3-)
4	52284	(3+)	45593	(1-)
5	53861	(3+)	46283	(1-)
6	55243	(3+)	46938	(1-)
7	57679	(3+)	47742	(1-)
8	58014	(3+)	48040	(1-)
9	60564	(3+)	48858	(1-)
10	61533	(3+)	49328	(1-)
11	63225	(3+)	50069	(1-)
12	64763	(3+)	50574	(1-)
13	65673	(3+)	51257	(1-)
14	69126	(3+)	52459	(1-)
15	71722	(3+)	53607	(1-)
16	74262	(3+)	54563	(1-)
17	74531	(3+)	54744	(1-)
18	74789	(3+)	54938	(1-)
19	75016	(3+)	55126	(1-)
TEB	75061	(3+)	55290	(1-)
TBW	75381	(3+)	55586	(1-)

Maximum and minimum rope tensions - DH

Rope point	Tmax [lb]	Lc	Tmin [lb]	Lc
BBW	48598	(1+)	43970	(1-)
BEB	48380	(1+)	43690	(4-)
1	48410	(4+)	43482	(4-)
2	48665	(4+)	43332	(4-)
3	49060	(4+)	43397	(4-)
4	50547	(4+)	44215	(1-)
5	51797	(4+)	44730	(1-)
6	52883	(4+)	45261	(1-)
7	55066	(4+)	45989	(1-)
8	55177	(4+)	46201	(1-)
9	57436	(4+)	46881	(1-)
10	58172	(4+)	47203	(1-)
11	59543	(4+)	47861	(1-)
12	60720	(4+)	48107	(1-)
13	61398	(4+)	48492	(1-)
14	64444	(4+)	49592	(1-)
15	66823	(4+)	50659	(1-)
16	69036	(4+)	51434	(1-)
17	68833	(4+)	51306	(1-)
18	68638	(4+)	51163	(1-)
19	68431	(4+)	51027	(1-)
TEB	68240	(4+)	51005	(1-)
TBW	67903	(4+)	50751	(1-)

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Maximum and minimum tower loads - UH

Rope point	Sheave assembly	Dmax Lc [lb]	Psi [°]	Alpha [°]	Dmin Lc [lb]	Psi [°]	Alpha [°]
BEB	3R-403	2272 (3+)	-1.32	2.65	44 (1-)	-0.03	0.06
1	6N-401	-5450 (1+)	3.20	-6.35	-2040 (3-)	5.06	-2.62
2	8N-401	-7191 (3+)	10.64	-8.35	-3691 (3-)	8.84	-4.74
3	4T-401	4382 (3-)	16.15	5.49	877 (1+)	14.82	1.01
4	6T-401	7331 (3+)	16.96	8.06	3293 (1-)	15.29	4.13
5	8T-401	7815 (3+)	15.27	8.37	3769 (1-)	13.79	4.66
6	6T-401	5431 (3-)	15.17	6.21	1013 (6)	13.75	1.11
7	6T-401	5617 (6)	14.37	6.23	1824 (6)	15.39	1.95
8	6T-401	5253 (6)	15.66	5.68	1514 (6)	14.63	1.56
9	8T-401	8164 (3+)	16.34	7.74	4008 (1-)	14.64	4.70
10	6T-401	5635 (3-)	14.06	5.69	1861 (6)	13.07	1.82
11	4T-401	4981 (6)	13.42	5.00	1181 (6)	14.38	1.13
12	12N-401	-11815 (6)	21.03	-11.02	-8502 (6)	22.42	-8.21
13	6T-401	5936 (3+)	24.69	5.29	2487 (1-)	26.04	2.78
14	4T-401	5070 (6)	27.64	4.53	1369 (6)	26.39	1.22
15	4T-401	4677 (6)	26.08	4.13	1574 (6)	27.03	1.33
16	6T-501	10243 (3+)	25.88	7.95	5877 (1-)	24.98	6.16
17	6T-501	11799 (3+)	17.35	9.25	6929 (1-)	18.36	7.24
18	6T-501	12032 (3+)	12.18	9.33	7030 (1-)	11.16	7.32
19	6T-501	11309 (3+)	5.41	8.79	6370 (1-)	4.29	6.61
TEB	3R-403	4326 (3+)	1.68	3.37	1196 (1-)	0.62	1.24

Maximum and minimum tower loads - DH

Rope point	Sheave assembly	Dmax Lc [lb]	Psi [°]	Alpha [°]	Dmin Lc [lb]	Psi [°]	Alpha [°]
BEB	3R-403	2850 (4+)	-1.69	3.38	60 (1-)	-0.04	0.08
1	6N-401	-5356 (1+)	3.18	-6.35	-1972 (4-)	1.30	-2.58
2	8N-401	-7039 (4+)	10.63	-8.36	-3561 (4-)	8.79	-4.68
3	4T-401	4410 (4-)	16.18	5.69	896 (1+)	14.81	1.07
4	6T-401	7294 (4+)	13.74	8.46	3267 (1-)	15.29	4.23
5	8T-401	7750 (4+)	12.10	8.78	3723 (1-)	13.78	4.76
6	6T-401	5510 (4-)	11.75	6.71	1083 (6)	13.74	1.24
7	6T-401	5620 (4-)	14.34	6.51	1829 (6)	15.42	2.04
8	6T-401	5272 (4-)	15.69	6.00	1532 (6)	14.60	1.66
9	8T-401	8041 (4+)	16.44	8.04	3935 (1-)	14.65	4.80
10	6T-401	5658 (4-)	14.09	6.06	1886 (6)	13.05	1.95
11	4T-401	5043 (4-)	13.37	5.38	1238 (6)	14.39	1.26
12	12N-401	-10962 (6)	21.04	-10.94	-7766 (4-)	19.70	-8.15
13	6T-401	5870 (4+)	24.56	5.61	2443 (1-)	26.02	2.88
14	4T-401	5138 (4-)	25.16	5.05	1434 (6)	26.39	1.37
15	4T-401	4695 (6)	26.02	4.44	1590 (6)	27.04	1.44
16	6T-501	9765 (4+)	25.96	8.15	5608 (1-)	25.00	6.24
17	6T-501	11136 (4+)	19.41	9.30	6475 (1-)	18.36	7.22
18	6T-501	11365 (4+)	12.31	9.60	6568 (1-)	11.16	7.35
19	6T-501	10623 (4+)	3.03	9.12	5911 (1-)	4.28	6.63
TEB	3R-403	4213 (4+)	1.81	3.61	1108 (1-)	0.62	1.24

Ropeline calculation

Gondola Section 1

8-MGD

Project no.: SAA0002176

Index Text

Date

Name

created: 2015-07-05 RUSSO

vis.:

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

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Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Key to symbols and abbreviations

Alpha	Deflection angle
Ao	Suspension above
Au	Suspension below
BW	Bullwheel
D	Tower load (more exact: resulting force in the rope point)
DCG	Distance CTF to ground
DS	Load per sheave
EB	Entrance beam
Eps	Tower angle
F2	Distance sheave assembly main axle - rope point
HA	Suspension height of sheave assembly
HC	Distance bottom of yoke to bottom of connection plate
HG	Grout (more exact: distance bottom of shaft to CTF)
HS	Shaft length
HY	Yoke height
IO	In Operation
Lc	Load configuration
OO	Out of Operation
Phi	Rope angle
PhiH	Horizontal lead-on/lead-off angle
Psi	Tower load reaction angle
Q	Cross inclination
RP	Rope point
T	Rope tension
T0	Basic tension force
T0+	Basic tension force with overtension
T0-	Basic tension force with undertension
W	Wind
WS	Wind load per sheave
XA,ZA	Coordinates of sheave assembly main axle
XF,ZF	Coordinates CTF
XR,ZR	Rope point coordinates
f	Sag
fy	Horizontal rope deflection
h	Elevation difference
l	Horizontal length
l*	Inclined length
q	Wind pressure
r	Rope radius

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]Location: Telluride, CO
Client: Telluride Transit Authorities**Main data**

Installation type:	8-MGD
Direction of rotation:	Counter-clockwise
Uphill transportation side:	Right
Uphill capacity:	100 %
Downhill capacity:	100 %
Drive location:	Top
Tension location:	Bottom
Tensioning type:	Hydraulic tensioning
Tensioning force:	92846 lb
Basic tension force:	46423 lb
Regulation:	US design parameters <modified>
Haul rope:	44 mm 6x25 FW 2160 Fatzer
Carriers:	8 Omega IV-LWI CWAA108C
Number of carriers:	56 (on line: 44.4; in stations: 12.0)
Spacing:	266.67 ft

Geometrical data

Horizontal length:	5610.84 ft
Vertical rise:	1775.02 ft
Average grade:	31.64 %
Inclined length:	5914.58 ft
Transportation length:	6055.66 ft
Spliced rope length:	12056.55 ft
Rope gauge:	16.73 ft

Speed/Capacity

Main drive:	Rope speed:	1000 fpm	(backwards: -984 fpm)
	Capacity:	1800 persons/h	
	Interval:	16.00 s	
	Trip time:	7.51 min	
Evac drive:	Rope speed:	197 fpm	
	Capacity:	354 persons/h	
	Interval:	81.28 s	
	Trip time:	31.64 min	

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Carrier

Manufacturer: CWA
 Type: 8 Omega IV-LWI
 Dead weight: 1592 lb
 Pay load: 1358 lb or 8 persons
 Total weight: 2950 lb
 Length (L): 6.17 ft
 Dist. CL rope to top of seat (HS): 13.52 ft
 Dist. CL rope to bottom of carrier (H0): 13.52 ft
 Dist. CL rope to bottom of carrier (Hmax): 13.71 ft
 Projection area, in line: 49.5 sqft (Cf: 1.00)
 Projection area, perpendicular: 43.1 sqft (Cf: 1.00)
 Side swing 0.00 rad: Yi0 / Yo0: 3.97 / 3.38 ft
 Side swing 0.20 rad: Yi20 / Yo20: 5.94 / 5.54 ft
 Side swing 0.34 rad: Yi34 / Yo34: 7.55 / 7.12 ft
 Grip: A108C
 Grip pulling force: 5845.0 lb

Haul rope data

Manufacturer: Fatzer
 Rope construction: 6x25 FW compact type
 Diameter: 1.73 in
 Dia. outer wire: 0.116 in
 Metallic area: 1.349 sqin
 Rope weight: 5.09 lb/ft
 Tensile strength: 313281 psi
 Calculated breaking strength: 422416 lb
 Minimum breaking strength: 363516 lb
 Elastic modulus: 1450368 psi
 Thermal expansion factor: 1.2e-05 1/K
 Permanent elongation: 1.50 ‰
 Spliced rope length: 12056.6 ft
 Order length: 12329 ft (incl. Splice length: 173 ft; Surcharge: 98 ft)

Rope sheaves

Sheave	Side plate diameter	Base of Groove diameter
401	412 mm	382 mm
403	430 mm	400 mm
501	530 mm	485 mm

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]
Station: Bottom station

Type: Custom Return movable
 Entrance beam: XR: 96.50 ft ZR: 8769.51 ft
 Gauge at entrance beam: 16.73 ft
 Rope angle at entrance beam: 0.00 °
 Distance EB to loading/unloading: 70.54 ft
 Rope length in station: 114.73 ft
 Distance EB to grip checking: 0.46 ft
 Min. distance EB to mid 1. line sheave T; N; T/N: 0.00 ft; 0.00 ft; 0.00 ft
 Driving speed: 68.90 fpm
 Deceleration anti-collision monitoring: 3.28 ft/s²
 Station dwell time: 85.25 s (1426 Impulses)
 Tire conveyor friction: 224.8 lb
 Rope deflections per side: 5.00° vertical, 4 Sheaves (501)
 Return bullwheel: Ø = 16.73 ft; I = 16500 kgm²
 Tmax: 49170 lb (Lc: 3+)
 Tmin: 43690 lb (Lc: 4-)

Station: Top station

Type: Custom Drive fixed
 Entrance beam: XR: 5707.34 ft ZR: 10544.53 ft
 Gauge at entrance beam: 16.73 ft
 Rope angle at entrance beam: 0.00 °
 Distance EB to loading/unloading: 70.54 ft
 Rope length in station: 112.66 ft
 Distance EB to grip checking: 0.46 ft
 Min. distance EB to mid 1. line sheave T; N; T/N: 0.00 ft; 0.00 ft; 0.00 ft
 Driving speed: 68.90 fpm
 Deceleration anti-collision monitoring: 3.28 ft/s²
 Station dwell time: 106.30 s (1778 Impulses)
 Tire conveyor friction: 224.8 lb
 Rope deflections per side: 5.00° vertical, 3 Sheaves (501)
 1.35° horizontal, 2 Sheaves (501)
 Min. wrap angle: 177.30 °
 Drive bullwheel: Ø = 15.75 ft; I = 11000 kgm²
 Tmax: 82377 lb (Lc: 3+)
 Tmin: 50754 lb (Lc: 1-)

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]**Main drive:**

Total inertia: 11.87 lbft²
Acceleration: 0.49 ft/s²
Deceleration: -1.31 ft/s²

Gearbox:

Manufacturer: Doppelmayr-Lohmann
Type: GPW 245
Nominal torque: 150831.5 ftlb
Max. torque: 211164.0 ftlb
Ratio: 94.25
Efficiency: 0.93
Inertia: 11.87 lbft²
Req. input speed: 1906 rpm
Utilisation: 90 %
Req. torque continuous: 135623.6 ftlb (Lc: 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO)
Req. torque maximum: 139820.4 ftlb (Lc: 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO)

Electric motor:

Manufacturer:
Type:
Rating: 0 HP
Speed: 0 rpm
Inertia: 0.00 lbft²
Req. power continuous: 561 HP (Lc: 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO)

Service brake(s):

Inertia: 0.00 lbft²
Max. braking torque: 0.00 ftlb (0.00 kN @ 0.00 m)
Req. braking torque: 981.83 ftlb (Lc: 0% UH Load, 100% DH Load; Braking (0.50m/s²); T0)

Emergency brake(s):

Max. braking torque: 0.0 ftlb (0.00 kN @ 0.00 m)
Req. braking torque: 92537.9 ftlb (Lc: 0% UH Load, 100% DH Load; Braking (0.50m/s²); T0)

Evac drive:

System: rim gear on bullwheel
Diesel engine:
Efficiency: continuous, starting: 0.80 braking: 1.00
Req. power continuous: 138 HP (Lc: 100% UH Load, 0% DH Load; Forward operation (1.0m/s); T0+; IO)
Req. power starting: 0 HP (Lc:)
Req. power braking: 0 HP (Lc:)
Req. M_{dmax} bullwheel: 0.0 ftlb (Lc:)

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Rope point coordinates - UH

Rope point	XR[ft]	ZR[ft]	Psi[°]	Alpha[°]	F2[ft]
BEB	96.50	8769.51	-0.7	1.4	0.00
1	109.89	8769.51	4.1	4.5	-1.21
2	124.04	8771.10	9.7	6.5	-1.34
3	212.21	8794.74	13.4	3.7	0.97
4	707.12	8934.71	16.1	6.9	1.18
5	1163.86	9054.25	14.5	7.2	1.20
6	1643.09	9164.50	12.8	4.6	1.04
7	2219.29	9316.72	16.6	4.4	1.03
8	2404.71	9366.19	14.9	4.0	1.02
9	2964.02	9519.27	15.5	6.7	1.18
10	3273.80	9592.00	13.5	4.2	1.03
11	3838.80	9728.32	15.3	3.6	1.00
12	4192.39	9820.92	21.8	9.4	-2.87
13	4364.48	9908.40	26.4	4.1	1.01
14	4824.82	10132.51	27.0	3.2	1.00
15	5257.18	10350.55	27.6	3.1	0.99
16	5608.31	10530.36	25.5	7.2	1.33
17	5625.03	10537.09	17.9	8.2	1.39
18	5640.31	10541.12	10.7	8.3	1.39
19	5658.99	10543.59	3.8	7.6	1.35
TEB	5707.34	10544.53	1.1	2.2	0.00

Rope point coordinates - DH

Rope point	XR[ft]	ZR[ft]	Psi[°]	Alpha[°]	F2[ft]
BEB	96.50	8769.51	-0.9	1.7	0.00
1	109.86	8769.50	2.3	4.6	-1.21
2	124.08	8771.11	11.6	6.4	-1.33
3	212.22	8794.74	13.4	3.8	0.97
4	707.12	8934.72	16.1	7.1	1.19
5	1163.86	9054.26	14.5	7.4	1.21
6	1643.09	9164.51	12.7	4.9	1.05
7	2219.29	9316.72	16.7	4.6	1.04
8	2404.70	9366.20	14.9	4.2	1.03
9	2964.02	9519.28	15.6	6.9	1.19
10	3273.82	9592.02	12.0	4.6	1.04
11	3838.82	9728.34	13.9	4.0	1.02
12	4192.31	9820.95	20.6	9.3	-2.81
13	4364.48	9908.40	26.4	4.4	1.02
14	4824.82	10132.52	27.0	3.6	1.01
15	5257.19	10350.56	26.5	3.4	1.00
16	5608.31	10530.36	25.6	7.4	1.34
17	5625.00	10537.08	18.8	8.2	1.39
18	5640.31	10541.12	10.6	8.4	1.40
19	5658.99	10543.60	3.7	7.7	1.36
TEB	5707.34	10544.53	1.2	2.3	0.00

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Span geometry - UH

Span	l[ft]	h[ft]	l*[ft]	Chord angle[°]	Chord angle[%]
BEB - 1	13.39	-0.00	13.39	-0.01	-0.02
1 - 2	14.14	1.59	14.23	6.42	11.26
2 - 3	88.18	23.64	91.29	15.01	26.82
3 - 4	494.91	139.96	514.32	15.79	28.28
4 - 5	456.74	119.54	472.12	14.67	26.17
5 - 6	479.23	110.25	491.75	12.96	23.01
6 - 7	576.20	152.22	595.97	14.80	26.42
7 - 8	185.42	49.47	191.90	14.94	26.68
8 - 9	559.31	153.09	579.89	15.31	27.37
9 - 10	309.78	72.73	318.20	13.21	23.48
10 - 11	564.99	136.32	581.21	13.57	24.13
11 - 12	353.59	92.59	365.51	14.67	26.19
12 - 13	172.09	87.48	193.05	26.95	50.83
13 - 14	460.34	224.11	511.99	25.96	48.68
14 - 15	432.36	218.04	484.22	26.76	50.43
15 - 16	351.13	179.81	394.49	27.12	51.21
16 - 17	16.72	6.73	18.02	21.94	40.28
17 - 18	15.28	4.03	15.80	14.77	26.37
18 - 19	18.68	2.47	18.84	7.54	13.23
19 - TEB	48.35	0.94	48.36	1.11	1.94

Span geometry - DH

Span	l[ft]	h[ft]	l*[ft]	Chord angle[°]	Chord angle[%]
BEB - 1	13.36	-0.01	13.36	-0.03	-0.06
1 - 2	14.22	1.61	14.31	6.45	11.31
2 - 3	88.14	23.63	91.25	15.01	26.82
3 - 4	494.90	139.97	514.32	15.79	28.28
4 - 5	456.74	119.54	472.12	14.67	26.17
5 - 6	479.23	110.25	491.75	12.96	23.01
6 - 7	576.20	152.21	595.97	14.80	26.42
7 - 8	185.42	49.47	191.90	14.94	26.68
8 - 9	559.31	153.09	579.89	15.31	27.37
9 - 10	309.81	72.73	318.23	13.21	23.48
10 - 11	564.99	136.32	581.20	13.57	24.13
11 - 12	353.49	92.61	365.42	14.68	26.20
12 - 13	172.17	87.45	193.11	26.93	50.80
13 - 14	460.34	224.11	511.99	25.96	48.68
14 - 15	432.37	218.05	484.24	26.76	50.43
15 - 16	351.11	179.80	394.47	27.12	51.21
16 - 17	16.70	6.72	18.00	21.93	40.25
17 - 18	15.30	4.04	15.83	14.79	26.40
18 - 19	18.68	2.47	18.84	7.54	13.24
19 - TEB	48.35	0.93	48.36	1.11	1.93

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Calculation mode and load configurations

Calculation method: Point load (increments of 2.67 ft)

Calculated load configurations:

- | | |
|----|--|
| 1 | Rope Only; Forward operation (5.1m/s); T0; IO |
| 1+ | Rope Only; Forward operation (5.1m/s); T0+; IO |
| 1- | Rope Only; Forward operation (5.1m/s); T0-; IO |
| 2 | Rope + Empty Carriers; Forward operation (5.1m/s); T0; IO |
| 2+ | Rope + Empty Carriers; Forward operation (5.1m/s); T0+; IO |
| 2- | Rope + Empty Carriers; Forward operation (5.1m/s); T0-; IO |
| 3 | 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0; IO |
| 3+ | 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO |
| 3- | 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0-; IO |
| 4 | 100% UH+DH Load; Forward operation (5.1m/s); T0; IO |
| 4+ | 100% UH+DH Load; Forward operation (5.1m/s); T0+; IO |
| 4- | 100% UH+DH Load; Forward operation (5.1m/s); T0-; IO |
| 5 | 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0; IO |
| 5+ | 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0+; IO |
| 5- | 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0-; IO |
| 6 | 100% UH+DH Load, 2 Spans Rope Only; Forward operation (5.1m/s); T0; IO |

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Rope tension extremes, rope design factors, shear ratio

Rope tension extremes at nominal tension without dynamics

Tmax = 79995 lb (Rp: TBW; Lc: 3)

Tmin = 45632 lb (Rp: 2; Lc: 4)

Rope tension extremes at nominal tension with dynamics

Tmax = 79995 lb (Rp: TBW; Lc: 3)

Tmin = 45632 lb (Rp: 2; Lc: 4)

Rope tension extremes without dynamics

Tmax = 82377 lb (Rp: TBW; Lc: 3+)

Tmin = 43333 lb (Rp: 2; Lc: 4-)

Rope tension extremes with dynamics

Tmax = 82377 lb (Rp: TBW; Lc: 3+)

Tmin = 43333 lb (Rp: 2; Lc: 4-)

Rope design factor: min.: 4.54 max.: 8.39

Minimum shear ratio carrier: 15.47

Specific weights: rope only: 5.0868 lb/ft
 rope + empty carriers: 11.0558 lb/ft
 rope + loaded carriers UH: 16.1485 lb/ft
 rope + loaded carriers DH: 16.1485 lb/ft

Max. angle change: 0.0759 rad; 4.35° (Rp: 3; to top; DH)

Greatest rope inclination: 60.12%; 31.01° (Rp: 15; to bottom; DH; Lc: 6)

Required min. pull force: 4560 lb

Bullwheels - SumTmax and Mdmax, extreme values with dynamic

	SumTmax [lb]	(Lc)	Md [lbft]	Mdmax [lbft]	(Lc)	SumT [lb]
Drive bullwheel	156093	(4+)	75826	139820	(3+)	147496
Return bullwheel	97488	(1+)				
	pmax [lb/in²]	(Lc)				
Drive bullwheel	715	(4+)				
Return bullwheel	420	(1+)				

Bullwheels - SumTmax and Mdmax, values at nominal tension without dynamic

	SumTmax [lb]	(Lc)	Md [lbft]	Mdmax [lbft]	(Lc)	SumT [lb]
Drive bullwheel	151477	(4)	69739	134332	(3)	142872
Return bullwheel	92846	(1)				

Clearance envelope

Required line width: 46.94 ft

Required line width - UH - Right: 23.40 ft (RP: 6; Lc: Rope + Empty Carriers; Forward operation; T0-; IO)

Required line width - DH - Left: 23.54 ft (RP: 6; Lc: Rope + Empty Carriers; Forward operation; T0-; IO)

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Circumferential forces, torques, required power (min ... geo. mean ... max)

Lc	Circumferential force [lb]	torque [ftlb]	required power [HP]	μ nec
1	4977	41811	173	0.029
1+	5125	43082	178	0.029
1-	4830	40540	168	0.029
2	6560 ... 6826 ... 7102	54770 ... 56867 ... 59039	227 ... 235 ... 244	0.035
2+	6714 ... 6973 ... 7243	56094 ... 58136 ... 60262	232 ... 240 ... 249	0.034
2-	6405 ... 6679 ... 6961	53442 ... 55598 ... 57819	221 ... 230 ... 239	0.035
3	16186 ... 16634 ... 17182	130790 ... 134332 ... 138661	541 ... 556 ... 574	0.078
3+	16355 ... 16784 ... 17315	132236 ... 135624 ... 139820	547 ... 561 ... 578	0.076
3-	16010 ... 16484 ... 17050	129289 ... 133041 ... 137511	535 ... 550 ... 569	0.080
4	7782 ... 8407 ... 9034	64820 ... 69739 ... 74680	268 ... 288 ... 309	0.039
4+	7946 ... 8554 ... 9166	66223 ... 71005 ... 75826	274 ... 294 ... 314	0.038
4-	7616 ... 8260 ... 8904	63406 ... 68473 ... 73542	262 ... 283 ... 304	0.040
5	-2037 ... -1401 ... -737	-12707 ... -7716 ... -2507	-45 ... -28 ... -9	0.009
5+	-1874 ... -1257 ... -621	-11316 ... -6468 ... -1476	-40 ... -23 ... -5	0.008
5-	-2201 ... -1545 ... -851	-14106 ... -8961 ... -3515	-50 ... -32 ... -13	0.010
6	2179 ... 8430 ... 13961	20567 ... 69795 ... 113358	85 ... 289 ... 469	0.062

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Maximum and minimum rope tensions - UH

Rope point	Tmax [lb]	Lc	Tmin [lb]	Lc
BBW	48890	(1+)	44234	(1-)
BEB	49170	(3+)	44443	(1-)
1	49263	(3+)	44442	(3-)
2	49768	(3+)	44559	(3-)
3	50531	(3+)	44840	(3-)
4	53214	(3+)	45590	(1-)
5	55411	(3+)	46281	(1-)
6	56876	(3+)	46936	(1-)
7	60011	(3+)	47740	(1-)
8	60618	(3+)	48037	(1-)
9	63716	(3+)	48855	(1-)
10	64988	(3+)	49325	(1-)
11	67285	(3+)	50067	(1-)
12	68877	(3+)	50572	(1-)
13	70668	(3+)	51255	(1-)
14	74558	(3+)	52457	(1-)
15	78525	(3+)	53604	(1-)
16	81030	(3+)	54561	(1-)
17	81320	(3+)	54742	(1-)
18	81710	(3+)	54936	(1-)
19	81956	(3+)	55124	(1-)
TEB	82038	(3+)	55288	(1-)
TBW	82377	(3+)	55584	(1-)

Maximum and minimum rope tensions - DH

Rope point	Tmax [lb]	Lc	Tmin [lb]	Lc
BBW	48598	(1+)	43970	(1-)
BEB	48380	(1+)	43690	(4-)
1	48431	(4+)	43480	(4-)
2	48671	(4+)	43333	(4-)
3	49057	(4+)	43367	(4-)
4	51452	(4+)	44218	(1-)
5	53364	(4+)	44733	(1-)
6	54363	(4+)	45263	(1-)
7	57246	(4+)	45992	(1-)
8	57447	(4+)	46203	(1-)
9	60308	(4+)	46884	(1-)
10	61185	(4+)	47205	(1-)
11	63173	(4+)	47864	(1-)
12	64229	(4+)	48111	(1-)
13	65754	(4+)	48495	(1-)
14	69341	(4+)	49595	(1-)
15	73008	(4+)	50663	(1-)
16	75123	(4+)	51437	(1-)
17	74979	(4+)	51309	(1-)
18	74715	(4+)	51166	(1-)
19	74489	(4+)	51031	(1-)
TEB	74327	(4+)	51008	(1-)
TBW	73924	(4+)	50754	(1-)

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Maximum and minimum tower loads - UH

Rope point	Sheave assembly	Dmax Lc [lb]	Psi [°]	Alpha [°]	Dmin Lc [lb]	Psi [°]	Alpha [°]
BEB	3R-403	2408 (3+)	-1.40	2.81	44 (1-)	-0.03	0.06
1	6N-401	-5455 (3+)	3.20	-6.36	-1998 (3-)	5.09	-2.57
2	8N-401	-7133 (3+)	10.61	-8.28	-3632 (3-)	8.80	-4.66
3	6T-401	5061 (3-)	12.07	6.46	805 (1+)	14.80	0.93
4	8T-401	8917 (3+)	16.89	9.63	3317 (1-)	15.30	4.16
5	8T-401	9443 (3+)	15.23	9.79	3765 (1-)	13.78	4.66
6	6T-401	7122 (6)	11.80	8.16	961 (6)	13.75	1.03
7	6T-401	6548 (6)	17.88	6.87	1807 (6)	15.39	1.87
8	6T-401	6146 (6)	15.19	6.43	1517 (6)	14.65	1.50
9	8T-401	9621 (3+)	16.40	8.67	4004 (1-)	14.64	4.69
10	6T-401	6981 (6)	13.83	6.76	1842 (6)	13.09	1.71
11	6T-401	6604 (6)	16.21	6.21	1114 (6)	14.36	1.01
12	12N-401	-12683 (6)	21.01	-11.10	-8529 (6)	22.67	-7.76
13	6T-401	6696 (3+)	26.70	5.49	2489 (1-)	26.05	2.78
14	6T-401	6424 (6)	27.54	5.40	1298 (6)	26.39	1.07
15	6T-401	6227 (6)	28.22	5.00	1542 (6)	27.02	1.21
16	6T-501	11718 (3+)	26.07	8.32	5876 (1-)	24.98	6.16
17	6T-501	12868 (3+)	17.37	9.19	6935 (1-)	18.36	7.25
18	6T-501	13096 (3+)	10.15	9.29	7033 (1-)	11.15	7.33
19	6T-501	12131 (3+)	3.26	8.62	6362 (1-)	4.28	6.60
TEB	3R-403	4494 (3+)	1.58	3.16	1194 (1-)	0.62	1.24

Maximum and minimum tower loads - DH

Rope point	Sheave assembly	Dmax Lc [lb]	Psi [°]	Alpha [°]	Dmin Lc [lb]	Psi [°]	Alpha [°]
BEB	3R-403	2851 (4+)	-1.69	3.38	59 (1-)	-0.04	0.08
1	6N-401	-5397 (3+)	3.21	-6.40	-2104 (4-)	1.39	-2.76
2	8N-401	-6947 (4+)	10.62	-8.25	-3441 (4-)	12.48	-4.48
3	6T-401	5086 (4-)	11.96	6.69	823 (1+)	14.79	0.98
4	8T-401	8874 (4+)	16.94	9.91	3291 (1-)	15.30	4.26
5	8T-401	9375 (4+)	15.27	10.10	3719 (1-)	13.78	4.76
6	6T-401	7193 (6)	11.70	8.57	1031 (6)	13.74	1.15
7	6T-401	6547 (6)	18.02	7.19	1811 (6)	15.41	1.97
8	6T-401	6167 (6)	15.19	6.77	1538 (6)	14.63	1.60
9	8T-401	9483 (4+)	16.52	9.03	3931 (1-)	14.65	4.80
10	6T-401	7018 (6)	10.86	7.33	1865 (6)	13.07	1.84
11	6T-401	6664 (6)	13.47	6.79	1181 (6)	14.38	1.14
12	12N-401	-11711 (6)	21.02	-10.99	-7684 (6)	20.12	-7.57
13	6T-401	6591 (4+)	26.72	5.84	2427 (1-)	26.01	2.86
14	6T-401	6493 (6)	27.64	5.90	1367 (6)	26.39	1.21
15	6T-401	6261 (6)	26.02	5.55	1576 (6)	27.03	1.33
16	6T-501	11137 (4+)	26.17	8.55	5602 (1-)	25.00	6.23
17	6T-501	11903 (4+)	19.32	9.12	6482 (1-)	18.36	7.23
18	6T-501	12119 (4+)	10.11	9.42	6567 (1-)	11.16	7.34
19	6T-501	11243 (4+)	3.16	8.83	5907 (1-)	4.28	6.62
TEB	3R-403	4382 (4+)	1.70	3.41	1108 (1-)	0.62	1.24

Ropeline calculation

Gondola Section 1

8-MGD

Project no.: SAA0002176

Index Text

Date

Name

created: 2015-07-05 RUSSO

vis.:

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

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Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Key to symbols and abbreviations

Alpha	Deflection angle
Ao	Suspension above
Au	Suspension below
BW	Bullwheel
D	Tower load (more exact: resulting force in the rope point)
DCG	Distance CTF to ground
DS	Load per sheave
EB	Entrance beam
Eps	Tower angle
F2	Distance sheave assembly main axle - rope point
HA	Suspension height of sheave assembly
HC	Distance bottom of yoke to bottom of connection plate
HG	Grout (more exact: distance bottom of shaft to CTF)
HS	Shaft length
HY	Yoke height
IO	In Operation
Lc	Load configuration
OO	Out of Operation
Phi	Rope angle
PhiH	Horizontal lead-on/lead-off angle
Psi	Tower load reaction angle
Q	Cross inclination
RP	Rope point
T	Rope tension
T0	Basic tension force
T0+	Basic tension force with overtension
T0-	Basic tension force with undertension
W	Wind
WS	Wind load per sheave
XA,ZA	Coordinates of sheave assembly main axle
XF,ZF	Coordinates CTF
XR,ZR	Rope point coordinates
f	Sag
fy	Horizontal rope deflection
h	Elevation difference
l	Horizontal length
l*	Inclined length
q	Wind pressure
r	Rope radius

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]Location: Telluride, CO
Client: Telluride Transit Authorities**Main data**

Installation type:	8-MGD
Direction of rotation:	Counter-clockwise
Uphill transportation side:	Right
Uphill capacity:	100 %
Downhill capacity:	100 %
Drive location:	Top
Tension location:	Bottom
Tensioning type:	Hydraulic tensioning
Tensioning force:	92846 lb
Basic tension force:	46423 lb
Regulation:	US design parameters <modified>
Haul rope:	47 mm 6x25 FW 2160 Fatzer
Carriers:	8 Omega IV-LWI CWAA108C
Number of carriers:	75 (on line: 59.1; in stations: 16.0)
Spacing:	200.00 ft

Geometrical data

Horizontal length:	5610.84 ft
Vertical rise:	1775.02 ft
Average grade:	31.64 %
Inclined length:	5914.68 ft
Transportation length:	6055.76 ft
Spliced rope length:	12056.75 ft
Rope gauge:	16.73 ft

Speed/Capacity

Main drive:	Rope speed:	1000 fpm	(backwards: -984 fpm)
	Capacity:	2400 persons/h	
	Interval:	12.00 s	
	Trip time:	7.51 min	
Evac drive:	Rope speed:	197 fpm	
	Capacity:	472 persons/h	
	Interval:	60.96 s	
	Trip time:	31.64 min	

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Carrier

Manufacturer: CWA
 Type: 8 Omega IV-LWI
 Dead weight: 1592 lb
 Pay load: 1358 lb or 8 persons
 Total weight: 2950 lb
 Length (L): 6.17 ft
 Dist. CL rope to top of seat (HS): 13.52 ft
 Dist. CL rope to bottom of carrier (H0): 13.52 ft
 Dist. CL rope to bottom of carrier (Hmax): 13.71 ft
 Projection area, in line: 49.5 sqft (Cf: 1.00)
 Projection area, perpendicular: 43.1 sqft (Cf: 1.00)
 Side swing 0.00 rad: Yi0 / Yo0: 3.97 / 3.38 ft
 Side swing 0.20 rad: Yi20 / Yo20: 5.94 / 5.54 ft
 Side swing 0.34 rad: Yi34 / Yo34: 7.55 / 7.12 ft
 Grip: A108C
 Grip pulling force: 5845.0 lb

Haul rope data

Manufacturer: Fatzer
 Rope construction: 6x25 FW compact type
 Diameter: 1.85 in
 Dia. outer wire: 0.124 in
 Metallic area: 1.531 sqin
 Rope weight: 5.78 lb/ft
 Tensile strength: 313281 psi
 Calculated breaking strength: 479742 lb
 Minimum breaking strength: 412749 lb
 Elastic modulus: 1450368 psi
 Thermal expansion factor: 1.2e-05 1/K
 Permanent elongation: 1.50 ‰
 Stranding formula: FEC + 6 x (1 + 6 + [6] + 12)
 Spliced rope length: 12056.8 ft
 Order length: 12343 ft (incl. Splice length: 185 ft; Surcharge: 98 ft)

Rope sheaves

Sheave	Side plate diameter	Base of Groove diameter
403	430 mm	400 mm
420C.1	455 mm	420 mm
501	530 mm	485 mm

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Station: Bottom station

Type: Custom Return movable
Entrance beam: XR: 96.50 ft ZR: 8769.51 ft
Gauge at entrance beam: 16.73 ft
Rope angle at entrance beam: 0.00 °
Distance EB to loading/unloading: 70.54 ft
Rope length in station: 114.73 ft
Distance EB to grip checking: 0.46 ft
Min. distance EB to mid 1. line sheave T; N; T/N: 0.00 ft; 0.00 ft; 0.00 ft
Driving speed: 68.90 fpm
Deceleration anti-collision monitoring: 3.28 ft/s²
Station dwell time: 85.25 s (1426 Impulses)
Tire conveyor friction: 224.8 lb
Rope deflections per side: 5.00° vertical, 4 Sheaves (501)
Return bullwheel: Ø = 16.73 ft; I = 16500 kgm²
Tmax: 49178 lb (Lc: 3+)
Tmin: 43687 lb (Lc: 4-)

Station: Top station

Type: Custom Drive fixed
Entrance beam: XR: 5707.34 ft ZR: 10544.53 ft
Gauge at entrance beam: 16.73 ft
Rope angle at entrance beam: 0.00 °
Distance EB to loading/unloading: 70.54 ft
Rope length in station: 112.66 ft
Distance EB to grip checking: 0.46 ft
Min. distance EB to mid 1. line sheave T; N; T/N: 0.00 ft; 0.00 ft; 0.00 ft
Driving speed: 68.90 fpm
Deceleration anti-collision monitoring: 3.28 ft/s²
Station dwell time: 106.30 s (1778 Impulses)
Tire conveyor friction: 224.8 lb
Rope deflections per side: 5.00° vertical, 3 Sheaves (501)
1.35° horizontal, 2 Sheaves (501)
Min. wrap angle: 177.30 °
Drive bullwheel: Ø = 15.75 ft; I = 11000 kgm²
Tmax: 90828 lb (Lc: 3+)
Tmin: 51848 lb (Lc: 1-)

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]**Main drive:**

Total inertia: 11.87 lbft²
Acceleration: 0.49 ft/s²
Deceleration: -1.31 ft/s²

Gearbox:

Manufacturer: Doppelmayr-Lohmann
Type: GPW 270
Nominal torque: 200321.9 ftlb
Max. torque: 280421.1 ftlb
Ratio: 94.25
Efficiency: 0.93
Inertia: 11.87 lbft²
Req. input speed: 1906 rpm
Utilisation: 84 %
Req. torque continuous: 168703.5 ftlb (Lc: 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO)
Req. torque maximum: 173679.8 ftlb (Lc: 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO)

Electric motor:

Manufacturer:
Type:
Rating: 0 HP
Speed: 0 rpm
Inertia: 0.00 lbft²
Req. power continuous: 698 HP (Lc: 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO)

Service brake(s):

Inertia: 0.00 lbft²
Max. braking torque: 0.00 ftlb (0.00 kN @ 0.00 m)
Req. braking torque: 1353.86 ftlb (Lc: 0% UH Load, 100% DH Load; Braking (0.50m/s²); T0)

Emergency brake(s):

Max. braking torque: 0.0 ftlb (0.00 kN @ 0.00 m)
Req. braking torque: 127601.3 ftlb (Lc: 0% UH Load, 100% DH Load; Braking (0.50m/s²); T0)

Evac drive:

System: rim gear on bullwheel
Diesel engine:
Efficiency: continuous, starting: 0.80 braking: 1.00
Req. power continuous: 172 HP (Lc: 100% UH Load, 0% DH Load; Forward operation (1.0m/s); T0+; IO)
Req. power starting: 0 HP (Lc:)
Req. power braking: 0 HP (Lc:)
Req. M_{dmax} bullwheel: 0.0 ftlb (Lc:)

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Rope point coordinates - UH

Rope point	XR[ft]	ZR[ft]	Psi[°]	Alpha[°]	F2[ft]
BEB	96.50	8769.51	-0.9	1.8	0.00
1	109.86	8769.45	2.1	4.5	-1.36
2	124.07	8770.99	9.7	6.9	-1.52
3	212.18	8795.02	13.1	4.8	1.15
4	707.06	8934.81	16.1	8.2	1.39
5	1163.81	9054.34	14.5	8.4	1.40
6	1643.02	9164.69	12.8	5.9	1.33
7	2219.24	9316.78	16.9	5.0	1.20
8	2404.67	9366.35	14.6	4.6	1.18
9	2963.96	9519.40	15.6	7.5	1.41
10	3273.79	9592.07	12.0	5.1	1.20
11	3838.75	9728.39	15.3	4.3	1.17
12	4192.56	9820.48	21.8	9.5	-3.34
13	4364.43	9908.45	26.2	4.7	1.18
14	4824.77	10132.55	27.0	3.9	1.16
15	5257.15	10350.59	26.7	3.6	1.15
16	5608.37	10530.06	25.7	6.9	1.33
17	5625.05	10536.91	18.7	7.3	1.35
18	5640.26	10541.27	11.1	9.9	1.60
19	5658.97	10543.59	4.5	7.0	1.35
TEB	5707.34	10544.53	1.1	2.2	0.00

Rope point coordinates - DH

Rope point	XR[ft]	ZR[ft]	Psi[°]	Alpha[°]	F2[ft]
BEB	96.50	8769.51	-1.0	1.9	0.00
1	109.90	8769.46	3.9	4.5	-1.36
2	124.07	8770.99	9.7	6.9	-1.52
3	212.13	8795.06	15.0	5.0	1.20
4	707.05	8934.85	16.1	8.4	1.44
5	1163.85	9054.35	13.0	8.8	1.40
6	1643.03	9164.69	12.7	6.3	1.33
7	2219.23	9316.79	17.0	5.3	1.21
8	2404.67	9366.36	14.6	4.9	1.19
9	2963.99	9519.41	14.4	7.9	1.41
10	3273.79	9592.08	11.9	5.4	1.21
11	3838.77	9728.41	14.0	4.8	1.19
12	4192.57	9820.48	21.9	9.3	-3.34
13	4364.43	9908.46	26.2	4.9	1.19
14	4824.78	10132.57	25.9	4.3	1.17
15	5257.14	10350.60	26.6	3.9	1.16
16	5608.36	10530.06	25.8	7.0	1.33
17	5625.02	10536.93	19.5	7.7	1.38
18	5640.25	10541.20	12.0	9.5	1.53
19	5658.99	10543.60	3.6	7.4	1.36
TEB	5707.34	10544.53	1.1	2.2	0.00

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Span geometry - UH

Span	l[ft]	h[ft]	l*[ft]	Chord angle[°]	Chord angle[%]
BEB - 1	13.36	-0.06	13.36	-0.24	-0.42
1 - 2	14.21	1.54	14.29	6.18	10.83
2 - 3	88.11	24.03	91.33	15.25	27.27
3 - 4	494.88	139.79	514.25	15.77	28.25
4 - 5	456.75	119.54	472.13	14.67	26.17
5 - 6	479.21	110.35	491.75	12.97	23.03
6 - 7	576.21	152.09	595.95	14.79	26.39
7 - 8	185.43	49.57	191.94	14.97	26.73
8 - 9	559.29	153.06	579.85	15.30	27.37
9 - 10	309.83	72.67	318.24	13.20	23.45
10 - 11	564.96	136.32	581.17	13.57	24.13
11 - 12	353.81	92.09	365.60	14.59	26.03
12 - 13	171.87	87.97	193.08	27.11	51.19
13 - 14	460.34	224.10	511.99	25.96	48.68
14 - 15	432.38	218.04	484.24	26.76	50.43
15 - 16	351.22	179.47	394.42	27.07	51.10
16 - 17	16.68	6.85	18.03	22.32	41.05
17 - 18	15.21	4.36	15.82	16.00	28.68
18 - 19	18.71	2.32	18.85	7.07	12.40
19 - TEB	48.37	0.94	48.38	1.11	1.94

Span geometry - DH

Span	l[ft]	h[ft]	l*[ft]	Chord angle[°]	Chord angle[%]
BEB - 1	13.40	-0.05	13.40	-0.23	-0.40
1 - 2	14.17	1.54	14.25	6.19	10.85
2 - 3	88.06	24.07	91.29	15.29	27.33
3 - 4	494.92	139.79	514.28	15.77	28.25
4 - 5	456.80	119.50	472.17	14.66	26.16
5 - 6	479.18	110.34	491.72	12.97	23.03
6 - 7	576.21	152.09	595.94	14.79	26.40
7 - 8	185.44	49.57	191.95	14.97	26.73
8 - 9	559.32	153.05	579.88	15.30	27.36
9 - 10	309.80	72.67	318.21	13.20	23.46
10 - 11	564.98	136.33	581.20	13.57	24.13
11 - 12	353.79	92.07	365.58	14.59	26.02
12 - 13	171.86	87.98	193.07	27.11	51.19
13 - 14	460.36	224.12	512.01	25.96	48.68
14 - 15	432.36	218.03	484.22	26.76	50.43
15 - 16	351.22	179.46	394.42	27.07	51.10
16 - 17	16.66	6.86	18.02	22.38	41.18
17 - 18	15.22	4.28	15.81	15.69	28.09
18 - 19	18.74	2.39	18.89	7.28	12.77
19 - TEB	48.35	0.93	48.36	1.11	1.93

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Calculation mode and load configurations

Calculation method: Point load (increments of 2.00 ft)

Calculated load configurations:

- | | |
|----|--|
| 1 | Rope Only; Forward operation (5.1m/s); T0; IO |
| 1+ | Rope Only; Forward operation (5.1m/s); T0+; IO |
| 1- | Rope Only; Forward operation (5.1m/s); T0-; IO |
| 2 | Rope + Empty Carriers; Forward operation (5.1m/s); T0; IO |
| 2+ | Rope + Empty Carriers; Forward operation (5.1m/s); T0+; IO |
| 2- | Rope + Empty Carriers; Forward operation (5.1m/s); T0-; IO |
| 3 | 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0; IO |
| 3+ | 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO |
| 3- | 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0-; IO |
| 4 | 100% UH+DH Load; Forward operation (5.1m/s); T0; IO |
| 4+ | 100% UH+DH Load; Forward operation (5.1m/s); T0+; IO |
| 4- | 100% UH+DH Load; Forward operation (5.1m/s); T0-; IO |
| 5 | 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0; IO |
| 5+ | 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0+; IO |
| 5- | 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0-; IO |
| 6 | 100% UH+DH Load, 2 Spans Rope Only; Forward operation (5.1m/s); T0; IO |

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Rope tension extremes, rope design factors, shear ratio

Rope tension extremes at nominal tension without dynamics

Tmax = 88415 lb (Rp: TBW; Lc: 3)

Tmin = 45614 lb (Rp: 2; Lc: 4)

Rope tension extremes at nominal tension with dynamics

Tmax = 88415 lb (Rp: TBW; Lc: 3)

Tmin = 45614 lb (Rp: 2; Lc: 4)

Rope tension extremes without dynamics

Tmax = 90828 lb (Rp: TBW; Lc: 3+)

Tmin = 43317 lb (Rp: 2; Lc: 4-)

Rope tension extremes with dynamics

Tmax = 90828 lb (Rp: TBW; Lc: 3+)

Tmin = 43317 lb (Rp: 2; Lc: 4-)

Rope design factor: min.: 4.67 max.: 9.53

Minimum shear ratio carrier: 15.46

Specific weights:

rope only:	5.7789 lb/ft
rope + empty carriers:	13.7376 lb/ft
rope + loaded carriers UH:	20.5279 lb/ft
rope + loaded carriers DH:	20.5279 lb/ft

Max. angle change: 0.0862 rad; 4.94° (Rp: 3; to top; DH)

Greatest rope inclination: 61.13%; 31.44° (Rp: 15; to bottom; DH; Lc: 6)

Required min. pull force: 4615 lb

Bullwheels - SumTmax and Mdmax, extreme values with dynamic

	SumTmax [lb]	(Lc)	Md [lbft]	Mdmax [lbft]	(Lc)	SumT [lb]
Drive bullwheel	171241	(4+)	86878	173680	(3+)	160078
Return bullwheel	97488	(1+)				
	pmax [lb/in²]	(Lc)				
Drive bullwheel	735	(4+)				
Return bullwheel	394	(1+)				

Bullwheels - SumTmax and Mdmax, values at nominal tension without dynamic

	SumTmax [lb]	(Lc)	Md [lbft]	Mdmax [lbft]	(Lc)	SumT [lb]
Drive bullwheel	166586	(4)	81367	167377	(3)	155422
Return bullwheel	92846	(1)				

Clearance envelope

Required line width: 48.11 ft

Required line width - UH - Right: 23.96 ft (RP: 6; Lc: Rope + Empty Carriers; Forward operation; T0-; IO)

Required line width - DH - Left: 24.14 ft (RP: 6; Lc: Rope + Empty Carriers; Forward operation; T0-; IO)

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Circumferential forces, torques, required power (min ... geo. mean ... max)

Lc	Circumferential force [lb]	torque [ftlb]	required power [HP]	μ nec
1	5254	44049	182	0.030
1+	5404	45343	188	0.030
1-	5103	42755	177	0.030
2	7294 ... 7726 ... 8089	60794 ... 64189 ... 67047	251 ... 266 ... 277	0.037
2+	7435 ... 7876 ... 8249	62009 ... 65483 ... 68420	257 ... 271 ... 283	0.036
2-	7155 ... 7575 ... 7927	59585 ... 62895 ... 65662	246 ... 260 ... 272	0.037
3	20425 ... 20792 ... 21407	164475 ... 167377 ... 172232	680 ... 692 ... 712	0.090
3+	20598 ... 20947 ... 21577	165948 ... 168704 ... 173680	686 ... 698 ... 718	0.088
3-	20245 ... 20637 ... 21236	162947 ... 166045 ... 170774	674 ... 687 ... 706	0.092
4	9205 ... 9835 ... 10423	76419 ... 81367 ... 85995	316 ... 337 ... 356	0.041
4+	9333 ... 9985 ... 10521	77532 ... 82659 ... 86878	321 ... 342 ... 359	0.040
4-	9064 ... 9685 ... 10330	75186 ... 80069 ... 85151	311 ... 331 ... 352	0.041
5	-3965 ... -3232 ... -2789	-27599 ... -21838 ... -18368	-99 ... -78 ... -66	0.017
5+	-3835 ... -3086 ... -2658	-26463 ... -20580 ... -17227	-95 ... -74 ... -62	0.016
5-	-4094 ... -3377 ... -2918	-28723 ... -23092 ... -19495	-103 ... -83 ... -70	0.018
6	2070 ... 9861 ... 16624	20062 ... 81417 ... 134666	83 ... 337 ... 557	0.068

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Maximum and minimum rope tensions - UH

Rope point	Tmax [lb]	Lc	Tmin [lb]	Lc
BBW	48890	(1+)	44234	(1-)
BEB	49178	(3+)	44443	(1-)
1	49253	(2+)	44428	(3-)
2	49762	(3+)	44549	(3-)
3	50615	(3+)	44857	(3-)
4	54002	(3+)	45728	(1-)
5	56626	(3+)	46510	(1-)
6	58896	(3+)	47249	(1-)
7	62321	(3+)	48167	(1-)
8	63547	(3+)	48505	(1-)
9	67053	(3+)	49436	(1-)
10	68779	(3+)	49964	(1-)
11	71725	(3+)	50807	(1-)
12	73944	(3+)	51383	(1-)
13	75783	(3+)	52134	(1-)
14	80898	(3+)	53501	(1-)
15	85243	(3+)	54807	(1-)
16	89311	(3+)	55893	(1-)
17	89781	(3+)	56076	(1-)
18	90159	(3+)	56258	(1-)
19	90412	(3+)	56493	(1-)
TEB	90465	(3+)	56651	(1-)
TBW	90828	(3+)	56951	(1-)

Maximum and minimum rope tensions - DH

Rope point	Tmax [lb]	Lc	Tmin [lb]	Lc
BBW	48598	(1+)	43970	(1-)
BEB	48380	(1+)	43687	(4-)
1	48445	(4+)	43472	(4-)
2	48653	(4+)	43317	(4-)
3	49079	(4+)	43326	(4-)
4	52251	(4+)	44302	(1-)
5	54275	(4+)	44892	(1-)
6	55925	(4+)	45490	(1-)
7	58963	(4+)	46317	(1-)
8	59814	(4+)	46557	(1-)
9	62841	(4+)	47335	(1-)
10	64121	(4+)	47699	(1-)
11	66648	(4+)	48442	(1-)
12	68525	(4+)	48745	(1-)
13	69755	(4+)	49184	(1-)
14	74447	(4+)	50432	(1-)
15	78467	(4+)	51642	(1-)
16	82034	(4+)	52545	(1-)
17	82124	(4+)	52429	(1-)
18	81841	(4+)	52259	(1-)
19	81553	(4+)	52128	(1-)
TEB	81334	(4+)	52104	(1-)
TBW	80954	(4+)	51848	(1-)

Project [-no.]: 8-MGD Gondola Section 1 [SAA0002176]

Maximum and minimum tower loads - UH

Rope point	Sheave assembly	Dmax Lc [lb]	Psi [°]	Alpha [°]	Dmin Lc [lb]	Psi [°]	Alpha [°]
BEB	3R-403	2754 (3+)	-1.61	3.21	226 (1-)	-0.15	0.29
1	6N-420C	-5434 (1+)	2.97	-6.33	-2117 (3-)	1.17	-2.73
2	8N-420C	-7514 (2+)	10.59	-8.73	-3937 (3-)	8.76	-5.05
3	6T-420C	6460 (3-)	11.46	8.24	1231 (1+)	14.84	1.42
4	6T-501	11088 (3+)	16.95	11.81	3636 (1-)	15.30	4.55
5	6T-501	11556 (3+)	15.16	11.74	4086 (1-)	13.78	5.03
6	6T-501	9365 (6)	11.81	10.53	1307 (6)	13.73	1.36
7	6T-420C	7743 (6)	18.41	7.99	2026 (6)	15.44	2.02
8	6T-420C	7403 (6)	14.63	7.56	1791 (6)	14.62	1.70
9	6T-501	11650 (3+)	16.51	9.99	4336 (1-)	14.67	5.02
10	6T-420C	8837 (6)	10.92	8.27	2109 (6)	13.06	1.86
11	6T-420C	8456 (6)	16.28	7.44	1460 (6)	14.34	1.24
12	12N-420C	-13764 (6)	21.06	-11.27	-8843 (6)	22.58	-7.70
13	6T-420C	8147 (3+)	26.38	6.18	2867 (1-)	26.08	3.15
14	6T-420C	8412 (6)	27.51	6.56	1534 (6)	26.39	1.18
15	6T-420C	7786 (6)	26.30	5.82	1842 (6)	27.00	1.32
16	6T-501	12331 (3+)	26.27	7.97	5708 (1-)	25.19	5.84
17	6T-501	12590 (3+)	18.27	8.15	6278 (1-)	19.16	6.41
18	8T-501	16738 (3+)	10.62	10.83	8881 (1-)	11.53	9.03
19	6T-501	12389 (3+)	4.99	7.93	6075 (1-)	4.05	6.15
TEB	3R-403	4791 (3+)	1.53	3.06	1239 (1-)	0.63	1.25

Maximum and minimum tower loads - DH

Rope point	Sheave assembly	Dmax Lc [lb]	Psi [°]	Alpha [°]	Dmin Lc [lb]	Psi [°]	Alpha [°]
BEB	3R-403	2980 (4+)	-1.77	3.53	216 (1-)	-0.14	0.28
1	6N-420C	-5335 (3+)	2.98	-6.33	-1983 (4-)	4.84	-2.59
2	8N-420C	-7366 (4+)	10.61	-8.74	-3814 (4-)	8.75	-5.01
3	6T-420C	6511 (4-)	15.08	8.40	1273 (1+)	14.83	1.51
4	6T-501	11048 (4+)	17.00	12.17	3613 (1-)	15.29	4.67
5	6T-501	11472 (4+)	12.15	12.42	4033 (1-)	13.77	5.14
6	6T-501	9449 (6)	11.71	11.10	1385 (6)	13.73	1.51
7	6T-420C	7743 (6)	18.57	8.38	2035 (6)	15.47	2.14
8	6T-420C	7427 (6)	14.61	7.99	1814 (6)	14.59	1.83
9	6T-501	11466 (4+)	14.03	10.67	4255 (1-)	14.69	5.14
10	6T-420C	8879 (6)	10.74	8.87	2140 (6)	13.04	2.02
11	6T-420C	8525 (6)	13.72	8.16	1546 (6)	14.36	1.42
12	12N-420C	-12575 (6)	21.08	-11.17	-7779 (6)	22.69	-7.33
13	6T-420C	8036 (4+)	26.37	6.61	2811 (1-)	26.05	3.27
14	6T-420C	8496 (6)	25.35	7.32	1620 (6)	26.40	1.35
15	6T-420C	7821 (6)	26.25	6.33	1876 (6)	27.01	1.47
16	6T-501	11639 (4+)	26.44	8.18	5365 (1-)	25.25	5.84
17	6T-501	12278 (4+)	19.95	8.59	6222 (1-)	19.04	6.79
18	8T-501	14755 (4+)	12.45	10.42	7782 (1-)	11.48	8.52
19	6T-501	11750 (4+)	3.12	8.39	5816 (1-)	4.14	6.39
TEB	3R-403	4535 (4+)	1.62	3.23	1146 (1-)	0.63	1.26

Ropeline calculation

Gondola Section 2

8-MGD

Project no.: SAP0000008

Index Text

Date

Name

created: 2015-07-05 RUSSO

vis.:

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

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Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Key to symbols and abbreviations

Alpha	Deflection angle
Ao	Suspension above
Au	Suspension below
BW	Bullwheel
D	Tower load (more exact: resulting force in the rope point)
DCG	Distance CTF to ground
DS	Load per sheave
EB	Entrance beam
Eps	Tower angle
F2	Distance sheave assembly main axle - rope point
HA	Suspension height of sheave assembly
HC	Distance bottom of yoke to bottom of connection plate
HG	Grout (more exact: distance bottom of shaft to CTF)
HS	Shaft length
HY	Yoke height
IO	In Operation
Lc	Load configuration
OO	Out of Operation
Phi	Rope angle
PhiH	Horizontal lead-on/lead-off angle
Psi	Tower load reaction angle
Q	Cross inclination
RP	Rope point
T	Rope tension
T0	Basic tension force
T0+	Basic tension force with overtension
T0-	Basic tension force with undertension
W	Wind
WS	Wind load per sheave
XA,ZA	Coordinates of sheave assembly main axle
XF,ZF	Coordinates CTF
XR,ZR	Rope point coordinates
f	Sag
fy	Horizontal rope deflection
h	Elevation difference
l	Horizontal length
l*	Inclined length
q	Wind pressure
r	Rope radius

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]Location: Telluride, CO
Client: Telluride Transit Authorities**Main data**

Installation type:	8-MGD
Direction of rotation:	Counter-clockwise
Uphill transportation side:	Right
Uphill capacity:	100 %
Downhill capacity:	100 %
Drive location:	Top
Tension location:	Bottom
Tensioning type:	Hydraulic tensioning
Tensioning force:	92846 lb
Basic tension force:	46423 lb
Regulation:	US design parameters <modified>
Haul rope:	44 mm 6x25 FW 1960 Fatzer
Carriers:	8 Omega IV-LWI CWAA108C
Number of carriers:	27 (on line: 19.6; in stations: 8.0)
Spacing:	400.00 ft

Geometrical data

Horizontal length:	3774.35 ft
Vertical rise:	995.04 ft
Average grade:	26.36 %
Inclined length:	3919.51 ft
Transportation length:	4060.58 ft
Spliced rope length:	8066.41 ft
Rope gauge:	16.73 ft

Speed/Capacity

Main drive:	Rope speed:	1000 fpm	(backwards: -984 fpm)
	Capacity:	1200 persons/h	
	Interval:	24.00 s	
	Trip time:	5.52 min	
Evac drive:	Rope speed:	197 fpm	
	Capacity:	236 persons/h	
	Interval:	121.92 s	
	Trip time:	21.51 min	

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Carrier

Manufacturer: CWA
 Type: 8 Omega IV-LWI
 Dead weight: 1592 lb
 Pay load: 1358 lb or 8 persons
 Total weight: 2950 lb
 Length (L): 6.17 ft
 Dist. CL rope to top of seat (HS): 13.52 ft
 Dist. CL rope to bottom of carrier (H0): 13.52 ft
 Dist. CL rope to bottom of carrier (Hmax): 13.71 ft
 Projection area, in line: 49.5 sqft (Cf: 1.00)
 Projection area, perpendicular: 43.1 sqft (Cf: 1.00)
 Side swing 0.00 rad: Yi0 / Yo0: 3.97 / 3.38 ft
 Side swing 0.20 rad: Yi20 / Yo20: 5.94 / 5.54 ft
 Side swing 0.34 rad: Yi34 / Yo34: 7.55 / 7.12 ft
 Grip: A108C
 Grip pulling force: 5845.0 lb

Haul rope data

Manufacturer: Fatzer
 Rope construction: 6x25 FW compact type
 Diameter: 1.73 in
 Dia. outer wire: 0.116 in
 Metallic area: 1.349 sqin
 Rope weight: 5.09 lb/ft
 Tensile strength: 284274 psi
 Calculated breaking strength: 383299 lb
 Minimum breaking strength: 333616 lb
 Elastic modulus: 1450368 psi
 Thermal expansion factor: 1.2e-05 1/K
 Permanent elongation: 1.50 ‰
 Spliced rope length: 8066.4 ft
 Order length: 8340 ft (incl. Splice length: 173 ft; Surcharge: 98 ft)

Rope sheaves

Sheave	Side plate diameter	Base of Groove diameter
401	412 mm	382 mm
403	430 mm	400 mm
501	530 mm	485 mm

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Station: Bottom station

Type: Custom Return movable
Entrance beam: XR: -9689.63 ft ZR: 9549.48 ft
Gauge at entrance beam: 16.73 ft
Rope angle at entrance beam: 0.00 °
Distance EB to loading/unloading: 70.54 ft
Rope length in station: 114.73 ft
Distance EB to grip checking: 0.46 ft
Min. distance EB to mid 1. line sheave T; N; T/N: 0.00 ft; 0.00 ft; 0.00 ft
Driving speed: 68.90 fpm
Deceleration anti-collision monitoring: 3.28 ft/s²
Station dwell time: 85.19 s (1425 Impulses)
Tire conveyor friction: 224.8 lb
Rope deflections per side: 5.00° vertical, 4 Sheaves (501)
Return bullwheel: Ø = 16.73 ft; I = 16500 kgm²
Tmax: 49177 lb (Lc: 3+)
Tmin: 43684 lb (Lc: 4-)

Station: Top station

Type: Custom Drive fixed
Entrance beam: XR: -5915.28 ft ZR: 10544.52 ft
Gauge at entrance beam: 16.73 ft
Rope angle at entrance beam: 0.00 °
Distance EB to loading/unloading: 70.54 ft
Rope length in station: 112.66 ft
Distance EB to grip checking: 0.46 ft
Min. distance EB to mid 1. line sheave T; N; T/N: 0.00 ft; 0.00 ft; 0.00 ft
Driving speed: 68.90 fpm
Deceleration anti-collision monitoring: 3.28 ft/s²
Station dwell time: 106.30 s (1778 Impulses)
Tire conveyor friction: 224.8 lb
Rope deflections per side: 5.00° vertical, 3 Sheaves (501)
1.35° horizontal, 2 Sheaves (501)
Min. wrap angle: 177.30 °
Drive bullwheel: Ø = 15.75 ft; I = 11000 kgm²
Tmax: 64700 lb (Lc: 3+)
Tmin: 47224 lb (Lc: 1-)

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Main drive:

Total inertia: 11.87 lbft²
Acceleration: 0.49 ft/s²
Deceleration: -1.31 ft/s²

Gearbox:

Manufacturer: Doppelmayr-Lohmann
Type: GPW 215
Nominal torque: 104512.6 ftlb
Max. torque: 146332.3 ftlb
Ratio: 89.58
Efficiency: 0.93
Inertia: 11.87 lbft²
Req. input speed: 1811 rpm
Utilisation: 68 %
Req. torque continuous: 71222.3 ftlb (Lc: 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO)
Req. torque maximum: 79272.9 ftlb (Lc: 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO)

Electric motor:

Manufacturer:
Type:
Rating: 0 HP
Speed: 0 rpm
Inertia: 0.00 lbft²
Req. power continuous: 295 HP (Lc: 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO)

Service brake(s):

Inertia: 0.00 lbft²
Max. braking torque: 0.00 ftlb (0.00 kN @ 0.00 m)
Req. braking torque: 351.27 ftlb (Lc: 0% UH Load, 100% DH Load; Braking (0.50m/s²); T0)

Emergency brake(s):

Max. braking torque: 0.0 ftlb (0.00 kN @ 0.00 m)
Req. braking torque: 31466.7 ftlb (Lc: 0% UH Load, 100% DH Load; Braking (0.50m/s²); T0)

Evac drive:

System: rim gear on bullwheel
Diesel engine:
Efficiency: continuous, starting: 0.80 braking: 1.00
Req. power continuous: 72 HP (Lc: 100% UH Load, 0% DH Load; Forward operation (1.0m/s); T0+; IO)
Req. power starting: 0 HP (Lc:)
Req. power braking: 0 HP (Lc:)
Req. M_{dmax} bullwheel: 0.0 ftlb (Lc:)

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Rope point coordinates - UH

Rope point	XR[ft]	ZR[ft]	Psi[°]	Alpha[°]	F2[ft]
BEB	-9689.63	9549.48	-0.9	1.7	0.00
1	-9671.53	9549.39	5.1	7.2	-1.41
2	-9654.82	9551.96	14.6	11.7	-2.96
3	-9498.57	9617.28	21.5	6.4	1.10
4	-9017.08	9793.02	21.5	4.0	0.97
5	-8608.88	9948.43	19.1	6.2	1.08
6	-8148.44	10107.19	17.8	8.3	1.24
7	-7634.02	10248.48	14.4	8.2	1.24
8	-7273.75	10319.90	10.6	5.8	1.14
9	-6698.87	10417.81	9.4	4.4	1.02
10	-6148.59	10519.75	11.0	5.2	1.06
11	-5988.05	10544.51	5.2	10.7	1.47
TEB	-5915.28	10544.52	0.8	1.5	0.00

Rope point coordinates - DH

Rope point	XR[ft]	ZR[ft]	Psi[°]	Alpha[°]	F2[ft]
BEB	-9689.63	9549.48	-1.0	2.0	0.00
1	-9671.52	9549.40	5.1	7.1	-1.41
2	-9654.83	9551.96	14.6	11.6	-2.95
3	-9498.57	9617.29	21.5	6.6	1.11
4	-9017.07	9793.09	19.8	4.3	1.03
5	-8608.88	9948.44	19.0	6.4	1.09
6	-8148.44	10107.20	17.8	8.6	1.25
7	-7633.99	10248.49	12.9	8.5	1.25
8	-7273.72	10319.92	9.2	6.1	1.15
9	-6698.87	10417.82	9.3	4.8	1.04
10	-6148.60	10519.76	11.1	5.5	1.07
11	-5988.05	10544.52	5.2	10.8	1.48
TEB	-5915.28	10544.52	0.8	1.6	0.00

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Span geometry - UH

Span	l[ft]	h[ft]	l*[ft]	Chord angle[°]	Chord angle[%]
BEB - 1	18.10	-0.09	18.10	-0.27	-0.48
1 - 2	16.70	2.57	16.90	8.74	15.37
2 - 3	156.25	65.32	169.36	22.69	41.81
3 - 4	481.50	175.74	512.57	20.05	36.50
4 - 5	408.19	155.41	436.78	20.84	38.07
5 - 6	460.44	158.76	487.05	19.02	34.48
6 - 7	514.42	141.29	533.47	15.36	27.47
7 - 8	360.27	71.43	367.28	11.21	19.83
8 - 9	574.88	97.91	583.16	9.67	17.03
9 - 10	550.28	101.94	559.64	10.50	18.53
10 - 11	160.54	24.76	162.44	8.77	15.42
11 - TEB	72.77	0.01	72.77	0.01	0.01

Span geometry - DH

Span	l[ft]	h[ft]	l*[ft]	Chord angle[°]	Chord angle[%]
BEB - 1	18.11	-0.08	18.11	-0.27	-0.47
1 - 2	16.70	2.57	16.89	8.75	15.38
2 - 3	156.25	65.32	169.36	22.69	41.81
3 - 4	481.51	175.80	512.59	20.06	36.51
4 - 5	408.18	155.35	436.75	20.84	38.06
5 - 6	460.44	158.76	487.04	19.02	34.48
6 - 7	514.45	141.29	533.50	15.36	27.46
7 - 8	360.27	71.43	367.28	11.21	19.83
8 - 9	574.85	97.90	583.13	9.67	17.03
9 - 10	550.27	101.94	559.64	10.49	18.52
10 - 11	160.54	24.76	162.44	8.77	15.42
11 - TEB	72.77	0.00	72.77	0.00	0.00

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Calculation mode and load configurations

Calculation method: Point load (increments of 3.28 ft)

Calculated load configurations:

- | | |
|----|--|
| 1 | Rope Only; Forward operation (5.1m/s); T0; IO |
| 1+ | Rope Only; Forward operation (5.1m/s); T0+; IO |
| 1- | Rope Only; Forward operation (5.1m/s); T0-; IO |
| 2 | Rope + Empty Carriers; Forward operation (5.1m/s); T0; IO |
| 2+ | Rope + Empty Carriers; Forward operation (5.1m/s); T0+; IO |
| 2- | Rope + Empty Carriers; Forward operation (5.1m/s); T0-; IO |
| 3 | 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0; IO |
| 3+ | 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO |
| 3- | 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0-; IO |
| 4 | 100% UH+DH Load; Forward operation (5.1m/s); T0; IO |
| 4+ | 100% UH+DH Load; Forward operation (5.1m/s); T0+; IO |
| 4- | 100% UH+DH Load; Forward operation (5.1m/s); T0-; IO |
| 5 | 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0; IO |
| 5+ | 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0+; IO |
| 5- | 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0-; IO |
| 6 | 100% UH+DH Load, 2 Spans Rope Only; Forward operation (5.1m/s); T0; IO |

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Rope tension extremes, rope design factors, shear ratio

Rope tension extremes at nominal tension without dynamics

Tmax = 62314 lb (Rp: TBW; Lc: 3)

Tmin = 45462 lb (Rp: 2; Lc: 4)

Rope tension extremes at nominal tension with dynamics

Tmax = 62314 lb (Rp: TBW; Lc: 3)

Tmin = 45462 lb (Rp: 2; Lc: 4)

Rope tension extremes without dynamics

Tmax = 64700 lb (Rp: TBW; Lc: 3+)

Tmin = 43171 lb (Rp: 2; Lc: 4-)

Rope tension extremes with dynamics

Tmax = 64700 lb (Rp: TBW; Lc: 3+)

Tmin = 43171 lb (Rp: 2; Lc: 4-)

Rope design factor: min.: 5.35 max.: 7.73

Minimum shear ratio carrier: 15.41

Specific weights:

rope only:	5.0868 lb/ft
rope + empty carriers:	9.0661 lb/ft
rope + loaded carriers UH:	12.4613 lb/ft
rope + loaded carriers DH:	12.4613 lb/ft

Max. angle change: 0.0666 rad; 3.81° (Rp: 3; to top; DH)

Greatest rope inclination: 50.21%; 26.66° (Rp: 3; to bottom; DH; Lc: 4-)

Required min. pull force: 3971 lb

Bullwheels - SumTmax and Mdmax, extreme values with dynamic

	SumTmax [lb]	(Lc)	Md [lbft]	Mdmax [lbft]	(Lc)	SumT [lb]
Drive bullwheel	122906	(4+)	58023	79273	(3+)	119825
Return bullwheel	97488	(1+)				
	pmax [lb/in²]	(Lc)				
Drive bullwheel	563	(4+)				
Return bullwheel	420	(1+)				

Bullwheels - SumTmax and Mdmax, values at nominal tension without dynamic

	SumTmax [lb]	(Lc)	Md [lbft]	Mdmax [lbft]	(Lc)	SumT [lb]
Drive bullwheel	118267	(4)	46280	70084	(3)	115186
Return bullwheel	92846	(1)				

Clearance envelope

Required line width: 46.01 ft

Required line width - UH - Right: 22.92 ft (RP: 8; Lc: Rope + Empty Carriers; Forward operation; T0-; IO)

Required line width - DH - Left: 23.08 ft (RP: 8; Lc: Rope + Empty Carriers; Forward operation; T0-; IO)

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Circumferential forces, torques, required power (min ... geo. mean ... max)

Lc	Circumferential force [lb]	torque [ftlb]	required power [HP]	μ nec
1	4056	34368	142	0.025
1+	4186	35500	147	0.025
1-	3926	33236	137	0.026
2	4042 ... 4847 ... 5541	34452 ... 40792 ... 46251	143 ... 169 ... 191	0.032
2+	4178 ... 4977 ... 5665	35629 ... 41923 ... 47338	147 ... 173 ... 196	0.032
2-	3906 ... 4718 ... 5417	33269 ... 39660 ... 45169	138 ... 164 ... 187	0.033
3	7397 ... 8556 ... 9585	60948 ... 70084 ... 78187	252 ... 290 ... 323	0.054
3+	7540 ... 8687 ... 9709	62181 ... 71222 ... 79273	257 ... 295 ... 328	0.053
3-	7253 ... 8426 ... 9461	59706 ... 68945 ... 77106	247 ... 285 ... 319	0.056
4	3953 ... 5524 ... 6885	33910 ... 46280 ... 56993	140 ... 191 ... 236	0.038
4+	4096 ... 5654 ... 7002	35148 ... 47410 ... 58023	145 ... 196 ... 240	0.037
4-	3809 ... 5395 ... 6770	32665 ... 45149 ... 55975	135 ... 187 ... 232	0.039
5	598 ... 1815 ... 2841	7414 ... 16988 ... 25057	31 ... 70 ... 104	0.016
5+	734 ... 1944 ... 2958	8596 ... 18112 ... 26088	36 ... 75 ... 108	0.016
5-	461 ... 1686 ... 2726	6228 ... 15865 ... 24038	26 ... 66 ... 99	0.016
6	888 ... 5544 ... 9464	9703 ... 46368 ... 77234	40 ... 192 ... 319	0.053

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Maximum and minimum rope tensions - UH

Rope point	Tmax [lb]	Lc	Tmin [lb]	Lc
BBW	48890	(1+)	44234	(1-)
BEB	49177	(3+)	44443	(1-)
1	49308	(2+)	44431	(3-)
2	50053	(3+)	44604	(3-)
3	51354	(3+)	45197	(3-)
4	53451	(3+)	46215	(1-)
5	55507	(3+)	47046	(1-)
6	58081	(3+)	47946	(1-)
7	60048	(3+)	48803	(1-)
8	61126	(3+)	49311	(1-)
9	62314	(3+)	49902	(1-)
10	64018	(3+)	50474	(1-)
11	64405	(3+)	50683	(1-)
TEB	64410	(3+)	50892	(1-)
TBW	64700	(3+)	51150	(1-)

Maximum and minimum rope tensions - DH

Rope point	Tmax [lb]	Lc	Tmin [lb]	Lc
BBW	48598	(1+)	43970	(1-)
BEB	48380	(1+)	43684	(4-)
1	48434	(4+)	43410	(4-)
2	48725	(4+)	43171	(4-)
3	49484	(4+)	43371	(4-)
4	51288	(4+)	44440	(1-)
5	53050	(4+)	45139	(1-)
6	55399	(4+)	45811	(1-)
7	56718	(4+)	46390	(1-)
8	57476	(4+)	46663	(1-)
9	58287	(4+)	47106	(1-)
10	59779	(4+)	47544	(1-)
11	59733	(4+)	47473	(1-)
TEB	59490	(4+)	47468	(1-)
TBW	59208	(4+)	47224	(1-)

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Maximum and minimum tower loads - UH

Rope point	Sheave assembly	Dmax Lc [lb]	Psi [°]	Alpha [°]	Dmin Lc [lb]	Psi [°]	Alpha [°]
BEB	3R-403	2695 (3+)	-1.57	3.14	258 (1-)	-0.17	0.33
1	8N-401	-7644 (1+)	4.23	-8.91	-4242 (3-)	5.95	-5.45
2	12N-401	-11583 (2+)	15.51	-13.44	-7747 (3-)	13.76	-9.93
3	6T-401	7295 (3+)	22.09	8.16	3701 (1-)	20.85	4.68
4	6T-401	5212 (3-)	22.42	6.16	1561 (1+)	20.56	1.75
5	6T-401	7431 (3+)	18.25	7.91	3704 (1-)	19.86	4.51
6	8T-401	10179 (3+)	18.40	10.07	5551 (1-)	17.11	6.63
7	8T-401	10122 (3+)	15.18	9.69	5762 (1-)	13.52	6.76
8	8T-401	7833 (3+)	11.12	7.36	3711 (1-)	10.13	4.31
9	6T-401	6703 (6)	8.66	6.94	2004 (6)	10.11	1.94
10	6T-401	7450 (3+)	11.75	6.70	3335 (1-)	10.20	3.78
11	6T-501	13226 (3+)	5.80	11.91	8353 (1-)	4.51	9.43
TEB	3R-403	3095 (3+)	1.40	2.79	191 (1-)	0.11	0.22

Maximum and minimum tower loads - DH

Rope point	Sheave assembly	Dmax Lc [lb]	Psi [°]	Alpha [°]	Dmin Lc [lb]	Psi [°]	Alpha [°]
BEB	3R-403	3103 (4+)	-1.84	3.68	250 (1-)	-0.16	0.33
1	8N-401	-7501 (3+)	4.24	-8.91	-4047 (4-)	6.04	-5.30
2	12N-401	-11262 (4+)	15.51	-13.42	-7436 (4-)	13.70	-9.80
3	6T-401	7204 (4+)	22.11	8.36	3623 (1-)	20.83	4.76
4	6T-401	5244 (4-)	19.08	6.64	1598 (1+)	20.56	1.87
5	6T-401	7356 (4+)	18.16	8.19	3639 (1-)	19.85	4.61
6	8T-401	10000 (4+)	18.45	10.38	5415 (1-)	17.11	6.76
7	8T-401	9858 (4+)	12.36	10.16	5586 (1-)	13.53	6.89
8	8T-401	7737 (4+)	8.22	7.83	3640 (1-)	10.11	4.46
9	6T-401	6762 (6)	8.56	7.49	2059 (6)	10.11	2.12
10	6T-401	7313 (4+)	11.90	7.05	3246 (1-)	10.23	3.91
11	6T-501	12481 (4+)	5.91	12.17	7863 (1-)	4.52	9.48
TEB	3R-403	3106 (4+)	1.52	3.05	187 (1-)	0.11	0.23

Ropeline calculation

Gondola Section 2

8-MGD

Project no.: SAP0000008

Index Text

Date

Name

created: 2015-07-05 RUSSO

vis.:

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]**Contents**

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Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Key to symbols and abbreviations

Alpha	Deflection angle
Ao	Suspension above
Au	Suspension below
BW	Bullwheel
D	Tower load (more exact: resulting force in the rope point)
DCG	Distance CTF to ground
DS	Load per sheave
EB	Entrance beam
Eps	Tower angle
F2	Distance sheave assembly main axle - rope point
HA	Suspension height of sheave assembly
HC	Distance bottom of yoke to bottom of connection plate
HG	Grout (more exact: distance bottom of shaft to CTF)
HS	Shaft length
HY	Yoke height
IO	In Operation
Lc	Load configuration
OO	Out of Operation
Phi	Rope angle
PhiH	Horizontal lead-on/lead-off angle
Psi	Tower load reaction angle
Q	Cross inclination
RP	Rope point
T	Rope tension
T0	Basic tension force
T0+	Basic tension force with overtension
T0-	Basic tension force with undertension
W	Wind
WS	Wind load per sheave
XA,ZA	Coordinates of sheave assembly main axle
XF,ZF	Coordinates CTF
XR,ZR	Rope point coordinates
f	Sag
fy	Horizontal rope deflection
h	Elevation difference
l	Horizontal length
l*	Inclined length
q	Wind pressure
r	Rope radius

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]Location: Telluride, CO
Client: Telluride Transit Authorities**Main data**

Installation type:	8-MGD
Direction of rotation:	Counter-clockwise
Uphill transportation side:	Right
Uphill capacity:	100 %
Downhill capacity:	100 %
Drive location:	Top
Tension location:	Bottom
Tensioning type:	Hydraulic tensioning
Tensioning force:	92846 lb
Basic tension force:	46423 lb
Regulation:	US design parameters <modified>
Haul rope:	44 mm 6x25 FW 1960 Fatzer
Carriers:	8 Omega IV-LWI CWAA108C
Number of carriers:	41 (on line: 29.4; in stations: 12.0)
Spacing:	266.67 ft

Geometrical data

Horizontal length:	3774.35 ft
Vertical rise:	995.04 ft
Average grade:	26.36 %
Inclined length:	3919.60 ft
Transportation length:	4060.67 ft
Spliced rope length:	8066.59 ft
Rope gauge:	16.73 ft

Speed/Capacity

Main drive:	Rope speed:	1000 fpm	(backwards: -984 fpm)
	Capacity:	1800 persons/h	
	Interval:	16.00 s	
	Trip time:	5.51 min	
Evac drive:	Rope speed:	197 fpm	
	Capacity:	354 persons/h	
	Interval:	81.28 s	
	Trip time:	21.51 min	

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Carrier

Manufacturer: CWA
 Type: 8 Omega IV-LWI
 Dead weight: 1592 lb
 Pay load: 1358 lb or 8 persons
 Total weight: 2950 lb
 Length (L): 6.17 ft
 Dist. CL rope to top of seat (HS): 13.52 ft
 Dist. CL rope to bottom of carrier (H0): 13.52 ft
 Dist. CL rope to bottom of carrier (Hmax): 13.71 ft
 Projection area, in line: 49.5 sqft (Cf: 1.00)
 Projection area, perpendicular: 43.1 sqft (Cf: 1.00)
 Side swing 0.00 rad: Yi0 / Yo0: 3.97 / 3.38 ft
 Side swing 0.20 rad: Yi20 / Yo20: 5.94 / 5.54 ft
 Side swing 0.34 rad: Yi34 / Yo34: 7.55 / 7.12 ft
 Grip: A108C
 Grip pulling force: 5845.0 lb

Haul rope data

Manufacturer: Fatzer
 Rope construction: 6x25 FW compact type
 Diameter: 1.73 in
 Dia. outer wire: 0.116 in
 Metallic area: 1.349 sqin
 Rope weight: 5.09 lb/ft
 Tensile strength: 284274 psi
 Calculated breaking strength: 383299 lb
 Minimum breaking strength: 333616 lb
 Elastic modulus: 1450368 psi
 Thermal expansion factor: 1.2e-05 1/K
 Permanent elongation: 1.50 ‰
 Spliced rope length: 8066.6 ft
 Order length: 8340 ft (incl. Splice length: 173 ft; Surcharge: 98 ft)

Rope sheaves

Sheave	Side plate diameter	Base of Groove diameter
401	412 mm	382 mm
403	430 mm	400 mm
501	530 mm	485 mm

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Station: Bottom station

Type: Custom Return movable
Entrance beam: XR: -9689.63 ft ZR: 9549.48 ft
Gauge at entrance beam: 16.73 ft
Rope angle at entrance beam: 0.00 °
Distance EB to loading/unloading: 70.54 ft
Rope length in station: 114.73 ft
Distance EB to grip checking: 0.46 ft
Min. distance EB to mid 1. line sheave T; N; T/N: 0.00 ft; 0.00 ft; 0.00 ft
Driving speed: 68.90 fpm
Deceleration anti-collision monitoring: 3.28 ft/s²
Station dwell time: 85.19 s (1425 Impulses)
Tire conveyor friction: 224.8 lb
Rope deflections per side: 5.00° vertical, 4 Sheaves (501)
Return bullwheel: Ø = 16.73 ft; I = 16500 kgm²
Tmax: 49180 lb (Lc: 3+)
Tmin: 43684 lb (Lc: 4-)

Station: Top station

Type: Custom Drive fixed
Entrance beam: XR: -5915.28 ft ZR: 10544.52 ft
Gauge at entrance beam: 16.73 ft
Rope angle at entrance beam: 0.00 °
Distance EB to loading/unloading: 70.54 ft
Rope length in station: 112.66 ft
Distance EB to grip checking: 0.46 ft
Min. distance EB to mid 1. line sheave T; N; T/N: 0.00 ft; 0.00 ft; 0.00 ft
Driving speed: 68.90 fpm
Deceleration anti-collision monitoring: 3.28 ft/s²
Station dwell time: 106.18 s (1776 Impulses)
Tire conveyor friction: 224.8 lb
Rope deflections per side: 5.00° vertical, 3 Sheaves (501)
1.35° horizontal, 2 Sheaves (501)
Min. wrap angle: 177.30 °
Drive bullwheel: Ø = 15.75 ft; I = 11000 kgm²
Tmax: 69105 lb (Lc: 3+)
Tmin: 47223 lb (Lc: 1-)

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Main drive:

Total inertia: 11.87 lbft²
Acceleration: 0.49 ft/s²
Deceleration: -1.31 ft/s²

Gearbox:

Manufacturer: Doppelmayr-Lohmann
Type: GPW 215
Nominal torque: 104512.6 ftlb
Max. torque: 146332.3 ftlb
Ratio: 89.58
Efficiency: 0.93
Inertia: 11.87 lbft²
Req. input speed: 1811 rpm
Utilisation: 85 %
Req. torque continuous: 88997.8 ftlb (Lc: 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO)
Req. torque maximum: 100152.3 ftlb (Lc: 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO)

Electric motor:

Manufacturer:
Type:
Rating: 0 HP
Speed: 0 rpm
Inertia: 0.00 lbft²
Req. power continuous: 368 HP (Lc: 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO)

Service brake(s):

Inertia: 0.00 lbft²
Max. braking torque: 0.00 ftlb (0.00 kN @ 0.00 m)
Req. braking torque: 563.44 ftlb (Lc: 0% UH Load, 100% DH Load; Braking (0.50m/s²); T0)

Emergency brake(s):

Max. braking torque: 0.0 ftlb (0.00 kN @ 0.00 m)
Req. braking torque: 50473.0 ftlb (Lc: 0% UH Load, 100% DH Load; Braking (0.50m/s²); T0)

Evac drive:

System: rim gear on bullwheel
Diesel engine:
Efficiency: continuous, starting: 0.80 braking: 1.00
Req. power continuous: 91 HP (Lc: 100% UH Load, 0% DH Load; Forward operation (1.0m/s); T0+; IO)
Req. power starting: 0 HP (Lc:)
Req. power braking: 0 HP (Lc:)
Req. Mmax bullwheel: 0.0 ftlb (Lc:)

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Rope point coordinates - UH

Rope point	XR[ft]	ZR[ft]	Psi[°]	Alpha[°]	F2[ft]
BEB	-9689.63	9549.48	-0.9	1.8	0.00
1	-9671.57	9549.39	3.3	7.1	-1.41
2	-9654.73	9551.98	16.4	11.6	-2.96
3	-9498.60	9617.36	21.3	6.9	1.18
4	-9017.10	9793.09	21.4	4.8	1.04
5	-8608.94	9948.52	20.5	6.8	1.18
6	-8148.49	10107.26	17.7	8.9	1.42
7	-7634.05	10248.53	14.3	8.8	1.39
8	-7273.75	10319.90	10.6	6.5	1.14
9	-6698.91	10417.89	11.0	4.8	1.11
10	-6148.60	10521.82	10.8	6.4	1.13
11	-5988.04	10544.51	4.7	9.8	1.47
TEB	-5915.28	10544.52	0.7	1.4	0.00

Rope point coordinates - DH

Rope point	XR[ft]	ZR[ft]	Psi[°]	Alpha[°]	F2[ft]
BEB	-9689.63	9549.48	-1.0	2.0	0.00
1	-9671.52	9549.40	5.1	7.1	-1.41
2	-9654.83	9551.96	14.6	11.6	-2.95
3	-9498.57	9617.39	19.6	7.1	1.20
4	-9017.07	9793.09	19.8	5.1	1.03
5	-8608.92	9948.54	19.1	7.1	1.20
6	-8148.46	10107.28	16.3	9.3	1.43
7	-7634.06	10248.54	14.3	9.0	1.40
8	-7273.72	10319.92	9.2	6.8	1.15
9	-6698.89	10417.91	9.5	5.3	1.13
10	-6148.58	10521.84	9.6	6.7	1.14
11	-5988.04	10544.51	4.8	10.0	1.48
TEB	-5915.28	10544.52	0.8	1.5	0.00

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Span geometry - UH

Span	l[ft]	h[ft]	l*[ft]	Chord angle[°]	Chord angle[%]
BEB - 1	18.06	-0.09	18.06	-0.28	-0.49
1 - 2	16.83	2.59	17.03	8.76	15.41
2 - 3	156.13	65.38	169.27	22.72	41.87
3 - 4	481.50	175.73	512.56	20.05	36.50
4 - 5	408.16	155.43	436.75	20.85	38.08
5 - 6	460.45	158.74	487.05	19.02	34.48
6 - 7	514.44	141.27	533.48	15.36	27.46
7 - 8	360.30	71.37	367.30	11.20	19.81
8 - 9	574.84	97.98	583.13	9.67	17.05
9 - 10	550.31	103.93	560.04	10.70	18.89
10 - 11	160.56	22.69	162.16	8.04	14.13
11 - TEB	72.76	0.01	72.76	0.01	0.02

Span geometry - DH

Span	l[ft]	h[ft]	l*[ft]	Chord angle[°]	Chord angle[%]
BEB - 1	18.11	-0.08	18.11	-0.27	-0.47
1 - 2	16.70	2.57	16.90	8.75	15.39
2 - 3	156.25	65.42	169.40	22.72	41.87
3 - 4	481.50	175.70	512.56	20.05	36.49
4 - 5	408.15	155.46	436.75	20.85	38.09
5 - 6	460.46	158.74	487.05	19.02	34.47
6 - 7	514.40	141.26	533.45	15.36	27.46
7 - 8	360.33	71.38	367.33	11.20	19.81
8 - 9	574.84	97.99	583.13	9.67	17.05
9 - 10	550.31	103.93	560.03	10.69	18.89
10 - 11	160.54	22.68	162.13	8.04	14.12
11 - TEB	72.76	0.01	72.76	0.00	0.01

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Calculation mode and load configurations

Calculation method: Point load (increments of 2.67 ft)

Calculated load configurations:

- | | |
|----|--|
| 1 | Rope Only; Forward operation (5.1m/s); T0; IO |
| 1+ | Rope Only; Forward operation (5.1m/s); T0+; IO |
| 1- | Rope Only; Forward operation (5.1m/s); T0-; IO |
| 2 | Rope + Empty Carriers; Forward operation (5.1m/s); T0; IO |
| 2+ | Rope + Empty Carriers; Forward operation (5.1m/s); T0+; IO |
| 2- | Rope + Empty Carriers; Forward operation (5.1m/s); T0-; IO |
| 3 | 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0; IO |
| 3+ | 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO |
| 3- | 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0-; IO |
| 4 | 100% UH+DH Load; Forward operation (5.1m/s); T0; IO |
| 4+ | 100% UH+DH Load; Forward operation (5.1m/s); T0+; IO |
| 4- | 100% UH+DH Load; Forward operation (5.1m/s); T0-; IO |
| 5 | 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0; IO |
| 5+ | 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0+; IO |
| 5- | 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0-; IO |
| 6 | 100% UH+DH Load, 2 Spans Rope Only; Forward operation (5.1m/s); T0; IO |

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Rope tension extremes, rope design factors, shear ratio

Rope tension extremes at nominal tension without dynamics

Tmax = 66728 lb (Rp: TBW; Lc: 3)

Tmin = 45462 lb (Rp: 2; Lc: 4)

Rope tension extremes at nominal tension with dynamics

Tmax = 66728 lb (Rp: TBW; Lc: 3)

Tmin = 45462 lb (Rp: 2; Lc: 4)

Rope tension extremes without dynamics

Tmax = 69105 lb (Rp: TBW; Lc: 3+)

Tmin = 43172 lb (Rp: 2; Lc: 4-)

Rope tension extremes with dynamics

Tmax = 69105 lb (Rp: TBW; Lc: 3+)

Tmin = 43172 lb (Rp: 2; Lc: 4-)

Rope design factor: min.: 5.00 max.: 7.73

Minimum shear ratio carrier: 15.41

Specific weights:

rope only:	5.0868 lb/ft
rope + empty carriers:	11.0558 lb/ft
rope + loaded carriers UH:	16.1485 lb/ft
rope + loaded carriers DH:	16.1485 lb/ft

Max. angle change: 0.0743 rad; 4.26° (Rp: 3; to top; DH)

Greatest rope inclination: 50.25%; 26.68° (Rp: 3; to bottom; DH; Lc: 4-)

Required min. pull force: 3973 lb

Bullwheels - SumTmax and Mdmax, extreme values with dynamic

	SumTmax [lb]	(Lc)	Md [lbft]	Mdmax [lbft]	(Lc)	SumT [lb]
Drive bullwheel	130303	(4+)	67453	100152	(3+)	125869
Return bullwheel	97488	(1+)				
	pmax [lb/in²]	(Lc)				
Drive bullwheel	597	(4+)				
Return bullwheel	420	(1+)				

Bullwheels - SumTmax and Mdmax, values at nominal tension without dynamic

	SumTmax [lb]	(Lc)	Md [lbft]	Mdmax [lbft]	(Lc)	SumT [lb]
Drive bullwheel	125671	(4)	52188	87851	(3)	121228
Return bullwheel	92846	(1)				

Clearance envelope

Required line width: 46.24 ft

Required line width - UH - Right: 23.03 ft (RP: 8; Lc: Rope + Empty Carriers; Forward operation; T0-; IO)

Required line width - DH - Left: 23.21 ft (RP: 8; Lc: Rope + Empty Carriers; Forward operation; T0-; IO)

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Circumferential forces, torques, required power (min ... geo. mean ... max)

Lc	Circumferential force [lb]	torque [ftlb]	required power [HP]	μ nec
1	4059	34393	142	0.025
1+	4189	35526	147	0.025
1-	3929	33259	138	0.026
2	4341 ... 5241 ... 6118	36901 ... 43988 ... 50893	153 ... 182 ... 211	0.034
2+	4482 ... 5371 ... 6236	38120 ... 45121 ... 51933	158 ... 187 ... 215	0.034
2-	4199 ... 5111 ... 6001	35677 ... 42855 ... 49861	148 ... 177 ... 206	0.035
3	9536 ... 10795 ... 12229	77928 ... 87851 ... 99155	322 ... 363 ... 410	0.065
3+	9687 ... 10926 ... 12342	79226 ... 88998 ... 100152	328 ... 368 ... 414	0.064
3-	9383 ... 10663 ... 12118	76617 ... 86702 ... 98168	317 ... 359 ... 406	0.067
4	4400 ... 6251 ... 8075	37613 ... 52188 ... 66546	156 ... 216 ... 275	0.042
4+	4556 ... 6381 ... 8176	38951 ... 53319 ... 67453	161 ... 221 ... 279	0.041
4-	4242 ... 6122 ... 7976	36256 ... 51057 ... 65658	150 ... 211 ... 272	0.043
5	-795 ... 698 ... 1964	-3414 ... 8325 ... 18284	-12 ... 34 ... 76	0.011
5+	-648 ... 826 ... 2071	-2155 ... 9442 ... 19234	-8 ... 39 ... 80	0.011
5-	-942 ... 570 ... 1860	-4684 ... 7210 ... 17351	-17 ... 30 ... 72	0.010
6	38 ... 6280 ... 11785	3163 ... 52320 ... 95668	13 ... 216 ... 396	0.063

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Maximum and minimum rope tensions - UH

Rope point	Tmax [lb]	Lc	Tmin [lb]	Lc
BBW	48890	(1+)	44234	(1-)
BEB	49180	(3+)	44443	(1-)
1	49309	(2+)	44426	(3-)
2	50074	(3+)	44602	(3-)
3	51375	(3+)	45196	(3-)
4	54611	(3+)	46217	(1-)
5	57542	(3+)	47048	(1-)
6	60498	(3+)	47948	(1-)
7	63262	(3+)	48806	(1-)
8	64471	(3+)	49313	(1-)
9	66165	(3+)	49904	(1-)
10	67966	(3+)	50482	(1-)
11	68797	(3+)	50701	(1-)
TEB	68802	(3+)	50894	(1-)
TBW	69105	(3+)	51152	(1-)

Maximum and minimum rope tensions - DH

Rope point	Tmax [lb]	Lc	Tmin [lb]	Lc
BBW	48598	(1+)	43970	(1-)
BEB	48380	(1+)	43684	(4-)
1	48432	(4+)	43411	(4-)
2	48722	(4+)	43172	(4-)
3	49485	(4+)	43328	(4-)
4	52390	(4+)	44439	(1-)
5	55047	(4+)	45138	(1-)
6	57527	(4+)	45810	(1-)
7	59710	(4+)	46389	(1-)
8	60406	(4+)	46662	(1-)
9	61678	(4+)	47110	(1-)
10	63052	(4+)	47538	(1-)
11	63370	(4+)	47472	(1-)
TEB	63095	(4+)	47467	(1-)
TBW	62803	(4+)	47223	(1-)

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Maximum and minimum tower loads - UH

Rope point	Sheave assembly	Dmax Lc [lb]	Psi [°]	Alpha [°]	Dmin Lc [lb]	Psi [°]	Alpha [°]
BEB	3R-403	2803 (3+)	-1.63	3.27	266 (1-)	-0.17	0.34
1	8N-401	-7675 (1+)	4.24	-8.94	-4140 (3-)	2.44	-5.33
2	12N-401	-11590 (2+)	15.54	-13.45	-7734 (3-)	17.32	-9.80
3	8T-401	8035 (3+)	21.69	8.98	3728 (1-)	20.87	4.72
4	6T-401	6750 (3-)	22.30	7.76	1555 (1+)	20.56	1.75
5	8T-401	9126 (3+)	21.15	9.11	3710 (1-)	19.86	4.51
6	6T-501	11815 (3+)	18.38	11.23	5551 (1-)	17.11	6.63
7	6T-501	11865 (3+)	15.04	10.78	5769 (1-)	13.51	6.76
8	8T-401	9592 (3+)	11.04	8.63	3695 (1-)	10.12	4.29
9	8T-401	8213 (6)	11.71	8.01	1747 (6)	10.22	1.60
10	8T-401	9441 (3+)	11.75	8.11	4151 (1-)	9.93	4.71
11	6T-501	13016 (3+)	5.34	10.98	7712 (1-)	4.15	8.70
TEB	3R-403	3057 (3+)	1.29	2.58	193 (1-)	0.11	0.22

Maximum and minimum tower loads - DH

Rope point	Sheave assembly	Dmax Lc [lb]	Psi [°]	Alpha [°]	Dmin Lc [lb]	Psi [°]	Alpha [°]
BEB	3R-403	3097 (4+)	-1.83	3.67	250 (1-)	-0.16	0.33
1	8N-401	-7501 (1+)	4.24	-8.91	-4064 (4-)	6.03	-5.32
2	12N-401	-11287 (4+)	15.52	-13.45	-7441 (4-)	13.71	-9.81
3	8T-401	7955 (4+)	18.45	9.47	3654 (1-)	20.84	4.80
4	6T-401	6775 (4-)	19.04	8.35	1576 (1+)	20.56	1.84
5	8T-401	9047 (4+)	18.29	9.66	3653 (1-)	19.85	4.63
6	6T-501	11612 (4+)	15.58	11.85	5414 (1-)	17.10	6.76
7	6T-501	11584 (4+)	15.11	11.16	5592 (1-)	13.52	6.90
8	8T-401	9484 (4+)	8.25	9.24	3625 (1-)	10.10	4.45
9	8T-401	8290 (6)	8.78	8.79	1822 (6)	10.22	1.78
10	8T-401	9202 (4+)	9.26	8.66	4015 (1-)	9.96	4.83
11	6T-501	12250 (4+)	5.45	11.24	7258 (1-)	4.16	8.75
TEB	3R-403	3080 (4+)	1.42	2.85	189 (1-)	0.11	0.23

Ropeline calculation

Gondola Section 2

8-MGD

Project no.: SAP0000008

Index Text

Date

Name

created: 2015-07-05 RUSSO

vis.:

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

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Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Key to symbols and abbreviations

Alpha	Deflection angle
Ao	Suspension above
Au	Suspension below
BW	Bullwheel
D	Tower load (more exact: resulting force in the rope point)
DCG	Distance CTF to ground
DS	Load per sheave
EB	Entrance beam
Eps	Tower angle
F2	Distance sheave assembly main axle - rope point
HA	Suspension height of sheave assembly
HC	Distance bottom of yoke to bottom of connection plate
HG	Grout (more exact: distance bottom of shaft to CTF)
HS	Shaft length
HY	Yoke height
IO	In Operation
Lc	Load configuration
OO	Out of Operation
Phi	Rope angle
PhiH	Horizontal lead-on/lead-off angle
Psi	Tower load reaction angle
Q	Cross inclination
RP	Rope point
T	Rope tension
T0	Basic tension force
T0+	Basic tension force with overtension
T0-	Basic tension force with undertension
W	Wind
WS	Wind load per sheave
XA,ZA	Coordinates of sheave assembly main axle
XF,ZF	Coordinates CTF
XR,ZR	Rope point coordinates
f	Sag
fy	Horizontal rope deflection
h	Elevation difference
l	Horizontal length
l*	Inclined length
q	Wind pressure
r	Rope radius

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]Location: Telluride, CO
Client: Telluride Transit Authorities**Main data**

Installation type:	8-MGD
Direction of rotation:	Counter-clockwise
Uphill transportation side:	Right
Uphill capacity:	100 %
Downhill capacity:	100 %
Drive location:	Top
Tension location:	Bottom
Tensioning type:	Hydraulic tensioning
Tensioning force:	92846 lb
Basic tension force:	46423 lb
Regulation:	US design parameters <modified>
Haul rope:	47 mm 6x36 WS 1770 Fatzer
Carriers:	8 Omega IV-LWI CWAA108C
Number of carriers:	55 (on line: 39.2; in stations: 15.9)
Spacing:	200.00 ft

Geometrical data

Horizontal length:	3774.35 ft
Vertical rise:	995.04 ft
Average grade:	26.36 %
Inclined length:	3919.49 ft
Transportation length:	4060.57 ft
Spliced rope length:	8066.37 ft
Rope gauge:	16.73 ft

Speed/Capacity

Main drive:	Rope speed:	1000 fpm	(backwards: -984 fpm)
	Capacity:	2400 persons/h	
	Interval:	12.00 s	
	Trip time:	5.51 min	
Evac drive:	Rope speed:	197 fpm	
	Capacity:	472 persons/h	
	Interval:	60.96 s	
	Trip time:	21.51 min	

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Carrier

Manufacturer: CWA
 Type: 8 Omega IV-LWI
 Dead weight: 1592 lb
 Pay load: 1358 lb or 8 persons
 Total weight: 2950 lb
 Length (L): 6.17 ft
 Dist. CL rope to top of seat (HS): 13.52 ft
 Dist. CL rope to bottom of carrier (H0): 13.52 ft
 Dist. CL rope to bottom of carrier (Hmax): 13.71 ft
 Projection area, in line: 49.5 sqft (Cf: 1.00)
 Projection area, perpendicular: 43.1 sqft (Cf: 1.00)
 Side swing 0.00 rad: Yi0 / Yo0: 3.97 / 3.38 ft
 Side swing 0.20 rad: Yi20 / Yo20: 5.94 / 5.54 ft
 Side swing 0.34 rad: Yi34 / Yo34: 7.55 / 7.12 ft
 Grip: A108C
 Grip pulling force: 5845.0 lb

Haul rope data

Manufacturer: Fatzer
 Rope construction: 6x36 WS compact type
 Diameter: 1.85 in
 Dia. outer wire: 0.109 in
 Metallic area: 1.538 sqin
 Rope weight: 5.80 lb/ft
 Tensile strength: 256717 psi
 Calculated breaking strength: 394540 lb
 Minimum breaking strength: 339461 lb
 Elastic modulus: 1450368 psi
 Thermal expansion factor: 1.2e-05 1/K
 Permanent elongation: 1.50 ‰
 Spliced rope length: 8066.4 ft
 Order length: 8353 ft (incl. Splice length: 185 ft; Surcharge: 98 ft)

Rope sheaves

Sheave	Side plate diameter	Base of Groove diameter
403	430 mm	400 mm
420C.1	455 mm	420 mm
501	530 mm	485 mm

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Station: Bottom station

Type: Custom Return movable
Entrance beam: XR: -9689.63 ft ZR: 9549.48 ft
Gauge at entrance beam: 16.73 ft
Rope angle at entrance beam: 0.00 °
Distance EB to loading/unloading: 70.54 ft
Rope length in station: 114.73 ft
Distance EB to grip checking: 0.46 ft
Min. distance EB to mid 1. line sheave T; N; T/N: 0.00 ft; 0.00 ft; 0.00 ft
Driving speed: 68.90 fpm
Deceleration anti-collision monitoring: 3.28 ft/s²
Station dwell time: 85.19 s (1425 Impulses)
Tire conveyor friction: 224.8 lb
Rope deflections per side: 5.00° vertical, 4 Sheaves (501)
Return bullwheel: Ø = 16.73 ft; I = 16500 kgm²
Tmax: 49179 lb (Lc: 3+)
Tmin: 43686 lb (Lc: 4-)

Station: Top station

Type: Custom Drive fixed
Entrance beam: XR: -5915.28 ft ZR: 10544.52 ft
Gauge at entrance beam: 16.73 ft
Rope angle at entrance beam: 0.00 °
Distance EB to loading/unloading: 70.54 ft
Rope length in station: 112.66 ft
Distance EB to grip checking: 0.46 ft
Min. distance EB to mid 1. line sheave T; N; T/N: 0.00 ft; 0.00 ft; 0.00 ft
Driving speed: 68.90 fpm
Deceleration anti-collision monitoring: 3.28 ft/s²
Station dwell time: 106.18 s (1776 Impulses)
Tire conveyor friction: 224.8 lb
Rope deflections per side: 5.00° vertical, 3 Sheaves (501)
1.35° horizontal, 2 Sheaves (501)
Min. wrap angle: 177.30 °
Drive bullwheel: Ø = 15.75 ft; I = 11000 kgm²
Tmax: 73681 lb (Lc: 3+)
Tmin: 47868 lb (Lc: 1-)

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Main drive:

Total inertia: 11.87 lbft²
Acceleration: 0.49 ft/s²
Deceleration: -1.31 ft/s²

Gearbox:

Manufacturer: Doppelmayr-Lohmann
Type: GPW 245
Nominal torque: 150831.5 ftlb
Max. torque: 211164.0 ftlb
Ratio: 94.25
Efficiency: 0.93
Inertia: 11.87 lbft²
Req. input speed: 1906 rpm
Utilisation: 71 %
Req. torque continuous: 107816.3 ftlb (Lc: 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO)
Req. torque maximum: 115811.8 ftlb (Lc: 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO)

Electric motor:

Manufacturer:
Type:
Rating: 0 HP
Speed: 0 rpm
Inertia: 0.00 lbft²
Req. power continuous: 446 HP (Lc: 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO)

Service brake(s):

Inertia: 0.00 lbft²
Max. braking torque: 0.00 ftlb (0.00 kN @ 0.00 m)
Req. braking torque: 763.76 ftlb (Lc: 0% UH Load, 100% DH Load; Braking (0.50m/s²); T0)

Emergency brake(s):

Max. braking torque: 0.0 ftlb (0.00 kN @ 0.00 m)
Req. braking torque: 71984.6 ftlb (Lc: 0% UH Load, 100% DH Load; Braking (0.50m/s²); T0)

Evac drive:

System: rim gear on bullwheel
Diesel engine:
Efficiency: continuous, starting: 0.80 braking: 1.00
Req. power continuous: 110 HP (Lc: 100% UH Load, 0% DH Load; Forward operation (1.0m/s); T0+; IO)
Req. power starting: 0 HP (Lc:)
Req. power braking: 0 HP (Lc:)
Req. Mmax bullwheel: 0.0 ftlb (Lc:)

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Rope point coordinates - UH

Rope point	XR[ft]	ZR[ft]	Psi[°]	Alpha[°]	F2[ft]
BEB	-9689.63	9549.48	-0.8	1.7	0.00
1	-9671.51	9549.45	5.3	7.0	-1.53
2	-9654.68	9552.08	14.7	11.5	-3.51
3	-9498.63	9617.36	20.9	7.6	1.29
4	-9017.17	9793.16	21.5	5.9	1.23
5	-8609.03	9948.65	20.4	7.8	1.42
6	-8148.54	10107.43	17.7	10.0	1.59
7	-7634.10	10248.71	14.4	9.7	1.57
8	-7273.79	10320.06	10.4	7.2	1.40
9	-6698.95	10418.01	10.8	6.0	1.33
10	-6148.64	10519.90	11.2	6.1	1.31
11	-5988.06	10544.26	5.1	10.2	1.61
TEB	-5915.28	10544.52	0.8	1.5	0.00

Rope point coordinates - DH

Rope point	XR[ft]	ZR[ft]	Psi[°]	Alpha[°]	F2[ft]
BEB	-9689.63	9549.48	-0.9	1.9	0.00
1	-9671.56	9549.44	3.5	7.1	-1.54
2	-9654.68	9552.10	14.7	11.3	-3.49
3	-9498.60	9617.38	19.3	8.0	1.29
4	-9017.18	9793.18	21.6	6.2	1.24
5	-8608.99	9948.66	19.0	8.2	1.42
6	-8148.54	10107.43	17.7	10.4	1.59
7	-7634.10	10248.71	14.5	10.0	1.57
8	-7273.76	10320.07	9.1	7.7	1.40
9	-6698.92	10418.02	9.4	6.6	1.33
10	-6148.62	10519.90	10.1	6.4	1.31
11	-5988.07	10544.26	5.2	10.4	1.61
TEB	-5915.28	10544.52	0.8	1.7	0.00

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Span geometry - UH

Span	l[ft]	h[ft]	l*[ft]	Chord angle[°]	Chord angle[%]
BEB - 1	18.12	-0.03	18.12	-0.10	-0.18
1 - 2	16.83	2.64	17.03	8.91	15.67
2 - 3	156.05	65.28	169.15	22.70	41.83
3 - 4	481.46	175.80	512.55	20.06	36.51
4 - 5	408.14	155.48	436.76	20.85	38.09
5 - 6	460.48	158.79	487.09	19.03	34.48
6 - 7	514.44	141.28	533.49	15.36	27.46
7 - 8	360.31	71.35	367.31	11.20	19.80
8 - 9	574.84	97.95	583.13	9.67	17.04
9 - 10	550.31	101.89	559.66	10.49	18.51
10 - 11	160.58	24.36	162.42	8.63	15.17
11 - TEB	72.78	0.26	72.78	0.21	0.36

Span geometry - DH

Span	l[ft]	h[ft]	l*[ft]	Chord angle[°]	Chord angle[%]
BEB - 1	18.07	-0.04	18.07	-0.14	-0.24
1 - 2	16.87	2.66	17.08	8.97	15.78
2 - 3	156.09	65.28	169.19	22.70	41.82
3 - 4	481.42	175.80	512.51	20.06	36.52
4 - 5	408.18	155.48	436.79	20.85	38.09
5 - 6	460.45	158.77	487.06	19.03	34.48
6 - 7	514.44	141.28	533.49	15.36	27.46
7 - 8	360.34	71.36	367.34	11.20	19.80
8 - 9	574.84	97.95	583.13	9.67	17.04
9 - 10	550.30	101.89	559.65	10.49	18.51
10 - 11	160.55	24.36	162.39	8.63	15.17
11 - TEB	72.79	0.26	72.79	0.21	0.36

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Calculation mode and load configurations

Calculation method: Point load (increments of 2.00 ft)

Calculated load configurations:

- | | |
|----|--|
| 1 | Rope Only; Forward operation (5.1m/s); T0; IO |
| 1+ | Rope Only; Forward operation (5.1m/s); T0+; IO |
| 1- | Rope Only; Forward operation (5.1m/s); T0-; IO |
| 2 | Rope + Empty Carriers; Forward operation (5.1m/s); T0; IO |
| 2+ | Rope + Empty Carriers; Forward operation (5.1m/s); T0+; IO |
| 2- | Rope + Empty Carriers; Forward operation (5.1m/s); T0-; IO |
| 3 | 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0; IO |
| 3+ | 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO |
| 3- | 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0-; IO |
| 4 | 100% UH+DH Load; Forward operation (5.1m/s); T0; IO |
| 4+ | 100% UH+DH Load; Forward operation (5.1m/s); T0+; IO |
| 4- | 100% UH+DH Load; Forward operation (5.1m/s); T0-; IO |
| 5 | 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0; IO |
| 5+ | 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0+; IO |
| 5- | 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0-; IO |
| 6 | 100% UH+DH Load, 2 Spans Rope Only; Forward operation (5.1m/s); T0; IO |

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Rope tension extremes, rope design factors, shear ratio

Rope tension extremes at nominal tension without dynamics

Tmax = 71294 lb (Rp: TBW; Lc: 3)

Tmin = 45474 lb (Rp: 2; Lc: 6)

Rope tension extremes at nominal tension with dynamics

Tmax = 71294 lb (Rp: TBW; Lc: 3)

Tmin = 45474 lb (Rp: 2; Lc: 6)

Rope tension extremes without dynamics

Tmax = 73681 lb (Rp: TBW; Lc: 3+)

Tmin = 43185 lb (Rp: 2; Lc: 4-)

Rope tension extremes with dynamics

Tmax = 73681 lb (Rp: TBW; Lc: 3+)

Tmin = 43185 lb (Rp: 2; Lc: 4-)

Rope design factor: min.: 4.76 max.: 7.86

Minimum shear ratio carrier: 15.42

Specific weights:

rope only:	5.7991 lb/ft
rope + empty carriers:	13.7578 lb/ft
rope + loaded carriers UH:	20.5480 lb/ft
rope + loaded carriers DH:	20.5480 lb/ft

Max. angle change: 0.0847 rad; 4.86° (Rp: 3; to top; DH)

Greatest rope inclination: 52.43%; 27.67° (Rp: 4; to bottom; DH; Lc: 4-)

Required min. pull force: 4109 lb

Bullwheels - SumTmax and Mdmax, extreme values with dynamic

	SumTmax [lb]	(Lc)	Md [lbft]	Mdmax [lbft]	(Lc)	SumT [lb]
Drive bullwheel	139455	(4+)	69715	115812	(3+)	133219
Return bullwheel	97488	(1+)				
	pmax [lb/in²]	(Lc)				
Drive bullwheel	598	(4+)				
Return bullwheel	394	(1+)				

Bullwheels - SumTmax and Mdmax, values at nominal tension without dynamic

	SumTmax [lb]	(Lc)	Md [lbft]	Mdmax [lbft]	(Lc)	SumT [lb]
Drive bullwheel	134845	(4)	59224	106666	(3)	128604
Return bullwheel	92846	(1)				

Clearance envelope

Required line width: 47.23 ft

Required line width - UH - Right: 23.51 ft (RP: 8; Lc: Rope + Empty Carriers; Forward operation; T0-; IO)

Required line width - DH - Left: 23.73 ft (RP: 8; Lc: Rope + Empty Carriers; Forward operation; T0-; IO)

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Circumferential forces, torques, required power (min ... geo. mean ... max)

Lc	Circumferential force [lb]	torque [ftlb]	required power [HP]	μ nec
1	4186	35424	147	0.026
1+	4315	36552	151	0.026
1-	4056	34296	142	0.026
2	5143 ... 5766 ... 6360	43343 ... 48250 ... 52930	179 ... 200 ... 219	0.034
2+	5278 ... 5895 ... 6483	44521 ... 49376 ... 54006	184 ... 204 ... 223	0.033
2-	5007 ... 5637 ... 6238	42161 ... 47123 ... 51856	174 ... 195 ... 214	0.035
3	12244 ... 13164 ... 14185	99406 ... 106666 ... 114726	411 ... 441 ... 475	0.072
3+	12397 ... 13296 ... 14309	100721 ... 107816 ... 115812	417 ... 446 ... 479	0.070
3-	12088 ... 13031 ... 14061	98072 ... 105513 ... 113642	406 ... 436 ... 470	0.074
4	5853 ... 7119 ... 8322	49263 ... 59224 ... 68695	204 ... 245 ... 284	0.040
4+	5996 ... 7247 ... 8437	50497 ... 60346 ... 69715	209 ... 250 ... 288	0.039
4-	5710 ... 6990 ... 8207	48022 ... 58103 ... 67682	199 ... 240 ... 280	0.041
5	-1368 ... -279 ... 682	-7769 ... 787 ... 8335	-28 ... 3 ... 34	0.007
5+	-1233 ... -154 ... 783	-6596 ... 1885 ... 9241	-24 ... 8 ... 38	0.006
5-	-1504 ... -404 ... 583	-8946 ... -308 ... 7448	-32 ... -1 ... 31	0.008
6	55 ... 7164 ... 13173	3468 ... 59458 ... 106781	14 ... 246 ... 442	0.066

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Maximum and minimum rope tensions - UH

Rope point	Tmax [lb]	Lc	Tmin [lb]	Lc
BBW	48890	(1+)	44234	(1-)
BEB	49179	(3+)	44443	(1-)
1	49316	(3+)	44428	(3-)
2	50067	(3+)	44607	(3-)
3	51440	(3+)	45254	(3-)
4	55738	(3+)	46387	(1-)
5	59144	(3+)	47337	(1-)
6	62911	(3+)	48358	(1-)
7	65882	(3+)	49326	(1-)
8	67834	(3+)	49892	(1-)
9	70366	(3+)	50562	(1-)
10	72516	(3+)	51216	(1-)
11	73351	(3+)	51451	(1-)
TEB	73365	(3+)	51658	(1-)
TBW	73681	(3+)	51924	(1-)

Maximum and minimum rope tensions - DH

Rope point	Tmax [lb]	Lc	Tmin [lb]	Lc
BBW	48598	(1+)	43970	(1-)
BEB	48380	(1+)	43686	(4-)
1	48440	(4+)	43414	(4-)
2	48748	(4+)	43185	(4-)
3	49538	(4+)	43416	(4-)
4	53489	(4+)	44607	(1-)
5	56416	(4+)	45409	(1-)
6	59628	(4+)	46185	(1-)
7	61858	(4+)	46856	(1-)
8	63198	(4+)	47171	(1-)
9	65197	(4+)	47674	(1-)
10	66821	(4+)	48174	(1-)
11	67143	(4+)	48121	(1-)
TEB	66812	(4+)	48113	(1-)
TBW	66503	(4+)	47868	(1-)

Project [-no.]: 8-MGD Gondola Section 2 [SAP0000008]

Maximum and minimum tower loads - UH

Rope point	Sheave assembly	Dmax Lc [lb]	Psi [°]	Alpha [°]	Dmin Lc [lb]	Psi [°]	Alpha [°]
BEB	3R-403	2763 (3+)	-1.61	3.22	131 (1-)	-0.08	0.17
1	8N-420C	-7630 (3+)	4.41	-8.89	-3974 (3-)	6.29	-5.11
2	12N-420C	-11388 (3+)	15.57	-13.21	-7557 (3-)	13.81	-9.69
3	6T-420C	9213 (3+)	21.10	10.30	3931 (1-)	20.79	4.97
4	6T-420C	8693 (3-)	22.48	9.79	1873 (1+)	20.58	2.10
5	6T-501	11013 (3+)	21.01	10.74	4031 (1-)	19.85	4.87
6	8T-501	14257 (3+)	18.25	13.05	5927 (1-)	17.11	7.01
7	8T-501	13943 (3+)	15.25	12.23	6122 (1-)	13.54	7.10
8	6T-501	11543 (3+)	10.76	9.80	4044 (1-)	10.08	4.64
9	6T-501	10816 (6)	11.53	9.98	2320 (6)	10.11	2.01
10	6T-501	10030 (3+)	12.23	7.94	3733 (1-)	10.19	4.17
11	8T-501	14245 (3+)	5.67	11.25	8246 (1-)	4.56	9.17
TEB	3R-403	3367 (3+)	1.32	2.65	398 (1-)	0.22	0.44

Maximum and minimum tower loads - DH

Rope point	Sheave assembly	Dmax Lc [lb]	Psi [°]	Alpha [°]	Dmin Lc [lb]	Psi [°]	Alpha [°]
BEB	3R-403	3028 (4+)	-1.79	3.59	156 (1-)	-0.10	0.20
1	8N-420C	-7564 (1+)	4.42	-8.98	-4023 (4-)	2.57	-5.27
2	12N-420C	-11025 (4+)	15.59	-13.13	-7213 (4-)	13.78	-9.51
3	6T-420C	9135 (4+)	17.79	10.89	3854 (1-)	20.76	5.06
4	6T-420C	8712 (4-)	22.55	10.23	1901 (1+)	20.58	2.21
5	6T-501	10929 (4+)	18.19	11.44	3969 (1-)	19.85	5.00
6	8T-501	14037 (4+)	18.29	13.56	5788 (1-)	17.10	7.17
7	8T-501	13610 (4+)	15.35	12.74	5941 (1-)	13.55	7.26
8	6T-501	11432 (4+)	8.11	10.54	3972 (1-)	10.06	4.82
9	6T-501	10874 (6)	8.74	10.93	2387 (6)	10.11	2.23
10	6T-501	9818 (4+)	9.95	8.54	3632 (1-)	10.23	4.32
11	8T-501	13326 (4+)	5.78	11.51	7755 (1-)	4.57	9.22
TEB	3R-403	3363 (4+)	1.45	2.91	385 (1-)	0.23	0.46

Ropeline calculation

Gondola Section 3

8-MGD

Project no.: SAA0002176

Index Text

Date

Name

created: 2015-07-05 RUSSO

vis.:

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

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Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Key to symbols and abbreviations

Alpha	Deflection angle
Ao	Suspension above
Au	Suspension below
BW	Bullwheel
D	Tower load (more exact: resulting force in the rope point)
DCG	Distance CTF to ground
DS	Load per sheave
EB	Entrance beam
Eps	Tower angle
F2	Distance sheave assembly main axle - rope point
HA	Suspension height of sheave assembly
HC	Distance bottom of yoke to bottom of connection plate
HG	Grout (more exact: distance bottom of shaft to CTF)
HS	Shaft length
HY	Yoke height
IO	In Operation
Lc	Load configuration
OO	Out of Operation
Phi	Rope angle
PhiH	Horizontal lead-on/lead-off angle
Psi	Tower load reaction angle
Q	Cross inclination
RP	Rope point
T	Rope tension
T0	Basic tension force
T0+	Basic tension force with overtension
T0-	Basic tension force with undertension
W	Wind
WS	Wind load per sheave
XA,ZA	Coordinates of sheave assembly main axle
XF,ZF	Coordinates CTF
XR,ZR	Rope point coordinates
f	Sag
fy	Horizontal rope deflection
h	Elevation difference
l	Horizontal length
l*	Inclined length
q	Wind pressure
r	Rope radius

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Location: Telluride, CO
Client: Telluride Transit Authorities

Main data

Installation type: 8-MGD
Direction of rotation: Counter-clockwise
Uphill transportation side: Right
Uphill capacity: 100 %
Downhill capacity: 100 %
Drive location: Top
Tension location: Bottom
Tensioning type: Hydraulic tensioning
Tensioning force: 92846 lb
Basic tension force: 46423 lb
Regulation: US design parameters <modified>
Haul rope: 44 mm 6x25 FW 1570 Fatzer
Carriers: 8 Omega IV-LWI CWAA108C
Number of carriers: 20 (on line: 13.2; in stations: 7.1)
Spacing: 400.00 ft

Geometrical data

Horizontal length: 2626.17 ft
Vertical rise: 4.53 ft
Average grade: 0.17 %
Inclined length: 2635.84 ft
Transportation length: 2776.92 ft
Spliced rope length: 5499.08 ft
Rope gauge: 16.73 ft

Speed/Capacity

Main drive:	Rope speed:	1000 fpm	(backwards: -984 fpm)
	Capacity:	1200 persons/h	
	Interval:	24.00 s	
	Trip time:	4.06 min	
Evac drive:	Rope speed:	197 fpm	
	Capacity:	236 persons/h	
	Interval:	121.92 s	
	Trip time:	14.81 min	

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Carrier

Manufacturer: CWA
 Type: 8 Omega IV-LWI
 Dead weight: 1592 lb
 Pay load: 1358 lb or 8 persons
 Total weight: 2950 lb
 Length (L): 6.17 ft
 Dist. CL rope to top of seat (HS): 13.52 ft
 Dist. CL rope to bottom of carrier (H0): 13.52 ft
 Dist. CL rope to bottom of carrier (Hmax): 13.71 ft
 Projection area, in line: 49.5 sqft (Cf: 1.00)
 Projection area, perpendicular: 43.1 sqft (Cf: 1.00)
 Side swing 0.00 rad: Yi0 / Yo0: 3.97 / 3.38 ft
 Side swing 0.20 rad: Yi20 / Yo20: 5.94 / 5.54 ft
 Side swing 0.34 rad: Yi34 / Yo34: 7.55 / 7.12 ft
 Grip: A108C
 Grip pulling force: 5845.0 lb

Haul rope data

Manufacturer: Fatzer
 Rope construction: 6x25 FW
 Diameter: 1.73 in
 Dia. outer wire: 0.112 in
 Metallic area: 1.245 sqin
 Rope weight: 4.74 lb/ft
 Tensile strength: 227709 psi
 Calculated breaking strength: 283259 lb
 Minimum breaking strength: 246391 lb
 Elastic modulus: 1450368 psi
 Thermal expansion factor: 1.2e-05 1/K
 Permanent elongation: 1.50 ‰
 Stranding formula: Einlage + 6 x (1 + 6 + 6F+ 12)
 Spliced rope length: 5499.1 ft
 Order length: 5771 ft (incl. Splice length: 173 ft; Surcharge: 98 ft)

Rope sheaves

Sheave	Side plate diameter	Base of Groove diameter
401	412 mm	382 mm
403	430 mm	400 mm

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]
Station: Bottom station

Type: Custom Return movable
Entrance beam: XR: 10060.62 ft ZR: 9549.48 ft
Gauge at entrance beam: 16.73 ft
Rope angle at entrance beam: 0.00 °
Distance EB to loading/unloading: 70.54 ft
Rope length in station: 114.73 ft
Distance EB to grip checking: 0.46 ft
Min. distance EB to mid 1. line sheave T; N; T/N: 0.00 ft; 0.00 ft; 0.00 ft
Driving speed: 68.90 fpm
Deceleration anti-collision monitoring: 3.28 ft/s²
Station dwell time: 85.19 s (1425 Impulses)
Tire conveyor friction: 224.8 lb
Rope deflections per side: 5.00° vertical, 4 Sheaves (501)
Return bullwheel: Ø = 16.73 ft; I = 16500 kgm²
Tmax: 49172 lb (Lc: 3+)
Tmin: 43697 lb (Lc: 4-)

Station: Top station

Type: Custom Drive fixed
Entrance beam: XR: 12686.79 ft ZR: 9554.01 ft
Gauge at entrance beam: 16.73 ft
Rope angle at entrance beam: 0.00 °
Distance EB to loading/unloading: 70.54 ft
Rope length in station: 112.66 ft
Distance EB to grip checking: 0.46 ft
Min. distance EB to mid 1. line sheave T; N; T/N: 0.00 ft; 0.00 ft; 0.00 ft
Driving speed: 68.90 fpm
Deceleration anti-collision monitoring: 3.28 ft/s²
Station dwell time: 85.19 s (1425 Impulses)
Tire conveyor friction: 224.8 lb
Rope deflections per side: 5.00° vertical, 3 Sheaves (501)
1.35° horizontal, 2 Sheaves (501)
Min. wrap angle: 177.30 °
Drive bullwheel: Ø = 15.75 ft; I = 11000 kgm²
Tmax: 51979 lb (Lc: 3+)
Tmin: 41487 lb (Lc: 4-)

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]**Main drive:**

Total inertia: 11.87 lbft²
Acceleration: 0.49 ft/s²
Deceleration: -1.31 ft/s²

Gearbox:

Manufacturer: Doppelmayr-Lohmann
Type: GPW 215
Nominal torque: 104512.6 ftlb
Max. torque: 146332.3 ftlb
Ratio: 89.58
Efficiency: 0.93
Inertia: 11.87 lbft²
Req. input speed: 1811 rpm
Utilisation: 32 %
Req. torque continuous: 33733.5 ftlb (Lc: 100% UH+DH Load; Forward operation (5.1m/s); T0+; IO)
Req. torque maximum: 48081.1 ftlb (Lc: 100% UH+DH Load, 2 Spans Rope Only; Forward operation (5.1m/s); T0; IO)

Electric motor:

Manufacturer:
Type:
Rating: 0 HP
Speed: 0 rpm
Inertia: 0.00 lbft²
Req. power continuous: 140 HP (Lc: 100% UH+DH Load; Forward operation (5.1m/s); T0+; IO)

Service brake(s):

Inertia: 0.00 lbft²
Max. braking torque: 0.00 ftlb (0.00 kN @ 0.00 m)
Req. braking torque: 23.11 ftlb (Lc: 0% UH Load, 100% DH Load; Braking (0.50m/s²); T0)

Emergency brake(s):

Max. braking torque: 0.0 ftlb (0.00 kN @ 0.00 m)
Req. braking torque: 2070.3 ftlb (Lc: 0% UH Load, 100% DH Load; Braking (0.50m/s²); T0)

Evac drive:

System: rim gear on bullwheel
Diesel engine:
Efficiency: continuous, starting: 0.80 braking: 1.00
Req. power continuous: 34 HP (Lc: 100% UH+DH Load; Forward operation (1.0m/s); T0+; IO)
Req. power starting: 0 HP (Lc:)
Req. power braking: 0 HP (Lc:)
Req. M_{dmax} bullwheel: 0.0 ftlb (Lc:)

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Rope point coordinates - UH

Rope point	XR[ft]	ZR[ft]	Psi[°]	Alpha[°]	F2[ft]
BEB	10060.62	9549.48	-0.8	1.6	0.00
1	10075.88	9549.43	2.6	5.4	-1.25
2	10244.61	9572.01	5.3	9.0	1.26
3	10514.33	9579.27	0.8	6.4	1.09
4	10944.50	9571.40	-3.0	10.0	1.31
5	11360.65	9522.58	-7.4	3.9	0.97
6	11682.70	9485.76	0.5	8.6	-1.39
7	12117.73	9533.50	5.1	8.5	1.24
8	12592.99	9553.81	2.8	5.9	1.06
TEB	12686.79	9554.01	1.0	2.0	0.00

Rope point coordinates - DH

Rope point	XR[ft]	ZR[ft]	Psi[°]	Alpha[°]	F2[ft]
BEB	10060.62	9549.48	-0.8	1.7	0.00
1	10075.88	9549.43	2.5	5.4	-1.25
2	10244.65	9572.01	3.6	9.1	1.26
3	10514.33	9579.28	0.8	6.5	1.10
4	10944.50	9571.40	-3.0	10.1	1.32
5	11360.66	9522.58	-7.4	4.1	0.97
6	11682.70	9485.77	0.5	8.4	-1.38
7	12117.73	9533.51	5.1	8.7	1.25
8	12592.99	9553.82	2.9	6.1	1.06
TEB	12686.79	9554.01	1.1	2.2	0.00

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Span geometry - UH

Span	l[ft]	h[ft]	l*[ft]	Chord angle[°]	Chord angle[%]
BEB - 1	15.26	-0.05	15.26	-0.19	-0.33
1 - 2	168.74	22.58	170.24	7.62	13.38
2 - 3	269.72	7.26	269.82	1.54	2.69
3 - 4	430.16	-7.88	430.24	-1.05	-1.83
4 - 5	416.16	-48.82	419.01	-6.69	-11.73
5 - 6	322.05	-36.81	324.15	-6.52	-11.43
6 - 7	435.03	47.74	437.64	6.26	10.97
7 - 8	475.26	20.31	475.69	2.45	4.27
8 - TEB	93.80	0.20	93.80	0.12	0.21

Span geometry - DH

Span	l[ft]	h[ft]	l*[ft]	Chord angle[°]	Chord angle[%]
BEB - 1	15.26	-0.05	15.26	-0.19	-0.33
1 - 2	168.78	22.59	170.28	7.62	13.38
2 - 3	269.68	7.26	269.78	1.54	2.69
3 - 4	430.16	-7.87	430.24	-1.05	-1.83
4 - 5	416.16	-48.82	419.01	-6.69	-11.73
5 - 6	322.05	-36.81	324.14	-6.52	-11.43
6 - 7	435.03	47.74	437.64	6.26	10.97
7 - 8	475.26	20.30	475.69	2.45	4.27
8 - TEB	93.80	0.19	93.80	0.12	0.20

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Calculation mode and load configurations

Calculation method: Point load (increments of 3.28 ft)

Calculated load configurations:

- | | |
|----|--|
| 1 | Rope Only; Forward operation (5.1m/s); T0; IO |
| 1+ | Rope Only; Forward operation (5.1m/s); T0+; IO |
| 1- | Rope Only; Forward operation (5.1m/s); T0-; IO |
| 2 | Rope + Empty Carriers; Forward operation (5.1m/s); T0; IO |
| 2+ | Rope + Empty Carriers; Forward operation (5.1m/s); T0+; IO |
| 2- | Rope + Empty Carriers; Forward operation (5.1m/s); T0-; IO |
| 3 | 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0; IO |
| 3+ | 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO |
| 3- | 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0-; IO |
| 4 | 100% UH+DH Load; Forward operation (5.1m/s); T0; IO |
| 4+ | 100% UH+DH Load; Forward operation (5.1m/s); T0+; IO |
| 4- | 100% UH+DH Load; Forward operation (5.1m/s); T0-; IO |
| 5 | 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0; IO |
| 5+ | 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0+; IO |
| 5- | 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0-; IO |
| 6 | 100% UH+DH Load, 2 Spans Rope Only; Forward operation (5.1m/s); T0; IO |

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Rope tension extremes, rope design factors, shear ratio

Rope tension extremes at nominal tension without dynamics

Tmax = 49774 lb (Rp: TBW; Lc: 6)

Tmin = 43515 lb (Rp: TBW; Lc: 6)

Rope tension extremes at nominal tension with dynamics

Tmax = 49774 lb (Rp: TBW; Lc: 6)

Tmin = 43515 lb (Rp: TBW; Lc: 6)

Rope tension extremes without dynamics

Tmax = 51979 lb (Rp: TBW; Lc: 3+)

Tmin = 41465 lb (Rp: 6; Lc: 4-)

Rope tension extremes with dynamics

Tmax = 51979 lb (Rp: TBW; Lc: 3+)

Tmin = 41465 lb (Rp: 6; Lc: 4-)

Rope design factor: min.: 4.95 max.: 5.94

Minimum shear ratio carrier: 14.81

Specific weights:

rope only:	4.7441 lb/ft
rope + empty carriers:	8.7234 lb/ft
rope + loaded carriers UH:	12.1186 lb/ft
rope + loaded carriers DH:	12.1186 lb/ft

Max. angle change: 0.0701 rad; 4.02° (Rp: 8; to bottom; DH)

Greatest rope inclination: 21.26%; 12.00° (Rp: 4; to top; DH; Lc: 4-)

Required min. pull force: 1841 lb

Bullwheels - SumTmax and Mdmax, extreme values with dynamic

	SumTmax [lb]	(Lc)	Md [lbft]	Mdmax [lbft]	(Lc)	SumT [lb]
Drive bullwheel	98725	(3+)	43532	48081	(6)	94550
Return bullwheel	97488	(1+)				
	pmax [lb/in²]	(Lc)				
Drive bullwheel	452	(3+)				
Return bullwheel	420	(1+)				

Bullwheels - SumTmax and Mdmax, values at nominal tension without dynamic

	SumTmax [lb]	(Lc)	Md [lbft]	Mdmax [lbft]	(Lc)	SumT [lb]
Drive bullwheel	94097	(3)	31704	32676	(4)	93814
Return bullwheel	92846	(1)				

Clearance envelope

Required line width: 45.30 ft

Required line width - UH - Right: 22.57 ft (RP: 7; Lc: Rope + Empty Carriers; Forward operation; T0-; IO)

Required line width - DH - Left: 22.72 ft (RP: 7; Lc: 100% UH+DH Load; Forward operation; T0-; IO)

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Circumferential forces, torques, required power (min ... geo. mean ... max)

Lc	Circumferential force [lb]	torque [ftlb]	required power [HP]	μ nec
1	3254	27816	115	0.023
1+	3374	28873	119	0.022
1-	3134	26759	111	0.023
2	2731 ... 3586 ... 4374	23707 ... 30441 ... 36645	98 ... 126 ... 152	0.030
2+	2863 ... 3706 ... 4482	24862 ... 31499 ... 37609	103 ... 130 ... 156	0.030
2-	2597 ... 3466 ... 4267	22542 ... 29384 ... 35691	93 ... 122 ... 148	0.031
3	2632 ... 3745 ... 5135	22927 ... 31704 ... 42659	95 ... 131 ... 176	0.035
3+	2771 ... 3866 ... 5232	24136 ... 32762 ... 43532	100 ... 136 ... 180	0.034
3-	2490 ... 3625 ... 5041	21702 ... 30646 ... 41804	90 ... 127 ... 173	0.036
4	2097 ... 3869 ... 5505	18728 ... 32676 ... 45562	77 ... 135 ... 188	0.038
4+	2249 ... 3989 ... 5596	20034 ... 33733 ... 46383	83 ... 140 ... 192	0.037
4-	1941 ... 3748 ... 5418	17394 ... 31619 ... 44767	72 ... 131 ... 185	0.039
5	2196 ... 3709 ... 4744	19507 ... 31414 ... 39549	81 ... 130 ... 164	0.033
5+	2341 ... 3830 ... 4846	20761 ... 32470 ... 40460	86 ... 134 ... 167	0.032
5-	2048 ... 3589 ... 4644	18234 ... 30357 ... 38654	75 ... 126 ... 160	0.034
6	1632 ... 4032 ... 5826	15052 ... 33957 ... 48081	62 ... 140 ... 199	0.040

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Maximum and minimum rope tensions - UH

Rope point	Tmax [lb]	Lc	Tmin [lb]	Lc
BBW	48890	(1+)	44234	(1-)
BEB	49172	(3+)	44443	(1-)
1	49273	(3+)	44438	(3-)
2	50107	(3+)	44671	(3-)
3	50291	(3+)	44875	(1-)
4	50521	(3+)	44878	(3-)
5	50159	(3+)	44460	(3-)
6	50147	(3+)	43991	(3-)
7	51253	(3+)	44628	(3-)
8	51714	(3+)	44976	(3-)
TEB	51723	(3+)	45115	(3-)
TBW	51979	(3+)	45396	(3-)

Maximum and minimum rope tensions - DH

Rope point	Tmax [lb]	Lc	Tmin [lb]	Lc
BBW	48598	(1+)	43970	(1-)
BEB	48380	(1+)	43697	(4-)
1	48413	(4+)	43499	(4-)
2	48829	(4+)	43450	(4-)
3	48785	(4+)	43340	(4-)
4	48648	(4+)	42948	(4-)
5	47941	(4+)	42173	(4-)
6	47662	(4+)	41465	(4-)
7	48397	(4+)	41759	(4-)
8	48496	(4+)	41756	(4-)
TEB	48354	(4+)	41715	(4-)
TBW	48100	(4+)	41487	(4-)

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Maximum and minimum tower loads - UH

Rope point	Sheave assembly	Dmax Lc [lb]	Psi [°]	Alpha [°]	Dmin Lc [lb]	Psi [°]	Alpha [°]
BEB	3R-403	2514 (3+)	-1.47	2.93	183 (1-)	-0.12	0.24
1	8N-401	-6269 (1+)	3.51	-7.30	-2725 (3-)	1.61	-3.51
2	8T-401	9237 (4+)	6.11	10.60	5789 (1-)	4.43	7.41
3	6T-401	7076 (4+)	1.57	8.09	3691 (1-)	0.01	4.71
4	8T-401	10229 (3+)	-2.12	11.71	6436 (1-)	-3.85	8.20
5	4T-401	4681 (3-)	-8.25	5.99	1605 (1+)	-6.48	1.85
6	8N-401	-9250 (1+)	-0.28	-10.71	-5045 (3-)	1.26	-6.45
7	8T-401	9193 (4+)	5.83	10.38	5170 (1-)	4.30	6.55
8	6T-401	6814 (3+)	3.74	7.74	3196 (1-)	1.86	4.03
TEB	3R-403	3192 (4+)	1.82	3.64	318 (3-)	0.20	0.40

Maximum and minimum tower loads - DH

Rope point	Sheave assembly	Dmax Lc [lb]	Psi [°]	Alpha [°]	Dmin Lc [lb]	Psi [°]	Alpha [°]
BEB	3R-403	2603 (4+)	-1.54	3.08	181 (1-)	-0.12	0.24
1	8N-401	-6145 (1+)	3.50	-7.29	-2608 (4-)	1.57	-3.42
2	8T-401	9094 (4+)	2.69	10.82	5671 (1-)	4.42	7.45
3	6T-401	7018 (4+)	-1.85	8.30	3631 (1-)	-0.00	4.77
4	8T-401	10021 (4+)	-2.07	11.93	6279 (1-)	-3.85	8.29
5	4T-401	4688 (4-)	-8.31	6.30	1610 (1+)	-6.47	1.93
6	8N-401	-8758 (1+)	-0.29	-10.61	-4505 (4-)	1.23	-6.11
7	8T-401	8987 (4+)	5.90	10.76	5013 (1-)	4.29	6.70
8	6T-401	6657 (4+)	3.90	8.11	3092 (1-)	1.89	4.13
TEB	3R-403	3265 (4+)	2.00	4.01	309 (4-)	0.21	0.42

Ropeline calculation

Gondola Section 3

8-MGD

Project no.: SAA0002176

Index Text

Date

Name

created: 2015-07-05 RUSSO

vis.:

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

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Key to symbols and abbreviations

Alpha	Deflection angle
Ao	Suspension above
Au	Suspension below
BW	Bullwheel
D	Tower load (more exact: resulting force in the rope point)
DCG	Distance CTF to ground
DS	Load per sheave
EB	Entrance beam
Eps	Tower angle
F2	Distance sheave assembly main axle - rope point
HA	Suspension height of sheave assembly
HC	Distance bottom of yoke to bottom of connection plate
HG	Grout (more exact: distance bottom of shaft to CTF)
HS	Shaft length
HY	Yoke height
IO	In Operation
Lc	Load configuration
OO	Out of Operation
Phi	Rope angle
PhiH	Horizontal lead-on/lead-off angle
Psi	Tower load reaction angle
Q	Cross inclination
RP	Rope point
T	Rope tension
T0	Basic tension force
T0+	Basic tension force with overtension
T0-	Basic tension force with undertension
W	Wind
WS	Wind load per sheave
XA,ZA	Coordinates of sheave assembly main axle
XF,ZF	Coordinates CTF
XR,ZR	Rope point coordinates
f	Sag
fy	Horizontal rope deflection
h	Elevation difference
l	Horizontal length
l*	Inclined length
q	Wind pressure
r	Rope radius

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]Location: Telluride, CO
Client: Telluride Transit Authorities**Main data**

Installation type:	8-MGD
Direction of rotation:	Counter-clockwise
Uphill transportation side:	Right
Uphill capacity:	100 %
Downhill capacity:	100 %
Drive location:	Top
Tension location:	Bottom
Tensioning type:	Hydraulic tensioning
Tensioning force:	92846 lb
Basic tension force:	46423 lb
Regulation:	US design parameters <modified>
Haul rope:	44 mm 6x25 FW 1570 Fatzer
Carriers:	8 Omega IV-LWI CWAA108C
Number of carriers:	30 (on line: 19.8; in stations: 10.6)
Spacing:	266.67 ft

Geometrical data

Horizontal length:	2626.17 ft
Vertical rise:	4.53 ft
Average grade:	0.17 %
Inclined length:	2635.88 ft
Transportation length:	2776.96 ft
Spliced rope length:	5499.16 ft
Rope gauge:	16.73 ft

Speed/Capacity

Main drive:	Rope speed:	1000 fpm	(backwards: -984 fpm)
	Capacity:	1800 persons/h	
	Interval:	16.00 s	
	Trip time:	4.06 min	
Evac drive:	Rope speed:	197 fpm	
	Capacity:	354 persons/h	
	Interval:	81.28 s	
	Trip time:	14.81 min	

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Carrier

Manufacturer: CWA
 Type: 8 Omega IV-LWI
 Dead weight: 1592 lb
 Pay load: 1358 lb or 8 persons
 Total weight: 2950 lb
 Length (L): 6.17 ft
 Dist. CL rope to top of seat (HS): 13.52 ft
 Dist. CL rope to bottom of carrier (H0): 13.52 ft
 Dist. CL rope to bottom of carrier (Hmax): 13.71 ft
 Projection area, in line: 49.5 sqft (Cf: 1.00)
 Projection area, perpendicular: 43.1 sqft (Cf: 1.00)
 Side swing 0.00 rad: Yi0 / Yo0: 3.97 / 3.38 ft
 Side swing 0.20 rad: Yi20 / Yo20: 5.94 / 5.54 ft
 Side swing 0.34 rad: Yi34 / Yo34: 7.55 / 7.12 ft
 Grip: A108C
 Grip pulling force: 5845.0 lb

Haul rope data

Manufacturer: Fatzer
 Rope construction: 6x25 FW
 Diameter: 1.73 in
 Dia. outer wire: 0.112 in
 Metallic area: 1.245 sqin
 Rope weight: 4.74 lb/ft
 Tensile strength: 227709 psi
 Calculated breaking strength: 283259 lb
 Minimum breaking strength: 246391 lb
 Elastic modulus: 1450368 psi
 Thermal expansion factor: 1.2e-05 1/K
 Permanent elongation: 1.50 ‰
 Stranding formula: Einlage + 6 x (1 + 6 + 6F+ 12)
 Spliced rope length: 5499.2 ft
 Order length: 5771 ft (incl. Splice length: 173 ft; Surcharge: 98 ft)

Rope sheaves

Sheave	Side plate diameter	Base of Groove diameter
401	412 mm	382 mm
403	430 mm	400 mm
420C.1	455 mm	420 mm

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]
Station: Bottom station

Type: Custom Return movable
Entrance beam: XR: 10060.62 ft ZR: 9549.48 ft
Gauge at entrance beam: 16.73 ft
Rope angle at entrance beam: 0.00 °
Distance EB to loading/unloading: 70.54 ft
Rope length in station: 114.73 ft
Distance EB to grip checking: 0.46 ft
Min. distance EB to mid 1. line sheave T; N; T/N: 0.00 ft; 0.00 ft; 0.00 ft
Driving speed: 68.90 fpm
Deceleration anti-collision monitoring: 3.28 ft/s²
Station dwell time: 85.19 s (1425 Impulses)
Tire conveyor friction: 224.8 lb
Rope deflections per side: 5.00° vertical, 4 Sheaves (501)
Return bullwheel: Ø = 16.73 ft; I = 16500 kgm²
Tmax: 49175 lb (Lc: 3+)
Tmin: 43688 lb (Lc: 4-)

Station: Top station

Type: Custom Drive fixed
Entrance beam: XR: 12686.79 ft ZR: 9554.01 ft
Gauge at entrance beam: 16.73 ft
Rope angle at entrance beam: 0.00 °
Distance EB to loading/unloading: 70.54 ft
Rope length in station: 112.66 ft
Distance EB to grip checking: 0.46 ft
Min. distance EB to mid 1. line sheave T; N; T/N: 0.00 ft; 0.00 ft; 0.00 ft
Driving speed: 68.90 fpm
Deceleration anti-collision monitoring: 3.28 ft/s²
Station dwell time: 85.19 s (1425 Impulses)
Tire conveyor friction: 224.8 lb
Rope deflections per side: 5.00° vertical, 3 Sheaves (501)
1.35° horizontal, 2 Sheaves (501)
Min. wrap angle: 177.30 °
Drive bullwheel: Ø = 15.75 ft; I = 11000 kgm²
Tmax: 51708 lb (Lc: 3+)
Tmin: 41181 lb (Lc: 4-)

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]**Main drive:**

Total inertia: 11.87 lbft²
Acceleration: 0.49 ft/s²
Deceleration: -1.31 ft/s²

Gearbox:

Manufacturer: Doppelmayr-Lohmann
Type: GPW 215
Nominal torque: 104512.6 ftlb
Max. torque: 146332.3 ftlb
Ratio: 89.58
Efficiency: 0.93
Inertia: 11.87 lbft²
Req. input speed: 1811 rpm
Utilisation: 35 %
Req. torque continuous: 36249.0 ftlb (Lc: 100% UH+DH Load; Forward operation (5.1m/s); T0+; IO)
Req. torque maximum: 52564.3 ftlb (Lc: 100% UH+DH Load, 2 Spans Rope Only; Forward operation (5.1m/s); T0; IO)

Electric motor:

Manufacturer:
Type:
Rating: 0 HP
Speed: 0 rpm
Inertia: 0.00 lbft²
Req. power continuous: 150 HP (Lc: 100% UH+DH Load; Forward operation (5.1m/s); T0+; IO)

Service brake(s):

Inertia: 0.00 lbft²
Max. braking torque: 0.00 ftlb (0.00 kN @ 0.00 m)
Req. braking torque: 78.07 ftlb (Lc: 0% UH Load, 100% DH Load; Braking (0.50m/s²); T0)

Emergency brake(s):

Max. braking torque: 0.0 ftlb (0.00 kN @ 0.00 m)
Req. braking torque: 6993.7 ftlb (Lc: 0% UH Load, 100% DH Load; Braking (0.50m/s²); T0)

Evac drive:

System: rim gear on bullwheel
Diesel engine:
Efficiency: continuous, starting: 0.80 braking: 1.00
Req. power continuous: 37 HP (Lc: 100% UH+DH Load; Forward operation (1.0m/s); T0+; IO)
Req. power starting: 0 HP (Lc:)
Req. power braking: 0 HP (Lc:)
Req. M_{dmax} bullwheel: 0.0 ftlb (Lc:)

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Rope point coordinates - UH

Rope point	XR[ft]	ZR[ft]	Psi[°]	Alpha[°]	F2[ft]
BEB	10060.62	9549.48	-0.8	1.7	0.00
1	10075.88	9549.43	2.6	5.4	-1.25
2	10244.61	9572.01	5.4	8.8	1.26
3	10514.37	9580.37	-1.1	7.4	1.19
4	10944.51	9571.16	-3.1	10.8	1.57
5	11360.63	9522.66	-5.3	4.9	1.05
6	11682.70	9485.76	0.4	7.7	-1.39
7	12117.71	9533.67	5.1	9.5	1.50
8	12592.98	9553.81	3.0	6.3	1.06
TEB	12686.79	9554.01	1.0	2.1	0.00

Rope point coordinates - DH

Rope point	XR[ft]	ZR[ft]	Psi[°]	Alpha[°]	F2[ft]
BEB	10060.62	9549.48	-0.9	1.9	0.00
1	10075.88	9549.43	2.5	5.4	-1.25
2	10244.61	9572.01	5.4	8.9	1.26
3	10514.37	9580.37	-1.2	7.5	1.19
4	10944.51	9571.17	-3.0	11.0	1.58
5	11360.66	9522.66	-7.3	5.2	1.05
6	11682.70	9485.77	0.5	7.4	-1.38
7	12117.75	9533.69	3.3	10.0	1.52
8	12593.02	9553.82	1.3	6.6	1.06
TEB	12686.79	9554.01	1.1	2.2	0.00

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Span geometry - UH

Span	l[ft]	h[ft]	l*[ft]	Chord angle[°]	Chord angle[%]
BEB - 1	15.26	-0.05	15.26	-0.19	-0.33
1 - 2	168.74	22.58	170.24	7.62	13.38
2 - 3	269.76	8.36	269.89	1.78	3.10
3 - 4	430.14	-9.22	430.24	-1.23	-2.14
4 - 5	416.11	-48.49	418.93	-6.65	-11.65
5 - 6	322.07	-36.90	324.18	-6.54	-11.46
6 - 7	435.01	47.90	437.64	6.28	11.01
7 - 8	475.28	20.14	475.70	2.43	4.24
8 - TEB	93.81	0.20	93.81	0.12	0.21

Span geometry - DH

Span	l[ft]	h[ft]	l*[ft]	Chord angle[°]	Chord angle[%]
BEB - 1	15.26	-0.05	15.26	-0.19	-0.33
1 - 2	168.74	22.58	170.24	7.62	13.38
2 - 3	269.76	8.36	269.89	1.78	3.10
3 - 4	430.14	-9.21	430.24	-1.23	-2.14
4 - 5	416.15	-48.51	418.97	-6.65	-11.66
5 - 6	322.04	-36.89	324.14	-6.53	-11.45
6 - 7	435.05	47.92	437.68	6.29	11.01
7 - 8	475.26	20.13	475.69	2.43	4.23
8 - TEB	93.77	0.19	93.78	0.12	0.20

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Calculation mode and load configurations

Calculation method: Point load (increments of 2.67 ft)

Calculated load configurations:

- | | |
|----|--|
| 1 | Rope Only; Forward operation (5.1m/s); T0; IO |
| 1+ | Rope Only; Forward operation (5.1m/s); T0+; IO |
| 1- | Rope Only; Forward operation (5.1m/s); T0-; IO |
| 2 | Rope + Empty Carriers; Forward operation (5.1m/s); T0; IO |
| 2+ | Rope + Empty Carriers; Forward operation (5.1m/s); T0+; IO |
| 2- | Rope + Empty Carriers; Forward operation (5.1m/s); T0-; IO |
| 3 | 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0; IO |
| 3+ | 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO |
| 3- | 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0-; IO |
| 4 | 100% UH+DH Load; Forward operation (5.1m/s); T0; IO |
| 4+ | 100% UH+DH Load; Forward operation (5.1m/s); T0+; IO |
| 4- | 100% UH+DH Load; Forward operation (5.1m/s); T0-; IO |
| 5 | 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0; IO |
| 5+ | 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0+; IO |
| 5- | 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0-; IO |
| 6 | 100% UH+DH Load, 2 Spans Rope Only; Forward operation (5.1m/s); T0; IO |

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Rope tension extremes, rope design factors, shear ratio

Rope tension extremes at nominal tension without dynamics

Tmax = 50185 lb (Rp: TBW; Lc: 6)

Tmin = 43197 lb (Rp: TBW; Lc: 6)

Rope tension extremes at nominal tension with dynamics

Tmax = 50185 lb (Rp: TBW; Lc: 6)

Tmin = 43197 lb (Rp: TBW; Lc: 6)

Rope tension extremes without dynamics

Tmax = 51708 lb (Rp: TBW; Lc: 3+)

Tmin = 41071 lb (Rp: 6; Lc: 4-)

Rope tension extremes with dynamics

Tmax = 51708 lb (Rp: TBW; Lc: 3+)

Tmin = 41071 lb (Rp: 6; Lc: 4-)

Rope design factor: min.: 4.91 max.: 6.00

Minimum shear ratio carrier: 14.69

Specific weights:

rope only:	4.7441 lb/ft
rope + empty carriers:	10.7131 lb/ft
rope + loaded carriers UH:	15.8058 lb/ft
rope + loaded carriers DH:	15.8058 lb/ft

Max. angle change: 0.0817 rad; 4.68° (Rp: 7; to top; DH)

Greatest rope inclination: 23.49%; 13.22° (Rp: 4; to top; DH; Lc: 4-)

Required min. pull force: 2024 lb

Bullwheels - SumTmax and Mdmax, extreme values with dynamic

	SumTmax [lb]	(Lc)	Md [lbft]	Mdmax [lbft]	(Lc)	SumT [lb]
Drive bullwheel	98718	(3+)	43721	52564	(6)	95357
Return bullwheel	97488	(1+)				
	pmax [lb/in²]	(Lc)				
Drive bullwheel	452	(3+)				
Return bullwheel	420	(1+)				

Bullwheels - SumTmax and Mdmax, values at nominal tension without dynamic

	SumTmax [lb]	(Lc)	Md [lbft]	Mdmax [lbft]	(Lc)	SumT [lb]
Drive bullwheel	94100	(3)	33685	35191	(4)	94047
Return bullwheel	92846	(1)				

Clearance envelope

Required line width: 45.25 ft

Required line width - UH - Right: 22.55 ft (RP: 7; Lc: Rope + Empty Carriers; Forward operation; T0-; IO)

Required line width - DH - Left: 22.70 ft (RP: 7; Lc: Rope + Empty Carriers; Forward operation; T0-; IO)

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Circumferential forces, torques, required power (min ... geo. mean ... max)

Lc	Circumferential force [lb]	torque [ftlb]	required power [HP]	μ nec
1	3257	27840	115	0.023
1+	3377	28898	120	0.022
1-	3136	26781	111	0.023
2	3032 ... 3760 ... 4541	26074 ... 31806 ... 37949	108 ... 132 ... 157	0.032
2+	3161 ... 3881 ... 4652	27199 ... 32864 ... 38935	113 ... 136 ... 161	0.031
2-	2903 ... 3640 ... 4430	24942 ... 30748 ... 36971	103 ... 127 ... 153	0.032
3	2780 ... 3999 ... 5154	24072 ... 33685 ... 42789	100 ... 139 ... 177	0.036
3+	2917 ... 4119 ... 5258	25266 ... 34744 ... 43721	105 ... 144 ... 181	0.035
3-	2640 ... 3878 ... 5051	22863 ... 32625 ... 41869	95 ... 135 ... 173	0.037
4	2684 ... 4190 ... 5808	23331 ... 35191 ... 47923	97 ... 146 ... 198	0.040
4+	2828 ... 4311 ... 5903	24573 ... 36249 ... 48782	102 ... 150 ... 202	0.039
4-	2538 ... 4070 ... 5715	22072 ... 34133 ... 47084	91 ... 141 ... 195	0.042
5	2827 ... 3952 ... 5195	24464 ... 33309 ... 43083	101 ... 138 ... 178	0.036
5+	2963 ... 4072 ... 5297	25641 ... 34365 ... 43997	106 ... 142 ... 182	0.035
5-	2691 ... 3832 ... 5095	23278 ... 32252 ... 42186	96 ... 133 ... 175	0.038
6	1866 ... 4435 ... 6395	16910 ... 37135 ... 52564	70 ... 154 ... 217	0.044

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Maximum and minimum rope tensions - UH

Rope point	Tmax [lb]	Lc	Tmin [lb]	Lc
BBW	48890	(1+)	44234	(1-)
BEB	49175	(3+)	44443	(1-)
1	49274	(2+)	44433	(3-)
2	50103	(3+)	44668	(3-)
3	50563	(3+)	44875	(1-)
4	50768	(3+)	44741	(3-)
5	50185	(3+)	44164	(3-)
6	49760	(3+)	43781	(3-)
7	51008	(3+)	44426	(3-)
8	51444	(3+)	44883	(3-)
TEB	51453	(3+)	45017	(3-)
TBW	51708	(3+)	45262	(3-)

Maximum and minimum rope tensions - DH

Rope point	Tmax [lb]	Lc	Tmin [lb]	Lc
BBW	48598	(1+)	43970	(1-)
BEB	48380	(1+)	43688	(4-)
1	48440	(4+)	43456	(4-)
2	48830	(4+)	43375	(4-)
3	48862	(4+)	43254	(4-)
4	48582	(4+)	42675	(4-)
5	47681	(4+)	41690	(4-)
6	47460	(1+)	41071	(4-)
7	47747	(4+)	41360	(4-)
8	47871	(4+)	41415	(4-)
TEB	47703	(4+)	41408	(4-)
TBW	47450	(4+)	41181	(4-)

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Maximum and minimum tower loads - UH

Rope point	Sheave assembly	Dmax Lc [lb]	Psi [°]	Alpha [°]	Dmin Lc [lb]	Psi [°]	Alpha [°]
BEB	3R-403	2632 (3+)	-1.53	3.07	183 (1-)	-0.12	0.24
1	8N-401	-6269 (1+)	3.51	-7.30	-2718 (3-)	1.61	-3.50
2	8T-401	9090 (4+)	6.20	10.43	5606 (1-)	4.55	7.18
3	8T-401	8364 (3+)	-2.26	9.65	4015 (1-)	0.03	5.12
4	8T-420C	11908 (3+)	-2.20	13.56	6262 (1-)	-3.91	7.98
5	6T-401	6189 (3-)	-4.21	7.97	1656 (1+)	-6.46	1.91
6	8N-401	-9281 (1+)	-0.28	-10.74	-3561 (3-)	1.16	-4.59
7	8T-420C	10879 (3+)	5.89	12.50	5203 (1-)	4.30	6.59
8	6T-401	7663 (3+)	4.16	8.57	3180 (1-)	1.85	4.01
TEB	3R-403	3245 (4+)	1.85	3.70	317 (3-)	0.20	0.40

Maximum and minimum tower loads - DH

Rope point	Sheave assembly	Dmax Lc [lb]	Psi [°]	Alpha [°]	Dmin Lc [lb]	Psi [°]	Alpha [°]
BEB	3R-403	2965 (4+)	-1.76	3.51	181 (1-)	-0.12	0.24
1	8N-401	-6146 (1+)	3.50	-7.29	-2626 (4-)	1.58	-3.44
2	8T-401	8954 (4+)	6.24	10.54	5494 (1-)	4.54	7.21
3	8T-401	8290 (4+)	-2.35	9.90	3944 (1-)	0.02	5.19
4	8T-420C	11685 (4+)	-2.17	13.87	6112 (1-)	-3.92	8.07
5	6T-401	6186 (4-)	-8.07	8.41	1657 (1+)	-6.46	1.99
6	8N-401	-8788 (1+)	-0.29	-10.65	-2995 (4-)	1.25	-4.09
7	8T-420C	10656 (4+)	2.37	13.25	5046 (1-)	4.29	6.75
8	6T-401	7495 (4+)	0.82	9.09	3077 (1-)	1.88	4.11
TEB	3R-403	3258 (4+)	2.01	4.02	307 (4-)	0.21	0.42

Ropeline calculation

Gondola Section 3

8-MGD

Project no.: SAA0002176

Index Text

Date

Name

created: 2015-07-05 RUSSO

vis.:

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

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Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Key to symbols and abbreviations

Alpha	Deflection angle
Ao	Suspension above
Au	Suspension below
BW	Bullwheel
D	Tower load (more exact: resulting force in the rope point)
DCG	Distance CTF to ground
DS	Load per sheave
EB	Entrance beam
Eps	Tower angle
F2	Distance sheave assembly main axle - rope point
HA	Suspension height of sheave assembly
HC	Distance bottom of yoke to bottom of connection plate
HG	Grout (more exact: distance bottom of shaft to CTF)
HS	Shaft length
HY	Yoke height
IO	In Operation
Lc	Load configuration
OO	Out of Operation
Phi	Rope angle
PhiH	Horizontal lead-on/lead-off angle
Psi	Tower load reaction angle
Q	Cross inclination
RP	Rope point
T	Rope tension
T0	Basic tension force
T0+	Basic tension force with overtension
T0-	Basic tension force with undertension
W	Wind
WS	Wind load per sheave
XA,ZA	Coordinates of sheave assembly main axle
XF,ZF	Coordinates CTF
XR,ZR	Rope point coordinates
f	Sag
fy	Horizontal rope deflection
h	Elevation difference
l	Horizontal length
l*	Inclined length
q	Wind pressure
r	Rope radius

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Location: Telluride, CO
Client: Telluride Transit Authorities

Main data

Installation type:	8-MGD
Direction of rotation:	Counter-clockwise
Uphill transportation side:	Right
Uphill capacity:	100 %
Downhill capacity:	100 %
Drive location:	Top
Tension location:	Bottom
Tensioning type:	Hydraulic tensioning
Tensioning force:	92846 lb
Basic tension force:	46423 lb
Regulation:	US design parameters <modified>
Haul rope:	44 mm 6x25 FW 1570 Fatzer
Carriers:	8 Omega IV-LWI CWAA108C
Number of carriers:	40 (on line: 26.4; in stations: 14.2)
Spacing:	200.00 ft

Geometrical data

Horizontal length:	2626.17 ft
Vertical rise:	4.53 ft
Average grade:	0.17 %
Inclined length:	2635.90 ft
Transportation length:	2776.98 ft
Spliced rope length:	5499.19 ft
Rope gauge:	16.73 ft

Speed/Capacity

Main drive:	Rope speed:	1000 fpm	(backwards: -984 fpm)
	Capacity:	2400 persons/h	
	Interval:	12.00 s	
	Trip time:	4.06 min	
Evac drive:	Rope speed:	197 fpm	
	Capacity:	472 persons/h	
	Interval:	60.96 s	
	Trip time:	14.81 min	

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Carrier

Manufacturer: CWA
 Type: 8 Omega IV-LWI
 Dead weight: 1592 lb
 Pay load: 1358 lb or 8 persons
 Total weight: 2950 lb
 Length (L): 6.17 ft
 Dist. CL rope to top of seat (HS): 13.52 ft
 Dist. CL rope to bottom of carrier (H0): 13.52 ft
 Dist. CL rope to bottom of carrier (Hmax): 13.71 ft
 Projection area, in line: 49.5 sqft (Cf: 1.00)
 Projection area, perpendicular: 43.1 sqft (Cf: 1.00)
 Side swing 0.00 rad: Yi0 / Yo0: 3.97 / 3.38 ft
 Side swing 0.20 rad: Yi20 / Yo20: 5.94 / 5.54 ft
 Side swing 0.34 rad: Yi34 / Yo34: 7.55 / 7.12 ft
 Grip: A108C
 Grip pulling force: 5845.0 lb

Haul rope data

Manufacturer: Fatzer
 Rope construction: 6x25 FW
 Diameter: 1.73 in
 Dia. outer wire: 0.112 in
 Metallic area: 1.245 sqin
 Rope weight: 4.74 lb/ft
 Tensile strength: 227709 psi
 Calculated breaking strength: 283259 lb
 Minimum breaking strength: 246391 lb
 Elastic modulus: 1450368 psi
 Thermal expansion factor: 1.2e-05 1/K
 Permanent elongation: 1.50 ‰
 Stranding formula: Einlage + 6 x (1 + 6 + 6F+ 12)
 Spliced rope length: 5499.2 ft
 Order length: 5771 ft (incl. Splice length: 173 ft; Surcharge: 98 ft)

Rope sheaves

Sheave	Side plate diameter	Base of Groove diameter
401	412 mm	382 mm
403	430 mm	400 mm
501	530 mm	485 mm

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]
Station: Bottom station

Type: Custom Return movable
 Entrance beam: XR: 10060.62 ft ZR: 9549.48 ft
 Gauge at entrance beam: 16.73 ft
 Rope angle at entrance beam: 0.00 °
 Distance EB to loading/unloading: 70.54 ft
 Rope length in station: 114.73 ft
 Distance EB to grip checking: 0.46 ft
 Min. distance EB to mid 1. line sheave T; N; T/N: 0.00 ft; 0.00 ft; 0.00 ft
 Driving speed: 68.90 fpm
 Deceleration anti-collision monitoring: 3.28 ft/s²
 Station dwell time: 85.19 s (1425 Impulses)
 Tire conveyor friction: 224.8 lb
 Rope deflections per side: 5.00° vertical, 4 Sheaves (501)
 Return bullwheel: Ø = 16.73 ft; I = 16500 kgm²
 Tmax: 49179 lb (Lc: 3+)
 Tmin: 43691 lb (Lc: 4-)

Station: Top station

Type: Custom Drive fixed
 Entrance beam: XR: 12686.79 ft ZR: 9554.01 ft
 Gauge at entrance beam: 16.73 ft
 Rope angle at entrance beam: 0.00 °
 Distance EB to loading/unloading: 70.54 ft
 Rope length in station: 112.66 ft
 Distance EB to grip checking: 0.46 ft
 Min. distance EB to mid 1. line sheave T; N; T/N: 0.00 ft; 0.00 ft; 0.00 ft
 Driving speed: 68.90 fpm
 Deceleration anti-collision monitoring: 3.28 ft/s²
 Station dwell time: 85.19 s (1425 Impulses)
 Tire conveyor friction: 224.8 lb
 Rope deflections per side: 5.00° vertical, 3 Sheaves (501)
 1.35° horizontal, 2 Sheaves (501)
 Min. wrap angle: 177.30 °
 Drive bullwheel: Ø = 15.75 ft; I = 11000 kgm²
 Tmax: 52163 lb (Lc: 3+)
 Tmin: 41297 lb (Lc: 4-)

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]**Main drive:**

Total inertia: 11.87 lbft²
Acceleration: 0.49 ft/s²
Deceleration: -1.31 ft/s²

Gearbox:

Manufacturer: Doppelmayr-Lohmann
Type: GPW 215
Nominal torque: 104512.6 ftlb
Max. torque: 146332.3 ftlb
Ratio: 89.58
Efficiency: 0.93
Inertia: 11.87 lbft²
Req. input speed: 1811 rpm
Utilisation: 37 %
Req. torque continuous: 38658.2 ftlb (Lc: 100% UH+DH Load; Forward operation (5.1m/s); T0+; IO)
Req. torque maximum: 53472.7 ftlb (Lc: 100% UH+DH Load, 2 Spans Rope Only; Forward operation (5.1m/s); T0; IO)

Electric motor:

Manufacturer:
Type:
Rating: 0 HP
Speed: 0 rpm
Inertia: 0.00 lbft²
Req. power continuous: 160 HP (Lc: 100% UH+DH Load; Forward operation (5.1m/s); T0+; IO)

Service brake(s):

Inertia: 0.00 lbft²
Max. braking torque: 0.00 ftlb (0.00 kN @ 0.00 m)
Req. braking torque: 132.34 ftlb (Lc: 0% UH Load, 100% DH Load; Braking (0.50m/s²); T0)

Emergency brake(s):

Max. braking torque: 0.0 ftlb (0.00 kN @ 0.00 m)
Req. braking torque: 11854.7 ftlb (Lc: 0% UH Load, 100% DH Load; Braking (0.50m/s²); T0)

Evac drive:

System: rim gear on bullwheel
Diesel engine:
Efficiency: continuous, starting: 0.80 braking: 1.00
Req. power continuous: 39 HP (Lc: 100% UH+DH Load; Forward operation (1.0m/s); T0+; IO)
Req. power starting: 0 HP (Lc:)
Req. power braking: 0 HP (Lc:)
Req. M_{dmax} bullwheel: 0.0 ftlb (Lc:)

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Rope point coordinates - UH

Rope point	XR[ft]	ZR[ft]	Psi[°]	Alpha[°]	F2[ft]
BEB	10060.62	9549.48	-0.9	1.7	0.00
1	10075.88	9549.43	2.6	5.4	-1.25
2	10244.62	9572.01	5.2	9.2	1.26
3	10514.34	9580.42	0.6	8.1	1.24
4	10944.52	9571.24	-3.0	11.5	1.66
5	11360.63	9522.66	-5.3	5.7	1.05
6	11682.70	9485.76	0.3	6.9	-1.39
7	12117.70	9533.77	5.0	10.4	1.60
8	12592.97	9553.94	3.2	6.9	1.19
TEB	12686.79	9554.01	1.0	1.9	0.00

Rope point coordinates - DH

Rope point	XR[ft]	ZR[ft]	Psi[°]	Alpha[°]	F2[ft]
BEB	10060.62	9549.48	-0.9	1.8	0.00
1	10075.88	9549.43	2.5	5.4	-1.25
2	10244.62	9572.01	5.2	9.3	1.26
3	10514.37	9580.43	-1.1	8.4	1.25
4	10944.52	9571.24	-3.0	11.8	1.66
5	11360.66	9522.66	-7.3	6.0	1.05
6	11682.69	9485.83	0.2	6.4	-1.33
7	12117.70	9533.77	5.1	10.8	1.60
8	12592.97	9553.95	3.4	7.3	1.20
TEB	12686.79	9554.01	1.0	2.1	0.00

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Span geometry - UH

Span	l[ft]	h[ft]	l*[ft]	Chord angle[°]	Chord angle[%]
BEB - 1	15.26	-0.05	15.26	-0.19	-0.33
1 - 2	168.74	22.58	170.25	7.62	13.38
2 - 3	269.72	8.41	269.85	1.79	3.12
3 - 4	430.18	-9.18	430.28	-1.22	-2.13
4 - 5	416.11	-48.58	418.94	-6.66	-11.67
5 - 6	322.07	-36.90	324.18	-6.54	-11.46
6 - 7	435.00	48.01	437.64	6.30	11.04
7 - 8	475.27	20.16	475.70	2.43	4.24
8 - TEB	93.82	0.07	93.82	0.05	0.08

Span geometry - DH

Span	l[ft]	h[ft]	l*[ft]	Chord angle[°]	Chord angle[%]
BEB - 1	15.26	-0.05	15.26	-0.19	-0.33
1 - 2	168.74	22.58	170.25	7.62	13.38
2 - 3	269.76	8.42	269.89	1.79	3.12
3 - 4	430.14	-9.19	430.24	-1.22	-2.14
4 - 5	416.14	-48.58	418.97	-6.66	-11.67
5 - 6	322.03	-36.84	324.13	-6.53	-11.44
6 - 7	435.00	47.95	437.64	6.29	11.02
7 - 8	475.27	20.18	475.70	2.43	4.25
8 - TEB	93.82	0.06	93.82	0.04	0.06

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Calculation mode and load configurations

Calculation method: Point load (increments of 2.00 ft)

Calculated load configurations:

- | | |
|----|--|
| 1 | Rope Only; Forward operation (5.1m/s); T0; IO |
| 1+ | Rope Only; Forward operation (5.1m/s); T0+; IO |
| 1- | Rope Only; Forward operation (5.1m/s); T0-; IO |
| 2 | Rope + Empty Carriers; Forward operation (5.1m/s); T0; IO |
| 2+ | Rope + Empty Carriers; Forward operation (5.1m/s); T0+; IO |
| 2- | Rope + Empty Carriers; Forward operation (5.1m/s); T0-; IO |
| 3 | 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0; IO |
| 3+ | 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0+; IO |
| 3- | 100% UH Load, 0% DH Load; Forward operation (5.1m/s); T0-; IO |
| 4 | 100% UH+DH Load; Forward operation (5.1m/s); T0; IO |
| 4+ | 100% UH+DH Load; Forward operation (5.1m/s); T0+; IO |
| 4- | 100% UH+DH Load; Forward operation (5.1m/s); T0-; IO |
| 5 | 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0; IO |
| 5+ | 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0+; IO |
| 5- | 0% UH Load, 100% DH Load; Forward operation (5.1m/s); T0-; IO |
| 6 | 100% UH+DH Load, 2 Spans Rope Only; Forward operation (5.1m/s); T0; IO |

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Rope tension extremes, rope design factors, shear ratio

Rope tension extremes at nominal tension without dynamics

Tmax = 50552 lb (Rp: TBW; Lc: 6)

Tmin = 42681 lb (Rp: 6; Lc: 6)

Rope tension extremes at nominal tension with dynamics

Tmax = 50552 lb (Rp: TBW; Lc: 6)

Tmin = 42681 lb (Rp: 6; Lc: 6)

Rope tension extremes without dynamics

Tmax = 52163 lb (Rp: TBW; Lc: 3+)

Tmin = 40775 lb (Rp: 6; Lc: 4-)

Rope tension extremes with dynamics

Tmax = 52163 lb (Rp: TBW; Lc: 3+)

Tmin = 40775 lb (Rp: 6; Lc: 4-)

Rope design factor: min.: 4.87 max.: 6.04

Minimum shear ratio carrier: 14.47

Specific weights:

rope only:	4.7441 lb/ft
rope + empty carriers:	12.7028 lb/ft
rope + loaded carriers UH:	19.4930 lb/ft
rope + loaded carriers DH:	19.4930 lb/ft

Max. angle change: 0.0895 rad; 5.13° (Rp: 7; to top; DH)

Greatest rope inclination: 25.13%; 14.10° (Rp: 4; to top; DH; Lc: 4-)

Required min. pull force: 2157 lb

Bullwheels - SumTmax and Mdmax, extreme values with dynamic

	SumTmax [lb]	(Lc)	Md [lbft]	Mdmax [lbft]	(Lc)	SumT [lb]
Drive bullwheel	99589	(4+)	44639	53473	(6)	96065
Return bullwheel	97488	(1+)				
	pmax [lb/in²]	(Lc)				
Drive bullwheel	456	(4+)				
Return bullwheel	420	(1+)				

Bullwheels - SumTmax and Mdmax, values at nominal tension without dynamic

	SumTmax [lb]	(Lc)	Md [lbft]	Mdmax [lbft]	(Lc)	SumT [lb]
Drive bullwheel	94978	(4)	37601	37601	(4)	94978
Return bullwheel	92846	(1)				

Clearance envelope

Required line width: 46.10 ft

Required line width - UH - Right: 22.94 ft (RP: 7; Lc: 100% UH Load, 0% DH Load; Forward operation; T0-; IO)

Required line width - DH - Left: 23.16 ft (RP: 7; Lc: 100% UH+DH Load; Forward operation; T0-; IO)

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Circumferential forces, torques, required power (min ... geo. mean ... max)

Lc	Circumferential force [lb]	torque [ftlb]	required power [HP]	μ nec
1	3257	27843	115	0.023
1+	3378	28901	120	0.022
1-	3137	26784	111	0.023
2	3692 ... 3926 ... 4192	31271 ... 33109 ... 35199	129 ... 137 ... 146	0.029
2+	3806 ... 4046 ... 4303	32277 ... 34168 ... 36185	134 ... 141 ... 150	0.029
2-	3579 ... 3806 ... 4088	30270 ... 32051 ... 34270	125 ... 133 ... 142	0.030
3	3629 ... 4243 ... 4825	30766 ... 35622 ... 40232	127 ... 147 ... 166	0.033
3+	3783 ... 4363 ... 4927	32083 ... 36681 ... 41141	133 ... 152 ... 170	0.032
3-	3472 ... 4122 ... 4725	29421 ... 34564 ... 39335	122 ... 143 ... 163	0.034
4	3810 ... 4498 ... 5300	32186 ... 37601 ... 43918	133 ... 156 ... 182	0.037
4+	3971 ... 4618 ... 5377	33564 ... 38658 ... 44639	139 ... 160 ... 185	0.036
4-	3645 ... 4378 ... 5227	30771 ... 36544 ... 43231	127 ... 151 ... 179	0.038
5	3392 ... 4181 ... 4866	28936 ... 35122 ... 40494	120 ... 145 ... 168	0.034
5+	3532 ... 4301 ... 4952	30148 ... 36178 ... 41279	125 ... 150 ... 171	0.033
5-	3250 ... 4061 ... 4785	27710 ... 34067 ... 39739	115 ... 141 ... 164	0.035
6	2370 ... 4824 ... 6510	20877 ... 40201 ... 53473	86 ... 166 ... 221	0.045

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Maximum and minimum rope tensions - UH

Rope point	Tmax [lb]	Lc	Tmin [lb]	Lc
BBW	48890	(1+)	44234	(1-)
BEB	49179	(3+)	44443	(1-)
1	49275	(3+)	44428	(3-)
2	50121	(3+)	44666	(3-)
3	50542	(3+)	44875	(1-)
4	50786	(3+)	44932	(1-)
5	50177	(3+)	44219	(3-)
6	49749	(3+)	43646	(3-)
7	51280	(3+)	44485	(3-)
8	51900	(3+)	45259	(3-)
TEB	51907	(3+)	45452	(3-)
TBW	52163	(3+)	45697	(3-)

Maximum and minimum rope tensions - DH

Rope point	Tmax [lb]	Lc	Tmin [lb]	Lc
BBW	48598	(1+)	43970	(1-)
BEB	48380	(1+)	43691	(4-)
1	48430	(4+)	43472	(4-)
2	48829	(4+)	43358	(4-)
3	48822	(4+)	43289	(4-)
4	48597	(4+)	42891	(4-)
5	47676	(1+)	41620	(4-)
6	47461	(1+)	40775	(4-)
7	47947	(4+)	41244	(4-)
8	48091	(4+)	41531	(4-)
TEB	47918	(4+)	41525	(4-)
TBW	47667	(4+)	41297	(4-)

Project [-no.]: 8-MGD Gondola Section 3 [SAA0002176]

Maximum and minimum tower loads - UH

Rope point	Sheave assembly	Dmax Lc [lb]	Psi [°]	Alpha [°]	Dmin Lc [lb]	Psi [°]	Alpha [°]
BEB	3R-403	2761 (3+)	-1.61	3.22	183 (1-)	-0.12	0.24
1	8N-401	-6269 (1+)	3.51	-7.30	-2718 (3-)	1.61	-3.50
2	8T-401	9806 (4+)	5.79	11.25	5598 (1-)	4.55	7.17
3	8T-401	9788 (4+)	1.15	11.16	4019 (1-)	0.04	5.13
4	8T-501	13159 (4+)	-2.18	15.04	6275 (1-)	-3.92	7.99
5	6T-401	7381 (3-)	-4.21	9.51	1646 (1+)	-6.47	1.90
6	8N-401	-9293 (1+)	-0.27	-10.76	-2277 (3-)	0.84	-2.95
7	8T-501	12533 (3+)	5.79	14.19	5211 (1-)	4.31	6.60
8	8T-401	8608 (3+)	4.66	9.73	3242 (1-)	1.81	4.09
TEB	3R-403	3158 (4+)	1.75	3.50	259 (3-)	0.16	0.33

Maximum and minimum tower loads - DH

Rope point	Sheave assembly	Dmax Lc [lb]	Psi [°]	Alpha [°]	Dmin Lc [lb]	Psi [°]	Alpha [°]
BEB	3R-403	2829 (4+)	-1.68	3.35	181 (1-)	-0.12	0.24
1	8N-401	-6146 (1+)	3.50	-7.29	-2603 (4-)	1.56	-3.41
2	8T-401	9662 (4+)	5.80	11.38	5485 (1-)	4.55	7.20
3	8T-401	9718 (4+)	-2.31	11.51	3951 (1-)	0.03	5.20
4	8T-501	12924 (4+)	-2.14	15.46	6122 (1-)	-3.92	8.08
5	6T-401	7377 (4-)	-8.07	10.08	1641 (1+)	-6.46	1.97
6	8N-401	-8784 (1+)	-0.28	-10.64	-1613 (4-)	0.64	-2.25
7	8T-501	12280 (4+)	5.85	14.89	5045 (1-)	4.29	6.75
8	8T-401	8422 (4+)	4.92	10.32	3141 (1-)	1.84	4.20
TEB	3R-403	3196 (4+)	1.92	3.84	250 (4-)	0.17	0.34

APPENDIX – D

CABINS AND CLEARANCE DRAWINGS

Exiting Cabins

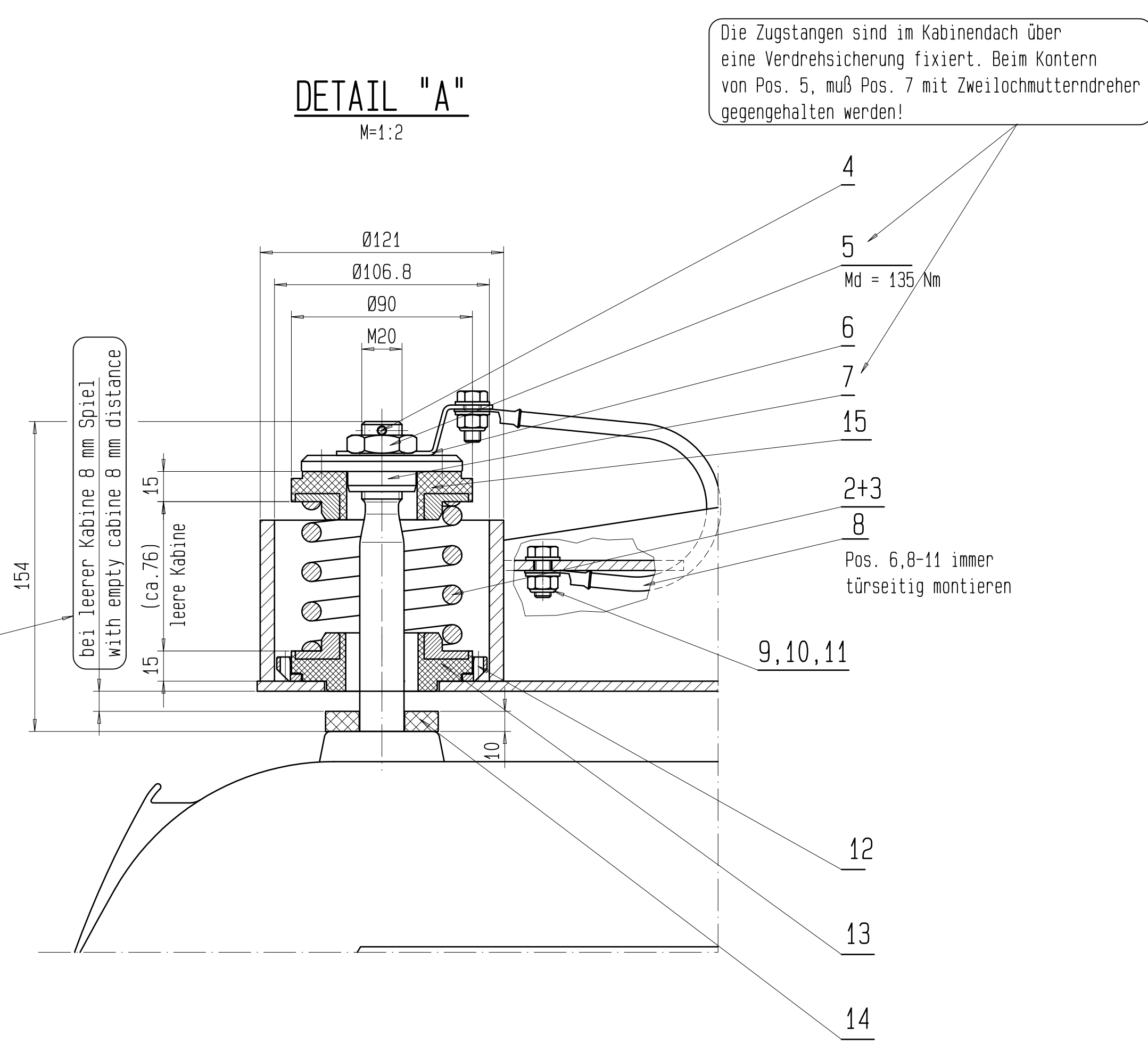
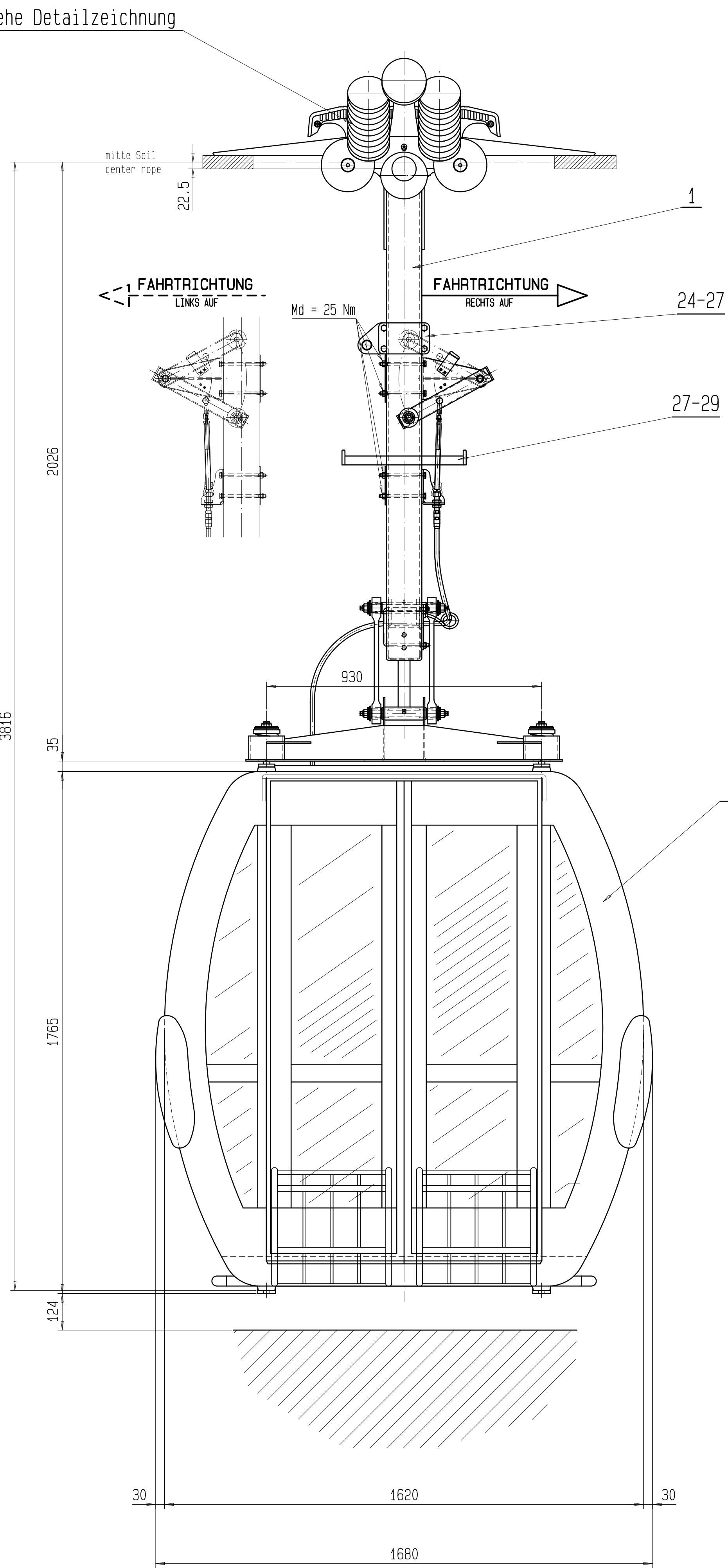
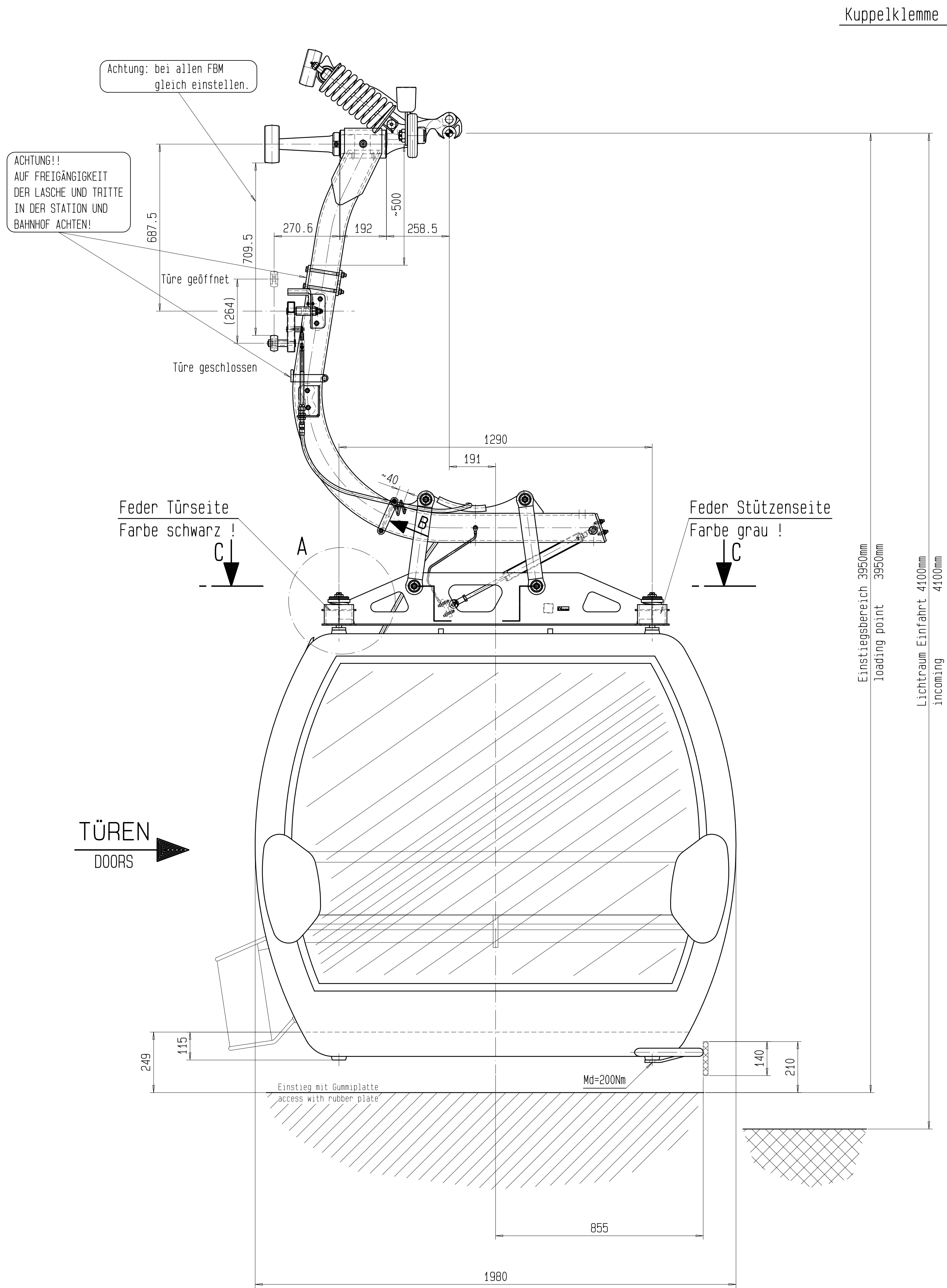
1. 20015699J113300.a CWA Omega
2. 20014909J113300 CWA Omega III
3. 20015685J113300.a CWA Omega III-XL

Proposed Cabins

4. 20014073J113300.c CWA Omega IV-LWI (Standard)
5. 20017143J113300.c CWA Omega IV-LWI, -220mm Suspension

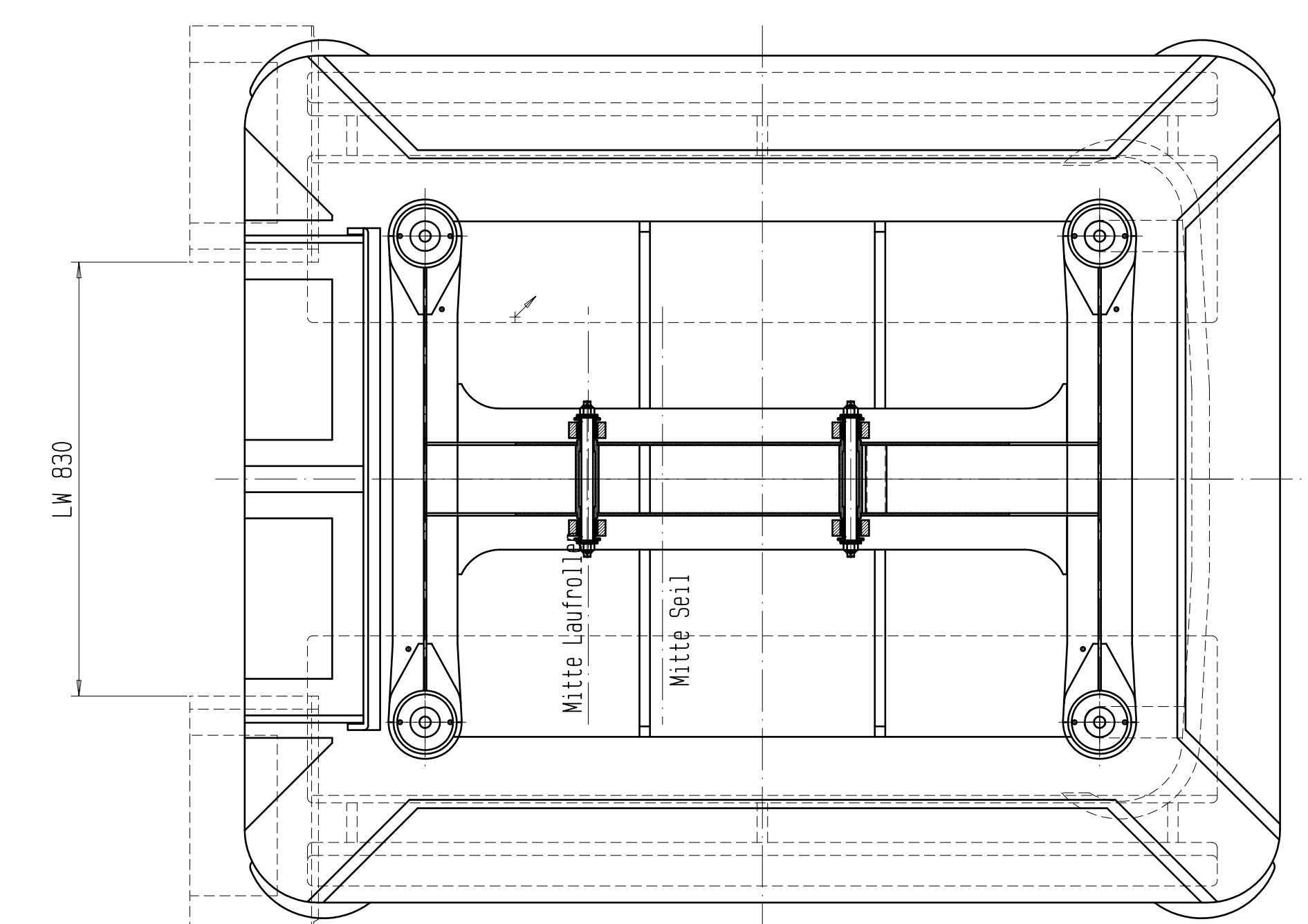
CWA Omega IV-LWI, w\Shortened Suspension (-220mm) Clearance Drawings

6. 90000215NAJ115301 Tower Swing Clearance
7. 90005513NAE001401 Terminal Clearance



Achtung: bei Montage muß das Maß 8 mm eingestellt und überprüft werden!

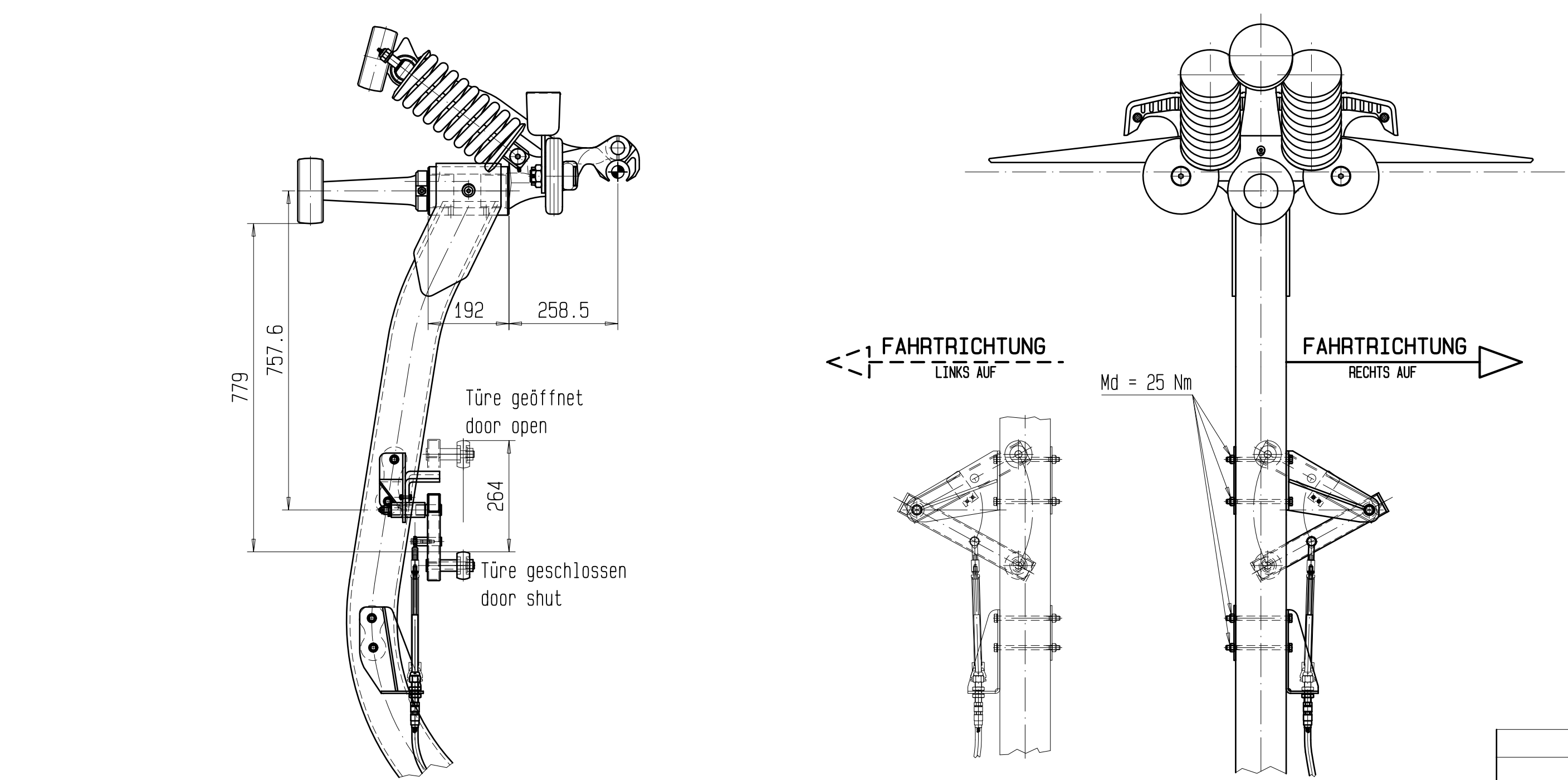
SNITT "C-C"
M=1:10



Fernbetätigung innen

ANSICHT B
M=1:5

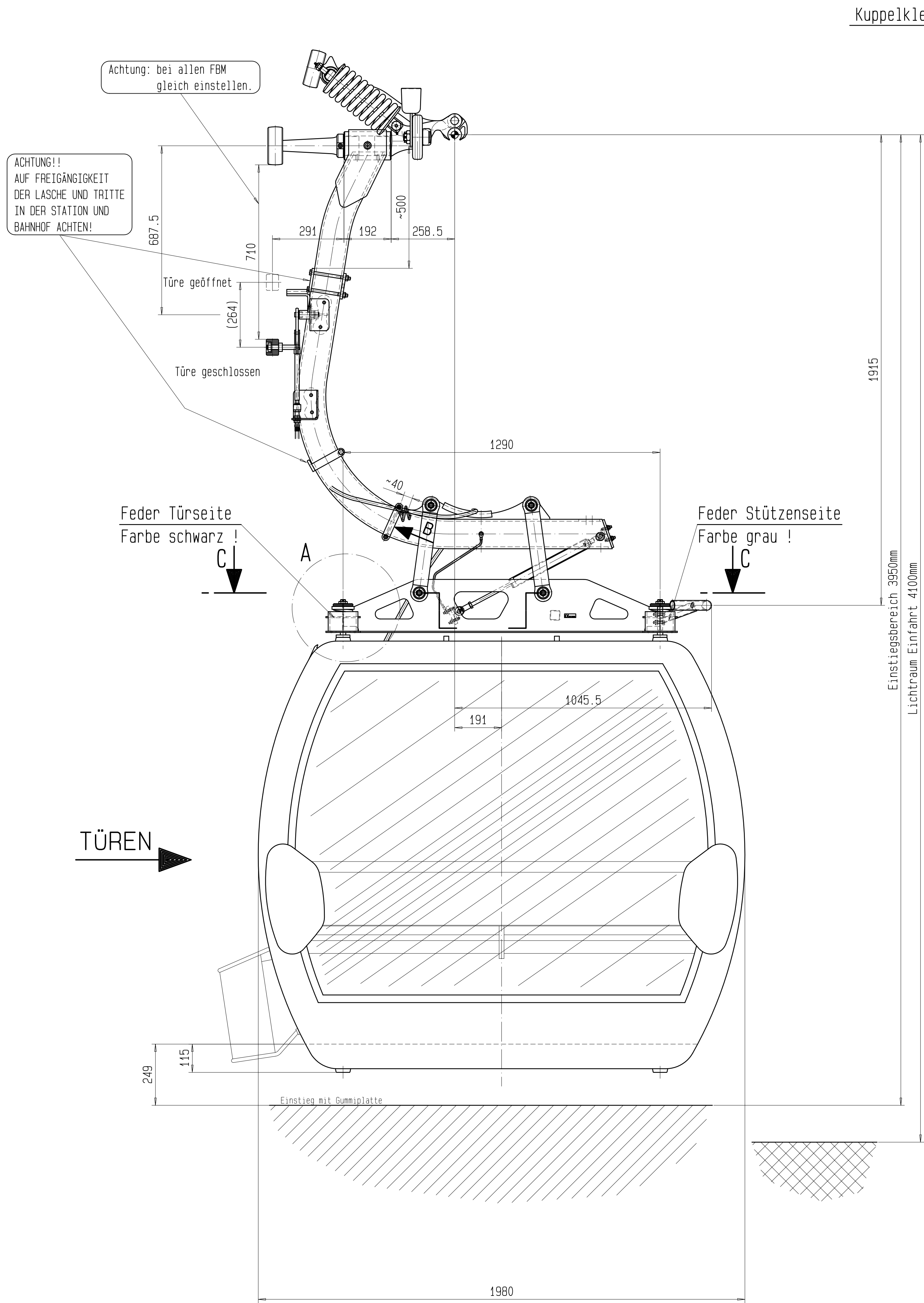
Melexspirale darf nur seitlich von der Gehängestange montiert werden



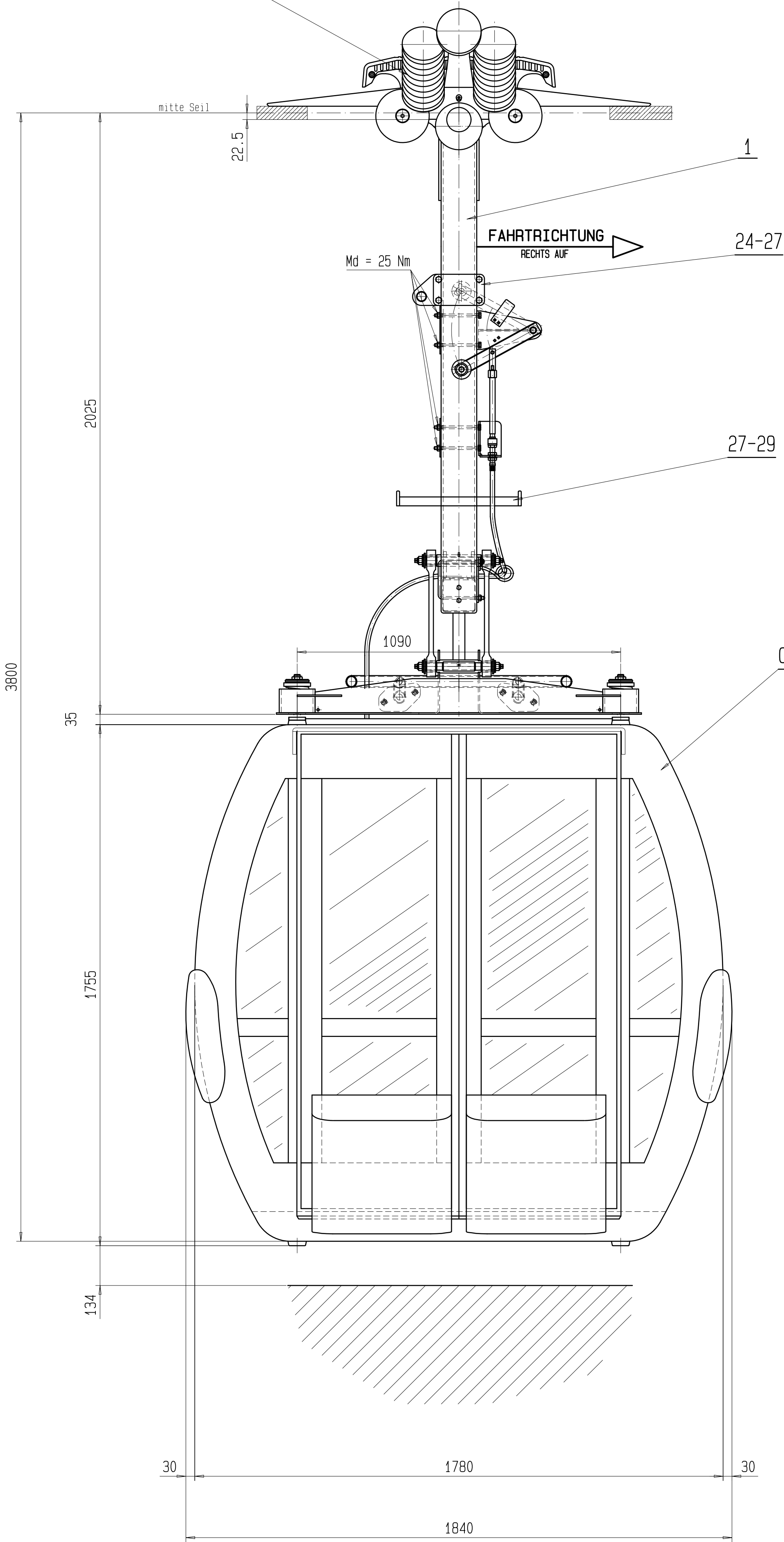
RECHTSAUF	LINKSAUF
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RECHTSAUF	LINKSAUF
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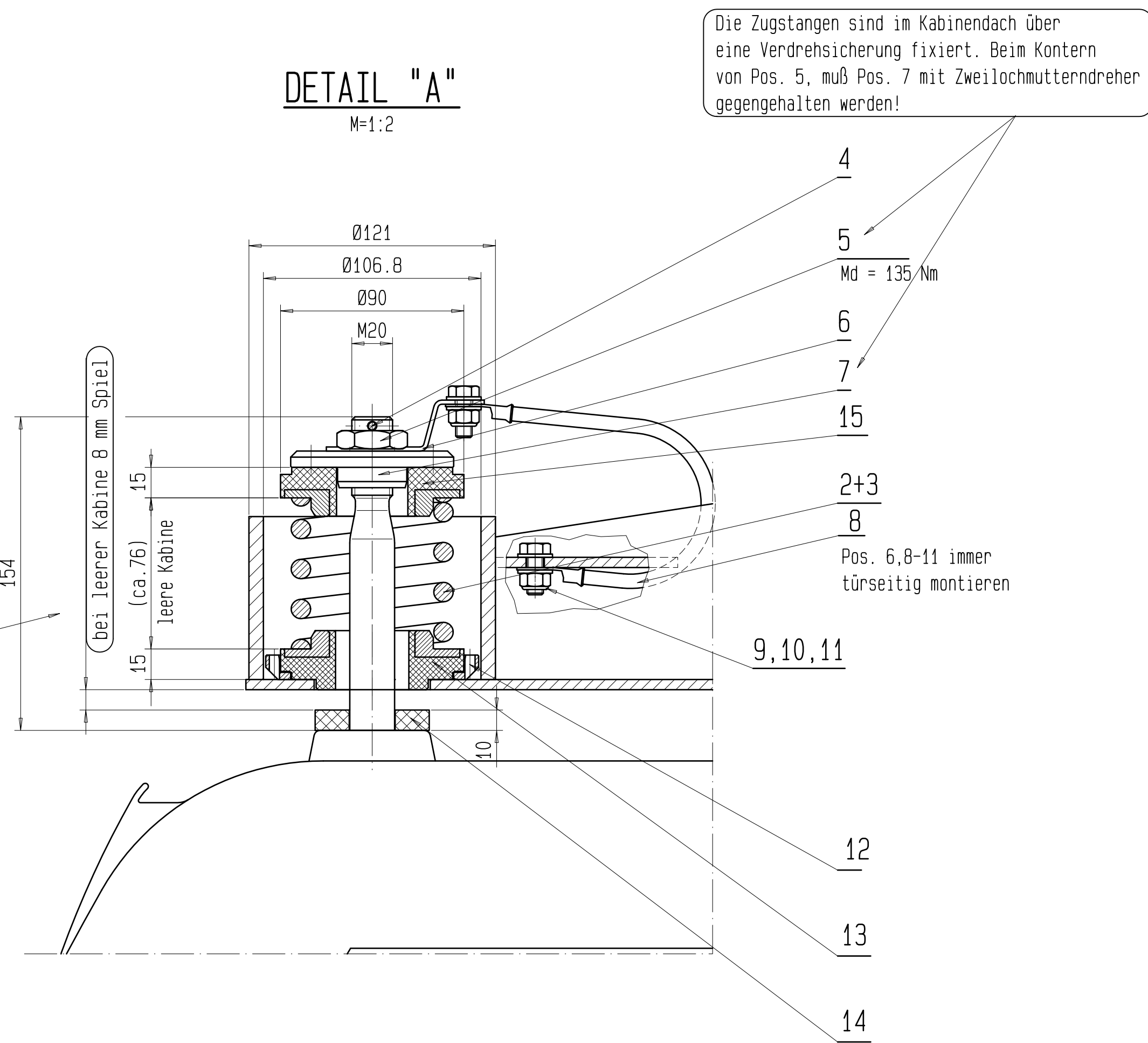
komplett		10686987
ohne Pos. 24 - 29		10686978
Zeichn. / Änderung	Bestell-Nr.	Best-Nr.
Doppelmayr	1700000000	1700000000
Aktuelle Ausführung	SB	1/10
Maße nach	kg	2015-09-02
Auftrag	Stück	X
FAHRZEUG 8MGD 2005		10673012
Kabine CWA Omega III		20012298/113300.a
A106C, A108-97, Aufhängung GCS		20014909/113300



Kuppelklemme DT108 siehe Detailzeichnung

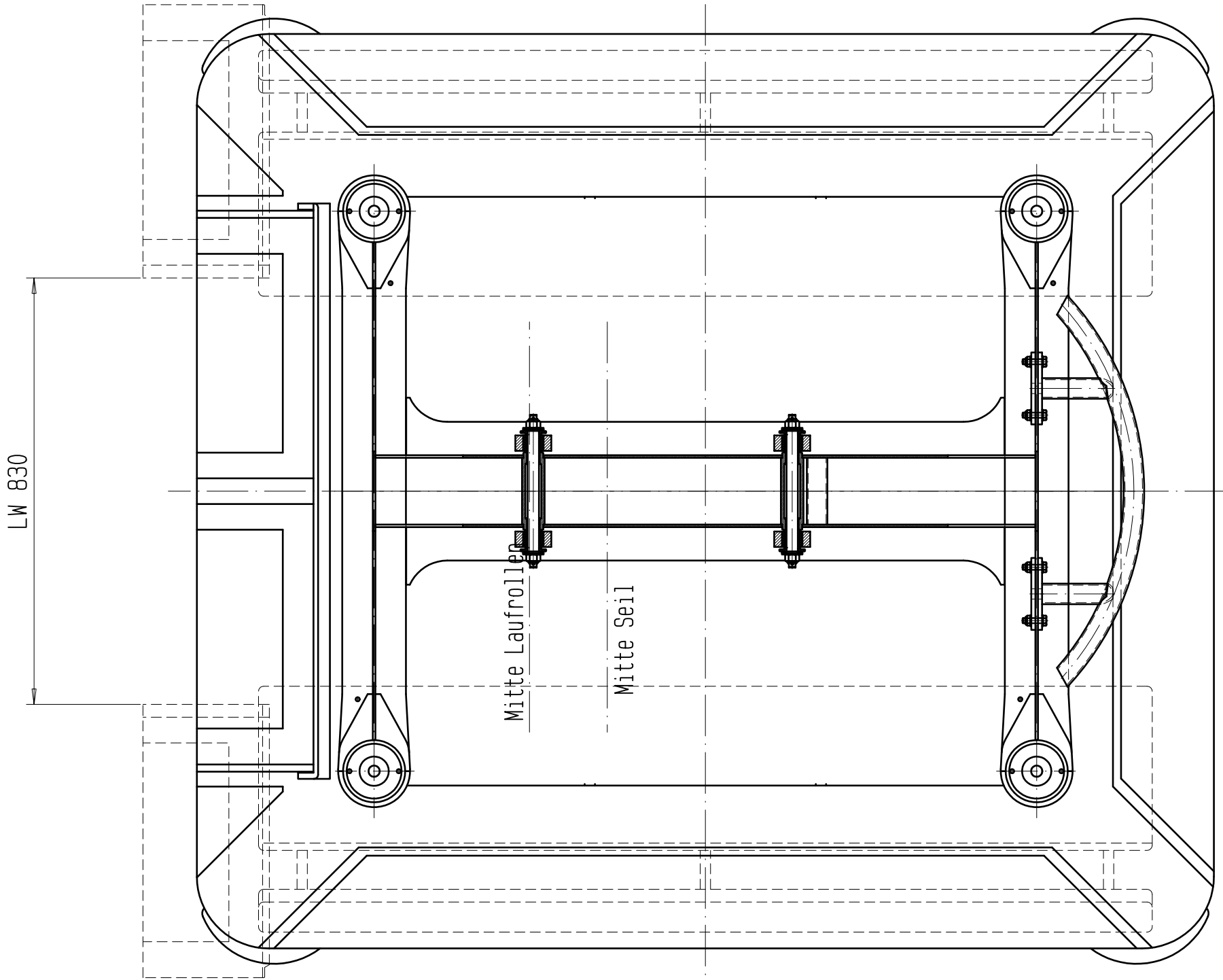


DETAIL "A"
M=1:2

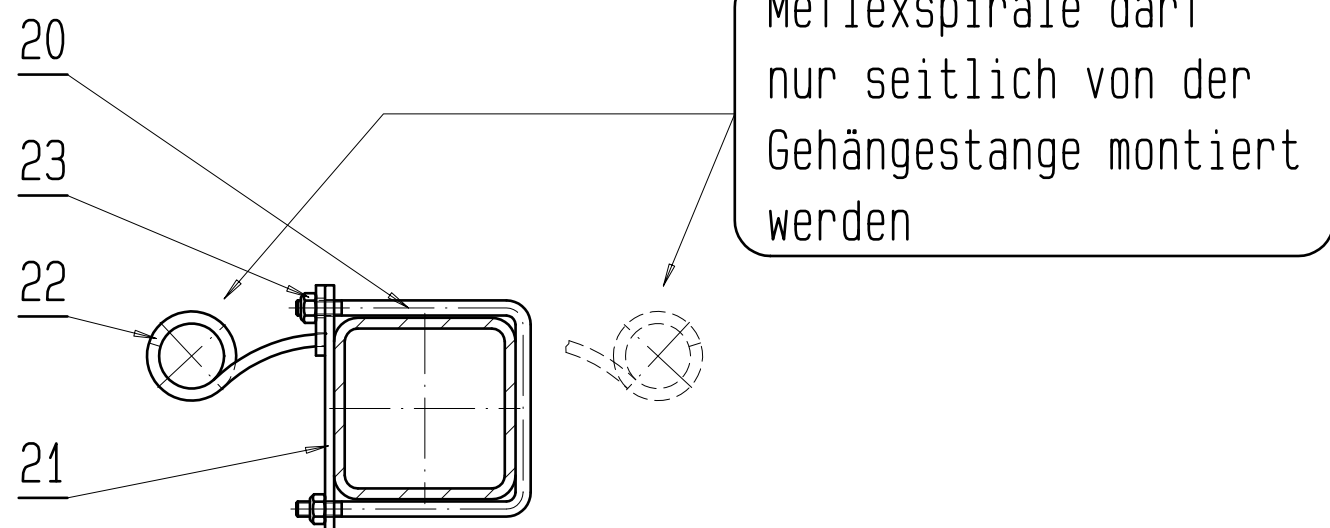


Achtung: bei Montage muß das Maß 8 mm eingestellt und überprüft werden!

SCHNITT "C-C"
M=1:10

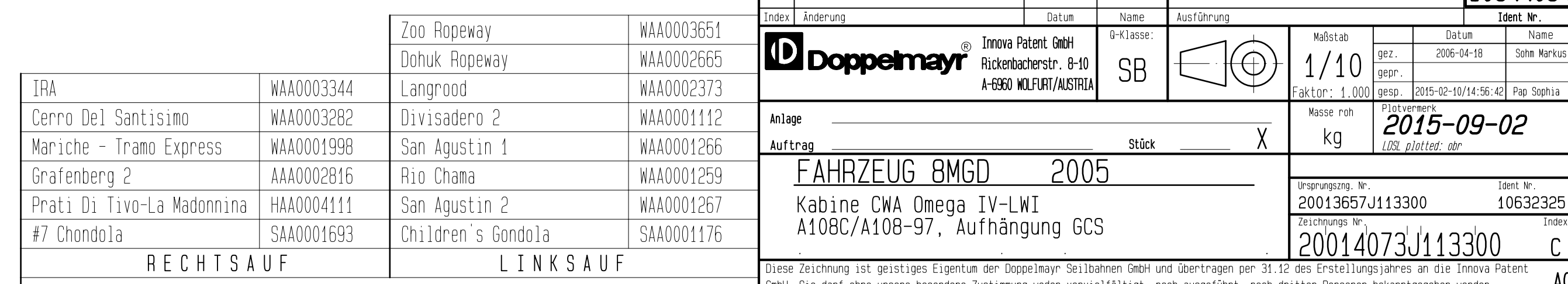


ANSICHT B
M=1:5



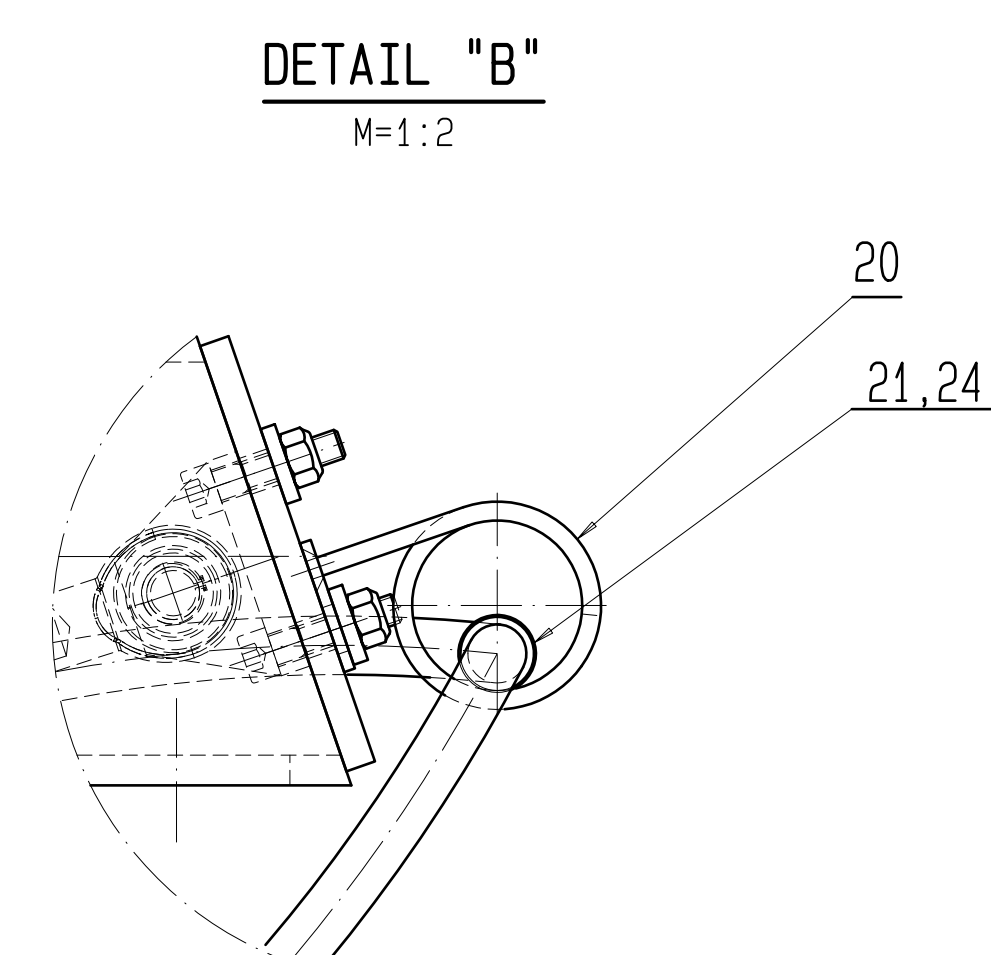
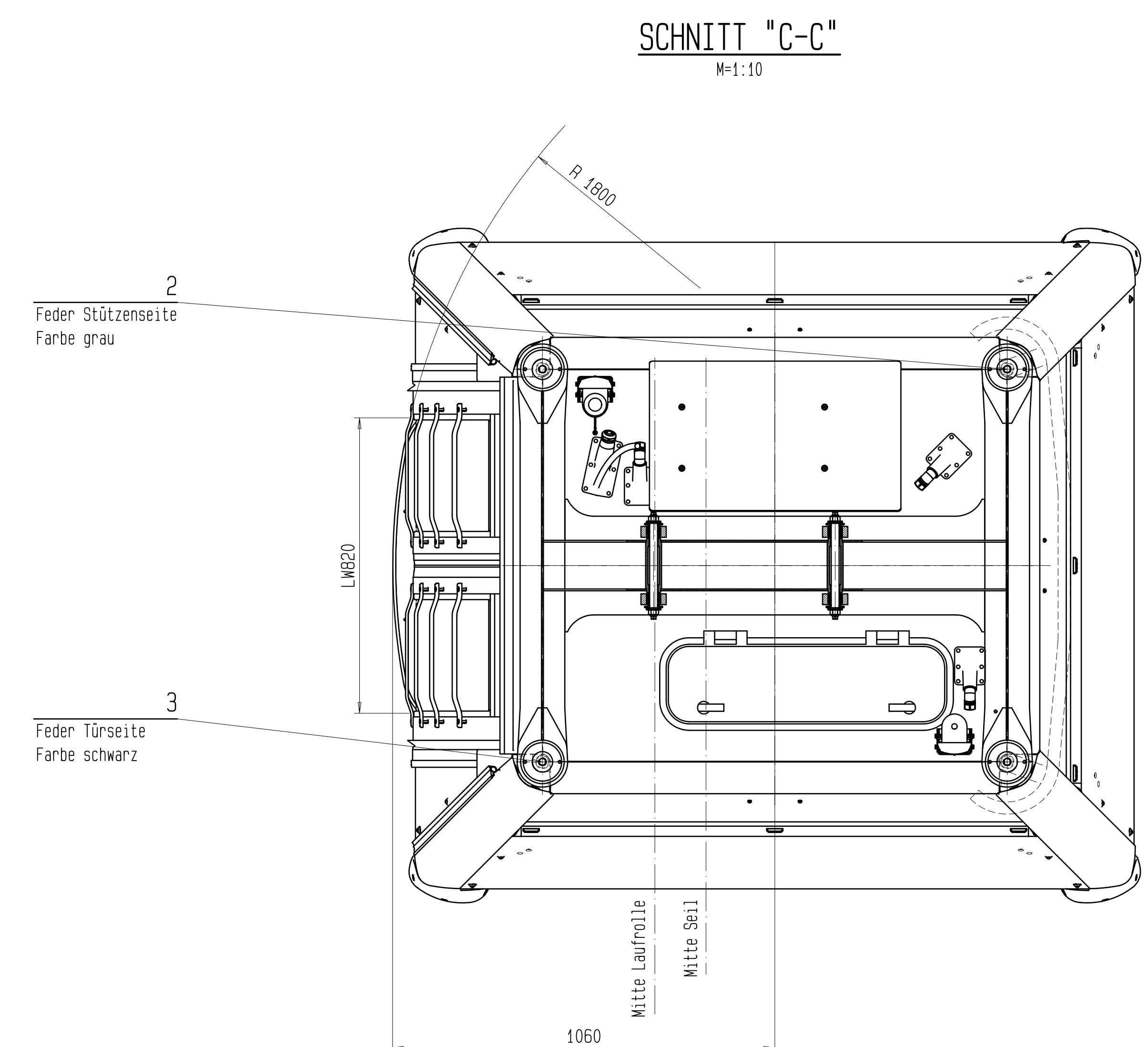
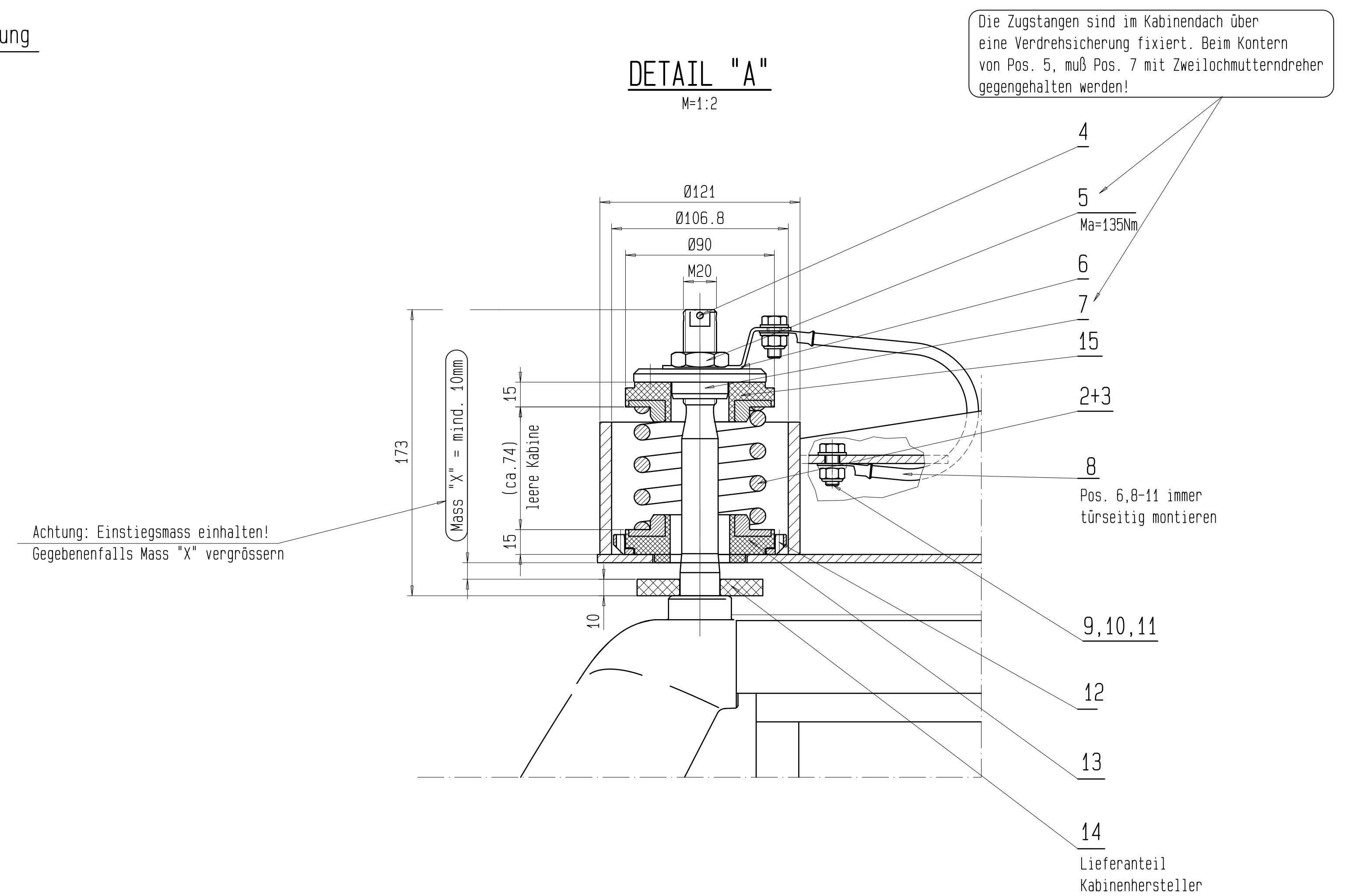
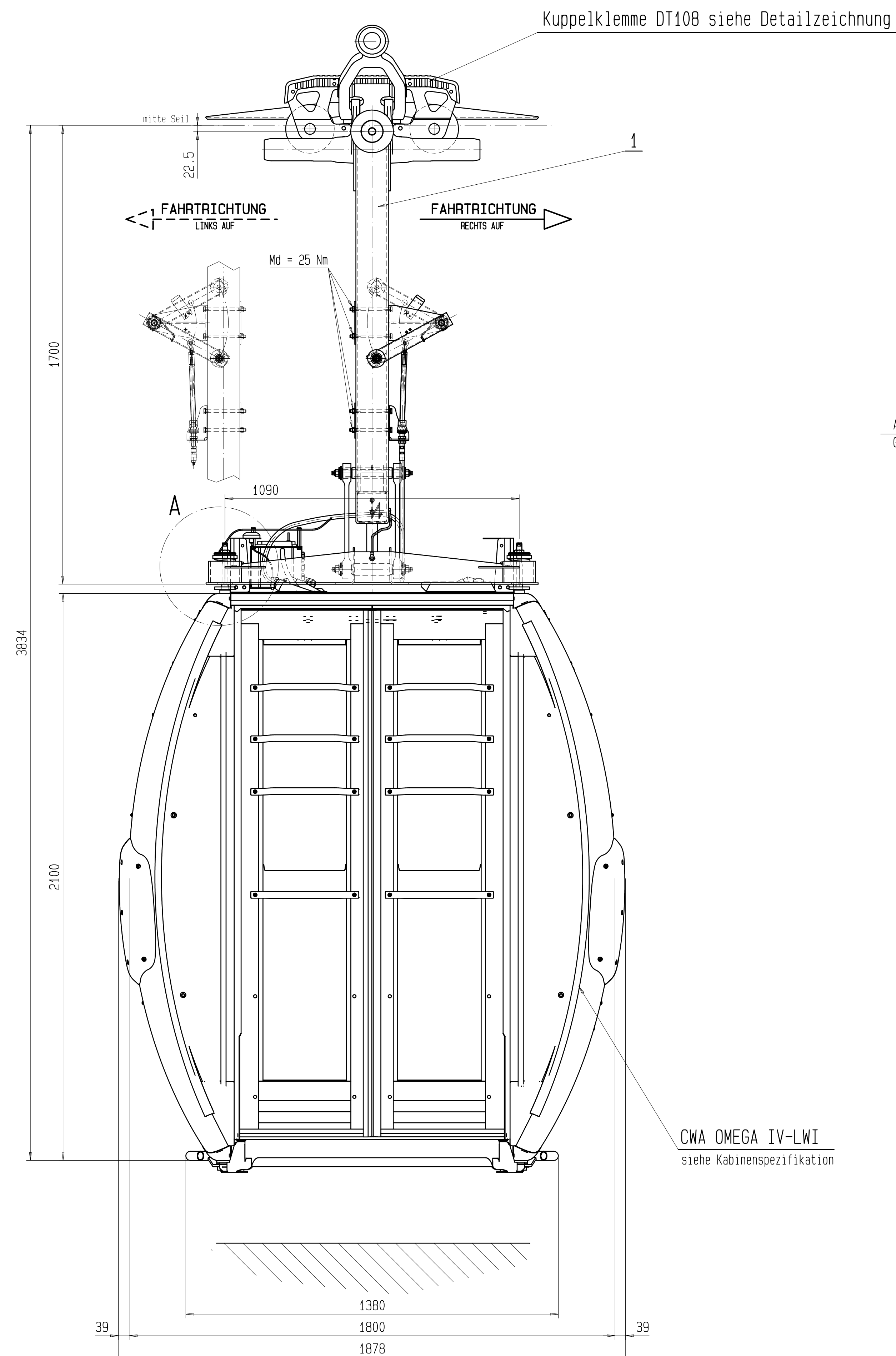
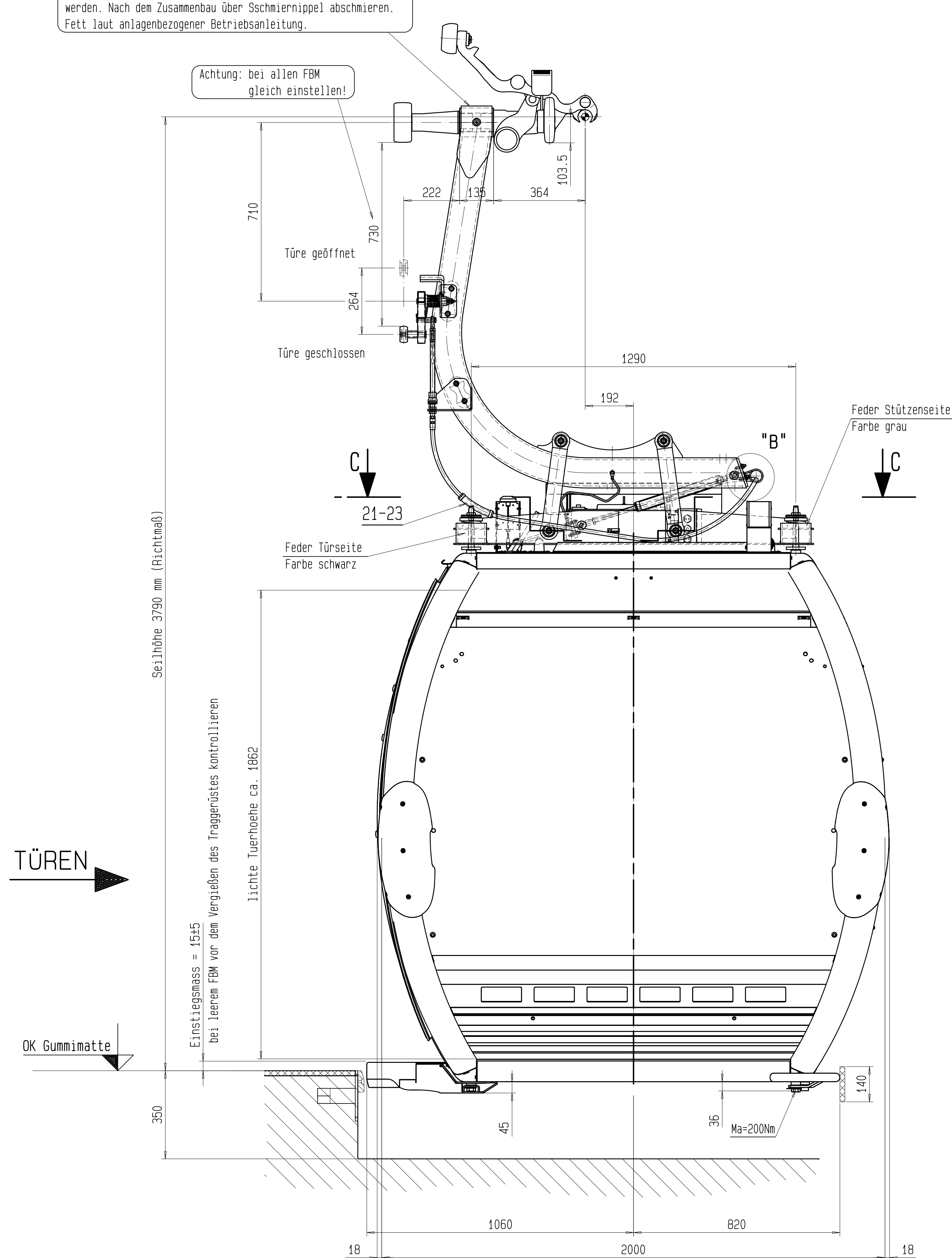
a Fernbetätigung angepasst		2007-09-12	Georen F.	10739721	
Zeichn. Übertrag	(Datum)	(Name)	(Ausführung)	(Menge)	(Stand)
ID Doppelmayer		Irrexa Patent GmbH	SB	1/10	2007-09-12
Auftrag		Mountain Village	Stück	X	2015-09-02
FAHRZEUG 8MGD		2005	10686987		
Kabinen CWA Omega III-XL		20014909/113300	10686987		
A106C, Aufhängung GCS		20015685/113300	a		

Verlauf und Führung des Meflex-Kabel

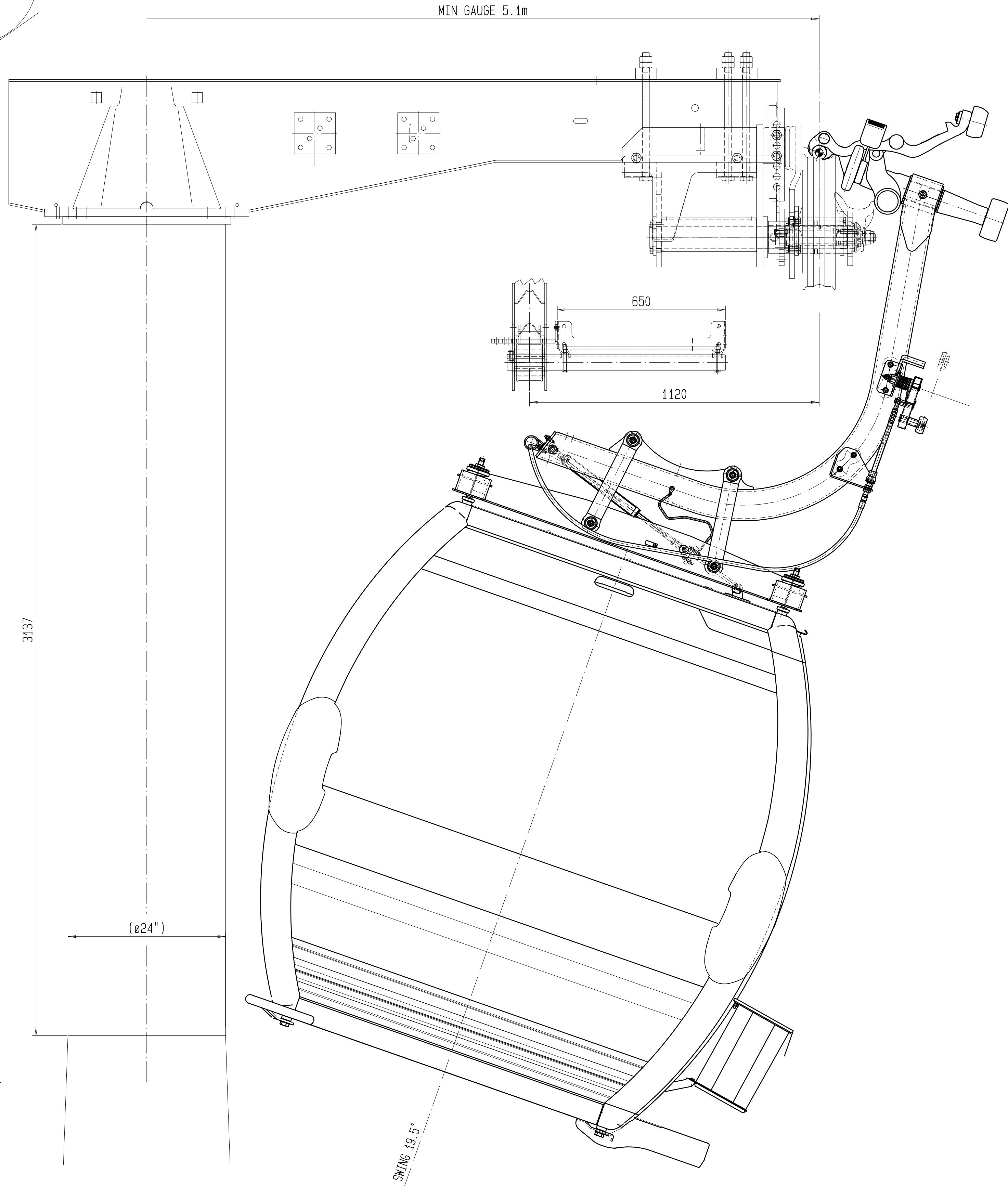
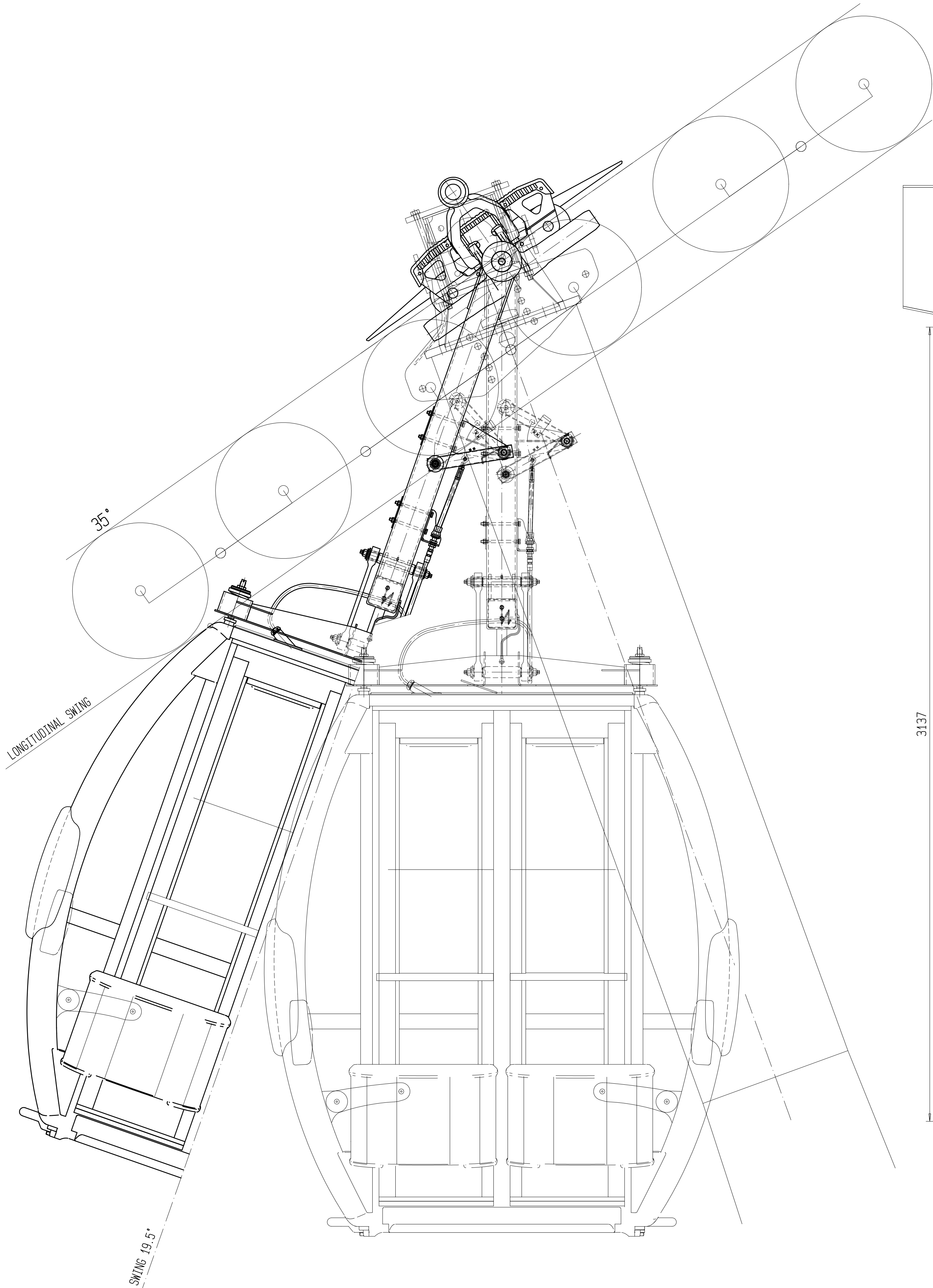


Schmieranweisung:
Reinigung der Lagerstelle. Diese muss vor dem Schmieren trocken sein. Vor dem Einbau der Klemme muss die Lagerstelle im Gehängekopf und die Gehängeachse der Klemme rundum geschmiert werden. Nach dem Zusammenbau über Sschiernippel abschmieren. Fett laut anlagenbezogener Betriebsanleitung.

Achtung: bei allen FBM
gleich einstellen!



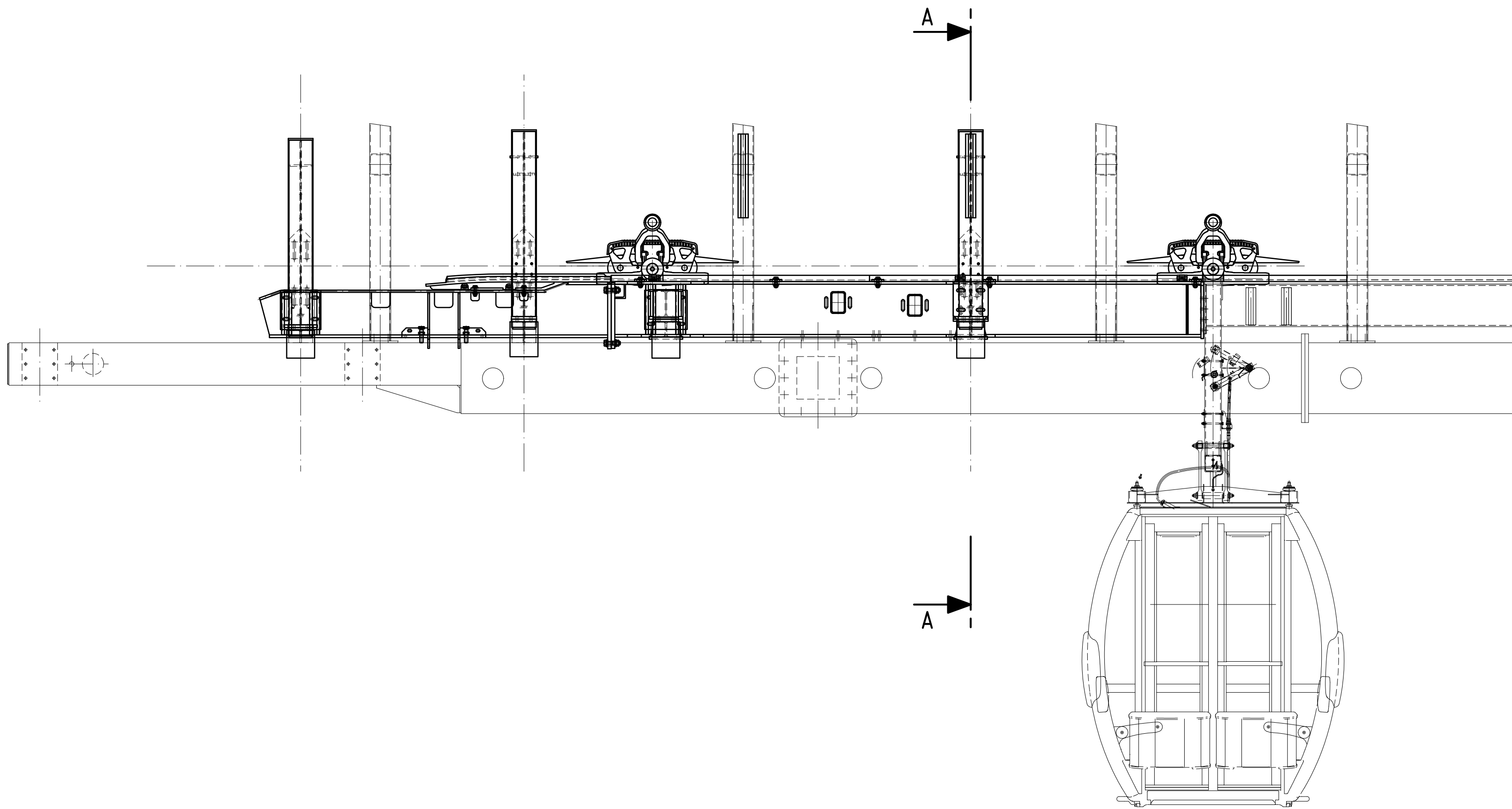
a	Hinweis: Schenkung Pkt. 21, 24 neu	2010-10-10	Barbara Ranta						
b	Bemassung Einstiegsraum anpos. 30 auf 14 Literanteil Kabinenh.	2010-11-30	Doreen F.	Duplex					10871829
c	Kabinenbauteile angepasst	2010-11-11	grf						10874503
Index	Abrechnung	Datum	Name	Anzahl/Anzahl					10874503
		Dr. rer. oec. Robert Götzel Rickenbachstr. 8-10 A-6800 MULLENBACH/NEUSTADT			1/10 Faktor: 1,000	Kalkulation 2009-01-01 - 2010-11-30 2010-11-30 bis 2011-10-31 2011-10-31 bis 2012-10-31	Datum 2010-11-30 2011-10-31 2012-10-31	Name Barbara Ranta Doreen F. grf	
Anlage					Masse / kg 1,000				2015-09-02
Auflauf				Stück X					US\$ 21.000,- net
FAHRZEUG BMGD 2005 -220									
Kabine CWA Omega IV-LW1 Klümme DT108, Aufhängung GCS									
					kg 130				
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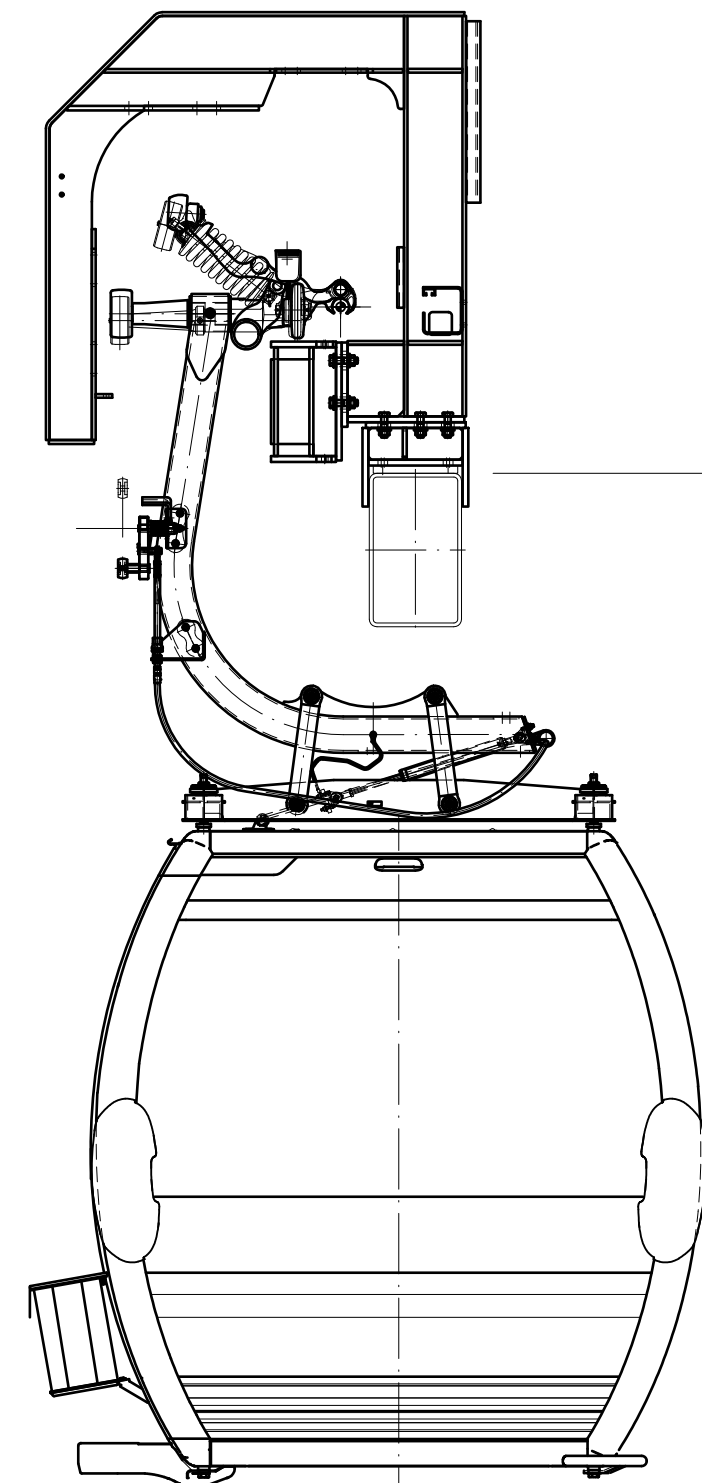
Zeichnung gilt auch für Fahrzeuge
mit A108C Klemme

INDEX		REVISION	DATE	NAME	CONFIGURATION	IDENT NO.				
Doppelmayr		Doppelmayr USA, Inc. 3160 West 500 South Salt Lake City UT 84104		0-CLASS: 0		SCALE 1/10 FACTOR: 1.000		DATE 2015-09-02	NAME Frenchik Jim	
INSTALLATION		ORDER				Mass kg	PLOTTED 2015-09-02 USER: plotter-admin	OVERLANDER		
COPIES						X				
SWING CLEARANCE - 8MGD/GCS-220										
Type 501C,CWA Omega IV-LWI GAUGE 5.1m, GRIP DT108										
ORIGINAL DRAWING NO.						20017145J115301		IDENT NO. 10847702		
DRAWING NO.						90000215NAJ115301		INDEX		
This drawing is intellectual property of Doppelmayr USA, Inc. This Drawing or information is our property and must not be copied nor utilized in whole or in part without permission and is subject to return upon request.										

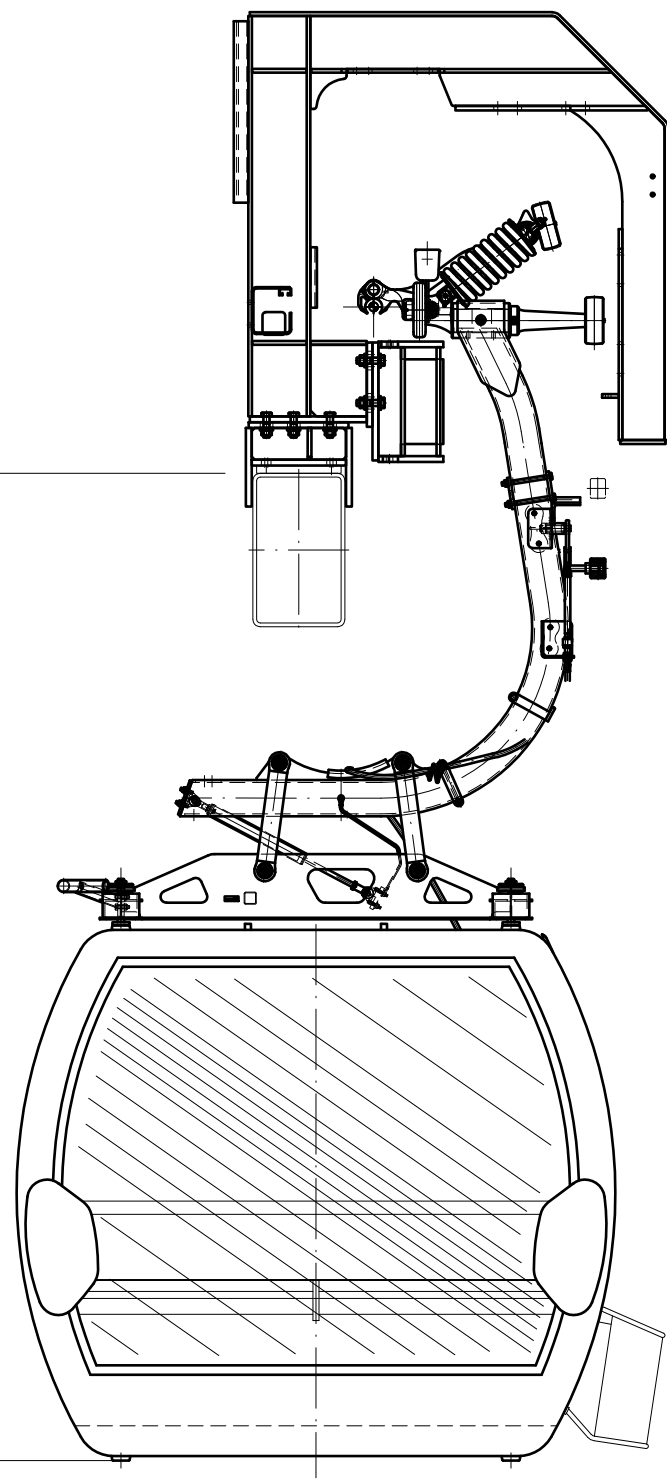
A1



Omega IV-LWI (-220mm)



Omega IV-LWI (-220mm)

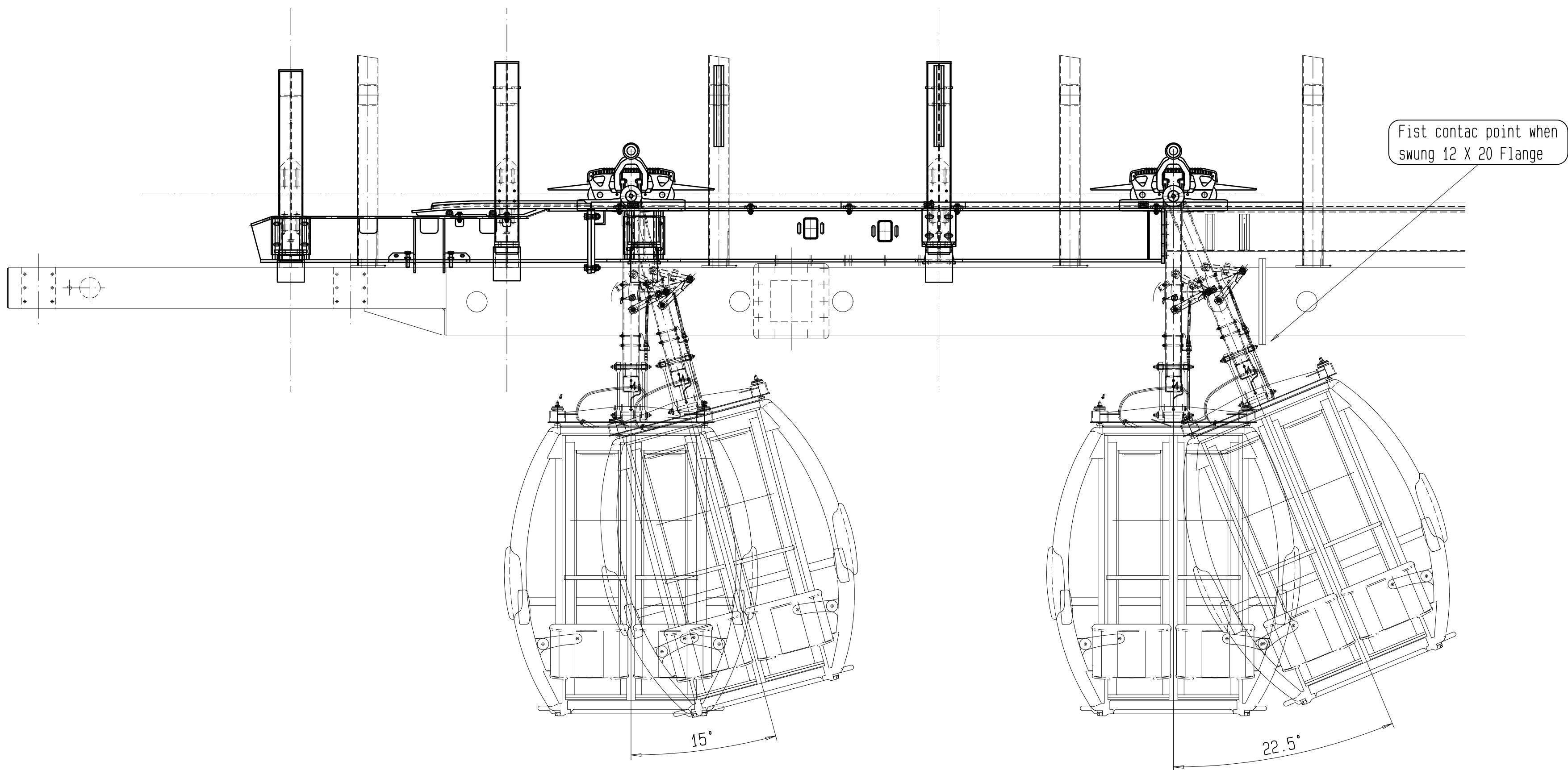


Existing Omega cabin

SECTION "A-A"
S=1:25

Proposed cabin 86 mm lower.

Height Difference



Minimum ANSI Required Swing

Omega IV-LWI (-220mm) shown

Allowable swing clearance

Omega IV-LWI (-220mm) shown

PRIME PAINTED				IDENT NO.	
INDEX	REVISION	DATE	NAME	CONFIGURATION	IDENT NO.
Doppelmayr Doppelmayr USA, Inc. 3160 West 500 South Salt Lake City UT 84104				0	
DETAILATION		MOUNTAIN VILLAGE GONDOLA	SCALE	1/25	DATE
ORDER		SAP0000008	FACTOR: 1.000	MASS	2015-09-02
				kg	2015-09-02/19:40:16
TERMINAL CLEARANCE				PLOTTED	2015-09-02
SECTIONS 1-2 & 3, EXISTING				LOSS	plotted: ddr
8-MGD, OMEG IV-LWI (-220mm)				ORIGINAL DRAWING NO.	IDENT NO.
				XXXXXXXXXX	XXXXXXXXXX
				DRAWING NO.	INDEX
				90005513NAE001401	

This drawing is intellectual property of Doppelmayr USA, Inc. This Drawing or information is our property and must not be copied nor utilized in whole or in part without permission and is subject to return upon request.

APPENDIX – E

MACHINERY

Drive Machinery

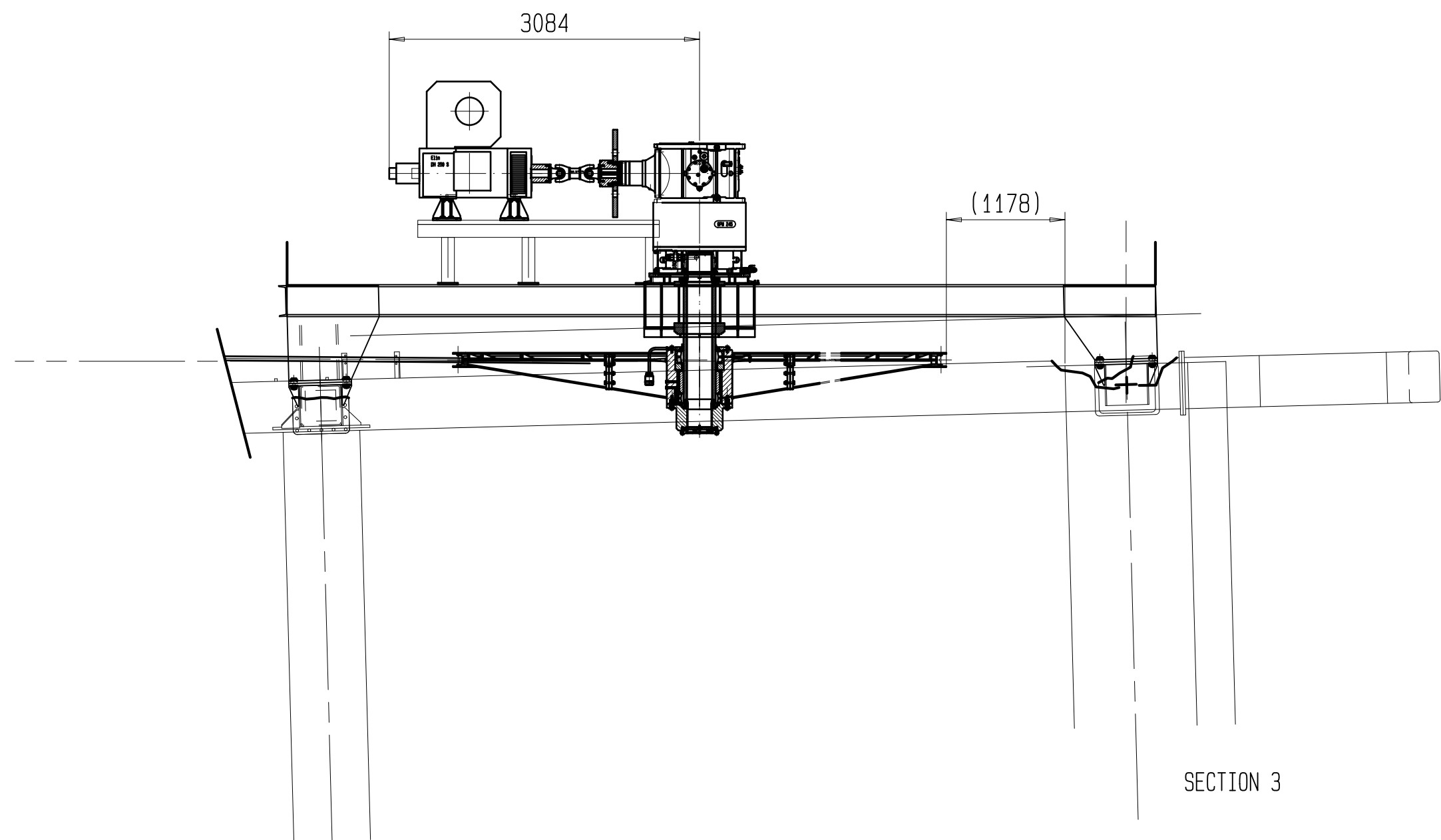
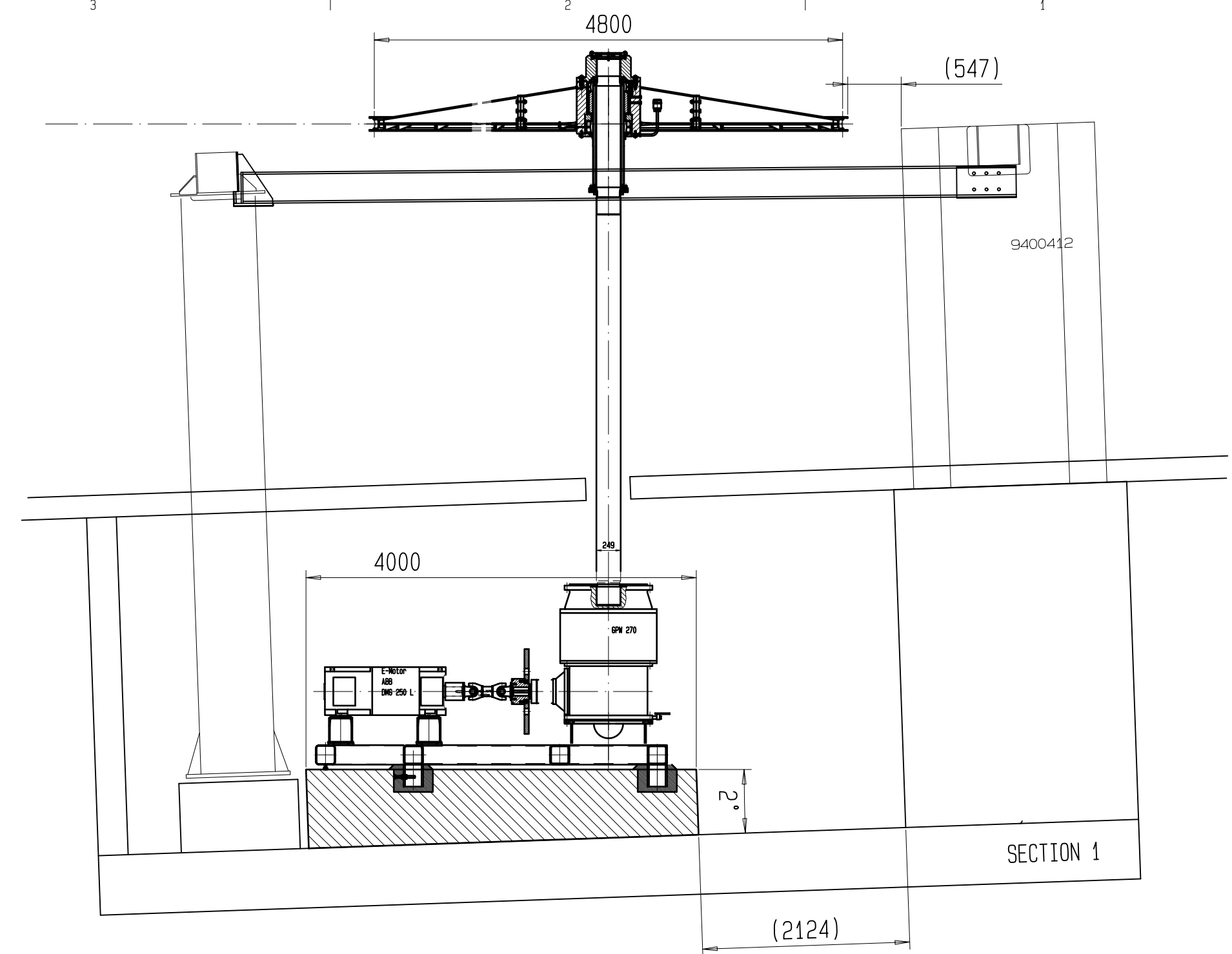
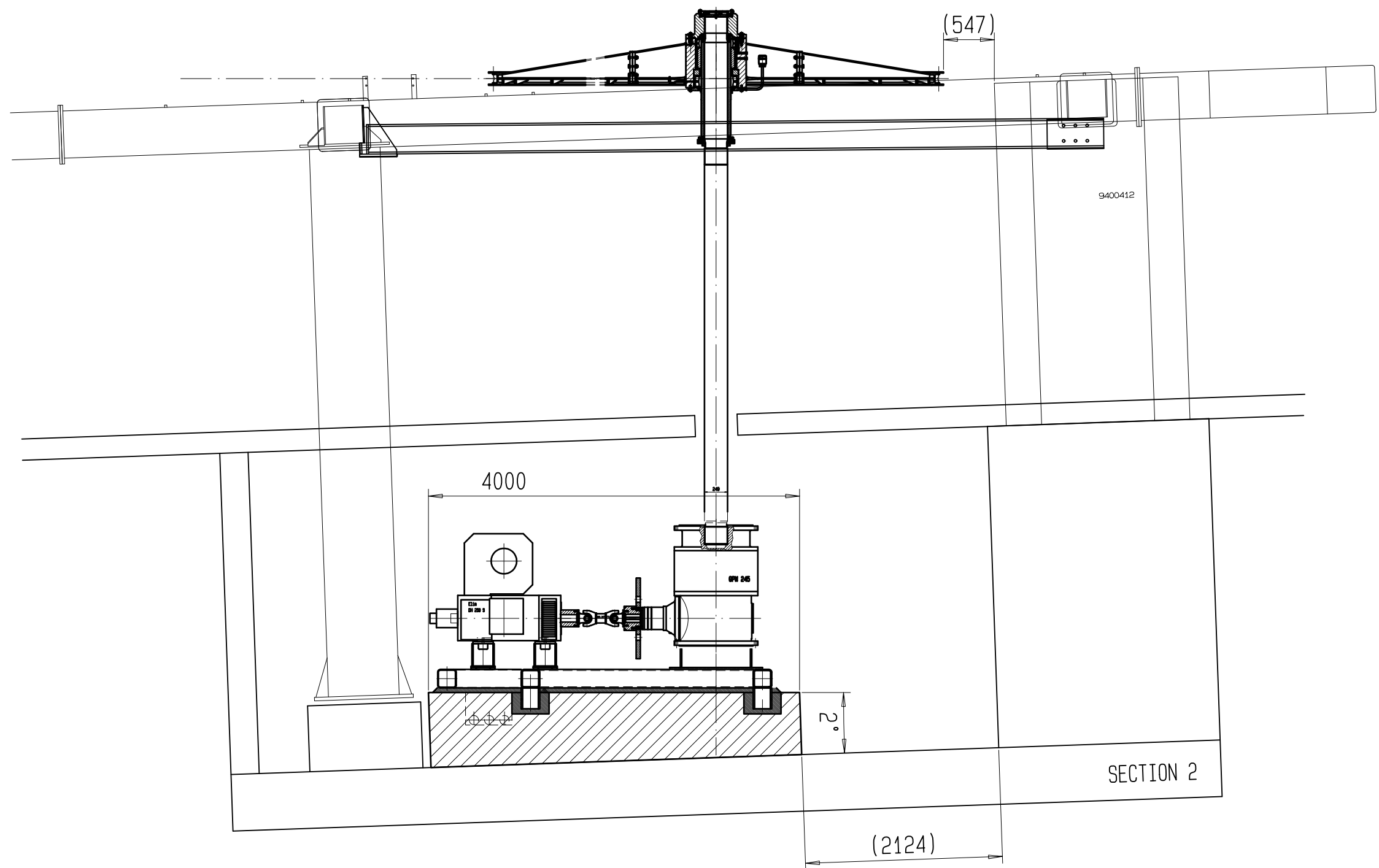
1. 90005501NAE001101 Section 1, 2 & 3 Drive Machinery Layout

Rope Deflections

2. 60014044NCE001201.a Horizontal Deflection Sheave Assembly
3. 60014043NCE001201 Deflection Sheave Detail

Emergency Equipment

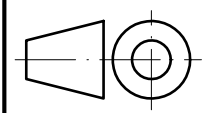
4. 60019684NCD000600 Evac Ring and Pinion
5. 60017359NCD003701 Evac Hydraulic Motor
6. 60017725NCD000801 Evac Drive Emergency Coupling
7. 60029070NCD000701 Evac Drive Bearing
8. 60026661NCD000701 Return Emergency Bearing

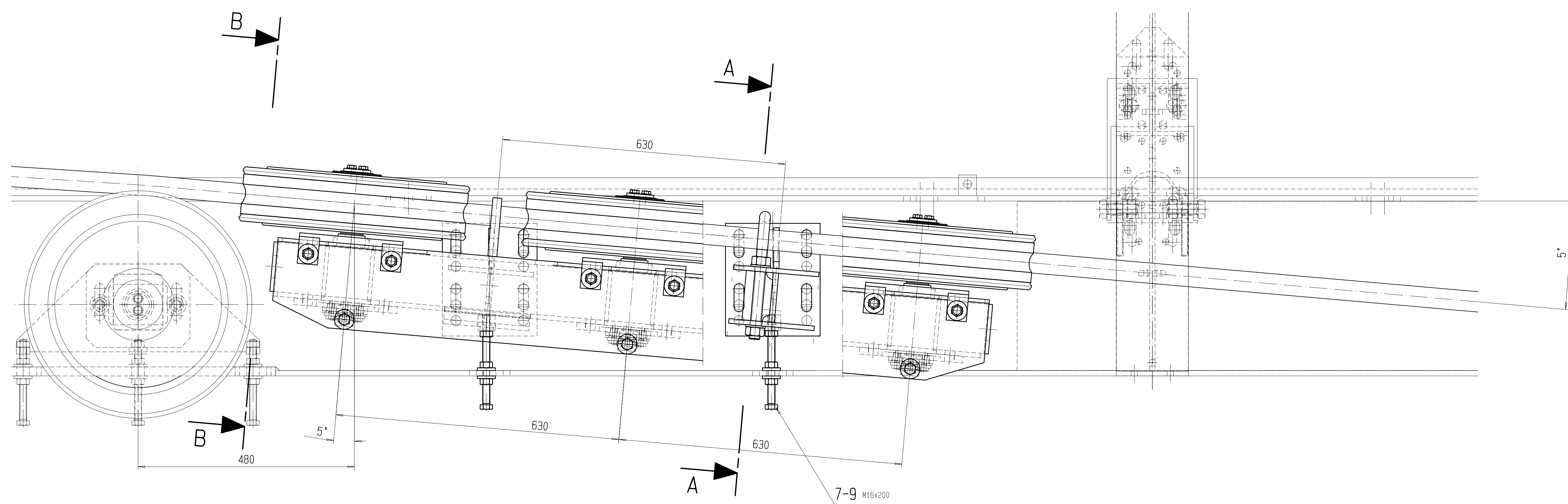


1200 PPH	1800 PPH	2400 PPH
SECTION 1) VAULT GPW 245 & 250kW / VAULT GPW 245 & 438kW / VAULT GPW 270 & 536kW		
SECTION 2) VAULT GPW 245 & 240kW / VAULT GPW 245 & 303kW / VAULT GPW 245 & 360kW		
SECTION 3) GPW 245 & 148kW / GPW 245 & 162kW / GPW 245 & 165kW		

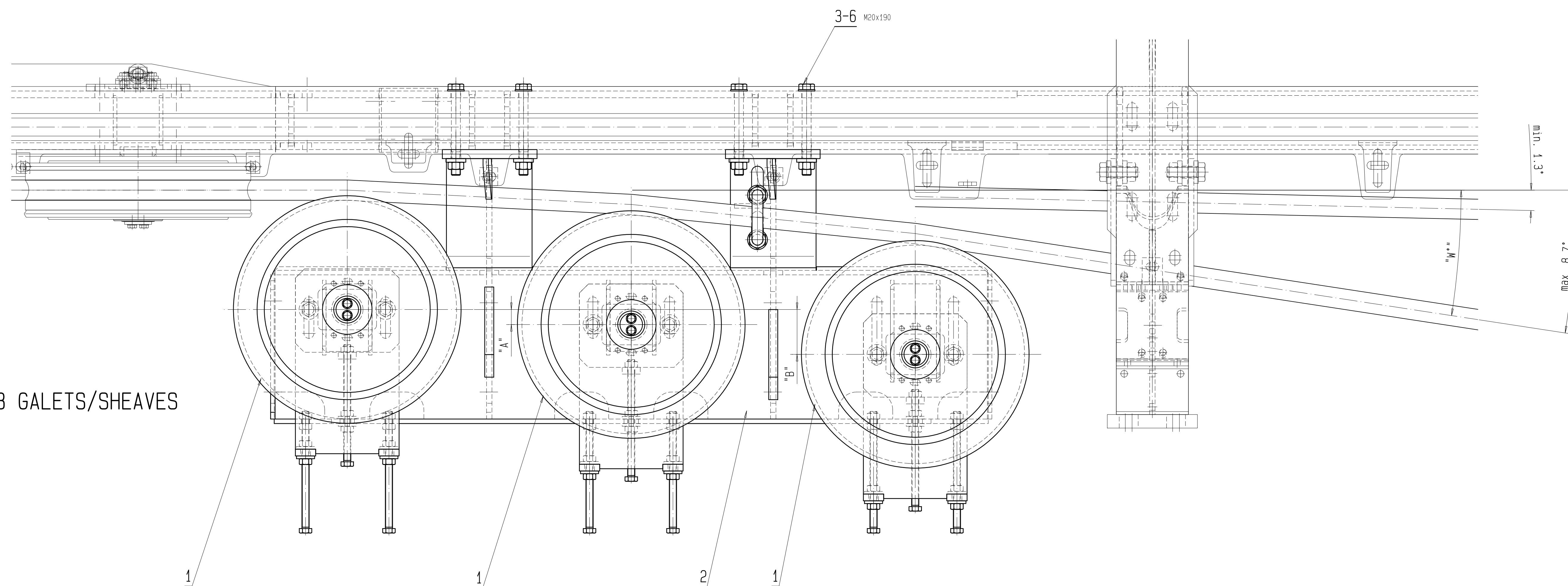
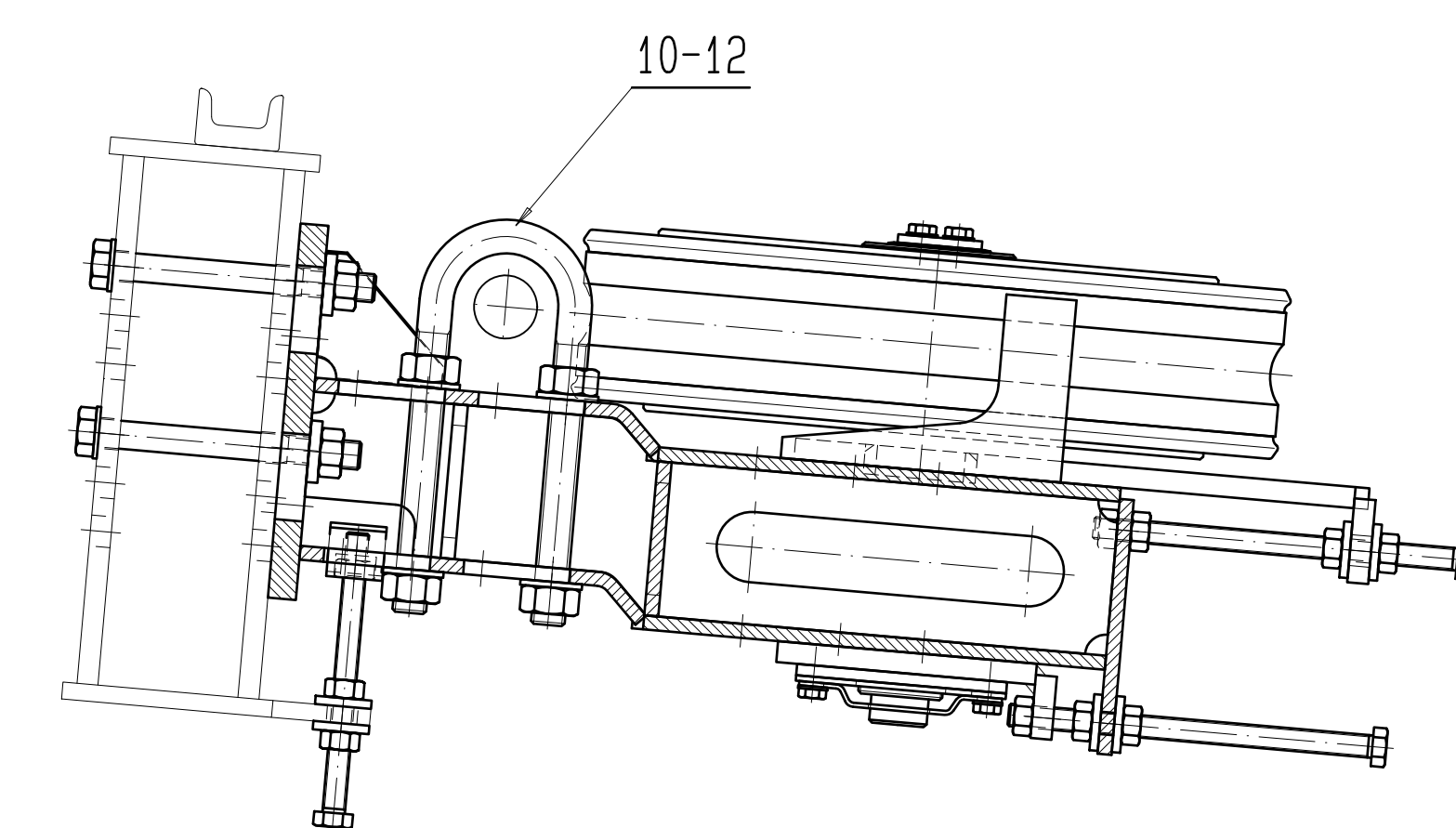
BULLWHEELS KEEP SAME FAR EDGE AS EXISTING 3500mm WHEELS
NEW BULLWHEELS SHOWN ARE 4800mm WHEELS

REFERENCE DRAWING FOR SECTIONS 1 & 2: 9950235A
REFERENCE DRAWING FOR SECTION 3: 9950234A

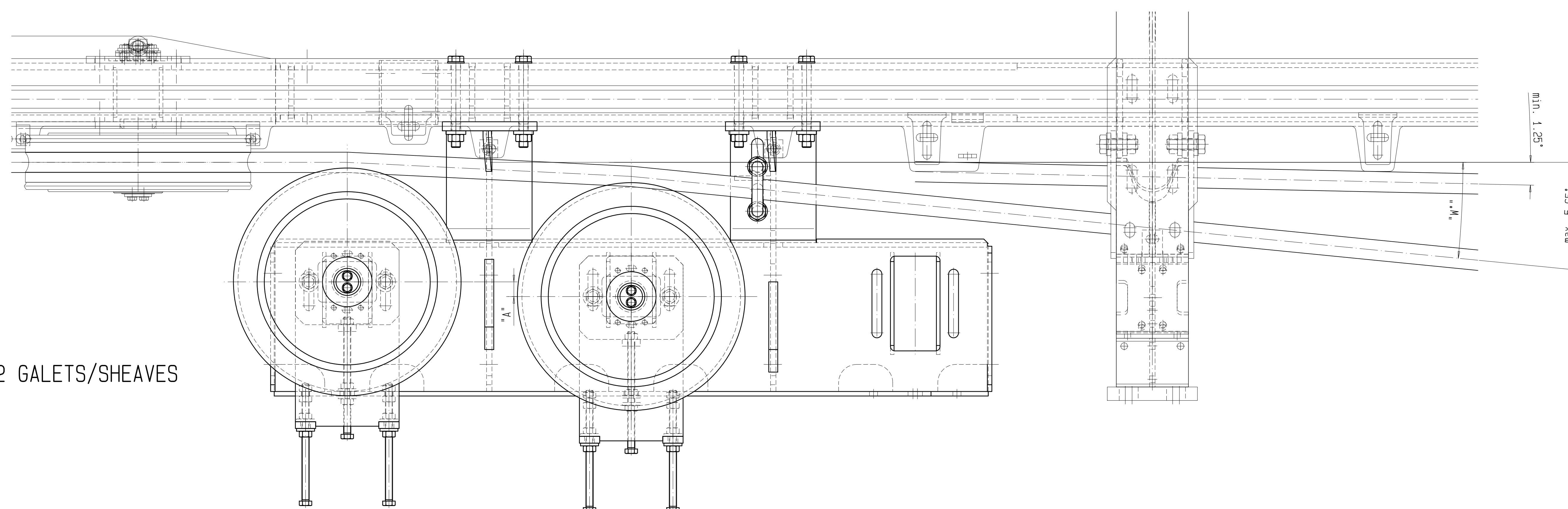
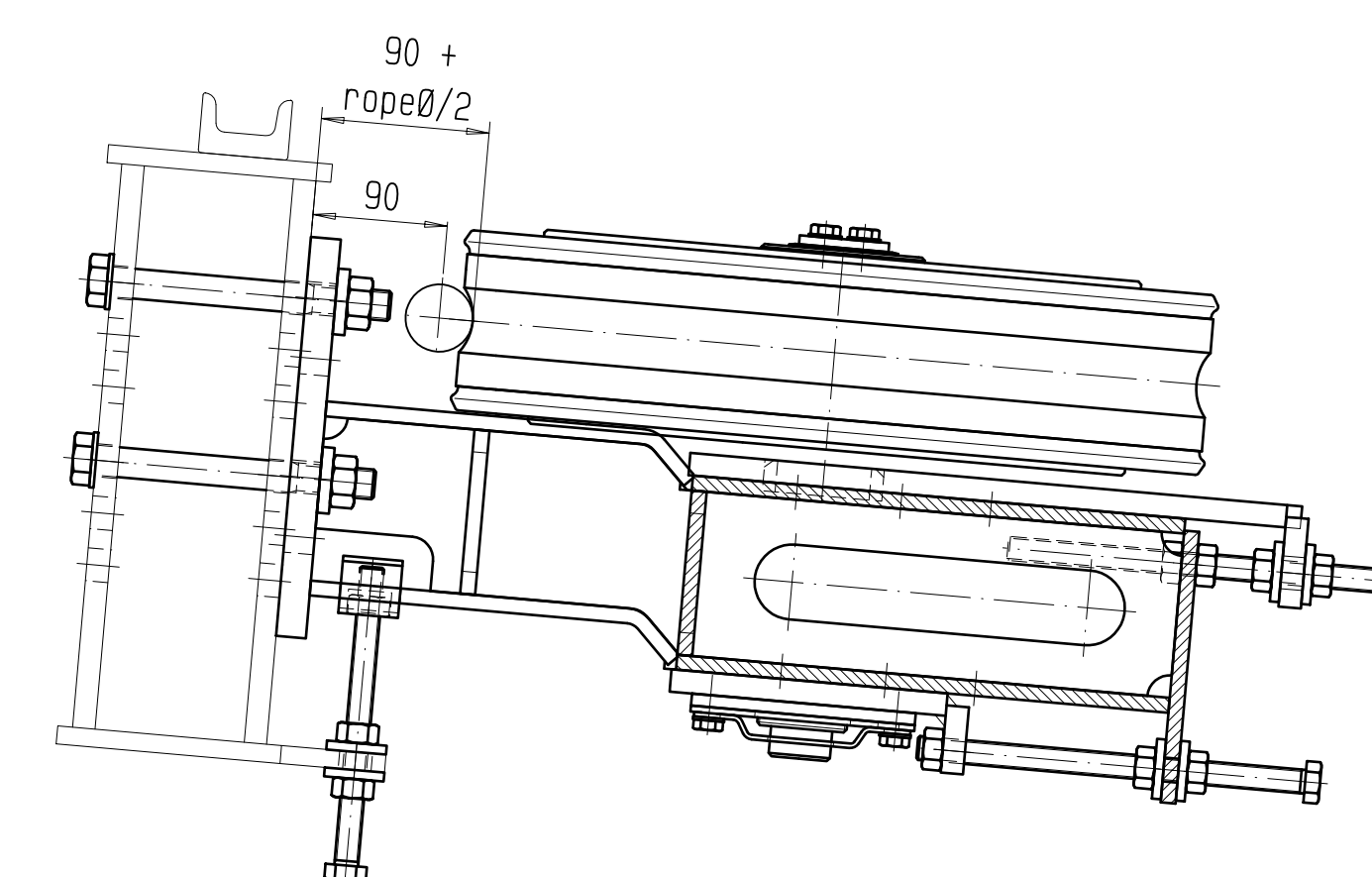
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				1/50	2015-09-01	Campanella M		
INSTALLATION		MOUNTAIN VILLAGE GONDOLA		FACTOR: 1.000	SAVED	2015-09-02/10:59:37 Campanella M		
ORDER				COPIES	X	PLOTTED		
DRIVE LAYOUTS		SECTION 1,2 & 3		Mass	kg	2015-09-02		
				ORIGINAL DRAWING NO.	IDENT NO.			
				DRAWING NO.	INDEX			
				90005501	NAE001101			
This drawing is intellectual property of Doppelmayr USA, Inc. This Drawing or information is our property and must not be copied nor utilized in whole or in part without permission and is subject to return upon request.								



SECTION A-A



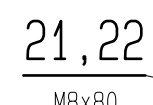
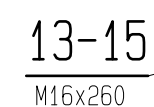
SECTION B-B



SE REFERER AU DESSIN DE MACHINERIE POUR LES DIMENSIONS A & B REFER TO MACHINERY LAYOUT FOR DIMENSIONS A & B
--

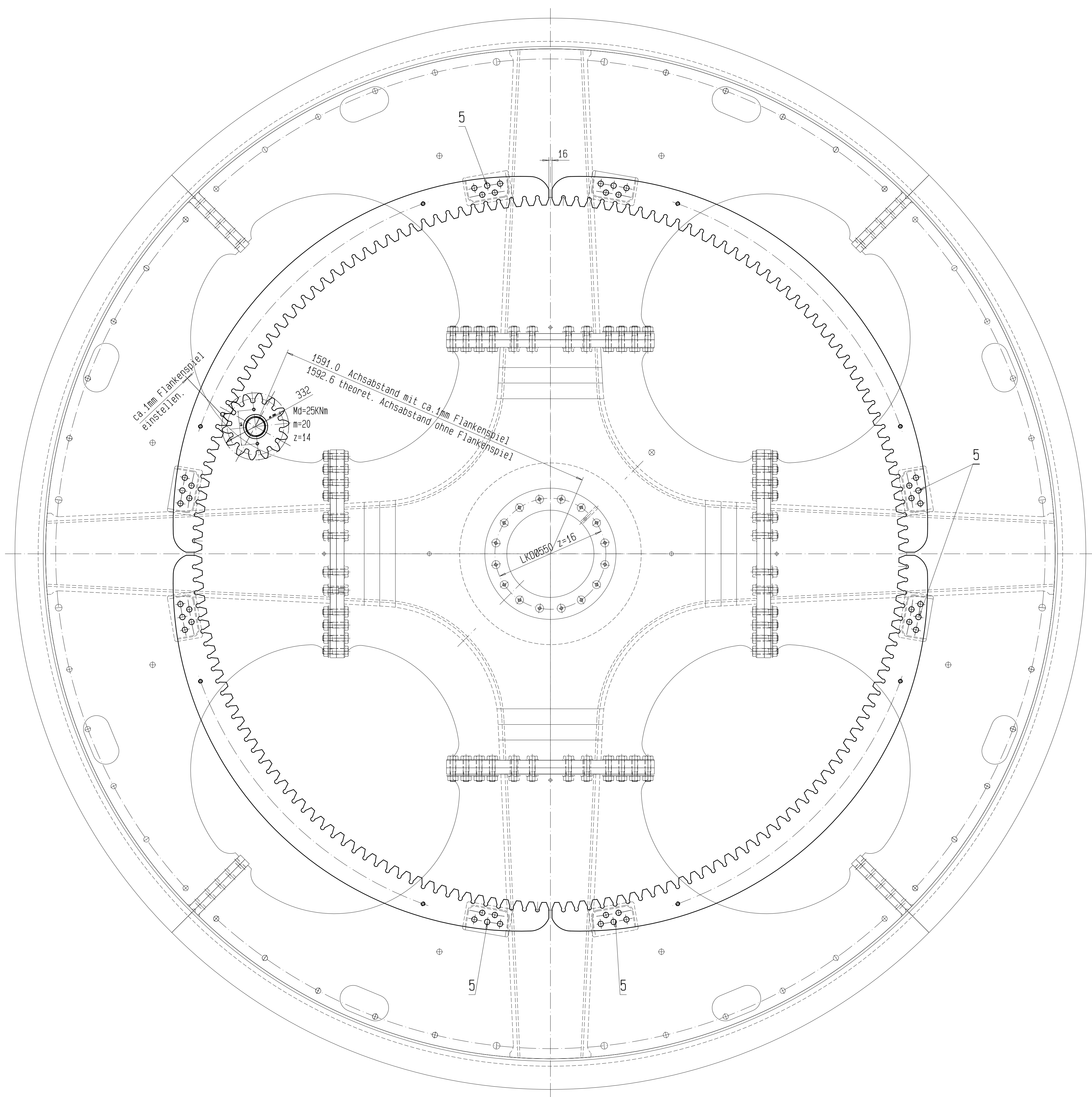
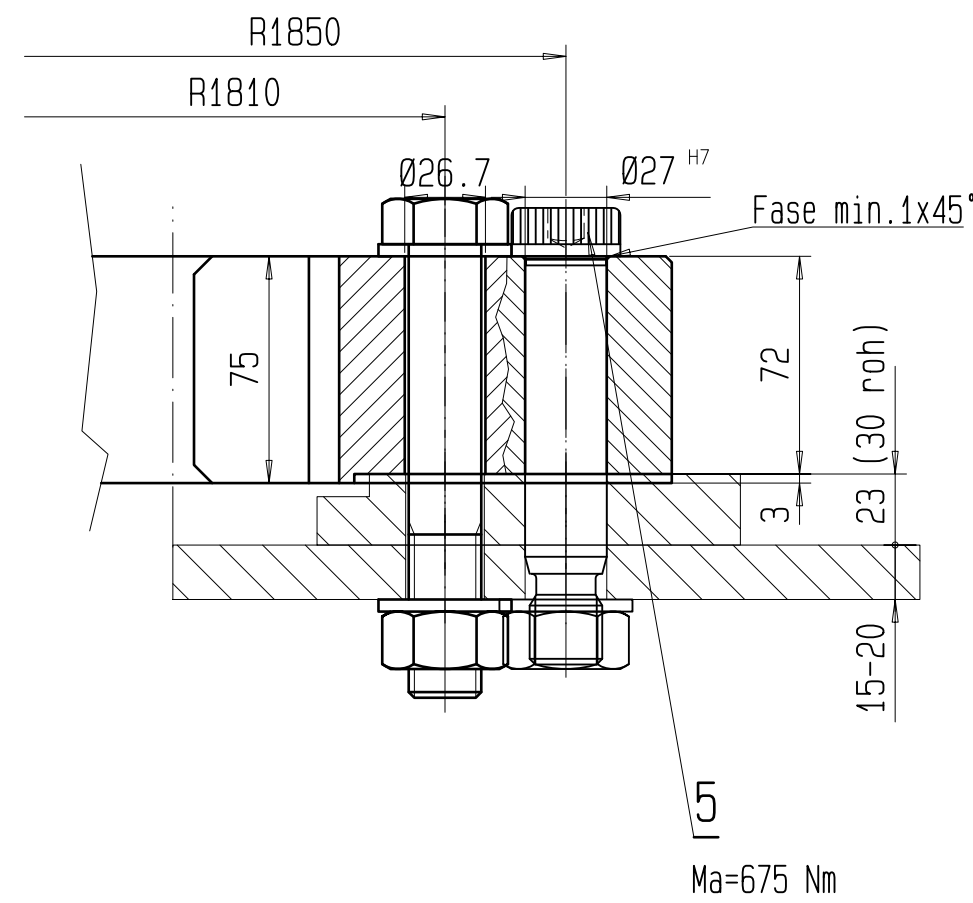
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	2 GAUETS DEFLECTION SELON DESSIN 2 GAUETS DEFLECTION SYMMETRICAL		
	2 GAUETS DEFLECTION SELON DESSIN 2 DEFLECTION SHEAVES SYMMETRICAL		
	2 GAUETS DEFLECTION SELON DESSIN 2 DEFLECTION SHEAVES SYMMETRICAL		
	2 DEFLECTION SHEAVES SYMMETRICAL		
a 02091973, 74, 75, 76 DEVIANT 0104581, 82, 83, 84		2007-01-21 Drawn Date	
Indical Multilanguage / Révisé	Date / N° de Version	Protection / Révision	Ident
Multilanguage Descriptions : EN 120 7860 (Symétrique / Symmetrical) Multilanguage Descriptions : EN 120 7860 (Symétrique / Symmetrical) Multilanguage Descriptions : EN 120 7860 (Symétrique / Symmetrical)	General Descriptions : EN 120 7860 (Symétrique / Symmetrical) General Descriptions : EN 120 7860 (Symétrique / Symmetrical) General Descriptions : EN 120 7860 (Symétrique / Symmetrical)	- - -	- - -
Multilanguage Descriptions : EN 120 7860 (Symétrique / Symmetrical) Multilanguage Descriptions : EN 120 7860 (Symétrique / Symmetrical)	Multilanguage Descriptions : EN 120 7860 (Symétrique / Symmetrical) Multilanguage Descriptions : EN 120 7860 (Symétrique / Symmetrical)	1/5	2007-01-24
Indication / N° d'ordonnée / Order	No. Drawn / Dates	Mode / Mode	Multilanguage / Tools
GAUET DE DEFLECTION DIA.500		(Drawn signature) / (Original drawing in)	
DEFLECTION SHEAVE DIA.500		2003158361021018	
UNIG		60014044NCE0012011	
(Drawn signature) / (Original drawing in)		2003158361021018	
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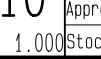
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A470	10401040
A434	10388082

-VUES EUROPEENNES -EUROPEAN VIEWS  	Indice Modifications / Revision		Date Nom/Name		Protection / Realization		Ident	
	Tolerances Generales DIN ISO 2768/1 (Longueurs et Angles) General tolerances as per DIN ISO 2768/1 (lengths a. angles)		Tolerances Generales DIN ISO 2768/2 (Forme et Position) General tolerances as per DIN ISO 2768/2 (form + position)		Echelle / Scale Dessiné drawn Approuvé approved		Date Nom/Name	
	Tolerances Generales ISO 13020 (Longueurs et Angles) General tolerances as per ISO 13020 (lengths a. angles)		Tolerances Generales ISO 13020 (Forme et Position) General tolerances as per ISO 13020 (form + position)		1/2.5		2002-10-24 am	
	Installation _____ N°Commande/Order _____ Nb.Copies / Copies _____ X		Moule / Model _____ Outillage / Tool _____		Dessin original / Original drawing no. 20040610E001201		Dessin n° / Drawing no. _____ Indice _____	
GALET DE DEFLECTION DIA.500 DEFLECTION SHEAVE DIA.500 UNIG		CAD		60014043NCE001201		60014043NCE001201		
Classe/ Class 		Ce Dessin est la propriété de Doppelmayr CTEC Ltée. et ne doit être ni copié ni utilisé en tout ou partie sans autorisation et doit être retourné sur demande. This drawing or information is the property of Doppelmayr CTEC Ltd. and must not be copied nor utilized in whole or in part without permission and is subject to return upon request.				Doppelmayr CTEC Ltée. 800, St. Nicolas St. Jerome, CANADA, J7Y4C8		A



10690756		Date article	
Index	Modifications	Date	Nom
Doppelmayr CTEC		Client	Location
			
Doppelmayr CTEC Ltd. 800 St. Andrew St. Jerome, Canada J7Y 0K3		1/10 10000	Date 2008-09-29 Descripteur 2015-09-02
Installation		Avis sur 1080 kg 2015-09-02 Avis, lettre de	
N° commande		X	
ROULE D'ENGRENAGE z=172, dt=3440, 4 PIÈCES RIM GEAR ASSEMBLY UNIS			
Date originel 2002782006060 60019684NC0000600		Date article 2002782006060 60019684NC0000600	
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21-23

17

4

16

14

11-13

24

19, 20

Ma=200Nm

18

15

1

5, 6

9

2

3

7, 8

4

180

N (W) 250x8x30x30 8e/9H
DIN 5480

1600

ø245

ø251

N (W) 250x8x30x30 8e/9H
DIN 5480

180

GPW 245 (270)

LKØ550
16xØ40H7

25

dkØ817

dtØ777.4

535

38

10

Indice

Modifications

Date

Nom

Exécution

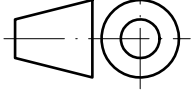
Code article

Doppelmayr CTEC

Doppelmayr CTEC Ltee.
800 St. Nicolas
St. Jerome, CANADA, J7Y4G8

Classe:

0



Echelle

1/2.5

Date

2006-05-19

Nom

Dienjen Eric

Installation

N° Commande

Nb. pièces

X

Poids brut

kg

Info plot

2015-09-02

Facteur: 1.00

Stocke

2015-09-02/16:45:03

Überlander Rg

ACCOUPEMENT A CHAINE

BW TYPE M / Ø250 / UNIG

CHAIN COUPLING

Dessin original N°

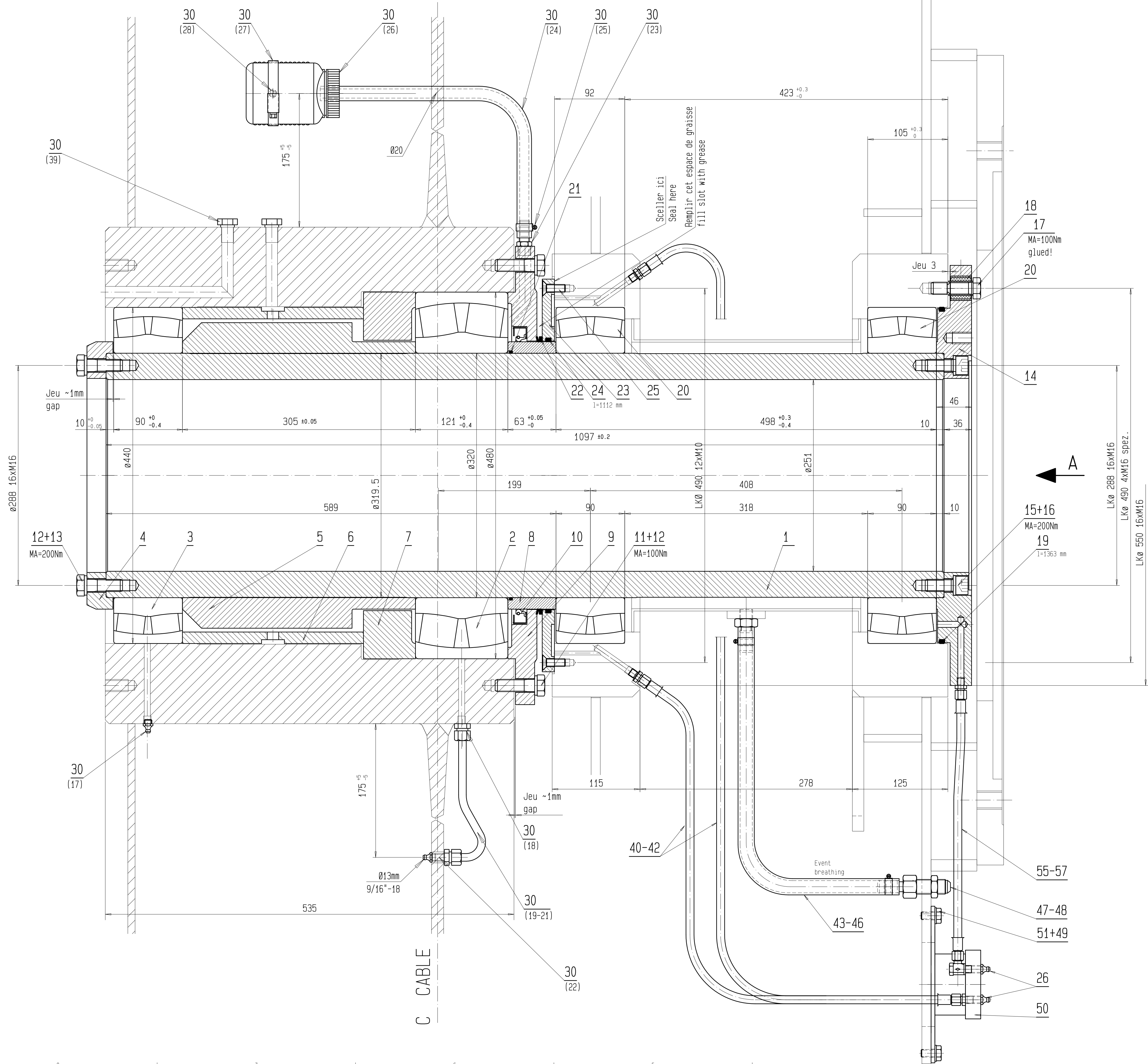
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Dessin N°

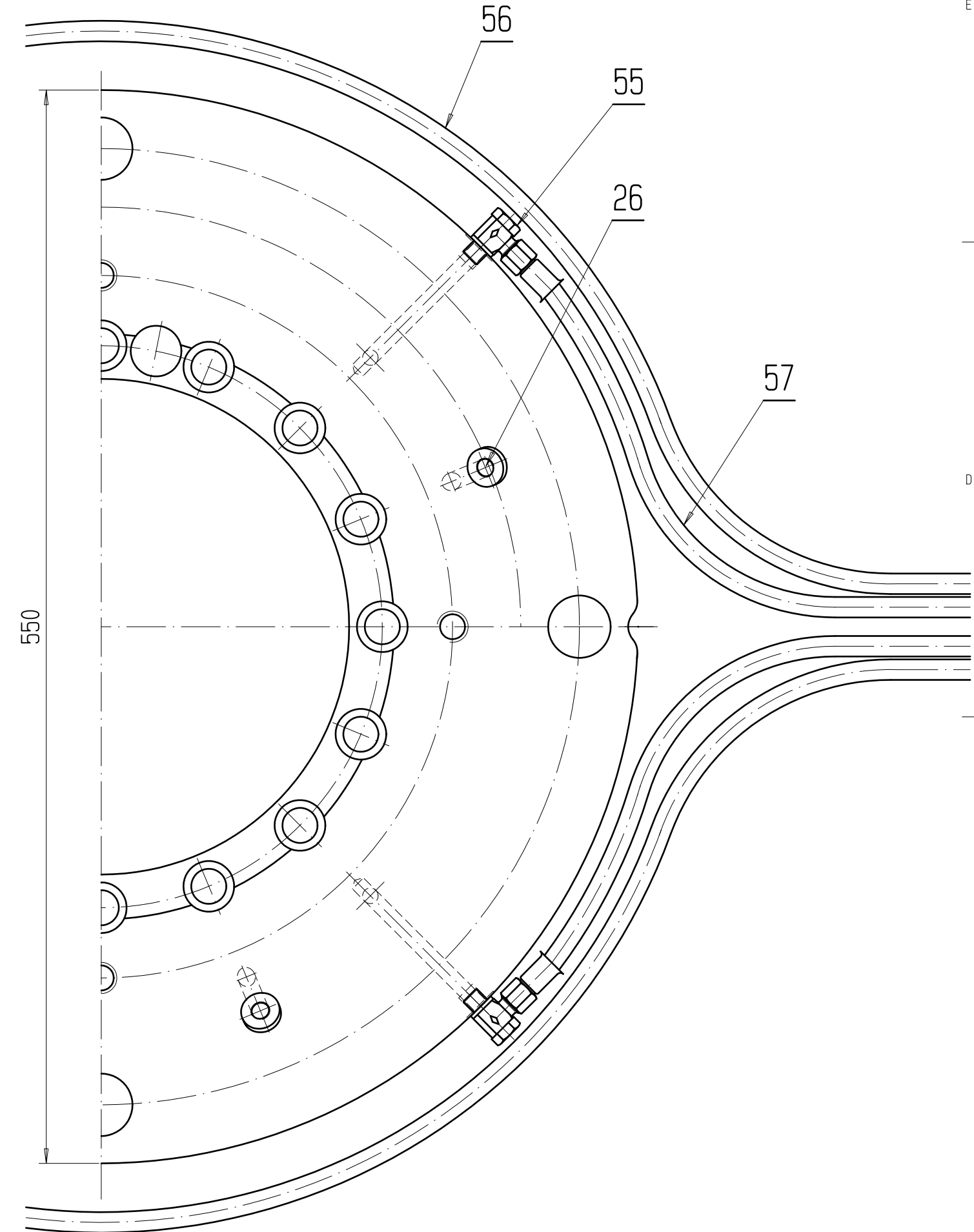
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A1

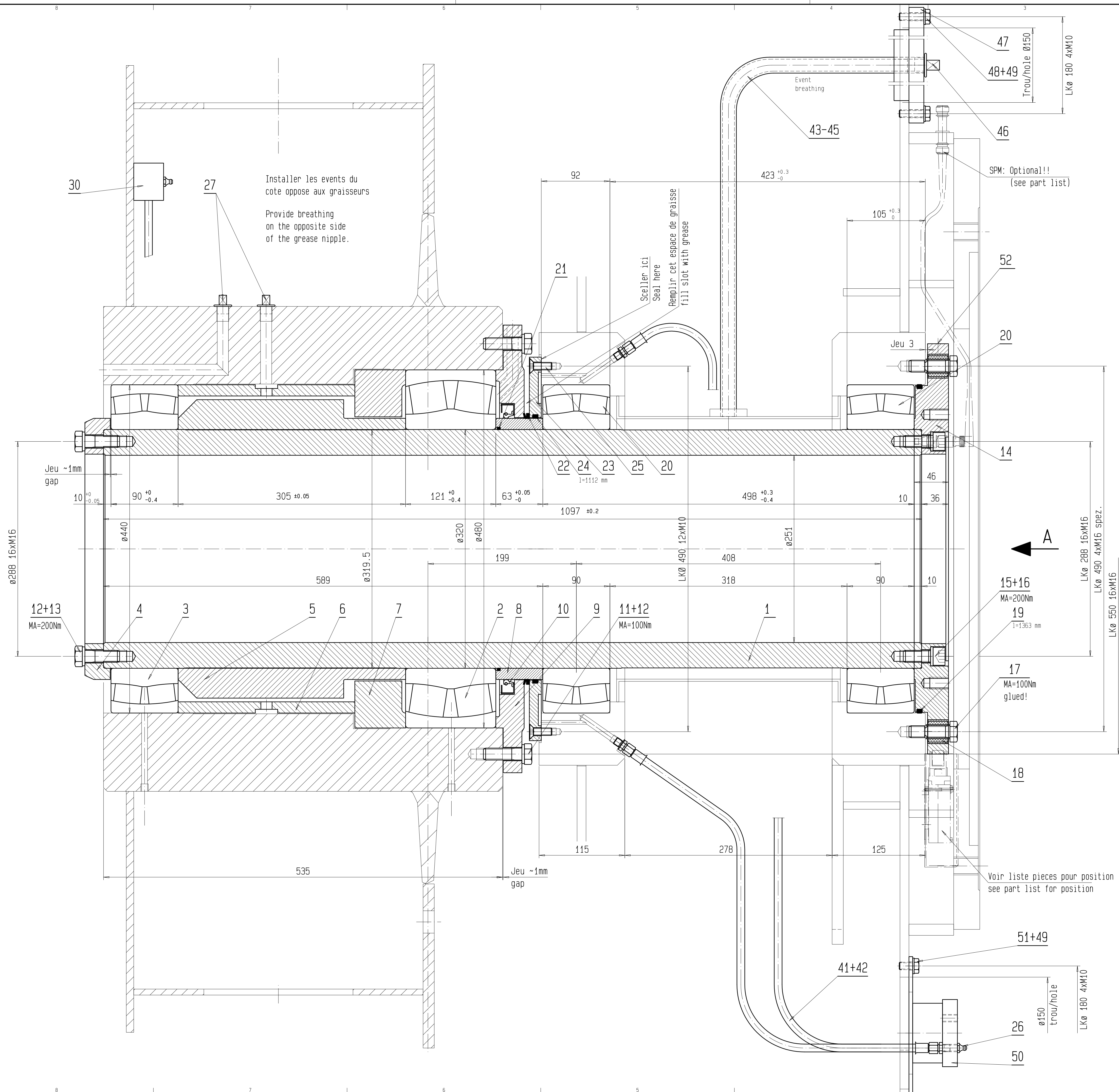


ANSICHT A
VIEW A



Coller item 8 avec Loctite
fix item 8 with Loctite

Index		Änderung	Datum	Name	Ausführung	Ident. Nr.	
Doppelmayr		Innova Patent GmbH Rickenbacherstr. 8-10 A-6560 WOLFRUT/ALSTRIA	SB			50053692	
Anlage							
Auftrag							
Stück							
X							
800							
kg							
2015-09-02							
60026661NCD000701							
60029070NCD000701							
ROULEMENT DE POULIE TYPE M							
ROULEMENTS D'URGENCE - UNIGS							
BULLWHEEL BEARINGS TYPE M							
WITH EMERGENCY BEARINGS							
Diese Zeichnung ist geistiges Eigentum der Doppelmayr Seilbahnen GmbH und übertragen per 31.12 des Erstellungsjahres an die Innova Patent GmbH. Sie darf ohne unsere besondere Zustimmung weder vervielfältigt, noch ausgeführt, noch dritten Personen bekanntgegeben werden.							

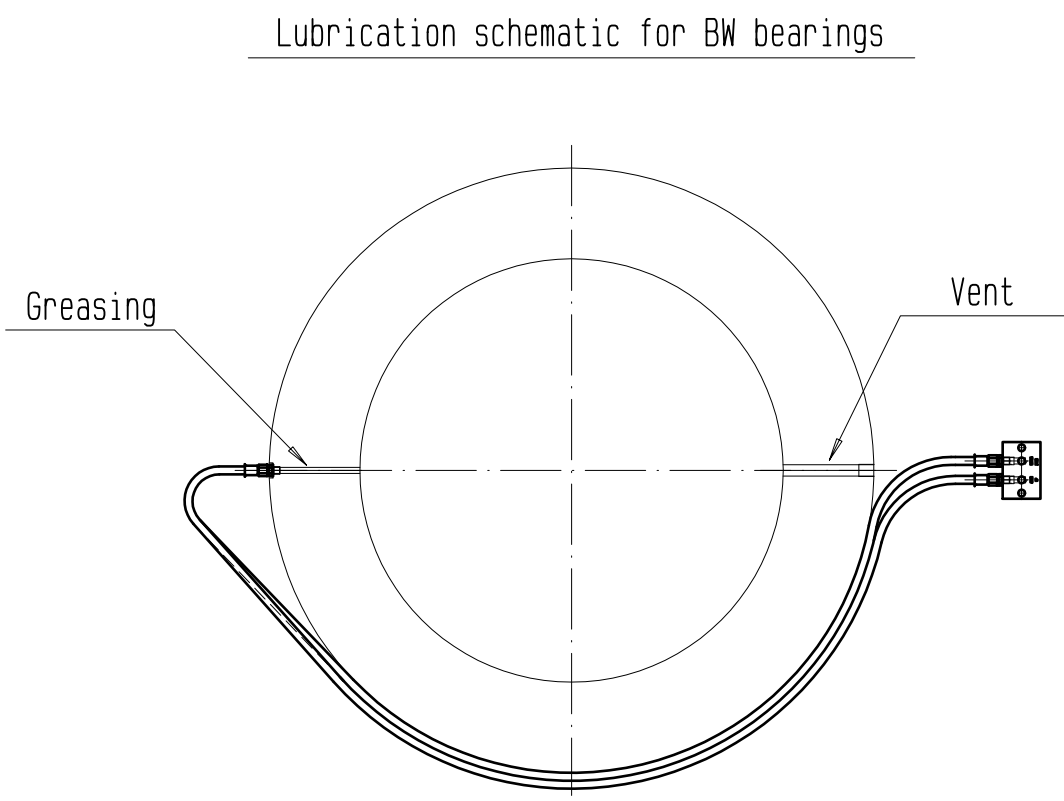


Installer les events du
cote oppose aux graisseurs

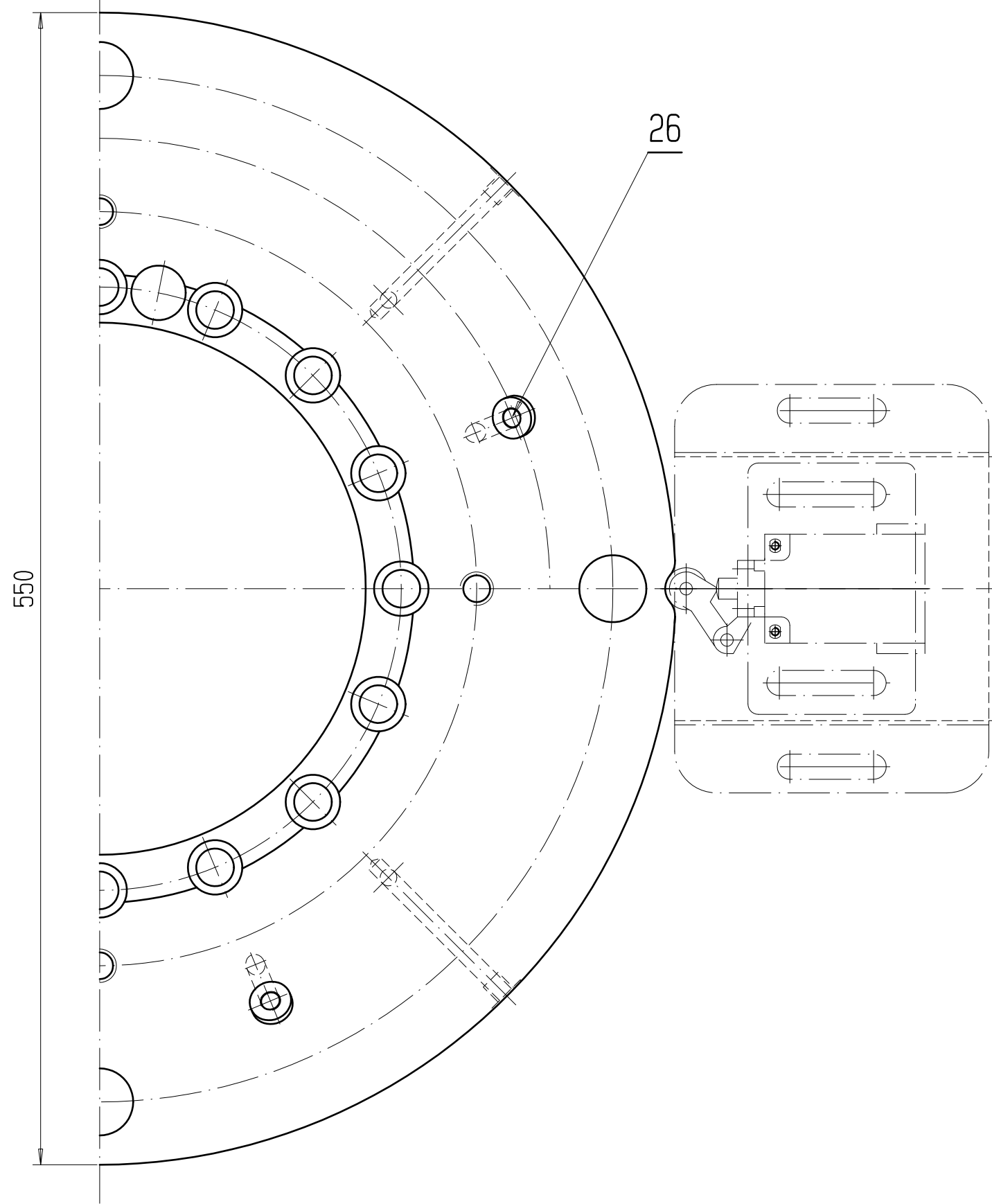
Provide breathing
on the opposite side
of the grease nipple.

Sealer here
Remplir cet espace de graisse
fill slot with grease

A

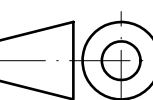


ANSICHT A
VIEW A



Coller item 8 avec Loctite 609
fix item 8 with Loctite 609

Voir liste pieces pour position
see part list for position

Index										Änderung										Datum										Name										Ausführung										Maßstab										gez.										Datum										Name										Ident. Nr.																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
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APPENDIX – F

FOUNDATION CALCULATIONS

TOWER FOUNDATIONS

- **Section 1**

1. Tower Foundations (S1) 1200pph
2. Tower Foundations (S1) 1800pph
3. Tower Foundations (S1) 2400pph

- **Section 2**

4. Tower Foundations (S2) 1200pph
5. Tower Foundations (S2) 1800pph
6. Tower Foundations (S2) 2400pph

- **Section 3**

7. Tower Foundations (S3) 1200pph
8. Tower Foundations (S3) 1800pph
9. Tower Foundations (S3) 2400pph

TERMINAL FOUNDATIONS

- **Section 1**

10. Tension Terminal Foundations (S1) @ Max Tension
11. Drive Terminal Foundations (S1) @ Max Tension

- **Section 2**

12. Tension Terminal Foundations (S2) @ Max Tension
13. Drive Terminal Foundations (S2) @ Max Tension

- **Section 3**

14. Tension Terminal Foundations (S3) @ Max Tension
15. Drive Terminal Foundations (S3) @ Max Tension

							Safety Factors		S.P.
TWR	Soil	Backfill	Materials	Pad (ft)	Column	Rebar	Slide	O.T.	(psf)
BEB No Footing	fv: psf fl: psf / ft Frict:	Ht: ft Wt: pcf	fy: ksi fc: psi	W: L: D:	Col W: in Col H: ft	Col: 0 # 0's P. Par: in^2 P. Perp: in^2	Opp Non Opp	Opp Non Opp	Opp Non Opp
1 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 8.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 8 # 8's P. Par: 3.46 in^2 P. Perp: 3.46 in^2	Opp 19.0 Non Opp 10.0	Opp 6.9 Non Opp 3.8	Opp 947 Non Opp 1,199
2 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 8.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 8 # 8's P. Par: 3.46 in^2 P. Perp: 3.46 in^2	Opp 6.7 Non Opp 5.0	Opp 2.2 Non Opp 1.7	Opp 1,508 Non Opp 2,025
3 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 12.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 5.18 in^2 P. Perp: 5.34 in^2	Opp 13.5 Non Opp 6.4	Opp 2.3 Non Opp 1.6	Opp 2,221 Non Opp 3,814
4 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 12.0 L: 9.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 5.18 in^2 P. Perp: 3.81 in^2	Opp 9.9 Non Opp 6.9	Opp 2.2 Non Opp 2.2	Opp 2,001 Non Opp 2,747
5 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 15.0 L: 9.0 D: 1.5	Square Column Col W: 60 in Col H: 4.0 ft	Col: 20 # 8's P. Par: 6.48 in^2 P. Perp: 5.87 in^2	Opp 12.5 Non Opp 7.0	Opp 2.1 Non Opp 1.9	Opp 2,188 Non Opp 2,839
6 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 15.0 L: 8.0 D: 1.5	Square Column Col W: 60 in Col H: 4.0 ft	Col: 20 # 8's P. Par: 6.48 in^2 P. Perp: 5.65 in^2	Opp 14.3 Non Opp 5.8	Opp 2.2 Non Opp 1.7	Opp 2,015 Non Opp 2,986
7 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 14.0 L: 9.0 D: 1.5	Square Column Col W: 60 in Col H: 4.0 ft	Col: 20 # 8's P. Par: 6.05 in^2 P. Perp: 4.97 in^2	Opp 13.6 Non Opp 7.2	Opp 2.2 Non Opp 1.8	Opp 2,035 Non Opp 2,994
8 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 16.0 L: 10.0 D: 1.5	Square Column Col W: 66 in Col H: 4.0 ft	Col: 24 # 9's P. Par: 6.91 in^2 P. Perp: 7.74 in^2	Opp 15.1 Non Opp 7.5	Opp 2.1 Non Opp 1.7	Opp 2,092 Non Opp 3,045
9 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 12.0 L: 10.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 5.18 in^2 P. Perp: 4.72 in^2	Opp 9.5 Non Opp 7.3	Opp 2.2 Non Opp 2.1	Opp 2,056 Non Opp 3,160
10 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 14.0 L: 8.0 D: 1.5	Square Column Col W: 60 in Col H: 4.0 ft	Col: 20 # 8's P. Par: 6.05 in^2 P. Perp: 4.40 in^2	Opp 13.0 Non Opp 6.5	Opp 2.0 Non Opp 1.8	Opp 2,225 Non Opp 2,875
11 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 14.0 L: 7.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 6.05 in^2 P. Perp: 5.94 in^2	Opp 14.5 Non Opp 5.4	Opp 2.4 Non Opp 1.8	Opp 2,110 Non Opp 3,119
12 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 10.0 L: 15.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 5.84 in^2 P. Perp: 6.48 in^2	Opp 5.7 Non Opp 4.9	Opp 2.2 Non Opp 1.9	Opp 1,547 Non Opp 2,738
13 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 11.0 L: 7.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 4.75 in^2 P. Perp: 2.82 in^2	Opp 6.2 Non Opp 6.0	Opp 3.8 Non Opp 1.9	Opp 1,599 Non Opp 2,758

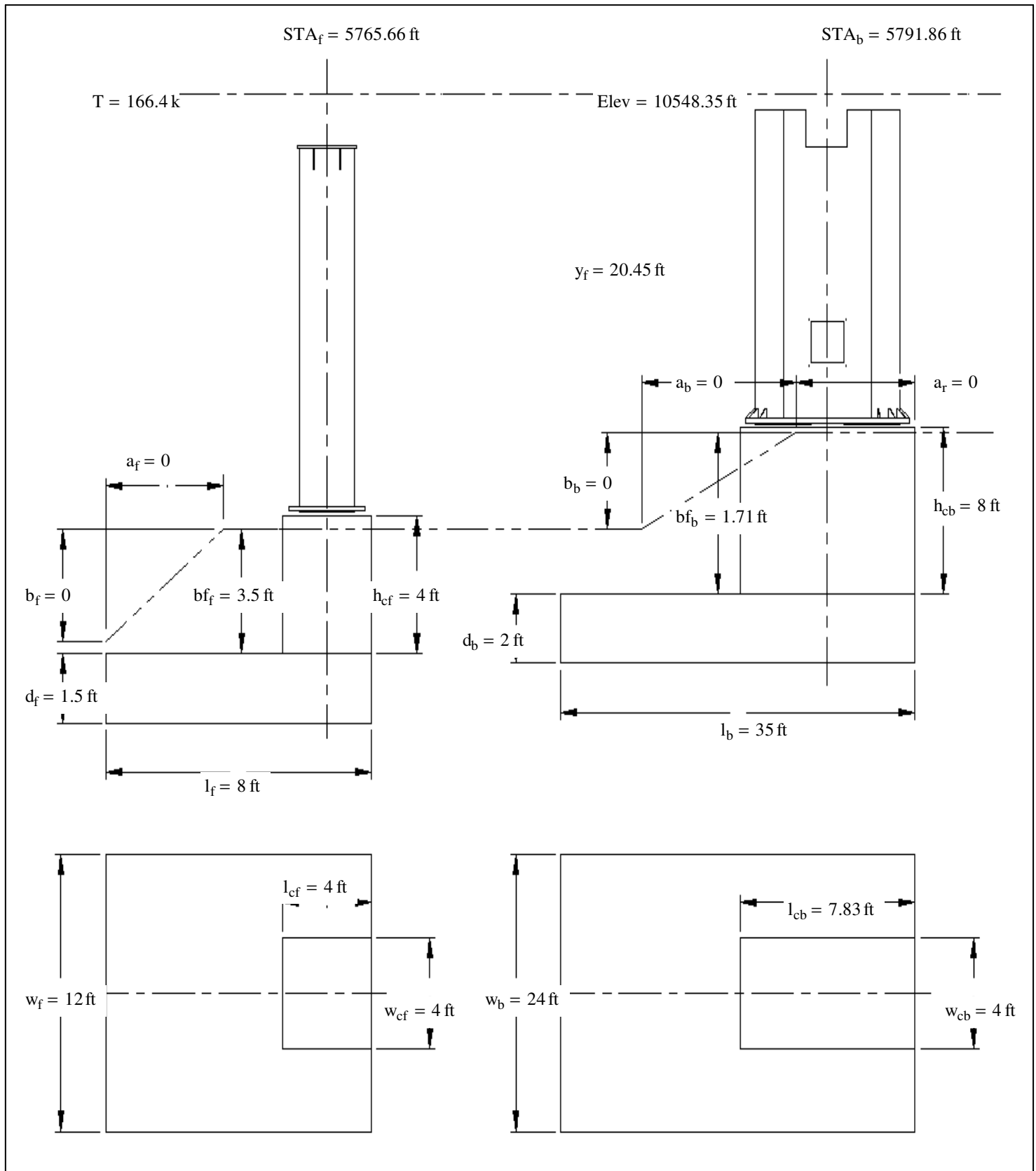
							Safety Factors		S.P.
TWR	Soil	Backfill	Materials	Pad (ft)	Column	Rebar	Slide	O.T.	(psf)
14 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 14.0 L: 7.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 6.05 in ² P. Perp: 5.89 in ²	Opp 8.5 Non Opp 5.3	Opp 3.1 Non Opp 1.7	Opp 1,820 Non Opp 3,319
15 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 16.0 L: 9.0 D: 1.5	Square Column Col W: 60 in Col H: 4.0 ft	Col: 20 # 8's P. Par: 6.91 in ² P. Perp: 7.99 in ²	Opp 11.7 Non Opp 6.5	Opp 2.5 Non Opp 1.6	Opp 1,904 Non Opp 3,451
16 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 8.0 L: 10.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 3.46 in ² P. Perp: 4.32 in ²	Opp 4.2 Non Opp 4.3	Opp 2.3 Non Opp 2.0	Opp 2,332 Non Opp 3,023
17 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 7.0 L: 7.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 3.02 in ² P. Perp: 3.02 in ²	Opp 4.1 Non Opp 4.0	Opp 3.1 Non Opp 2.3	Opp 2,407 Non Opp 3,033
18 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 7.0 L: 11.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 2.59 in ² P. Perp: 4.75 in ²	Opp 7.2 Non Opp 6.9	Opp 2.2 Non Opp 2.2	Opp 2,569 Non Opp 2,573
19 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 7.0 L: 7.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 8 # 8's P. Par: 3.02 in ² P. Perp: 3.02 in ²	Opp 13.7 Non Opp 9.6	Opp 2.8 Non Opp 2.0	Opp 2,613 Non Opp 3,304
TEB No Footing	fv: psf fl: psf / ft Frict:	Ht: ft Wt: pcf	fy: ksi fc: psi	W: L: D:	Col W: in Col H: ft	Col: 0 # 0's P. Par: in ² P. Perp: in ²	Opp Non Opp	Opp Non Opp	Opp Non Opp

							Safety Factors		S.P.
TWR	Soil	Backfill	Materials	Pad (ft)	Column	Rebar	Slide	O.T.	(psf)
BEB No Footing	fv: psf fl: psf / ft Frict:	Ht: ft Wt: pcf	fy: ksi fc: psi	W: L: D:	Col W: in Col H: ft	Col: 0 # 0's P. Par: in ² P. Perp: in ²	Opp Non Opp	Opp Non Opp	Opp Non Opp
1 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 8.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 8 # 8's P. Par: 3.46 in ² P. Perp: 3.46 in ²	Opp 19.3 Non Opp 10.1	Opp 7.1 Non Opp 3.8	Opp 944 Non Opp 1,199
2 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 8.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 8 # 8's P. Par: 3.46 in ² P. Perp: 3.46 in ²	Opp 6.6 Non Opp 4.9	Opp 2.2 Non Opp 1.7	Opp 1,511 Non Opp 2,035
3 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 13.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 5.62 in ² P. Perp: 4.59 in ²	Opp 14.3 Non Opp 7.0	Opp 2.4 Non Opp 2.0	Opp 1,850 Non Opp 2,588
4 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 13.0 L: 9.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 5.62 in ² P. Perp: 4.87 in ²	Opp 9.2 Non Opp 6.3	Opp 2.1 Non Opp 2.2	Opp 2,172 Non Opp 2,787
5 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 15.0 L: 10.0 D: 1.5	Square Column Col W: 60 in Col H: 4.0 ft	Col: 20 # 8's P. Par: 6.48 in ² P. Perp: 6.58 in ²	Opp 11.7 Non Opp 6.8	Opp 2.1 Non Opp 2.0	Opp 2,091 Non Opp 2,982
6 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 15.0 L: 9.0 D: 1.5	Square Column Col W: 60 in Col H: 4.0 ft	Col: 20 # 8's P. Par: 6.48 in ² P. Perp: 7.33 in ²	Opp 14.2 Non Opp 5.7	Opp 2.4 Non Opp 1.7	Opp 1,833 Non Opp 3,253
7 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 14.0 L: 10.0 D: 1.5	Square Column Col W: 60 in Col H: 4.0 ft	Col: 20 # 8's P. Par: 6.05 in ² P. Perp: 6.14 in ²	Opp 12.0 Non Opp 7.0	Opp 2.1 Non Opp 1.8	Opp 2,056 Non Opp 3,243
8 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 17.0 L: 10.0 D: 1.5	Square Column Col W: 66 in Col H: 4.0 ft	Col: 24 # 9's P. Par: 7.34 in ² P. Perp: 9.31 in ²	Opp 15.6 Non Opp 7.1	Opp 2.1 Non Opp 1.7	Opp 2,015 Non Opp 3,006
9 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 13.0 L: 10.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 5.62 in ² P. Perp: 5.63 in ²	Opp 8.8 Non Opp 6.9	Opp 2.0 Non Opp 2.1	Opp 2,197 Non Opp 3,079
10 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 14.0 L: 9.0 D: 1.5	Square Column Col W: 60 in Col H: 4.0 ft	Col: 20 # 8's P. Par: 6.05 in ² P. Perp: 4.98 in ²	Opp 13.4 Non Opp 6.3	Opp 2.3 Non Opp 1.8	Opp 1,933 Non Opp 2,969
11 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 14.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 6.05 in ² P. Perp: 5.67 in ²	Opp 11.0 Non Opp 5.8	Opp 2.1 Non Opp 1.9	Opp 2,089 Non Opp 2,691
12 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 11.0 L: 14.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 5.43 in ² P. Perp: 6.05 in ²	Opp 6.0 Non Opp 5.1	Opp 2.1 Non Opp 1.9	Opp 1,565 Non Opp 3,018
13 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 12.0 L: 7.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 5.18 in ² P. Perp: 3.52 in ²	Opp 5.8 Non Opp 6.0	Opp 2.9 Non Opp 1.9	Opp 1,763 Non Opp 2,613

							Safety Factors		S.P.
TWR	Soil	Backfill	Materials	Pad (ft)	Column	Rebar	Slide	O.T.	(psf)
14 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 15.0 L: 7.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 6.48 in ² P. Perp: 5.99 in ²	Opp 6.6 Non Opp 5.3	Opp 2.7 Non Opp 2.0	Opp 1,707 Non Opp 2,313
15 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 17.0 L: 8.0 D: 1.5	Square Column Col W: 60 in Col H: 4.0 ft	Col: 20 # 8's P. Par: 7.34 in ² P. Perp: 7.89 in ²	Opp 7.9 Non Opp 6.2	Opp 3.2 Non Opp 1.8	Opp 1,578 Non Opp 2,467
16 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 8.0 L: 11.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 3.46 in ² P. Perp: 4.75 in ²	Opp 4.0 Non Opp 4.3	Opp 2.2 Non Opp 1.9	Opp 2,308 Non Opp 2,854
17 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 8.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 3.46 in ² P. Perp: 3.46 in ²	Opp 4.3 Non Opp 4.4	Opp 3.5 Non Opp 3.0	Opp 2,192 Non Opp 2,269
18 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 7.0 L: 11.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 2.61 in ² P. Perp: 4.75 in ²	Opp 7.4 Non Opp 7.0	Opp 2.2 Non Opp 2.2	Opp 2,568 Non Opp 2,582
19 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 7.0 L: 7.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 8 # 8's P. Par: 3.02 in ² P. Perp: 3.02 in ²	Opp 13.1 Non Opp 9.5	Opp 2.7 Non Opp 2.0	Opp 2,856 Non Opp 3,366
TEB No Footing	fv: psf fl: psf / ft Frict:	Ht: ft Wt: pcf	fy: ksi fc: psi	W: L: D:	Col W: in Col H: ft	Col: 0 # 0's P. Par: in ² P. Perp: in ²	Opp Non Opp	Opp Non Opp	Opp Non Opp

							Safety Factors		S.P.
TWR	Soil	Backfill	Materials	Pad (ft)	Column	Rebar	Slide	O.T.	(psf)
BEB No Footing	fv: psf fl: psf / ft Frict:	Ht: ft Wt: pcf	fy: ksi fc: psi	W: L: D:	Col W: in Col H: ft	Col: 0 # 0's P. Par: in ² P. Perp: in ²	Opp Non Opp	Opp Non Opp	Opp Non Opp
1 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 9.0 L: 9.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 8 # 8's P. Par: 3.89 in ² P. Perp: 3.89 in ²	Opp 25.3 Non Opp 13.1	Opp 10.4 Non Opp 5.6	Opp 837 Non Opp 1,018
2 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 9.0 L: 9.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 8 # 8's P. Par: 3.89 in ² P. Perp: 3.89 in ²	Opp 8.2 Non Opp 6.2	Opp 3.1 Non Opp 2.4	Opp 1,242 Non Opp 1,438
3 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 13.0 L: 9.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 5.62 in ² P. Perp: 5.24 in ²	Opp 13.9 Non Opp 6.9	Opp 2.6 Non Opp 1.9	Opp 1,763 Non Opp 2,765
4 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 13.0 L: 10.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 5.62 in ² P. Perp: 5.43 in ²	Opp 8.3 Non Opp 6.2	Opp 2.1 Non Opp 2.3	Opp 2,159 Non Opp 2,913
5 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 15.0 L: 11.0 D: 1.5	Square Column Col W: 60 in Col H: 4.0 ft	Col: 20 # 8's P. Par: 6.48 in ² P. Perp: 7.71 in ²	Opp 10.7 Non Opp 6.7	Opp 2.1 Non Opp 1.9	Opp 2,070 Non Opp 3,147
6 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 16.0 L: 10.0 D: 1.5	Square Column Col W: 60 in Col H: 4.0 ft	Col: 20 # 8's P. Par: 6.91 in ² P. Perp: 8.22 in ²	Opp 13.4 Non Opp 5.8	Opp 2.5 Non Opp 1.8	Opp 1,777 Non Opp 2,891
7 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 14.0 L: 11.0 D: 1.5	Square Column Col W: 60 in Col H: 4.0 ft	Col: 20 # 8's P. Par: 6.05 in ² P. Perp: 7.13 in ²	Opp 11.3 Non Opp 6.8	Opp 2.1 Non Opp 1.7	Opp 2,014 Non Opp 3,587
8 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 17.0 L: 11.0 D: 1.5	Square Column Col W: 66 in Col H: 4.0 ft	Col: 24 # 9's P. Par: 7.34 in ² P. Perp: 1.92 in ²	Opp 15.4 Non Opp 7.0	Opp 2.3 Non Opp 1.7	Opp 1,896 Non Opp 3,315
9 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 13.0 L: 11.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 5.62 in ² P. Perp: 6.45 in ²	Opp 8.0 Non Opp 6.8	Opp 2.0 Non Opp 2.1	Opp 2,191 Non Opp 3,220
10 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 15.0 L: 9.0 D: 1.5	Square Column Col W: 60 in Col H: 4.0 ft	Col: 20 # 8's P. Par: 6.48 in ² P. Perp: 6.24 in ²	Opp 13.2 Non Opp 6.0	Opp 2.3 Non Opp 1.8	Opp 1,993 Non Opp 2,981
11 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 14.0 L: 10.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 6.05 in ² P. Perp: 6.73 in ²	Opp 10.1 Non Opp 6.1	Opp 2.3 Non Opp 2.0	Opp 1,837 Non Opp 2,602
12 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 11.0 L: 15.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 6.01 in ² P. Perp: 6.48 in ²	Opp 6.9 Non Opp 5.6	Opp 2.6 Non Opp 1.9	Opp 1,367 Non Opp 2,355
13 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 12.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 5.18 in ² P. Perp: 4.05 in ²	Opp 5.3 Non Opp 5.9	Opp 2.8 Non Opp 1.9	Opp 1,803 Non Opp 2,766

							Safety Factors		S.P.
TWR	Soil	Backfill	Materials	Pad (ft)	Column	Rebar	Slide	O.T.	(psf)
14 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 15.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 6.48 in ² P. Perp: 6.92 in ²	Opp 6.0 Non Opp 5.2	Opp 2.6 Non Opp 1.9	Opp 1,762 Non Opp 2,423
15 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 16.0 L: 10.0 D: 1.5	Square Column Col W: 60 in Col H: 4.0 ft	Col: 20 # 8's P. Par: 6.91 in ² P. Perp: 8.37 in ²	Opp 8.0 Non Opp 6.3	Opp 4.2 Non Opp 1.7	Opp 1,371 Non Opp 2,185
16 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 8.0 L: 12.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 2.97 in ² P. Perp: 5.18 in ²	Opp 4.0 Non Opp 4.5	Opp 2.4 Non Opp 1.8	Opp 2,122 Non Opp 2,766
17 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 8.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 3.46 in ² P. Perp: 3.46 in ²	Opp 4.4 Non Opp 4.5	Opp 3.6 Non Opp 3.0	Opp 2,052 Non Opp 2,249
18 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 7.0 L: 11.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 2.98 in ² P. Perp: 4.75 in ²	Opp 7.0 Non Opp 6.7	Opp 2.1 Non Opp 2.1	Opp 3,028 Non Opp 2,923
19 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 7.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 8 # 8's P. Par: 3.02 in ² P. Perp: 3.46 in ²	Opp 12.1 Non Opp 9.7	Opp 2.8 Non Opp 2.4	Opp 2,489 Non Opp 2,783
TEB No Footing	fv: psf fl: psf / ft Frict:	Ht: ft Wt: pcf	fy: ksi fc: psi	W: L: D:	Col W: in Col H: ft	Col: 0 # 0's P. Par: in ² P. Perp: in ²	Opp Non Opp	Opp Non Opp	Opp Non Opp



BACK PAD

INPUT AREA 1: Lift Properties

Rope Elevation Elev $\equiv$ 10548.35·ft Lift Tension T $\equiv$ 166.4·k Reaction Column Station STA_b $\equiv$ 5791.86·ft

INPUT AREA 2: Material Properties

Surcharge Weight $W_{\text{soil}} \equiv 145 \cdot \frac{\text{lb}}{\text{ft}^3}$ Conc. Strength $f_c \equiv 3000 \cdot \text{psi}$ $\phi \equiv .90$ $\rho_{\text{minslab}} \equiv .0018$
 Conc. Weight $W_{\text{conc}} \equiv 145 \cdot \frac{\text{lb}}{\text{ft}^3}$ Steel Strength $f_y \equiv 60000 \cdot \text{psi}$ $\beta \equiv .85$ $\rho_{\text{min}} \equiv \frac{200 \cdot \text{psi}}{f_y}$

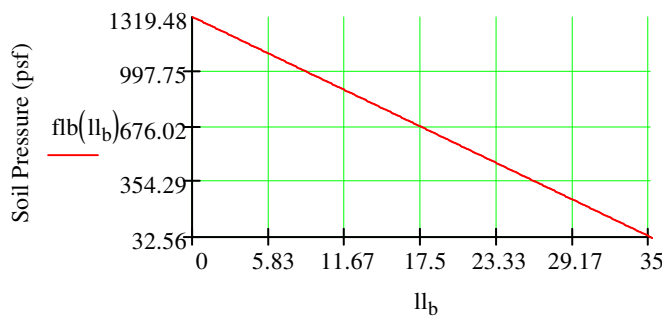
OUTPUT AREA 3: Safety Factors

Overturning: Parallel= $SF_{l_{\text{otb}}} = 2.47$
 Sliding: Perpendicular= $SF_{p_{\text{otb}}} = 23.81$
 Parallel= $SF_{l_{\text{sb}}} = 1.69$
 Perpendicular= $SF_{p_{\text{sb}}} = 29.73$

Soil Pressures:

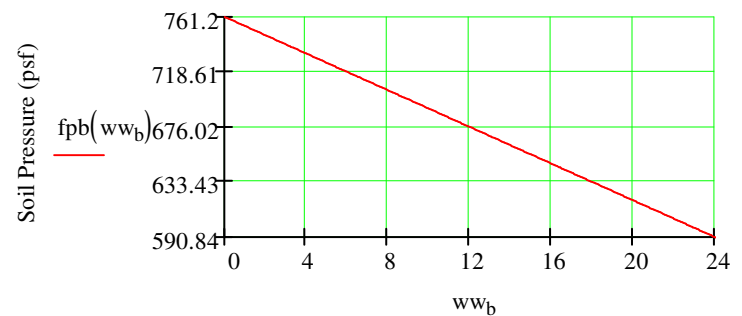
Maximum: Parallel= $\text{Max}_{l_b} = 1.32 \text{ ksf}$
 Minimum: Perpendicular= $\text{Min}_{p_b} = 0.59 \text{ ksf}$
 Parallel= $\text{Max}_{p_b} = 0.76 \text{ ksf}$
 Minimum= $\text{Min}_{l_b} = 0.03 \text{ ksf}$

Parallel Pressure Distribution



Pad Length (ft)

Perpendicular Pressure Distribution



Pad Width (ft)

Reinforcing Steel

Pad Parallel to Center Line= $A_{s1b} = 70.07 \text{ in}^2$ Pedestal Parallel to Center Line= $A_{s3b} = 15.83 \text{ in}^2$
 Pad Perpendicular to Center Line= $A_{s2b} = 14.74 \text{ in}^2$ Pedestal Perpendicular to Center Line= $A_{s4b} = 2.62 \text{ in}^2$

INPUT AREA 4:

MISC.:

Surcharge:

$bf_b \equiv (0.5 + 0.5 + 0.71) \cdot \text{ft}$
 Floor + Wall + Ceiling

COLUMN:

length: $l_{cb} \equiv 94 \cdot \text{in}$
 width: $w_{cb} \equiv 4 \cdot \text{ft}$
 height: $h_{cb} \equiv 8 \cdot \text{ft}$

SLAB:

length: $l_b \equiv 35 \cdot \text{ft}$
 width: $w_b \equiv 24 \cdot \text{ft}$
 depth: $d_b \equiv 2 \cdot \text{ft}$



**Telluride Mtn Gondola
Sect 1, Drive Terminal
Foundation Design**

Sheet: 3 of 3
Project No: SAP0000008

Effective M. Arm: $y_b \equiv 27.47 \cdot \text{ft}$

FRONT PAD

Column Station $\text{STA}_f \equiv 5726.37 \cdot \text{ft}$
 $\text{STA}_f \equiv 5765.66 \cdot \text{ft}$

OUTPUT AREA 5:

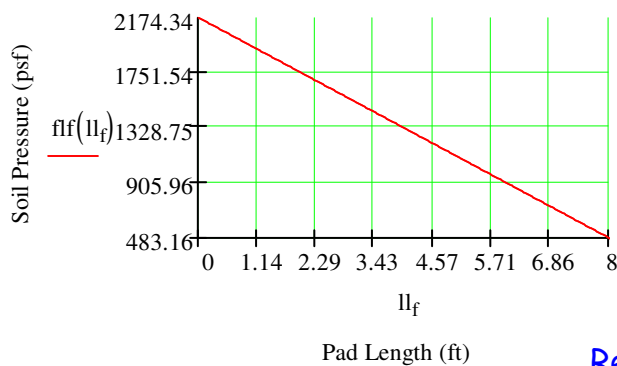
Safety Factors

Parallel= $\text{SF}_{l_{\text{otf}}} = 2.79$ $\text{SF}_{l_{\text{sf}}} = 5.85$
Perpendicular= $\text{SF}_{p_{\text{otf}}} = 5.29$ $\text{SF}_{p_{\text{sf}}} = 10.12$

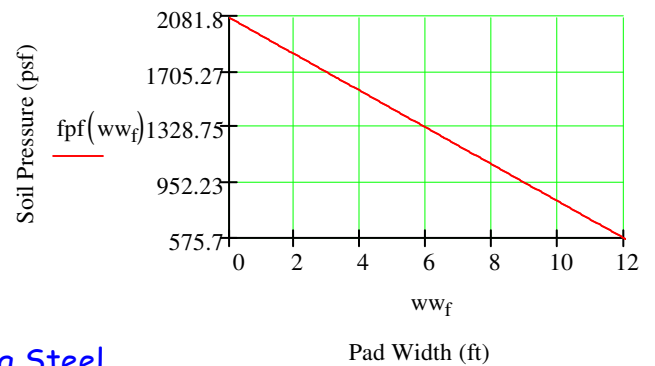
Soil Pressures:

Maximum: $\text{Max}_{lf} = 2.17 \text{ ksf}$ $\text{Min}_{lf} = 0.48 \text{ ksf}$
Minimum: $\text{Max}_{pf} = 2.08 \text{ ksf}$ $\text{Min}_{pf} = 0.58 \text{ ksf}$

Parallel Pressure Distribution



Perpendicular Pressure Distribution



Reinforcing Steel

Pad Parallel to Center Line= $A_{s1f} = 3.76 \text{ in}^2$
Pad Perpendicular to Center Line= $A_{s2f} = 2.33 \text{ in}^2$

Pedestal Parallel to Center Line= $A_{s3f} = 1.98 \text{ in}^2$
Pedestal Perpendicular to Center Line= $A_{s4f} = 1.28 \text{ in}^2$

INPUT AREA 6:

MISC.:

Soil Backfill: $bf_f \equiv 3.5 \cdot \text{ft}$
Void Length: $af_f \equiv 0 \cdot \text{ft}$
Void Height: $bf_f \equiv 0 \cdot \text{ft}$
Effective M. Arm: $y_f \equiv 20.45 \cdot \text{ft}$

COLUMN:

length: $l_{cf} \equiv 4 \cdot \text{ft}$
width: $w_{cf} \equiv 4 \cdot \text{ft}$
height: $h_{cf} \equiv 4 \cdot \text{ft}$

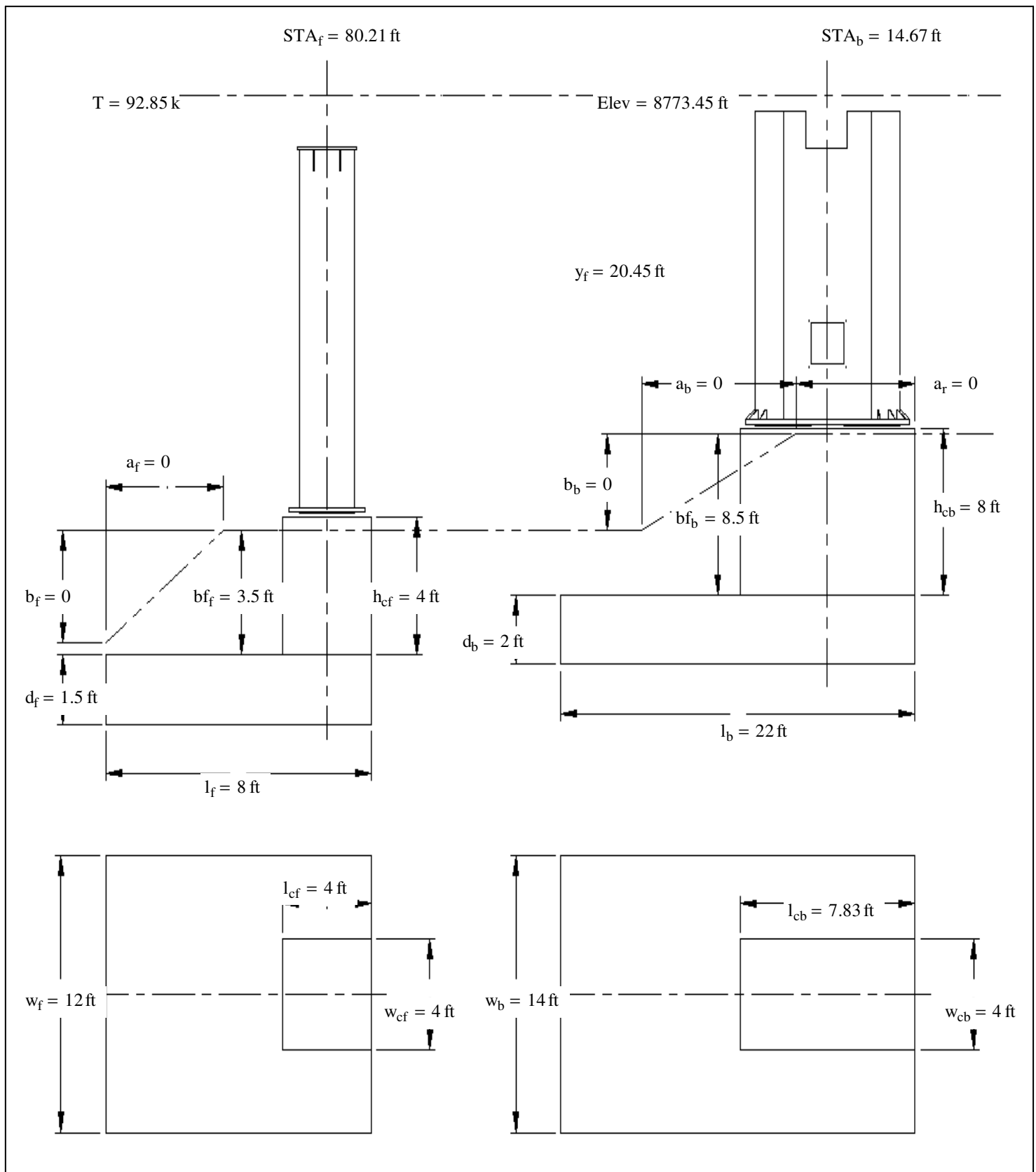
SLAB:

length: $l_f \equiv 8 \cdot \text{ft}$
width: $w_f \equiv 12 \cdot \text{ft}$
depth: $d_f \equiv 1.5 \cdot \text{ft}$

OUTPUT AREA 7

Volume of Concrete Pours:

First = 67.6 yd^3 Second = 17 yd^3 Total = 84.5 yd^3



BACK PAD

INPUT AREA 1: Lift Properties

Rope Elevation Elev $\equiv$ 8773.45·ft

Lift Tension T $\equiv$ 92.85·k

Reaction Column Station STA_b $\equiv$ 14.67·ft

INPUT AREA 2: Material Properties

Soil Weight $W_{\text{soil}} \equiv 110 \cdot \frac{\text{lb}}{\text{ft}^3}$

Conc. Strength $f_c \equiv 2500 \cdot \text{psi}$

$\phi \equiv .90$

$\rho_{\text{minslab}} \equiv .0018$

Conc. Weight $W_{\text{conc}} \equiv 145 \cdot \frac{\text{lb}}{\text{ft}^3}$

Steel Strength $f_y \equiv 60000 \cdot \text{psi}$

$\beta \equiv .85$

$\rho_{\text{min}} \equiv \frac{200 \cdot \text{psi}}{f_y}$

OUTPUT AREA 3:

Safety Factors

Overturning:

Parallel= $SF_{L_{\text{otb}}} = 2.59$

Sliding:

$SF_{L_{\text{sb}}} = 2.5$

Perpendicular= $SF_{P_{\text{otb}}} = 13.33$

$SF_{P_{\text{sb}}} = 24.7$

Soil Pressures:

Maximum:

Parallel= $\text{Max}_{l_b} = 3 \text{ ksf}$

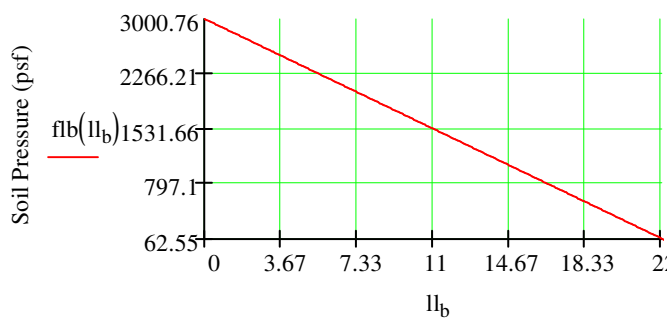
Minimum:

$\text{Min}_{l_b} = 0.06 \text{ ksf}$

Perpendicular= $\text{Max}_{p_b} = 1.88 \text{ ksf}$

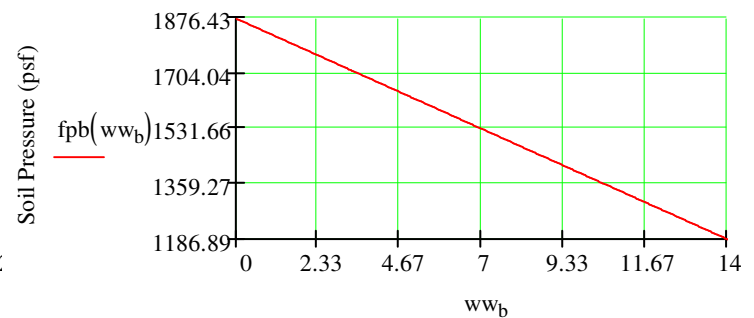
$\text{Min}_{p_b} = 1.19 \text{ ksf}$

Parallel Pressure Distribution



Pad Length (ft)

Perpendicular Pressure Distribution



Pad Width (ft)

Reinforcing Steel

Pad Parallel to Center Line= $A_{s1b} = 27.52 \text{ in}^2$

Pedestal Parallel to Center Line= $A_{s3b} = 9.8 \text{ in}^2$

Pad Perpendicular to Center Line= $A_{s2b} = 9.27 \text{ in}^2$

Pedestal Perpendicular to Center Line= $A_{s4b} = 2.25 \text{ in}^2$

INPUT AREA 4:

MISC.:

Soil Backfill: $bf_b \equiv 8.5 \cdot \text{ft}$

Void Length: $ab \equiv 0 \cdot \text{ft}$

Void Height: $bb \equiv 0 \cdot \text{ft}$

Ramp Length: $ar \equiv 0 \cdot \text{ft}$

COLUMN:

length: $l_{cb} \equiv 94 \cdot \text{in}$

width: $w_{cb} \equiv 4 \cdot \text{ft}$

height: $h_{cb} \equiv 8 \cdot \text{ft}$

SLAB:

length: $l_b \equiv 22 \cdot \text{ft}$

width: $w_b \equiv 14 \cdot \text{ft}$

depth: $d_b \equiv 2 \cdot \text{ft}$

Effective M. Arm: $y_b \equiv 23.445 \cdot \text{ft}$

Column Station $\text{STA}_f \equiv 47.43 \cdot \text{ft}$
 $\text{STA}_f \equiv 80.21 \cdot \text{ft}$

FRONT PADS

OUTPUT AREA 5:

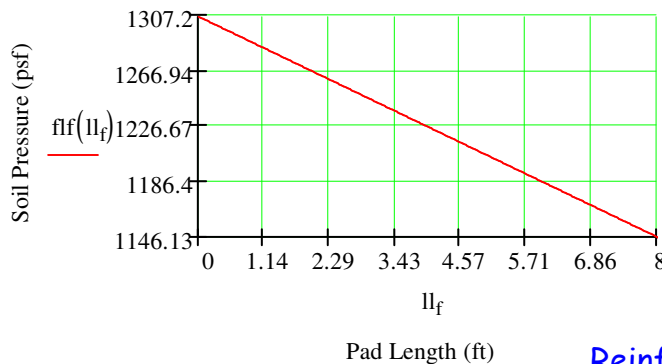
Safety Factors

Parallel= Perpendicular=	<u>Overturning:</u>	$\text{SF}_{l_{\text{otf}}} = 4.54$	<u>Sliding:</u>	$\text{SF}_{l_{\text{sf}}} = 9.34$
		$\text{SF}_{p_{\text{otf}}} = 4.89$		$\text{SF}_{p_{\text{sf}}} = 9.35$

Soil Pressures:

Parallel= Perpendicular=	<u>Maximum:</u>	$\text{Max}_{lf} = 1.31 \text{ ksf}$	<u>Minimum:</u>	$\text{Min}_{lf} = 1.15 \text{ ksf}$
		$\text{Max}_{pf} = 1.98 \text{ ksf}$		$\text{Min}_{pf} = 0.47 \text{ ksf}$

Parallel Pressure Distribution



Perpendicular Pressure Distribution



Reinforcing Steel

Pad Parallel to Center Line= $A_{s1f} = 3.76 \text{ in}^2$
 Pad Perpendicular to Center Line= $A_{s2f} = 2.33 \text{ in}^2$

Pedestal Parallel to Center Line= $A_{s3f} = 1.14 \text{ in}^2$
 Pedestal Perpendicular to Center Line= $A_{s4f} = 1.28 \text{ in}^2$

INPUT AREA 6:

MISC.:

Soil Backfill: $b_f \equiv 3.5 \cdot \text{ft}$
 Void Length: $a_f \equiv 0 \cdot \text{ft}$
 Void Height: $b_f \equiv 0 \cdot \text{ft}$
 Effective M. Arm: $y_f \equiv 20.45 \cdot \text{ft}$

COLUMN:

length: $l_{cf} \equiv 4 \cdot \text{ft}$
 width: $w_{cf} \equiv 4 \cdot \text{ft}$
 height: $h_{cf} \equiv 4 \cdot \text{ft}$

SLAB:

length: $l_f \equiv 8 \cdot \text{ft}$
 width: $w_f \equiv 12 \cdot \text{ft}$
 depth: $d_f \equiv 1.5 \cdot \text{ft}$

OUTPUT AREA 7

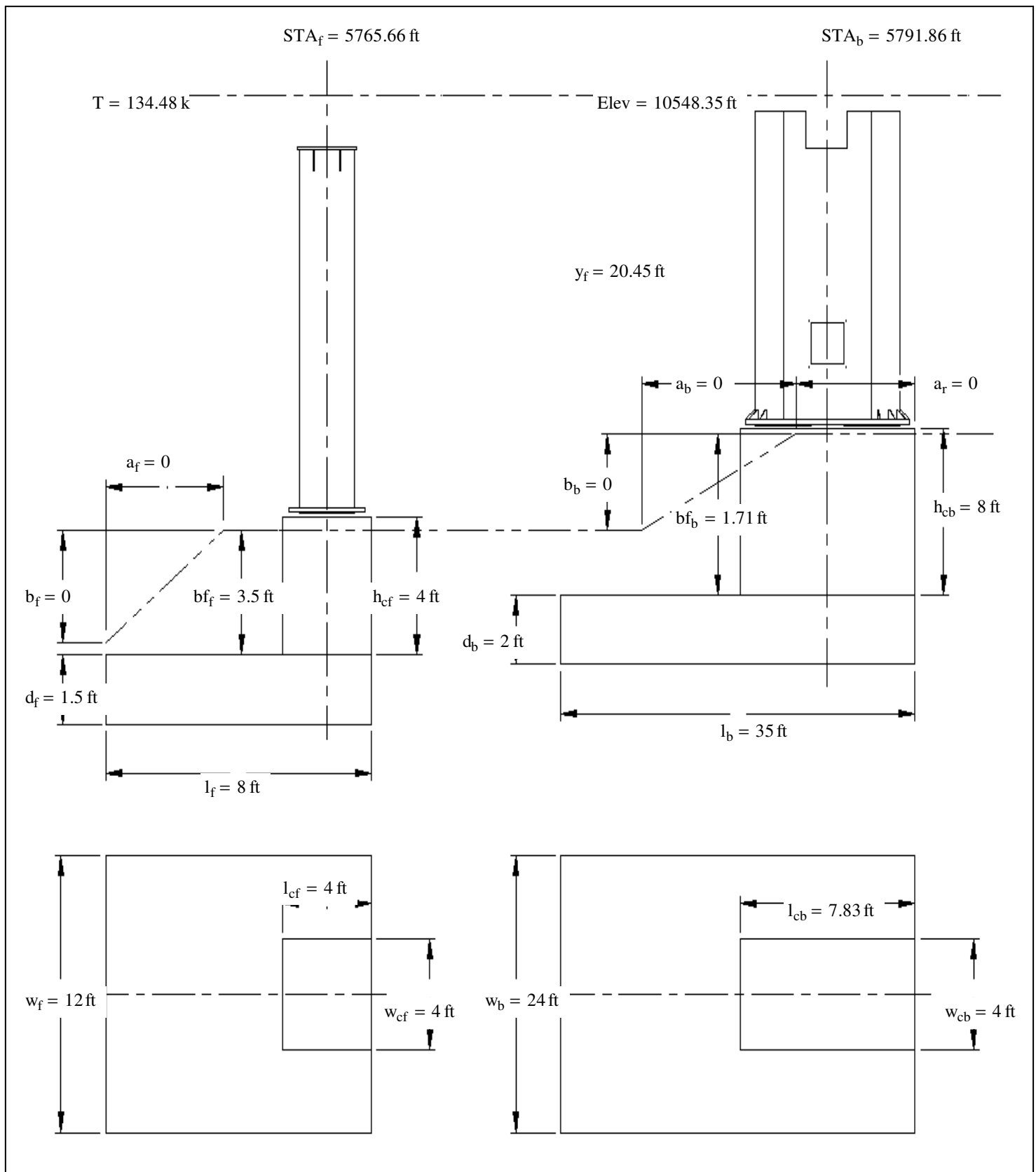
Volume of Concrete Pours:

First = 28.1 yd^3 Second = 17 yd^3 Total = 45.1 yd^3

							Safety Factors		S.P.
TWR	Soil	Backfill	Materials	Pad (ft)	Column	Rebar	Slide	O.T.	(psf)
BEB No Footing	fv: psf fl: psf / ft Frict:	Ht: ft Wt: pcf	fy: ksi fc: psi	W: L: D:	Col W: in Col H: ft	Col: 0 # 0's P. Par: in ² P. Perp: in ²	Opp Non Opp	Opp Non Opp	Opp Non Opp
1 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 8.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 8 # 8's P. Par: 3.46 in ² P. Perp: 3.46 in ²	Opp 12.3 Non Opp 7.7	Opp 5.3 Non Opp 3.4	Opp 935 Non Opp 1,155
2 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 8.0 L: 12.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 8 # 8's P. Par: 3.46 in ² P. Perp: 5.18 in ²	Opp 4.4 Non Opp 3.8	Opp 2.6 Non Opp 2.3	Opp 1,251 Non Opp 1,377
3 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 8.0 L: 11.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 8 # 8's P. Par: 3.46 in ² P. Perp: 4.75 in ²	Opp 6.9 Non Opp 6.5	Opp 2.2 Non Opp 2.0	Opp 2,100 Non Opp 2,995
4 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 13.0 L: 9.0 D: 1.5	Round Column Col W: 48 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 5.62 in ² P. Perp: 4.65 in ²	Opp 10.3 Non Opp 6.5	Opp 2.0 Non Opp 1.9	Opp 2,099 Non Opp 3,010
5 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 14.0 L: 10.0 D: 1.5	Round Column Col W: 48 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 6.05 in ² P. Perp: 5.50 in ²	Opp 10.3 Non Opp 7.9	Opp 2.2 Non Opp 2.1	Opp 1,913 Non Opp 2,525
6 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 13.0 L: 11.0 D: 1.5	Round Column Col W: 48 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 5.62 in ² P. Perp: 5.88 in ²	Opp 9.0 Non Opp 7.4	Opp 2.0 Non Opp 1.9	Opp 2,181 Non Opp 3,649
7 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 11.0 L: 10.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 4.75 in ² P. Perp: 3.82 in ²	Opp 8.3 Non Opp 7.4	Opp 2.1 Non Opp 2.1	Opp 2,216 Non Opp 3,550
8 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 13.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 5.62 in ² P. Perp: 5.07 in ²	Opp 12.2 Non Opp 6.3	Opp 2.2 Non Opp 1.9	Opp 2,125 Non Opp 3,124
9 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 15.0 L: 8.0 D: 1.5	Round Column Col W: 48 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 6.48 in ² P. Perp: 6.59 in ²	Opp 15.0 Non Opp 5.8	Opp 2.5 Non Opp 1.8	Opp 1,815 Non Opp 2,777
10 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 12.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 5.18 in ² P. Perp: 4.05 in ²	Opp 12.2 Non Opp 7.1	Opp 2.1 Non Opp 1.8	Opp 2,247 Non Opp 3,286
11 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 7.0 L: 9.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 8 # 8's P. Par: 3.02 in ² P. Perp: 3.89 in ²	Opp 11.3 Non Opp 9.2	Opp 2.4 Non Opp 2.1	Opp 2,689 Non Opp 3,064
TEB No Footing	fv: psf fl: psf / ft Frict:	Ht: ft Wt: pcf	fy: ksi fc: psi	W: L: D:	Col W: in Col H: ft	Col: 0 # 0's P. Par: in ² P. Perp: in ²	Opp Non Opp	Opp Non Opp	Opp Non Opp

							Safety Factors		S.P.
TWR	Soil	Backfill	Materials	Pad (ft)	Column	Rebar	Slide	O.T.	(psf)
BEB No Footing	fv: psf fl: psf / ft Frict:	Ht: ft Wt: pcf	fy: ksi fc: psi	W: L: D:	Col W: in Col H: ft	Col: 0 # 0's P. Par: in ² P. Perp: in ²	Opp Non Opp	Opp Non Opp	Opp Non Opp
1 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 8.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 8 # 8's P. Par: 3.46 in ² P. Perp: 3.46 in ²	Opp 12.2 Non Opp 7.7	Opp 5.2 Non Opp 3.3	Opp 936 Non Opp 1,163
2 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 8.0 L: 12.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 8 # 8's P. Par: 3.46 in ² P. Perp: 5.18 in ²	Opp 4.4 Non Opp 3.8	Opp 2.6 Non Opp 2.3	Opp 1,254 Non Opp 1,390
3 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 8.0 L: 12.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 3.00 in ² P. Perp: 5.18 in ²	Opp 7.0 Non Opp 7.1	Opp 2.5 Non Opp 1.8	Opp 1,947 Non Opp 3,000
4 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 12.0 L: 11.0 D: 1.5	Round Column Col W: 48 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 5.18 in ² P. Perp: 5.35 in ²	Opp 9.3 Non Opp 6.3	Opp 2.2 Non Opp 1.7	Opp 1,917 Non Opp 3,654
5 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 14.0 L: 11.0 D: 1.5	Round Column Col W: 48 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 6.05 in ² P. Perp: 6.34 in ²	Opp 8.9 Non Opp 7.3	Opp 2.1 Non Opp 2.2	Opp 2,025 Non Opp 2,814
6 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 14.0 L: 12.0 D: 1.5	Round Column Col W: 48 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 6.05 in ² P. Perp: 7.19 in ²	Opp 8.3 Non Opp 7.6	Opp 2.0 Non Opp 2.2	Opp 2,117 Non Opp 3,060
7 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 10.0 L: 12.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 3.73 in ² P. Perp: 5.18 in ²	Opp 7.8 Non Opp 7.2	Opp 2.4 Non Opp 1.9	Opp 2,004 Non Opp 3,200
8 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 13.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 5.62 in ² P. Perp: 6.28 in ²	Opp 11.9 Non Opp 5.6	Opp 2.1 Non Opp 1.7	Opp 2,222 Non Opp 3,799
9 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 16.0 L: 8.0 D: 1.5	Round Column Col W: 48 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 6.91 in ² P. Perp: 8.21 in ²	Opp 14.0 Non Opp 5.2	Opp 2.3 Non Opp 1.7	Opp 1,948 Non Opp 2,930
10 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 12.0 L: 9.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 5.18 in ² P. Perp: 5.43 in ²	Opp 11.5 Non Opp 6.8	Opp 2.1 Non Opp 1.7	Opp 2,241 Non Opp 3,870
11 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 7.0 L: 9.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 8 # 8's P. Par: 3.02 in ² P. Perp: 3.89 in ²	Opp 12.1 Non Opp 9.6	Opp 2.6 Non Opp 1.9	Opp 2,546 Non Opp 3,720
TEB No Footing	fv: psf fl: psf / ft Frict:	Ht: ft Wt: pcf	fy: ksi fc: psi	W: L: D:	Col W: in Col H: ft	Col: 0 # 0's P. Par: in ² P. Perp: in ²	Opp Non Opp	Opp Non Opp	Opp Non Opp

							Safety Factors		S.P.
TWR	Soil	Backfill	Materials	Pad (ft)	Column	Rebar	Slide	O.T.	(psf)
BEB No Footing	fv: psf fl: psf / ft Frict:	Ht: ft Wt: pcf	fy: ksi fc: psi	W: L: D:	Col W: in Col H: ft	Col: 0 # 0's P. Par: in ² P. Perp: in ²	Opp Non Opp	Opp Non Opp	Opp Non Opp
1 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 8.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 8 # 8's P. Par: 3.46 in ² P. Perp: 3.46 in ²	Opp 12.6 Non Opp 8.1	Opp 5.4 Non Opp 3.5	Opp 965 Non Opp 1,146
2 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 8.0 L: 12.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 8 # 8's P. Par: 3.46 in ² P. Perp: 5.18 in ²	Opp 4.7 Non Opp 4.0	Opp 2.8 Non Opp 2.4	Opp 1,274 Non Opp 1,417
3 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 9.0 L: 12.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 3.35 in ² P. Perp: 5.18 in ²	Opp 6.5 Non Opp 7.0	Opp 2.3 Non Opp 2.0	Opp 2,020 Non Opp 2,571
4 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 12.0 L: 14.0 D: 1.5	Round Column Col W: 48 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 4.89 in ² P. Perp: 6.17 in ²	Opp 9.2 Non Opp 6.8	Opp 2.7 Non Opp 1.8	Opp 1,606 Non Opp 3,221
5 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 12.0 L: 14.0 D: 1.5	Round Column Col W: 48 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 5.53 in ² P. Perp: 6.02 in ²	Opp 8.2 Non Opp 7.2	Opp 2.4 Non Opp 1.9	Opp 1,789 Non Opp 2,902
6 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 12.0 L: 15.0 D: 1.5	Square Column Col W: 60 in Col H: 4.0 ft	Col: 24 # 8's P. Par: 5.98 in ² P. Perp: 5.61 in ²	Opp 7.8 Non Opp 7.1	Opp 2.4 Non Opp 1.8	Opp 1,919 Non Opp 3,287
7 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 10.0 L: 13.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 4.68 in ² P. Perp: 4.57 in ²	Opp 7.3 Non Opp 6.6	Opp 2.4 Non Opp 1.7	Opp 2,018 Non Opp 3,712
8 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 13.0 L: 10.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 5.62 in ² P. Perp: 6.97 in ²	Opp 12.3 Non Opp 6.0	Opp 2.7 Non Opp 1.8	Opp 1,779 Non Opp 3,325
9 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 16.0 L: 9.0 D: 1.5	Square Column Col W: 60 in Col H: 4.0 ft	Col: 24 # 8's P. Par: 6.91 in ² P. Perp: 7.85 in ²	Opp 12.9 Non Opp 5.4	Opp 2.3 Non Opp 1.7	Opp 1,948 Non Opp 3,056
10 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 12.0 L: 10.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 5.18 in ² P. Perp: 5.74 in ²	Opp 10.2 Non Opp 6.7	Opp 2.1 Non Opp 1.7	Opp 2,148 Non Opp 3,800
11 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 9.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 8 # 8's P. Par: 3.89 in ² P. Perp: 3.46 in ²	Opp 12.2 Non Opp 10.1	Opp 2.3 Non Opp 2.1	Opp 2,688 Non Opp 3,938
TEB No Footing	fv: psf fl: psf / ft Frict:	Ht: ft Wt: pcf	fy: ksi fc: psi	W: L: D:	Col W: in Col H: ft	Col: 0 # 0's P. Par: in ² P. Perp: in ²	Opp Non Opp	Opp Non Opp	Opp Non Opp



BACK PAD

INPUT AREA 1: Lift Properties

Rope Elevation Elev $\equiv$ 10548.35·ft Lift Tension T $\equiv$ 134.48·k Reaction Column Station STA_b $\equiv$ 5791.86·ft

INPUT AREA 2: Material Properties

Surcharge Weight $W_{\text{soil}} \equiv 145 \cdot \frac{\text{lb}}{\text{ft}^3}$ Conc. Strength $f_c \equiv 3000 \cdot \text{psi}$ $\phi \equiv .90$ $\rho_{\text{minslab}} \equiv .0018$
 Conc. Weight $W_{\text{conc}} \equiv 145 \cdot \frac{\text{lb}}{\text{ft}^3}$ Steel Strength $f_y \equiv 60000 \cdot \text{psi}$ $\beta \equiv .85$ $\rho_{\text{min}} \equiv \frac{200 \cdot \text{psi}}{f_y}$

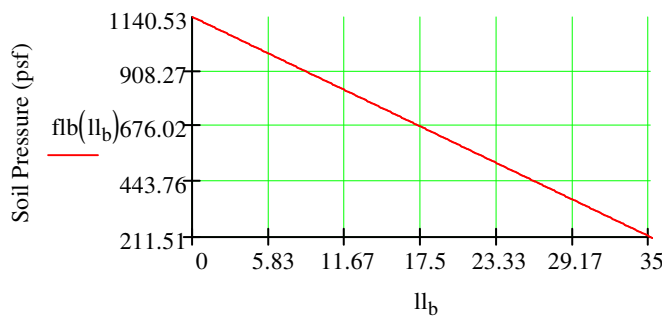
OUTPUT AREA 3: Safety Factors

Overturning: Sliding:
 Parallel= $\text{SF}_{\text{L_otb}} = 3.05$ $\text{SF}_{\text{L_sb}} = 2.09$
 Perpendicular= $\text{SF}_{\text{p_otb}} = 23.81$ $\text{SF}_{\text{p_sb}} = 29.73$

Soil Pressures:

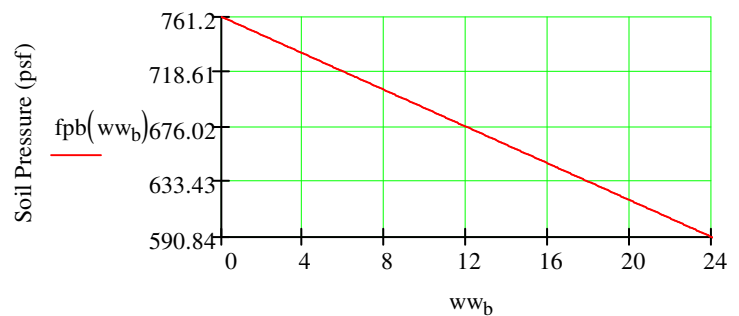
Maximum: Minimum:
 Parallel= $\text{Max}_{\text{lb}} = 1.14 \text{ ksf}$ $\text{Min}_{\text{lb}} = 0.21 \text{ ksf}$
 Perpendicular= $\text{Max}_{\text{pb}} = 0.76 \text{ ksf}$ $\text{Min}_{\text{pb}} = 0.59 \text{ ksf}$

Parallel Pressure Distribution



Pad Length (ft)

Perpendicular Pressure Distribution



Pad Width (ft)

Reinforcing Steel

Pad Parallel to Center Line= $A_{s1b} = 54.63 \text{ in}^2$ Pedestal Parallel to Center Line= $A_{s3b} = 14.08 \text{ in}^2$
 Pad Perpendicular to Center Line= $A_{s2b} = 14.74 \text{ in}^2$ Pedestal Perpendicular to Center Line= $A_{s4b} = 2.62 \text{ in}^2$

INPUT AREA 4:

MISC.:

Surcharge:

$bf_b \equiv (0.5 + 0.5 + 0.71) \cdot \text{ft}$
 Floor + Wall + Ceiling

COLUMN:

length: $l_{cb} \equiv 94 \cdot \text{in}$
 width: $w_{cb} \equiv 4 \cdot \text{ft}$
 height: $h_{cb} \equiv 8 \cdot \text{ft}$

SLAB:

length: $l_b \equiv 35 \cdot \text{ft}$
 width: $w_b \equiv 24 \cdot \text{ft}$
 depth: $d_b \equiv 2 \cdot \text{ft}$

Effective M. Arm: $y_b \equiv 27.47 \cdot \text{ft}$

FRONT PADS

Column Station $\text{STA}_f \equiv 5726.37 \cdot \text{ft}$
 $\text{STA}_f \equiv 5765.66 \cdot \text{ft}$

OUTPUT AREA 5:

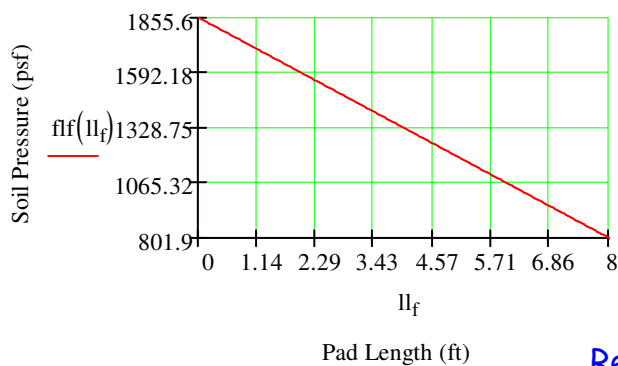
Safety Factors

	<u>Overturning:</u>	<u>Sliding:</u>
Parallel=	$\text{SF}_{l_{\text{otf}}} = 3.41$	$\text{SF}_{l_{\text{sf}}} = 7.16$
Perpendicular=	$\text{SF}_{p_{\text{otf}}} = 5.29$	$\text{SF}_{p_{\text{sf}}} = 10.12$

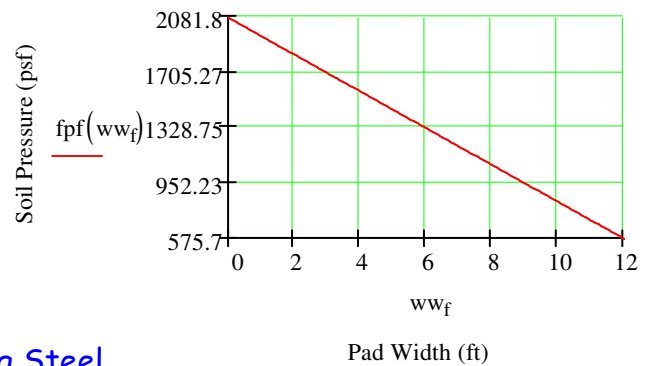
Soil Pressures:

	<u>Maximum:</u>	<u>Minimum:</u>
Parallel=	$\text{Max}_{lf} = 1.86 \text{ ksf}$	$\text{Min}_{lf} = 0.8 \text{ ksf}$
Perpendicular=	$\text{Max}_{pf} = 2.08 \text{ ksf}$	$\text{Min}_{pf} = 0.58 \text{ ksf}$

Parallel Pressure Distribution



Perpendicular Pressure Distribution



Reinforcing Steel

Pad Parallel to Center Line=	$A_{s1f} = 3.76 \text{ in}^2$	Pedestal Parallel to Center Line=	$A_{s3f} = 1.61 \text{ in}^2$
Pad Perpendicular to Center Line=	$A_{s2f} = 2.33 \text{ in}^2$	Pedestal Perpendicular to Center Line=	$A_{s4f} = 1.28 \text{ in}^2$

INPUT AREA 6:

MISC.:

Soil Backfill: $\text{bf}_f \equiv 3.5 \cdot \text{ft}$
 Void Length: $\text{a}_f \equiv 0 \cdot \text{ft}$
 Void Height: $\text{b}_f \equiv 0 \cdot \text{ft}$
 Effective M. Arm: $\text{y}_f \equiv 20.45 \cdot \text{ft}$

COLUMN:

length: $\text{l}_{cf} \equiv 4 \cdot \text{ft}$
 width: $\text{w}_{cf} \equiv 4 \cdot \text{ft}$
 height: $\text{h}_{cf} \equiv 4 \cdot \text{ft}$

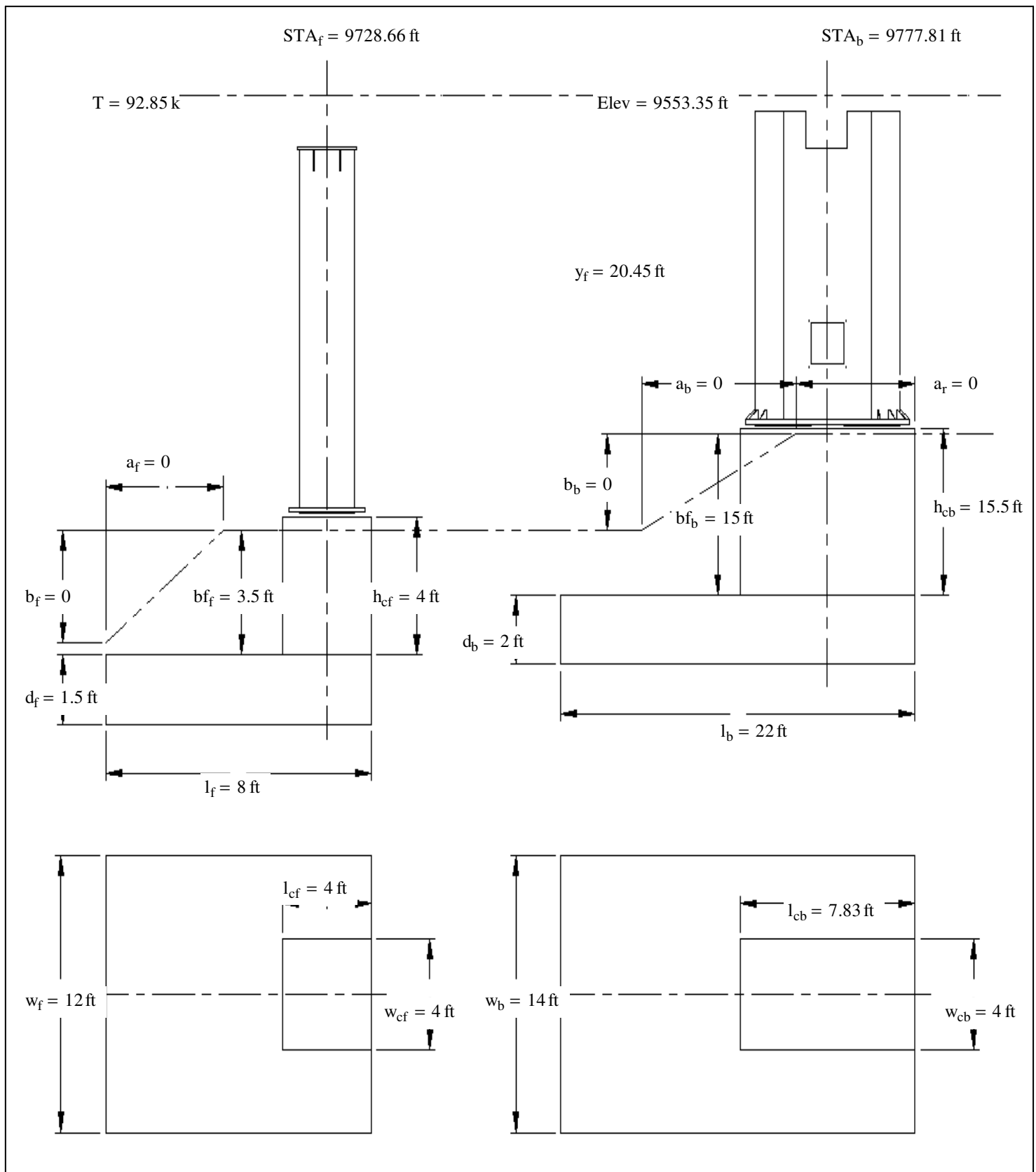
SLAB:

length: $\text{l}_f \equiv 8 \cdot \text{ft}$
 width: $\text{w}_f \equiv 12 \cdot \text{ft}$
 depth: $\text{d}_f \equiv 1.5 \cdot \text{ft}$

OUTPUT AREA 7

Volume of Concrete Pours:

First = 67.6 yd^3 Second = 17 yd^3 Total = 84.5 yd^3



BACK PAD

INPUT AREA 1: Lift Properties

Rope Elevation Elev $\equiv$ 9553.35·ft

Lift Tension T $\equiv$ 92.85·k

Reaction Column Station STA_b $\equiv$ 9777.81·ft

INPUT AREA 2: Material Properties

Soil Weight $W_{\text{soil}} \equiv 110 \cdot \frac{\text{lb}}{\text{ft}^3}$

Conc. Strength $f_c \equiv 3000 \cdot \text{psi}$

$\phi \equiv .90$

$\rho_{\text{minslab}} \equiv .0018$

Conc. Weight $W_{\text{conc}} \equiv 145 \cdot \frac{\text{lb}}{\text{ft}^3}$

Steel Strength $f_y \equiv 60000 \cdot \text{psi}$

$\beta \equiv .85$

$\rho_{\text{min}} \equiv \frac{200 \cdot \text{psi}}{f_y}$

OUTPUT AREA 3:

Safety Factors

Overturning:

Parallel= $SF_{l_{\text{otb}}} = 2.83$

Sliding:

$SF_{l_{\text{sb}}} = 3.73$

Perpendicular= $SF_{p_{\text{otb}}} = 15.24$

$SF_{p_{\text{sb}}} = 36.84$

Soil Pressures:

Maximum:

Parallel= $\text{Max}_{l_b} = 4.34 \text{ ksf}$

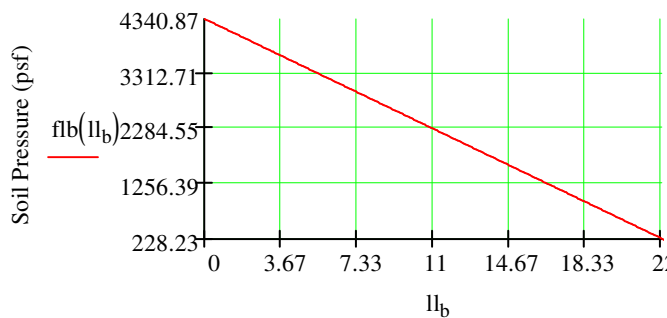
Minimum:

$\text{Min}_{l_b} = 0.23 \text{ ksf}$

Perpendicular= $\text{Max}_{p_b} = 2.73 \text{ ksf}$

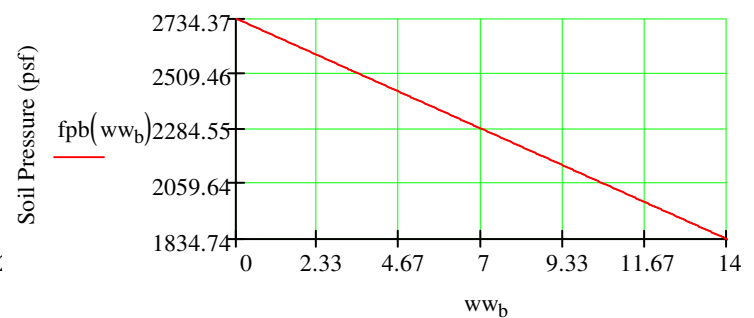
$\text{Min}_{p_b} = 1.83 \text{ ksf}$

Parallel Pressure Distribution



Pad Length (ft)

Perpendicular Pressure Distribution



Pad Width (ft)

Reinforcing Steel

Pad Parallel to Center Line= $A_{s1b} = 37.06 \text{ in}^2$

Pedestal Parallel to Center Line= $A_{s3b} = 13.45 \text{ in}^2$

Pad Perpendicular to Center Line= $A_{s2b} = 9.27 \text{ in}^2$

Pedestal Perpendicular to Center Line= $A_{s4b} = 2.99 \text{ in}^2$

INPUT AREA 4:

MISC.:

Soil Backfill: $bf_b \equiv 15 \cdot \text{ft}$

Void Length: $ab \equiv 0 \cdot \text{ft}$

Void Height: $bb \equiv 0 \cdot \text{ft}$

Ramp Length: $ar \equiv 0 \cdot \text{ft}$

COLUMN:

length: $l_{cb} \equiv 94 \cdot \text{in}$

width: $w_{cb} \equiv 4 \cdot \text{ft}$

height: $h_{cb} \equiv 15.5 \cdot \text{ft}$

SLAB:

length: $l_b \equiv 22 \cdot \text{ft}$

width: $w_b \equiv 14 \cdot \text{ft}$

depth: $db \equiv 2 \cdot \text{ft}$

Effective M. Arm: $y_b \equiv 31.35 \cdot \text{ft}$

Column Station $\text{STA}_f \equiv 9728.66 \cdot \text{ft}$

FRONT PAD

OUTPUT AREA 5:

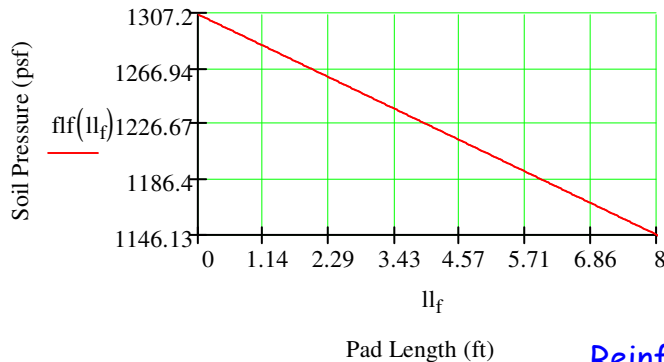
Safety Factors

	<u>Overtuning:</u>	<u>Sliding:</u>
Parallel=	$\text{SF}_{l_{\text{otf}}} = 4.54$	$\text{SF}_{l_{\text{sf}}} = 9.34$
Perpendicular=	$\text{SF}_{p_{\text{otf}}} = 4.89$	$\text{SF}_{p_{\text{sf}}} = 9.35$

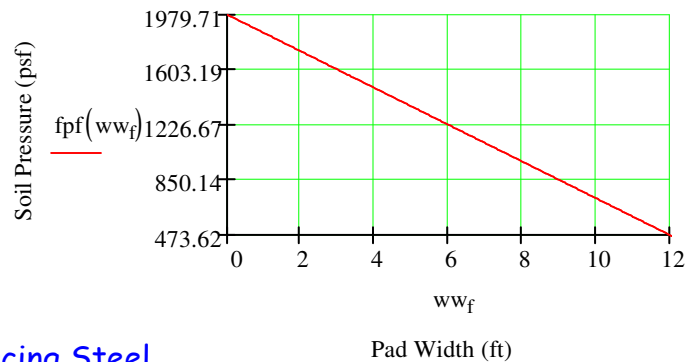
Soil Pressures:

	<u>Maximum:</u>	<u>Minimum:</u>
Parallel=	$\text{Max}_{lf} = 1.31 \text{ ksf}$	$\text{Min}_{lf} = 1.15 \text{ ksf}$
Perpendicular=	$\text{Max}_{pf} = 1.98 \text{ ksf}$	$\text{Min}_{pf} = 0.47 \text{ ksf}$

Parallel Pressure Distribution



Perpendicular Pressure Distribution



Reinforcing Steel

Pad Parallel to Center Line= $A_{s1f} = 3.76 \text{ in}^2$
 Pad Perpendicular to Center Line= $A_{s2f} = 2.33 \text{ in}^2$

Pedestal Parallel to Center Line= $A_{s3f} = 1.14 \text{ in}^2$
 Pedestal Perpendicular to Center Line= $A_{s4f} = 1.28 \text{ in}^2$

INPUT AREA 6:

MISC.:

Soil Backfill: $b_f \equiv 3.5 \cdot \text{ft}$
 Void Length: $a_f \equiv 0 \cdot \text{ft}$
 Void Height: $b_f \equiv 0 \cdot \text{ft}$
 Effective M. Arm: $y_f \equiv 20.45 \cdot \text{ft}$

COLUMN:

length: $l_{cf} \equiv 4 \cdot \text{ft}$
 width: $w_{cf} \equiv 4 \cdot \text{ft}$
 height: $h_{cf} \equiv 4 \cdot \text{ft}$

SLAB:

length: $l_f \equiv 8 \cdot \text{ft}$
 width: $w_f \equiv 12 \cdot \text{ft}$
 depth: $d_f \equiv 1.5 \cdot \text{ft}$

OUTPUT AREA 7

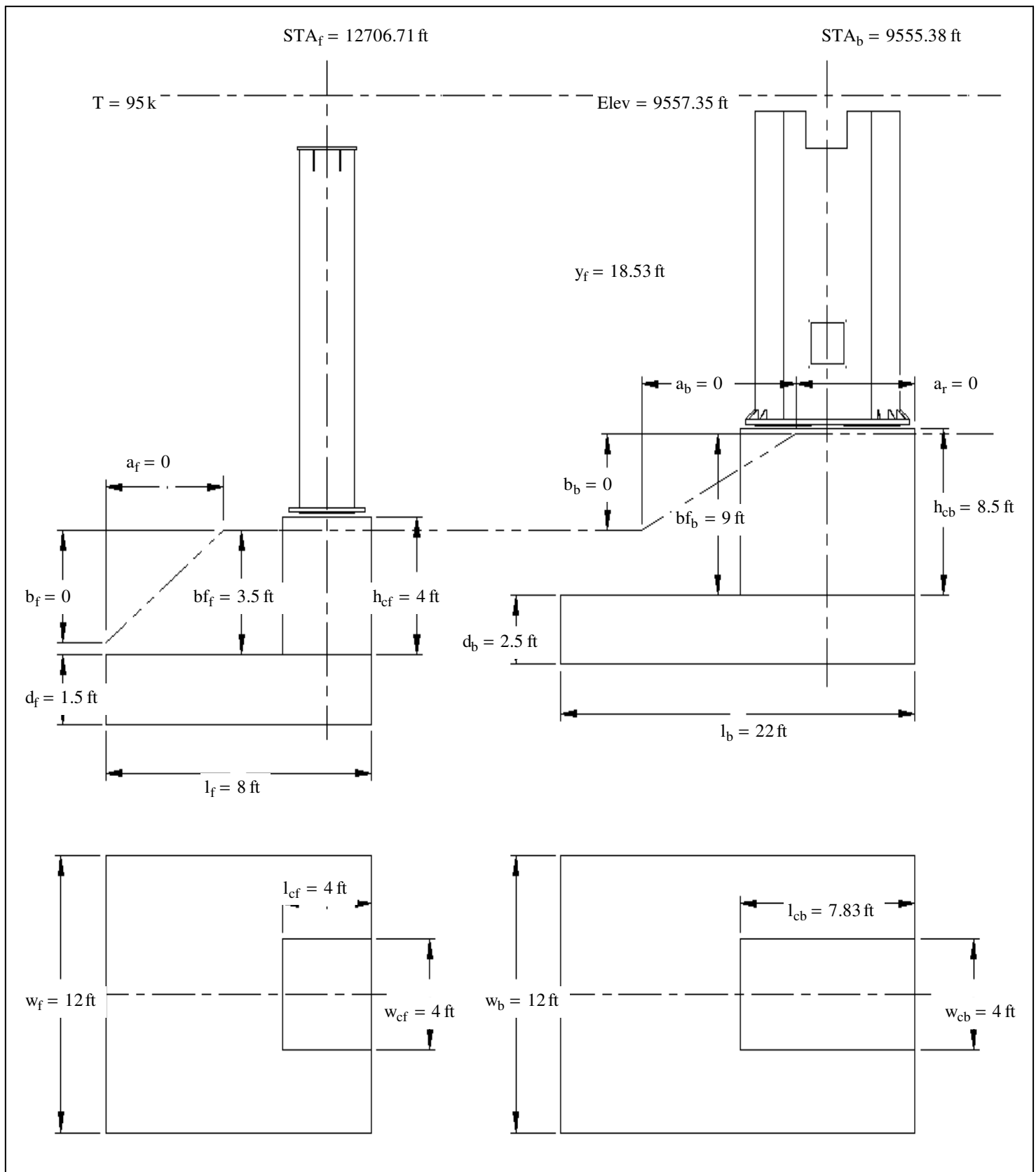
Volume of Concrete Pours:

First = 28.1 yd^3 Second = 25.7 yd^3 Total = 53.8 yd^3

							Safety Factors		S.P.
TWR	Soil	Backfill	Materials	Pad (ft)	Column	Rebar	Slide	O.T.	(psf)
BEB No Footing	fv: psf fl: psf / ft Frict:	Ht: ft Wt: pcf	fy: ksi fc: psi	W: L: D:	Col W: in Col H: ft	Col: 0 # 0's P. Par: in ² P. Perp: in ²	Opp Non Opp	Opp Non Opp	Opp Non Opp
1 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 8.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 8 # 8's P. Par: 3.46 in ² P. Perp: 3.46 in ²	Opp 15.3 Non Opp 7.6	Opp 3.9 Non Opp 2.0	Opp 1,191 Non Opp 2,215
2 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 11.0 L: 7.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 4.75 in ² P. Perp: 3.07 in ²	Opp 18.2 Non Opp 8.3	Opp 2.8 Non Opp 1.9	Opp 2,109 Non Opp 3,129
3 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 12.0 L: 7.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 5.18 in ² P. Perp: 3.94 in ²	Opp 31.5 Non Opp 6.5	Opp 5.2 Non Opp 1.7	Opp 1,453 Non Opp 2,521
4 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 13.0 L: 7.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 5.62 in ² P. Perp: 5.14 in ²	Opp 30.0 Non Opp 6.4	Opp 4.8 Non Opp 1.8	Opp 1,582 Non Opp 3,170
5 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 12.0 L: 7.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 5.18 in ² P. Perp: 4.12 in ²	Opp 19.3 Non Opp 5.8	Opp 3.5 Non Opp 1.8	Opp 1,812 Non Opp 3,211
6 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 14.0 L: 7.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 6.05 in ² P. Perp: 3.95 in ²	Opp 26.7 Non Opp 5.1	Opp 5.2 Non Opp 1.8	Opp 744 Non Opp 1,669
7 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 14.0 L: 7.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 6.05 in ² P. Perp: 5.97 in ²	Opp 17.1 Non Opp 6.1	Opp 2.5 Non Opp 1.8	Opp 2,060 Non Opp 3,129
8 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 12.0 L: 7.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 5.18 in ² P. Perp: 4.04 in ²	Opp 21.3 Non Opp 7.0	Opp 3.0 Non Opp 1.7	Opp 1,855 Non Opp 3,315
TEB No Footing	fv: psf fl: psf / ft Frict:	Ht: ft Wt: pcf	fy: ksi fc: psi	W: L: D:	Col W: in Col H: ft	Col: 0 # 0's P. Par: in ² P. Perp: in ²	Opp Non Opp	Opp Non Opp	Opp Non Opp

							Safety Factors		S.P.
TWR	Soil	Backfill	Materials	Pad (ft)	Column	Rebar	Slide	O.T.	(psf)
BEB No Footing	fv: psf fl: psf / ft Frict:	Ht: ft Wt: pcf	fy: ksi fc: psi	W: L: D:	Col W: in Col H: ft	Col: 0 # 0's P. Par: in^2 P. Perp: in^2	Opp Non Opp	Opp Non Opp	Opp Non Opp
1 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 8.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 8 # 8's P. Par: 3.46 in^2 P. Perp: 3.46 in^2	Opp 15.3 Non Opp 7.2	Opp 3.9 Non Opp 1.8	Opp 1,191 Non Opp 2,420
2 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 11.0 L: 7.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 4.75 in^2 P. Perp: 3.26 in^2	Opp 14.4 Non Opp 7.5	Opp 2.2 Non Opp 1.7	Opp 2,501 Non Opp 3,700
3 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 13.0 L: 7.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 5.62 in^2 P. Perp: 5.09 in^2	Opp 28.3 Non Opp 6.1	Opp 4.4 Non Opp 1.7	Opp 1,559 Non Opp 2,599
4 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 14.0 L: 7.0 D: 1.5	Round Column Col W: 48 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 6.05 in^2 P. Perp: 5.79 in^2	Opp 28.7 Non Opp 5.9	Opp 4.6 Non Opp 1.8	Opp 1,625 Non Opp 3,174
5 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 12.0 L: 7.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 5.18 in^2 P. Perp: 3.67 in^2	Opp 18.0 Non Opp 5.7	Opp 3.3 Non Opp 1.8	Opp 1,674 Non Opp 2,945
6 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 14.0 L: 7.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 6.05 in^2 P. Perp: 4.32 in^2	Opp 38.3 Non Opp 4.6	Opp 7.8 Non Opp 1.6	Opp 733 Non Opp 1,829
7 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 15.0 L: 7.0 D: 1.5	Round Column Col W: 48 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 6.48 in^2 P. Perp: 6.78 in^2	Opp 20.6 Non Opp 5.7	Opp 3.1 Non Opp 1.8	Opp 1,860 Non Opp 3,116
8 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 13.0 L: 7.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 5.62 in^2 P. Perp: 4.92 in^2	Opp 26.2 Non Opp 6.7	Opp 3.7 Non Opp 1.7	Opp 1,637 Non Opp 3,095
TEB No Footing	fv: psf fl: psf / ft Frict:	Ht: ft Wt: pcf	fy: ksi fc: psi	W: L: D:	Col W: in Col H: ft	Col: 0 # 0's P. Par: in^2 P. Perp: in^2	Opp Non Opp	Opp Non Opp	Opp Non Opp

							Safety Factors		S.P.
TWR	Soil	Backfill	Materials	Pad (ft)	Column	Rebar	Slide	O.T.	(psf)
BEB No Footing	fv: psf fl: psf / ft Frict:	Ht: ft Wt: pcf	fy: ksi fc: psi	W: L: D:	Col W: in Col H: ft	Col: 0 # 0's P. Par: in^2 P. Perp: in^2	Opp Non Opp	Opp Non Opp	Opp Non Opp
1 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 8.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 8 # 8's P. Par: 3.46 in^2 P. Perp: 3.46 in^2	Opp 15.3 Non Opp 6.7	Opp 3.9 Non Opp 1.7	Opp 1,191 Non Opp 2,645
2 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 11.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 4.75 in^2 P. Perp: 3.88 in^2	Opp 15.7 Non Opp 7.5	Opp 2.7 Non Opp 1.8	Opp 2,050 Non Opp 3,530
3 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 13.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 5.62 in^2 P. Perp: 6.45 in^2	Opp 46.6 Non Opp 6.1	Opp 9.0 Non Opp 1.7	Opp 1,205 Non Opp 3,038
4 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 14.0 L: 8.0 D: 1.5	Round Column Col W: 48 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 6.05 in^2 P. Perp: 6.44 in^2	Opp 30.1 Non Opp 5.9	Opp 5.4 Non Opp 1.8	Opp 1,467 Non Opp 3,234
5 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 13.0 L: 7.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 5.62 in^2 P. Perp: 4.48 in^2	Opp 17.2 Non Opp 5.5	Opp 3.1 Non Opp 1.8	Opp 1,705 Non Opp 2,801
6 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 14.0 L: 8.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 8's P. Par: 6.05 in^2 P. Perp: 4.91 in^2	Opp 38.7 Non Opp 4.8	Opp 8.8 Non Opp 1.7	Opp 751 Non Opp 1,817
7 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 15.0 L: 8.0 D: 1.5	Round Column Col W: 48 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 6.48 in^2 P. Perp: 7.58 in^2	Opp 16.2 Non Opp 5.7	Opp 2.7 Non Opp 1.7	Opp 1,947 Non Opp 3,187
8 Spread Footing	fv: 4,000 psf fl: 750 psf / ft Frict: 0.5	Ht: 3.5 ft Wt: 110 pcf	fy: 60 ksi fc: 3,000 psi	W: 15.0 L: 7.0 D: 1.5	Round Column Col W: 42 in Col H: 4.0 ft	Col: 12 # 9's P. Par: 6.48 in^2 P. Perp: 6.73 in^2	Opp 20.0 Non Opp 6.6	Opp 2.7 Non Opp 1.9	Opp 1,888 Non Opp 2,676
TEB No Footing	fv: psf fl: psf / ft Frict:	Ht: ft Wt: pcf	fy: ksi fc: psi	W: L: D:	Col W: in Col H: ft	Col: 0 # 0's P. Par: in^2 P. Perp: in^2	Opp Non Opp	Opp Non Opp	Opp Non Opp



BACK PAD

INPUT AREA 1: Lift Properties

Rope Elevation Elev $\equiv$ 9557.35·ft

Lift Tension T $\equiv$ 95·k

Reaction Column Station STA_b $\equiv$ 9555.38·ft

INPUT AREA 2: Material Properties

Surcharge Weight $W_{\text{soil}} \equiv 110 \cdot \frac{\text{lb}}{\text{ft}^3}$

Conc. Strength $f_c \equiv 3000 \cdot \text{psi}$

$\phi \equiv .90$

$\rho_{\text{minslab}} \equiv .0018$

Conc. Weight $W_{\text{conc}} \equiv 145 \cdot \frac{\text{lb}}{\text{ft}^3}$

Steel Strength $f_y \equiv 60000 \cdot \text{psi}$

$\beta \equiv .85$

$\rho_{\text{min}} \equiv \frac{200 \cdot \text{psi}}{f_y}$

OUTPUT AREA 3:

Safety Factors

Overturning:

Sliding:

Parallel= $SF_{l_{\text{otb}}} = 2.31$

$SF_{l_{\text{sb}}} = 2.34$

Perpendicular= $SF_{p_{\text{otb}}} = 10.39$

$SF_{p_{\text{sb}}} = 23.67$

Soil Pressures:

Maximum:

Minimum:

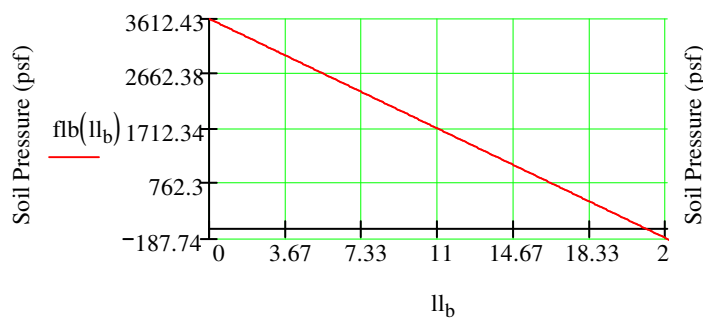
Parallel= $\text{Max}_{l_b} = 3.61 \text{ ksf}$

$\text{Min}_{l_b} = -0.19 \text{ ksf}$

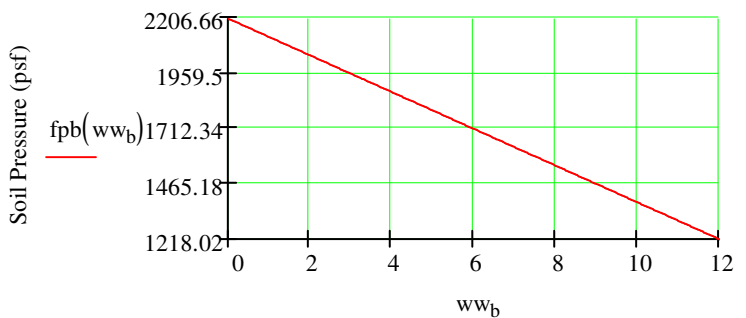
Perpendicular= $\text{Max}_{p_b} = 2.21 \text{ ksf}$

$\text{Min}_{p_b} = 1.22 \text{ ksf}$

Parallel Pressure Distribution



Perpendicular Pressure Distribution



Reinforcing Steel

Pad Parallel to Center Line= $A_{s1b} = 21.9 \text{ in}^2$

Pedestal Parallel to Center Line= $A_{s3b} = 10.4 \text{ in}^2$

Pad Perpendicular to Center Line= $A_{s2b} = 12.12 \text{ in}^2$

Pedestal Perpendicular to Center Line= $A_{s4b} = 2.33 \text{ in}^2$

INPUT AREA 4:

MISC.:

Surcharge:

$bf_b \equiv 9 \cdot \text{ft}$

Floor + Wall + Ceiling

COLUMN:

length: $l_{cb} \equiv 94 \cdot \text{in}$

width: $w_{cb} \equiv 4 \cdot \text{ft}$

height: $h_{cb} \equiv 8.5 \cdot \text{ft}$

SLAB:

length: $l_b \equiv 22 \cdot \text{ft}$

width: $w_b \equiv 12 \cdot \text{ft}$

depth: $d_b \equiv 2.5 \cdot \text{ft}$



Telluride Mtn Gondola *Sect 3, Drive Terminal* *Foundation Design*

Sheet:3 of 3

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Effective M. Arm: $y_b \equiv 24.83 \cdot \text{ft}$

FRONT PADS

Column Station $\text{STA}_f \equiv 12667.43 \cdot \text{ft}$
 $\text{STA}_f \equiv 12706.71 \cdot \text{ft}$

OUTPUT AREA 5:

Safety Factors

Parallel= $\text{SF}_{l_{\text{otf}}} = 4.9$
 Perpendicular= $\text{SF}_{p_{\text{otf}}} = 5.33$

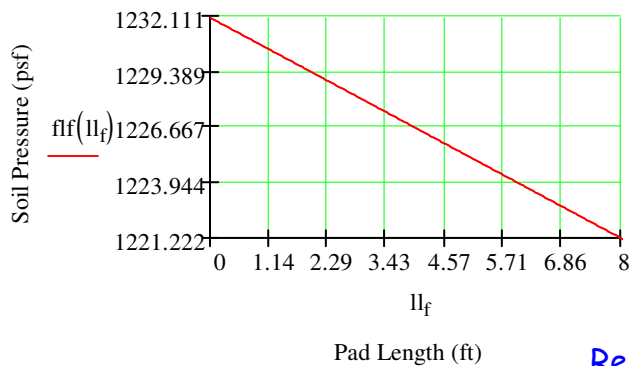
Parallel= $\text{SF}_{l_{\text{sf}}} = 9.15$
 Perpendicular= $\text{SF}_{p_{\text{sf}}} = 9.35$

Soil Pressures:

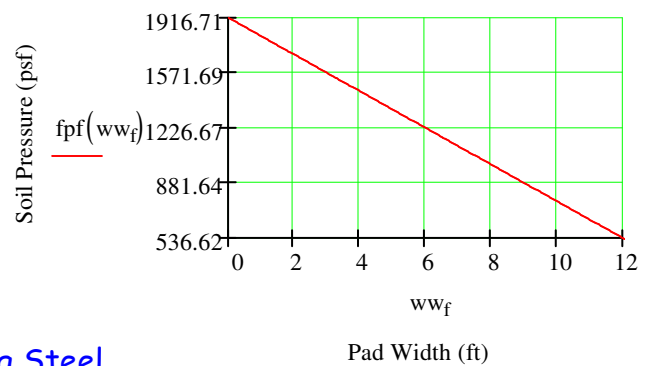
Parallel= $\text{Max}_{lf} = 1.23 \text{ ksf}$
 Perpendicular= $\text{Max}_{pf} = 1.92 \text{ ksf}$

Minimum:
 $\text{Min}_{lf} = 1.22 \text{ ksf}$
 $\text{Min}_{pf} = 0.54 \text{ ksf}$

Parallel Pressure Distribution



Perpendicular Pressure Distribution



Reinforcing Steel

Pad Parallel to Center Line= $A_{s1f} = 3.76 \text{ in}^2$
 Pad Perpendicular to Center Line= $A_{s2f} = 2.33 \text{ in}^2$

Pedestal Parallel to Center Line= $A_{s3f} = 1.05 \text{ in}^2$
 Pedestal Perpendicular to Center Line= $A_{s4f} = 1.17 \text{ in}^2$

INPUT AREA 6:

MISC.:

Soil Backfill: $\text{bf}_f \equiv 3.5 \cdot \text{ft}$
 Void Length: $\text{af}_f \equiv 0 \cdot \text{ft}$
 Void Height: $\text{bf}_f \equiv 0 \cdot \text{ft}$
 Effective M. Arm: $y_f \equiv 18.53 \cdot \text{ft}$

COLUMN:

length: $\text{l}_{cf} \equiv 4 \cdot \text{ft}$
 width: $\text{w}_{cf} \equiv 4 \cdot \text{ft}$
 height: $\text{h}_{cf} \equiv 4 \cdot \text{ft}$

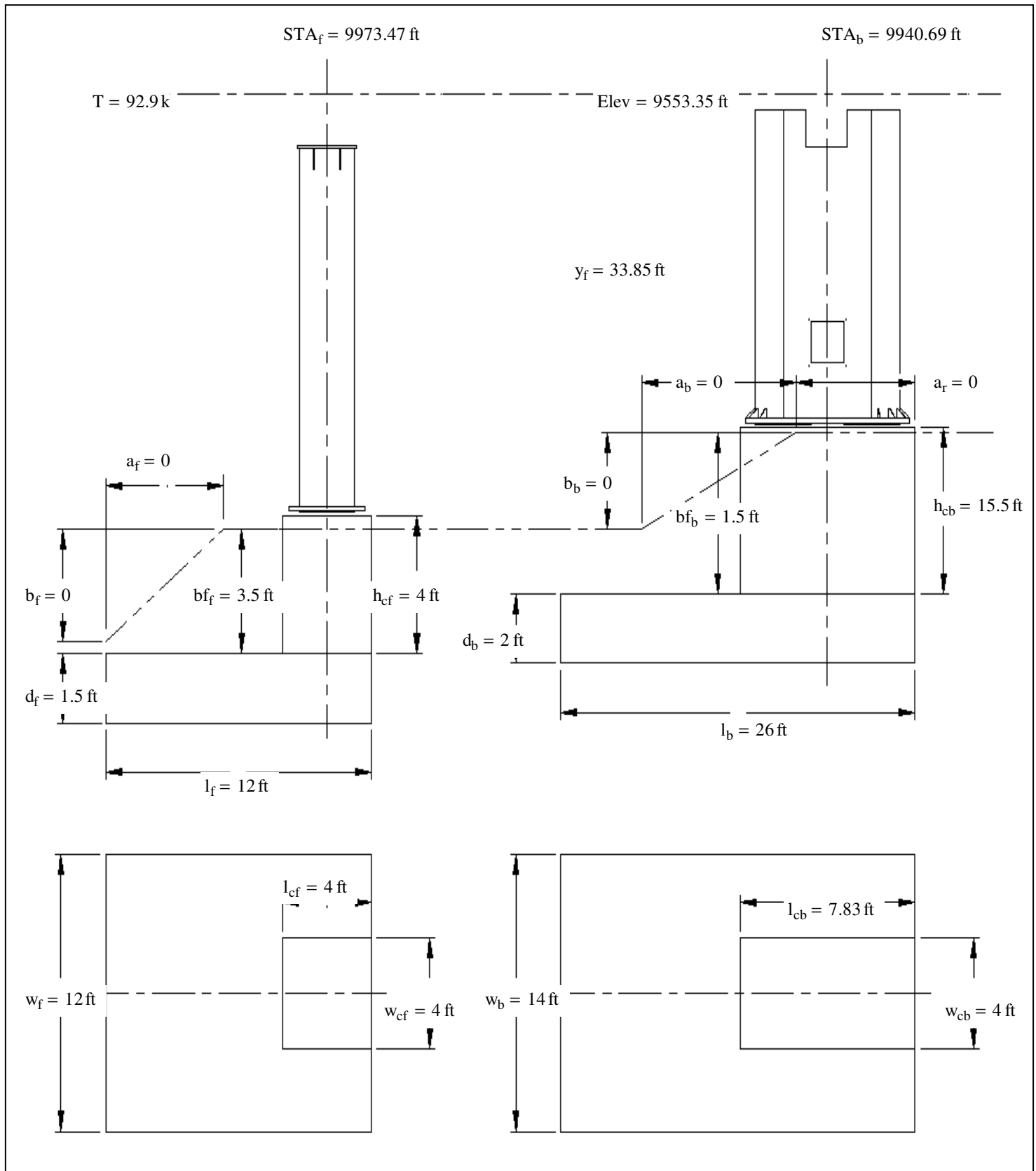
SLAB:

length: $\text{l}_f \equiv 8 \cdot \text{ft}$
 width: $\text{w}_f \equiv 12 \cdot \text{ft}$
 depth: $\text{d}_f \equiv 1.5 \cdot \text{ft}$

OUTPUT AREA 7

Volume of Concrete Pours:

First = 29.8 yd^3 Second = 17.5 yd^3 Total = 47.3 yd^3



BACK PAD

INPUT AREA 1: Lift Properties

Rope Elevation Elev $\equiv$ 9553.35·ft

Lift Tension T $\equiv$ 92.9·k

Reaction Column Station STA_b $\equiv$ 9940.69·ft

INPUT AREA 2: Material Properties

Soil Weight $W_{\text{soil}} \equiv 110 \cdot \frac{\text{lb}}{\text{ft}^3}$

Conc. Strength $f_c \equiv 2500 \cdot \text{psi}$

$\phi \equiv .90$

$\rho_{\text{minslab}} \equiv .0018$

Conc. Weight $W_{\text{conc}} \equiv 145 \cdot \frac{\text{lb}}{\text{ft}^3}$

Steel Strength $f_y \equiv 60000 \cdot \text{psi}$

$\beta \equiv .85$

$\rho_{\text{min}} \equiv \frac{200 \cdot \text{psi}}{f_y}$

OUTPUT AREA 3: Safety Factors

Overturning:

Sliding:

Parallel= $SF_{l_{\text{otb}}} = 1.83$

$SF_{l_{\text{sb}}} = 1.69$

Perpendicular= $SF_{p_{\text{otb}}} = 6.89$

$SF_{p_{\text{sb}}} = 16.66$

Soil Pressures:

Maximum:

Minimum:

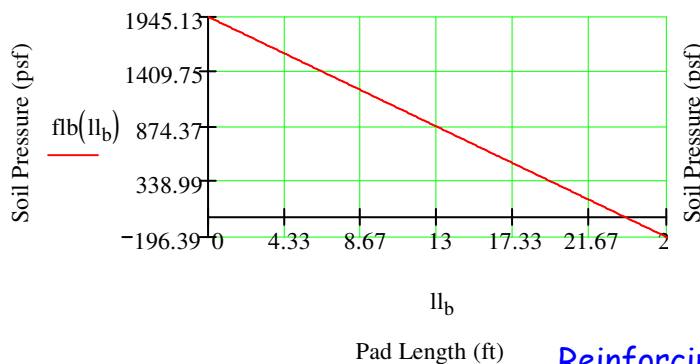
Parallel= $Max_{lb} = 1.95 \text{ ksf}$

$Min_{lb} = -0.2 \text{ ksf}$

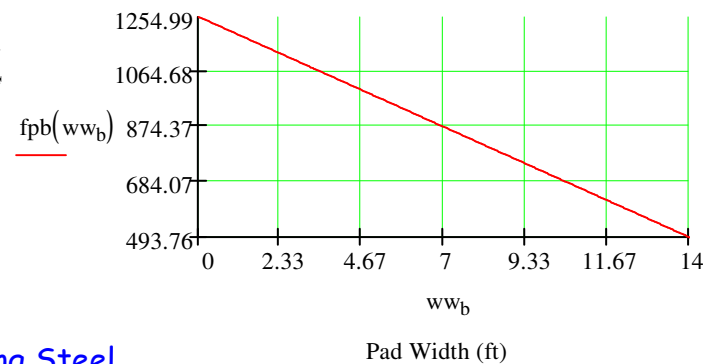
Perpendicular= $Max_{pb} = 1.25 \text{ ksf}$

$Min_{pb} = 0.49 \text{ ksf}$

Parallel Pressure Distribution



Perpendicular Pressure Distribution



Reinforcing Steel

Pad Parallel to Center Line= $A_{s1b} = 41.98 \text{ in}^2$

Pedestal Parallel to Center Line= $A_{s3b} = 13.54 \text{ in}^2$

Pad Perpendicular to Center Line= $A_{s2b} = 10.95 \text{ in}^2$

Pedestal Perpendicular to Center Line= $A_{s4b} = 2.99 \text{ in}^2$

INPUT AREA 4:

MISC.:

Soil Backfill: $bf_b \equiv 1.5 \cdot \text{ft}$

Void Length: $a_b \equiv 0 \cdot \text{ft}$

Void Height: $b_b \equiv 0 \cdot \text{ft}$

Ramp Length: $a_r \equiv 0 \cdot \text{ft}$

COLUMN:

length: $l_{cb} \equiv 94 \cdot \text{in}$

width: $w_{cb} \equiv 4 \cdot \text{ft}$

height: $h_{cb} \equiv 15.5 \cdot \text{ft}$

SLAB:

length: $l_b \equiv 26 \cdot \text{ft}$

width: $w_b \equiv 14 \cdot \text{ft}$

depth: $d_b \equiv 2 \cdot \text{ft}$

Effective M. Arm: $y_b \equiv 31.35 \cdot \text{ft}$

Column Station $\text{STA}_f \equiv 10006.25 \cdot \text{ft}$
 $\text{STA}_f \equiv 9973.47 \cdot \text{ft}$

FRONT PADS

OUTPUT AREA 5:

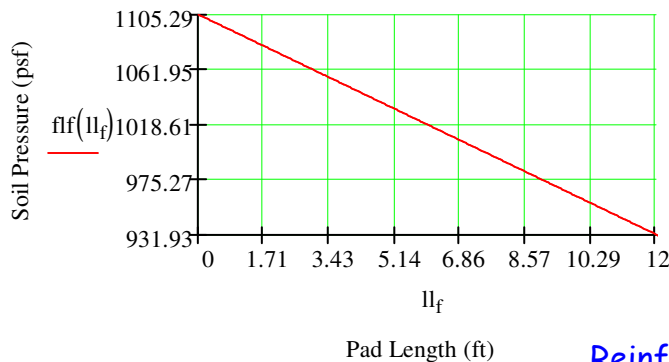
Safety Factors

Parallel= Perpendicular=	<u>Overturning:</u>	$\text{SF}_{l_{\text{otf}}} = 5.22$	<u>Sliding:</u>	$\text{SF}_{l_{\text{sf}}} = 11.63$
		$\text{SF}_{p_{\text{otf}}} = 3.84$		$\text{SF}_{p_{\text{sf}}} = 11.64$

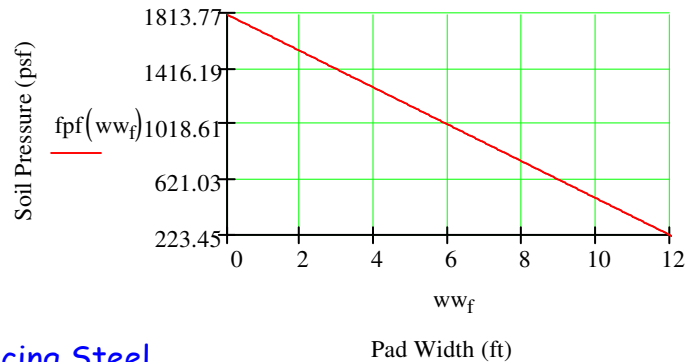
Soil Pressures:

Parallel= Perpendicular=	<u>Maximum:</u>	$\text{Max}_{lf} = 1.11 \text{ ksf}$	<u>Minimum:</u>	$\text{Min}_{lf} = 0.93 \text{ ksf}$
		$\text{Max}_{pf} = 1.81 \text{ ksf}$		$\text{Min}_{pf} = 0.22 \text{ ksf}$

Parallel Pressure Distribution



Perpendicular Pressure Distribution



Reinforcing Steel

Pad Parallel to Center Line= $A_{s1f} = 3.93 \text{ in}^2$
 Pad Perpendicular to Center Line= $A_{s2f} = 3.5 \text{ in}^2$

Pedestal Parallel to Center Line= $A_{s3f} = 1.95 \text{ in}^2$
 Pedestal Perpendicular to Center Line= $A_{s4f} = 2.09 \text{ in}^2$

INPUT AREA 6:

MISC.:

Soil Backfill: $b_f \equiv 3.5 \cdot \text{ft}$
 Void Length: $a_f \equiv 0 \cdot \text{ft}$
 Void Height: $b_f \equiv 0 \cdot \text{ft}$
 Effective M. Arm: $y_f \equiv 33.85 \cdot \text{ft}$

COLUMN:

length: $l_{cf} \equiv 4 \cdot \text{ft}$
 width: $w_{cf} \equiv 4 \cdot \text{ft}$
 height: $h_{cf} \equiv 4 \cdot \text{ft}$

SLAB:

length: $l_f \equiv 12 \cdot \text{ft}$
 width: $w_f \equiv 12 \cdot \text{ft}$
 depth: $d_f \equiv 1.5 \cdot \text{ft}$

OUTPUT AREA 7

Volume of Concrete Pours:

First = 35 yd^3

Second = 25.7 yd^3

Total = 60.6 yd^3